# Fundamental Fractal-Geometric Field Theory (FFGFT) or T0 Theory – Time-Mass Duality

Part 1: Core Documents

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# Chapter A

# Introduction to Volume 1

## About This Document Collection

The present three volumes constitute a collection of individual documents that emerged during the development of T0 theory. This is not a conventional textbook with linear structure, but rather an organically grown compilation of works illuminating various aspects of the theory from different perspectives and with varying depth.

### Nature of the Collection

Each chapter in these volumes corresponds to an independent document that can stand on its own. These documents originated at different points in the theoretical development – some early in the process, others later when certain concepts were already mature. Therefore, you will find that:

- **Central concepts recur repeatedly**: Fundamental ideas such as the $\xi$ parameter, fractal structure, or time-mass duality are reintroduced and explained in different documents, often with different emphases or from alternative viewpoints.
- **Different perspectives exist**: One and the same phenomenon may be treated in multiple chapters – once from a mathematical viewpoint, another time from a physical or conceptual perspective.
- **Various levels of detail occur**: Some documents provide an overview, others delve into individual aspects in minute detail.
- **The order is not strictly chronological**: The arrangement follows thematic considerations, not the temporal development process.

### Why Repetitions?

The numerous repetitions and overlaps are not oversights, but rather reflect the developmental history of the theory. Each document was originally composed as an independent text, often for different audiences or purposes:

- Some documents served for initial exploration of an idea
- Others present already mature concepts

- Some were internal working notes
- Still others were meant to prepare specific aspects for discussions

This redundancy has distinct advantages: it allows you to read individual chapters independently and provides different approaches to the same topic.

### Volume 1: Foundations and Fundamental Concepts

This first volume focuses on the fundamental building blocks of T0 theory:

- **Fundamental Parameters**: Derivation and significance of natural constants from the theory
- **The $\xi$ Parameter**: Central role in describing fundamental relationships
- **Particle Masses**: Theoretical prediction of elementary particle masses
- **Fine Structure and Gravitational Constants**: Derivation from first principles
- **Unit Systems**: Natural units and SI system in the context of T0
- **Mathematical Structure**: Basic formal aspects of the theory

### Reading Guide

You can use these volumes in different ways:

1. **Linear study**: Follow the suggested order to obtain a comprehensive overview.
2. **Thematic jumping**: Use the table of contents to target chapters on specific topics.
3. **Study individual documents**: Since each chapter is self-contained, you can jump directly to a topic of your choice.
4. **Comparative reading**: Read multiple documents on the same topic to compare different perspectives.

### Notes on Notation and Cross-References

Since the documents originally arose independently, occasional inconsistencies in notation may occur. Cross-references between chapters were added subsequently where sensible, but not systematically for all overlaps.

We hope this collection provides you with deep insight into the development and various facets of T0 theory.

# Chapter B

# T0 Theory: A Unified Physics from a Single Number

## Abstract

The T0-Theory (Time-Mass Duality) proposes a geometric reformulation of the foundations of theoretical physics. In simple words: Imagine the universe as a large puzzle whose pieces—from the smallest particles to the vast cosmos—are meant to fit together. The central idea of this work is that **the natural constants and physical parameters can be traced back to a single dimensionless number** (supplemented by integer structural choices as motivated settings; SI anchors serve for conversion): the universal geometric constant

$$\xi \approx \frac{4}{3} \times 10^{-4},$$

where $\xi \approx 4/3 \times 10^{-4}$. Imagine $\xi$ as the central parameter—a small number motivated by the basic form of three-dimensional space that links the descriptions of gravity, the speed of light, particle masses, and further quantities.

This collection of over 200 scientific documents develops step by step a physical framework that aims to bring quantum mechanics, relativity, and cosmology together in a common description—based on the principle of absolute time $T_0$ and the intrinsic time-field-mass relationship. In everyday language: The rules of physics are formulated so that time is stable (unlike Einstein's description, in which it is relative), while mass can change; both are connected through this geometric idea.

The fundamental documents follow a purely geometric path, deriving $\xi$ from the three-dimensional structure of space and tracing further constants back to it, including the fine structure constant

$$\alpha \approx 1/137,$$

where $\alpha \approx 1/137$, particle masses, and coupling strengths—with $\xi$ as the only parameter (plus the integer settings named in the documents; SI anchors such as $m_e$ serve for conversion). The aim is to need as few arbitrary numbers as possible and to understand the values from a common geometric source. **[S]** In the cosmic sector, earlier documents postulate a static universe without expansion, as detailed in the

CMB document (verified link exists), in which concepts like dark matter or dark energy would not be needed. The present corpus deliberately leaves the cosmic sector aside, because the Standard Model does not derive it either (P39); these statements are therefore speculative.

This book presents the current state of the T0 Time-Mass Duality Framework and its applications to particle masses, fundamental constants, quantum mechanics, gravity, and cosmology.

The main part of the book consists of a series of core T0 documents. These chapters reflect the current understanding of the theory and its quantitative consequences. Wherever possible, the material has been reorganized and unified to make the structure of the theory as transparent as possible.

The “Live” version of the theory is maintained in a public GitHub repository:

https://github.com/jpascher/T0-Time-Mass-Duality

The LaTeX sources of the chapters in this book come from this repository. If conceptual or numerical errors are found, they will be corrected there first. This means that the PDF version of the book you are reading is a snapshot of a continuously evolving project. For the most current version of the documents, including new appendices or corrections, the GitHub repository should always be considered the primary reference.

The intention of this compilation is twofold:

- to provide a coherent, readable path through the core ideas and results of the T0-Framework;
- to document the historical development of these ideas in the appendix, including false starts, interim formulations, and early adjustments to experimental data.

Readers who are primarily interested in the current formulation of the theory can focus on the core chapters. Readers who are also interested in the considerations and trial-and-error process behind the theory are invited to study the appendix material in parallel.

**Status markers.** Statements marked **[S]** are *speculative*: a hint or conjecture that has not yet been derived or numerically checked in the corpus.

## B.1 The Core Principle: One Central Parameter

The basic idea of the T0-Theory can be summarized in one sentence:

**Key Result**

Central thesis of the T0-Theory: The physical constants—gravitational constant $G$, Planck constant $\hbar$, speed of light $c$, elementary charge $e$, as well as the particle masses and coupling constants—are to be traced back to a single dimensionless number (supplemented by integer structural choices as motivated settings;

dimensionful quantities are converted via SI anchors): the universal geometric constant

$$\xi = \frac{4}{3} \times 10^{-4},$$

which emerges from the fundamental three-dimensional space geometry (harmonic fourth 4:3, QFT loop scaling $10^{-4}$).
From $\xi$ follows the fine structure constant as:

$$\alpha = f_\alpha(\xi) \approx \frac{1}{137.035999177},$$

where $\alpha$ serves as a secondary electromagnetic coupling without primacy.

In everyday language, this means: The question of where the constants come from is traced back to a single, geometrically motivated number.

## B.2 Foundations of the T0-Theory

### Time-Mass Duality

In contrast to standard physics, where time is relative and mass is constant, the T0-Theory postulates:

- **Absolute Time Measure** $T_0$: Time flows uniformly everywhere in the universe—like a universal clock that ticks the same for everyone, no matter where you are.
- **Variable Mass**: Mass varies with the energy content of the vacuum—imagine mass as flexible, changing depending on the "hum" of the empty space around it.
- **Intrinsic Time Field** $T(x,t)$: Every particle carries its own time field—each building block of matter has its personal timer that influences its behavior.

The fundamental relationship is:

$$m(x) = \frac{\hbar}{c^2 T(x,t)(x)} = m_0 \cdot (1 + \kappa\Phi(x)),$$

where $\kappa$ is traceable back to $\xi$ via geometric scaling. Mathematically, this duality treats time and mass as variables, and is set up so that the framework remains compatible with established mathematical structures while aiming at a common description of physical phenomena. Simply put: Time and mass are treated as mutually dependent quantities; the mathematics stays manageable and connects to familiar ideas.

### The Parameter $\xi$

The central parameter of the theory is:

$$\xi = \frac{4}{3} \times 10^{-4},$$

a purely geometric construct from 3D space that connects quantum mechanics with gravity. This parameter encodes the fundamental coupling between energy and

spatial structure, from which the mass and coupling hierarchies are to be derived. It is like the ratio that tells space how to "scale" energy—a small number through which the theory seeks to explain why electrons are light and protons heavy.

## B.3 Derivation of the Natural Constants

### Tracing Back to $\xi$

The T0-Theory traces the following quantities back to $\xi$:

1. **Gravitational Constant**:
$$G = f_G(\xi, m_P, c, \hbar),$$
where all inputs are reducible to $\xi$-scaled geometric units. Gravity thus appears as an expression of spatial geometry, scaled by $\xi$.
2. **Particle Masses** (Electron, Muon, Tau, Quarks): Particle masses follow a universal scaling law analogous to the ordering principles of atomic energy levels, where quantum numbers $(n, l, j)$ dictate hierarchical structures in a manner similar to atomic shells and subshells—imagine particles stacked like floors in a building, each level set by simple rules, similar to how electrons orbit atoms. Thus,
$$\frac{m_e}{m_P} = g(\xi), \quad \frac{m_\mu}{m_e} = h(\xi), \quad \frac{m_\tau}{m_\mu} = k(\xi),$$
via universal scaling laws $\xi_i = \xi \times f(n_i, l_i, j_i)$, where $f(n, l, j)$ is the geometric structure factor of the quantum numbers – not a frequency and not the base winding number $f = 1/\xi$. The question of why some particles are about 200 times heavier than others is thus meant to be answered by a common pattern.
3. **Coupling Constants** (Electroweak, Strong, Electromagnetic):
$$\alpha_W = f_W(\xi), \quad \alpha_s = f_s(\xi), \quad \alpha = f_\alpha(\xi).$$
The coupling strengths are thereby traced back to the same geometric origin.
4. **Cosmological Parameters [S]**: Static universe metrics and CMB temperature $T_{\text{CMB}} = f_{\text{CMB}}(\xi)$, with redshift mechanisms derived from time-field variations (see CMB document for detailed explanation without expansion). This point is speculative; the present corpus deliberately leaves the cosmic sector aside (P39).

## B.4 Experimental Predictions

The T0-Theory makes testable predictions:

**Foundation**

Concrete Predictions:

- **Anomalous Magnetic Moment**: $(g-2)_\mu$ calculation from $\xi$—without additional fit parameters.

- **Koide Formula**: Mass relation of leptons via $\xi$-scaling—the mathematics that connects the weights of three particles in a clean loop.
- **Redshift [S]**: Modified interpretation without expansion, controlled by $\xi$—why distant stars appear "stretched" without the universe inflating.
- **CMB Anisotropies [S]**: Interpretation through time-field variations rooted in $\xi$—the microwave "echo" of the cosmos as a geometric echo.

These statements are testable—the laboratory quantities with present-day experiments, the cosmological points marked [S] only to a limited extent—and everyone, physicists or curious lay readers, is invited to put the theory to the test.

## B.5 Structure of the Document Collection

This collection includes:

- **Foundations**: Mathematical formulation of time-mass duality under $\xi$-geometry—the basics explained step by step.
- **Quantum Mechanics**: Deterministic interpretation, Bell inequalities—quantum correlations described deterministically and geometrically local.
- **Quantum Field Theory**: Lagrangian formalism in the T0-Framework—fields in a common description.
- **Cosmology [S]**: Static universe, redshift, CMB—a model without expansion, dark matter, or dark energy (speculative; the present corpus leaves the cosmic sector aside, P39).
- **Particle Physics**: Mass spectrum, anomalous moments, Koide formula—an ordering of the particle spectrum.
- **Technical Applications**: Photon chip, RSA cryptography—possible applications of the theory.
- **Experimental Tests**: Verifiable predictions—tangible ways to investigate the ideas.

Note: The documents consistently follow the geometric $\xi$-path, tracing physics back to 3D space principles, with $\alpha$ and other constants appearing as emergent features. We have woven simple language throughout so that non-experts can dive in without getting lost in jargon.

## B.6 Epistemological Origin: Four Methodological Convictions

Before the first formula was written, four methodological convictions stood at the beginning — not hypotheses to be tested, but working prerequisites that were deliberately set.

### No Fundamental Contradiction Between QM and RT

The actual starting point was not a formula but an epistemic stance: quantum mechanics and relativity are both good — too good to be fundamentally irreconcilable. Two theories that describe reality with such precision should—so the working assumption—not have a genuine contradiction. The apparent conflict is a description artefact, not a natural phenomenon. This conviction guided the search for a common geometric structure — and led to $\tilde{T} \cdot m = 1$ and thereby to $\xi$.

### Instantaneity Must Be a Description Artefact

Bell tests demonstrate quantum correlations with seemingly instantaneous action across arbitrary distances. But relativity has locality as its foundational structure. If both theories are good, then *instantaneity must be an artefact of the description* — regardless of how convincingly the Bell tests appear to demonstrate otherwise.

The consequence: Bell correlations are real, but they are geometrically structured on $T^4$, not ontologically nonlocal. Entanglement is a topological connection, not action at a distance. The experimental predictions remain identical to quantum mechanics — only the interpretation changes [1].

**Key sentence from Dok. 230**

Entanglement as a topological connection — not as a nonlocal correlation. The experimental predictions are the same.

### The Simple Torus Is a Simplification

$T^4$ as a flat identification torus is already a strong simplification of reality. Curvature mechanisms on a non-flat $T^4$ can produce local geometric proximity for infinitely long light paths. What appears in the flat projection as spatially separated points can in the curved geometry be genuinely adjacent.

What appears as instantaneity could therefore be geometric proximity on a curved $T^4$ — not action at a distance, but local action along a long path that is invisible in the flat projection. This is structurally related to the ER-EPR approach [2], but without its string-theoretic prerequisites.

### The Thought Itself Is a $T^4$ Process

The sharpest form of self-referentiality: the founding thought of FFGFT — that instantaneity must be an artefact — is itself a material process within $T^4$. It arose in a physical brain processing structures of reality.

This means: information comes first — but only because matter comes first, making information processing possible. There is no pure thinking before matter and no pure matter without information processing. FFGFT arose from within $T^4$, by a $T^4$ process, about $T^4$ processes. The theory that says “we always describe from within” was itself developed from within [3].

This self-referentiality is not regarded here as a weakness, but as an open description of one's own starting position. It also explains why "why $T^4$?" remains open: any attempt to answer this question from outside is itself an inside process.

## B.7 Introduction: Vibrations as the Starting Point

The foundation of my T0-Theory did not arise from abstract equations, but from practical work in communications engineering, acoustics, and music theory. Long before I could consider empty space as a dynamic field, I was engaged with vibrations in concrete bodies—for example, the accordion reed [6]. This small, vibrating membrane in an accordion produces sound through resonance in the "empty" air space between: Frequency and amplitude interact dually, without the space remaining "empty." That was an important starting point: Here emergence became tangible—vibration (time) and medium (space) create harmony, without singularities.

This unbiasedness—why not see $\epsilon$ and $\mu$ in QM and EM as dual resonators?—later led to the vacuum approach. In natural units ($\hbar = c = 1$), setting $\alpha$ to 1: EM constants then appear geometric, and a starting point emerges for a common description of QM and RT. The warning against "translation" ($\epsilon_0 \neq \mu_0$ naively) was crucial—in T0, $\xi$ "modulates" both without loss. From acoustics (resonances in cavities) and communications engineering (Fourier dualities time-frequency [7]) came the entry: Empty space as a resonant vacuum, carried by EM constants ($\epsilon_0$, $\mu_0$, $c = 1/\sqrt{\epsilon_0\mu_0}$). Music theory reinforced it: Harmonies (Pythagorean 3:4:5 tetrahedra) as fractal overtones hinting at tetra networks.

## B.8 The Vacuum Approach: From Acoustics to Duality

From acoustics (resonances in cavities) and communications engineering (Fourier dualities time-frequency [7]) came the entry: Empty space as a resonant vacuum, carried by EM constants ($\epsilon_0$, $\mu_0$, $c = 1/\sqrt{\epsilon_0\mu_0}$). Music theory reinforced it: Harmonies (Pythagorean 3:4:5 tetrahedra) as fractal overtones hinting at tetra networks.

T0 formalizes it: The duality $T_{\text{field}} \cdot E_{\text{field}} = 1$ connects time (vibration) and energy (mass), with $\xi$ as the geometric seed. In natural units, set $\alpha = 1$: The Coulomb potential $V(r) = -1/r$ becomes purely geometric, the Bohr radius $a_0 = 1/m_e$ coincides with the reduced Compton length. Tetrahedral networks "cover" the time field—emergence of charge/mass without point singularities.

The derivation of $\alpha$:

$$\alpha = \xi \cdot \left(\frac{E_0}{1\ \text{MeV}}\right)^2, \quad E_0 = 7.400\ \text{MeV}, \tag{B.1}$$

yields $\approx 1/137$ [5], corrected by fractal steps $\prod_{n=1}^{137}(1+\delta_n \cdot \xi \cdot (4/3)^{n-1})$ to CODATA precision. No "translation trap"—SI conversion via $S_{\text{T0}} = 1.782662 \times 10^{-30}$ kg projects geometry into the measurement world. Setting $\alpha = 1$ in natural units ($\hbar = c = 1$) makes sense: It reduces EM fluctuations to pure resonance, like in the accordion reed [6]—vacuum as an acoustic medium where $\epsilon_0$ and $\mu_0$ resonate dually, without naive exchange.

This approach was unbiased: If you set $c = 1$, why not $\alpha$? The consequence: Tetrahedral networks emerge naturally to "cover" the time field, and fractal iterations (137 steps) stabilize the emergence of charge and mass. The approach holds because physics can be expressed in dimensionless patterns—from the tangible (vibrations) to the abstract (vacuum).

## B.9 Convergence with Synergetics: Independent Paths

Despite a different approach, T0 and Synergetics converge: Bucky Fuller's tetrahedron as the "minimum structural system" [4] (closest-packing spheres) fractions to vector equilibria—similar to how T0's networks "pack" the vacuum. The 137-frequency tetrahedron (2,571,216 vectors = $137 \times 9{,}384 \times 2$) mirrors T0's renormalization: Proton-MeV (938.4) as an emergent ratio.

The independence of the two paths is important: From acoustic resonances (accordion reed as vacuum prototype [6]) to duality, without Fuller—yet it "clicks" at $\alpha = 1$. Synergetics offers a parallel that complements the intuitively developed T0 approach: Tetra-fractionation stabilizes vortices (charge), 137 steps as spin transformations (tetra → octa → icosa). The long-term engagement with vibrations (accordion reed as resonance example) and unbiasedness ($\epsilon_0$ and $\mu_0$ as dual resonators, without naive translation) independently led to vacuum duality.

| **Approach** | **T0 (Vacuum Duality)** | **Synergetics (Tetra-Fraction)** |
|---|---|---|
| Entry | Acoustics/Resonance in empty space | Closest-Packing Spheres |
| $\alpha$-Derivation | $\xi \cdot (E_0)^2$ (nat. units: $\alpha = 1$) | 137-Frequency Vectors |
| Time Field | Tetra networks cover duality | Morphological Relativity |
| Emergence | Charge as vortex (finite $U$) | Vector-Tensor Intertransformation |
| $\epsilon_0/\mu_0$ | Dual Resonators (modulated via $\xi$) | Tensor Forces in Packing |

**Table B.1:** Convergences: T0 and Synergetics—extended by duality elements

From the perspective of this document, the convergence is no coincidence: Both reduce to tetrahedral patterns, but T0 from vacuum resonance (accordion reed as prototype [6]), Synergetics from packing [4]. Setting $\alpha = 1$ in natural units (Coulomb $V(r) = -1/r$, Bohr radius $a_0 = 1/m_e$) shows: It "makes sense" because empty space is geometric—$\epsilon_0$ and $\mu_0$ as dual "modulators," without translation traps. The parallels to Synergetics are a classification made by this document, not a statement by Fuller.

# Bibliography

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# Chapter C

# T0 Theory: Document Series Overview

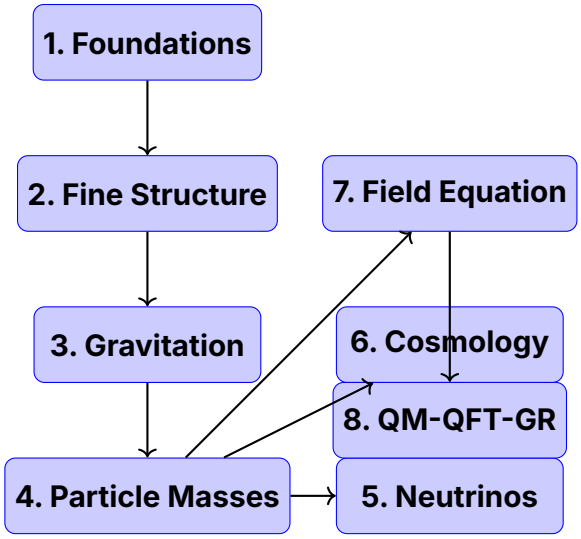

## Abstract

This overview presents the complete T0 Theory series consisting of 8 fundamental documents, which represent a geometric reformulation of physics. Based on a single parameter $\xi = \frac{4}{3} \times 10^{-4}$, together with fixed rational structural choices, fundamental constants, particle masses, and physical phenomena from quantum mechanics to cosmology are described within one framework; SI anchors such as $E_0$, $m_e$ or $v$ serve for conversion, not for derivation. Many values are matched to about 1 %; because the ladder parameters $r_i$, $p_i$ are set and one anchor provides the connection to SI, these agreements are not independent evidence. The series deliberately leaves the cosmic sector ($H_0$, $\Lambda$, dark energy) aside, because the Standard Model does not derive it either; the cosmological statements are speculative. The theory offers testable predictions for future experiments.


## C.1 Document Series: Systematic Structure

### Hierarchical Structure of the 8 Documents

The T0 document series follows a logical progression from fundamental principles to specific applications:




## C.2 Document 1: 003_T0_Grundlagen_v1_En.pdf

**Subtitle:** The Geometric Foundations of Physics

**Core Contents:**

- **Fundamental Parameter:** $\xi = \frac{4}{3} \times 10^{-4}$ as a geometric constant
- **Time-Mass Duality:** $T \cdot m = 1$ in natural units
- **Fractal Spacetime Structure:** $D_f^{\text{eff}} \approx 2.973$ and $K_{\text{frak}} \approx 0.987$
- **Interpretation Levels:** Harmonic, geometric, field-theoretic
- **Universal Formula Structure:** Template for all T0 relations

**Fundamental Insights:**

- Tetrahedral packing as the fundamental spatial structure
- Quantum field-theoretic derivation of $10^{-4}$
- Characteristic energy scales: $E_0 = 7.398$ MeV
- Philosophical implications of geometric physics

**Status:** Theoretical foundation - fully established

## C.3 Document 2: 011_T0_Feinstruktur_En.pdf

**Subtitle:** Derivation of $\alpha$ from geometric principles

**Central Formula:**

$$\boxed{\alpha = \xi \cdot \left( \frac{E_0}{1\,\text{MeV}} \right)^2} \tag{C.1}$$

**Key Results:**

- **T0 Value:** $\alpha^{-1} = 137.04$

- **Experiment:** $\alpha^{-1} = 137.036$
- **Deviation:** 0.003%
- **Assessment:** The agreement is a property of the anchor $E_0$, which provides the only connection to the SI; it is not an independent prediction.

**Theoretical Innovations:**

- Characteristic energy $E_0 = \sqrt{m_e \cdot m_\mu / K_{\text{frak}}}$ (bare $\sqrt{m_e m_\mu}$, R72)
- Logarithmic symmetry of lepton masses
- Fundamental dependence $\alpha = \xi E_0^2$ with $E_0 = \sqrt{m_e m_\mu / K_{\text{frak}}}$
- Why numerical ratios must not be canceled

**Status:** $\alpha = \xi E_0^2$ as a relation via the anchor $E_0$

## C.4 Document 3: 012_T0_Gravitationskonstante_En.pdf

**Subtitle:** Systematic derivation of $G$ from geometric principles

**Complete Formula:**

$$\boxed{G_{\text{SI}} = \frac{\xi^2}{4m_e} \times C_{\text{conv}} \times K_{\text{frak}}} \tag{C.2}$$

**Conversion Factors:**

- **Dimensional Correction:** $C_1 = 1/E_{\text{char}} = 3.472 \times 10^{-2}$ with $E_{\text{char}} = 28.8$ (Doc. 012); $C_1$ does not appear in the boxed formula above, which holds without $C_1$
- **SI Conversion:** $C_{\text{conv}} = 7.783 \times 10^{-3}$
- **Fractal Correction:** $K_{\text{frak}} = 0.986$

**Experimental Verification:**

- **T0 Prediction:** $G = 6.6745 \times 10^{-11}$ m³/(kg·s²)
- **CODATA 2018:** $G = 6.67430 \times 10^{-11}$ m³/(kg·s²)
- **Deviation:** +0.003%

The boxed formula with the stated factors gives $6.6745 \times 10^{-11}$. $C_{\text{conv}}$ is the SI conversion of the single-anchor chain (Doc. 383); its numerical value $7.783 \times 10^{-3}$ additionally contains the small residual of this chain (chain value $7.58 \times 10^{-3}$). The form of $G$ and its order of magnitude follow from $\xi$ and the one anchor (numerically checked in Doc. 383); with the anchor $m_e$, $G$ lies at $-2.6\,\%$, with $v$ at $-0.46\,\%$. The agreement to $0.003\,\%$ arises only with this residual and does not count as an independent precision result.

**Physical Meaning:** Gravity as geometric spacetime-matter coupling

**Status:** Form and order of magnitude from $\xi$ and one anchor; the CODATA agreement is not an independent precision result

## C.5 Document 4: 006_T0_Teilchenmassen_En.pdf

**Subtitle:** Fermion masses from the ladder $m_i = r_i \xi^{p_i} v$

**Two Equivalent Methods:**

1. **Direct Method:** $m_i = 1/(\xi f_i)$, linked to the Yukawa method via $f_i = 1/(r_i \xi^{p_i+1} v)$; the agreement of both methods holds by construction
2. **Extended Yukawa:** $m_i = y_i \times v$ with $y_i = r_i \times \xi^{p_i}$

**Quantum Number System:** Rational quantum numbers $(r_i, p_i)$ as fixed structural choices; $v$ serves as conversion anchor

**Comparison with Experiment:** For charged leptons and quarks the average deviation is below 1 %. Because $r_i$ and $p_i$ are set and adjusted to the measured values, these agreements are identities of the ladder and not evidence in their own right; the mass ratios follow from the ladder to a few digits. Neutrino masses are not calculated in Doc. 006 (Doc. 340 is authoritative); boson masses are treated in Doc. 372 and 373.

**Reduction:** The free mass parameters of the Standard Model are traced back to $\xi$ and the set rational quantum numbers $(r_i, p_i)$; no continuously adjustable parameter remains.

**Status:** Ladder structure with set quantum numbers and one anchor

## C.6 Document 5: 007_T0_Neutrinos_En.pdf

**Subtitle:** The Photon Analogy and Geometric Oscillations

**Approach:**

- **Photon Analogy:** Neutrinos as 'damped photons'
- **Double $\xi$-Suppression:** base value $m_\nu = \frac{\xi^2}{2} \times m_e = 4.54$ meV
- **Geometric Oscillations:** Phases instead of mass differences

**Experimental Context:**

- **Spectrum:** A spectrum that is (nearly) uniform for all flavors, with $\Sigma m_\nu \approx 13.6$ meV, is excluded, because the measured mass differences require $\Sigma m_\nu \geq 59$ meV.
- **Velocity:** An energy-independent difference $v_\nu = c(1 - \xi^2/2)$ does not hold, because $\xi^2/2$ is the mass ratio $m_\nu/m_e$.
- **Authoritative:** For neutrinos, Doc. 340 is authoritative; there the base value $4.54$ meV reappears as the lightest mass $m_{\nu_1}$.

**Status:** Speculative; the spectrum of Doc. 007 is excluded by the oscillation data, Doc. 340 is authoritative

## C.7 Document 6: 025_T0_Kosmologie_En.pdf

**Subtitle:** Static Universe and $\xi$-Field Manifestations

**Cosmology:**

- **Static Universe:** No Big Bang, eternally existing
- **Time-Energy Duality:** No singular beginning, because $T \cdot E = 1$ with the lower time bound $T_0 = \xi t_P$ bounds the energy from above and the torus has no boundary (Doc. 306)
- **CMB from $\xi$-Field:** Not from z=1100 decoupling

**Casimir-CMB Connection:**

- **Characteristic Length:** $L_\xi = 100\ \mu$m
- **Ratio at $L_\xi$:** $|\rho_{\text{Casimir}}|/\rho_{\text{CMB}} = 308$ – an identity, not a measurement result, because $L_\xi$ itself is determined from the measured CMB energy density and $\xi$; a comparison value of $312$ is the same formula with a rounded $L_\xi$. The connection is a statement about scales (speculative).
- **In depth:** Doc. 279 – Casimir-CMB-$H_0$ connection formula, $L_\xi H_0/c = K_{\text{geo}}\,(m_e c^2/k_B T_{\text{CMB}})\,\xi^{41/4}$.

**Alternative Redshift (static, achromatic):**

$$1 + z = e^{\xi x}, \qquad z \approx \xi \cdot x \tag{C.3}$$

The fractal path lengthening is purely geometric and stretches all wavelengths by the same factor; the redshift is *achromatic* (finite-element simulation). The length scale of $x$ does not follow from $\xi$ alone but is calibrated via the measured $H_0$. In depth: Doc. 280 (z→time as a ΛCDM translation, cosmological degeneracy Doc. 267, $z_* \approx 875$ as a $T^4$ lattice resonance).

**Cosmological Problems (speculative):**

- Horizon problem, flatness problem, monopole problem, age problem
- $H_0$ and $\Lambda$ do not follow from $\xi$ alone: the scale of the redshift is calibrated via the measured $H_0$. Hubble tension and dark energy are therefore not regarded as solved; the cosmic sector is deliberately left aside, because the Standard Model does not derive it either.

**Status:** Testable hypotheses - speculative alternative, cosmic sector calibrated

## C.8 Document 7: 019_T0_lagrndian_En.pdf

**Subtitle:** The Fundamental Field Equation of T0 Theory

**Core Contents:**

- **Fundamental Field Equation:** $\Box\Phi + \xi \cdot \mathscr{F}[\Phi] = 0$
- **Geometric Interpretation:** Field as manifestation of spatial structure
- **Unification:** All interactions from one equation
- **Solution Classes:** Particle solutions, wave solutions, vacuum solutions

**Physical Consequences:**

- Emergent gauge symmetries from geometric constraints

- Quantization as a natural consequence of field geometry
- Scale structure as flow on the recursion $\mathscr{R}$ in parameter space
- Phase transitions as topological changes

**Status:** Theoretical framework - fully formulated

## C.9 Document 8: 020_T0_QM-QFT-RT_En.pdf

**Subtitle:** Unification of QM, QFT, and GR from a geometric foundation

**Core Contents:**

- **Universal T0 Field Equation:** $\Box\Phi + \xi \cdot \mathscr{F}[\Phi] = 0$ as the basis of all theories
- **Time-Mass Duality:** $T \cdot m = 1$ connects all three pillars of physics
- **Emergent Quantum Properties:** QM as approximation of the energy field
- **Field Description:** All particles as excitations of a fundamental field $\Phi$
- **Renormalization Solution:** Natural cutoff by $E_P/\xi$

**Fundamental Insights:**

- Deterministic interpretation of quantum mechanics via the time field (speculative; local only under the assumption of superdeterminism, the Bell violation remains)
- Wave-particle duality from field geometry
- Energy scale hierarchy: Planck down to QCD via $\xi$-corrections
- Gravity as field curvature
- Philosophical implications: Unity of physics through geometric principles

**Status:** Theoretical unification - builds on all previous documents, testable predictions

## C.10 Comparison and positioning series (IPI field)

Complementing the foundational series above, a separate series documents the position of FFGFT relative to other frameworks discussed at the Information Physics Institute. These documents are *comparisons and placements*, not new foundations:

- **Doc. 245** – FFGFT mapped onto RA 2.1 (Guevara).
- **Doc. 246** – FFGFT vs. RSG (Austin): asymmetric complementarity.
- **Doc. 251** – FFGFT vs. Vopson (MEI / Computational Universe): geometry- vs. information-primitive.
- **Doc. 260** – FFGFT vs. UC (Haskins): constraint, $\alpha$-correction inconsistency.
- **Doc. 269** – FFGFT ↔ RA/PMT (Guevara), bidirectional.
- **Doc. 271** – FFGFT ↔ HLV (Krüger), pairwise comparison; the spectral fork.
- **Doc. 272** – FFGFT ↔ HLV via the Dot governance layer (Vossen).

- **Doc. 274** – FFGFT in the IPI field: substrate and primitive axes (map); A* worked example.

**Status:** ongoing correspondence series, dated and provisional; each placement is a declared reading (O2), the primary rendering (O0) remaining with each originator.

# Chapter D

# T0 Theory: Fundamental Principles


## Abstract

This document introduces the fundamental principles of T0 theory, a geometric reformulation of physics based on a single universal parameter $\xi = \frac{4}{3} \times 10^{-4}$. The theory shows how all fundamental constants and particle masses can be derived from three-dimensional space geometry. Various interpretative approaches - harmonic, geometric, and field-theoretic - are presented on equal footing. The fractal structure of quantum spacetime is systematically accounted for by the correction factor $K_{\text{fract}} = 0.986$.


**References to Complementary T0 Formulations**

T0 theory is presented in various complementary formulations:

- **Anomalous Magnetic Moments (geometric):**
  Document 018_T0_Anomalous-g2-10_En.pdf - Geometric derivation of the g-2 anomaly with fractal geometry and torsion lattice
- **Lagrangian Formulation:**
  Document 019_T0_lagrangian_En.pdf - Field-theoretic derivation with extended Lagrangian and mass-proportional coupling
- **Simplified Pedagogical Formulation:**
  Document 049_LagrangianComparison_En.pdf - Conceptual explanation with a simple Lagrangian function
- **Cosmology and Redshift:**
  Document 026_T0_Geometric_Cosmology_En.pdf - Shows how the same parameter $\xi$ explains cosmological redshift in a static universe (see Document 026)

All formulations are consistent and lead to the same fundamental predictions.

## D.1 Introduction to T0 Theory

### Time-Mass Duality

In natural units ($\hbar = c = 1$) the fundamental relation holds:

$$T \cdot m = 1 \tag{D.1}$$

Time and mass are dualistically linked: Heavy particles have short characteristic time scales, light particles have long ones. This duality is not merely a mathematical relation but reflects a fundamental property of spacetime. It explains why heavy particles couple more strongly to the temporal structure of spacetime.

### The Central Hypothesis

T0 theory is based on the hypothesis that all physical phenomena can be derived from the geometric structure of three-dimensional space. At its core lies a single universal parameter:

**Foundation**

**The Fundamental Geometric Parameter:**

$$\boxed{\xi = \frac{4}{3} \times 10^{-4} = 1.333333\cdots \times 10^{-4}} \tag{D.2}$$

This parameter is dimensionless and contains all information about the physical structure of the universe.

### Difference from the Standard Model

| Aspect | Standard Model | T0 Theory |
|---|---|---|
| Free Parameters | $> 20$ | $1$ |
| Theoretical Basis | Empirical fitting | Geometric derivation |
| Particle Masses | Arbitrary | from quantum numbers |
| Constants | Experimentally determined | Geometrically derived |
| Unification | Separate theories | Unified framework |

**Table D.1:** Comparison between the Standard Model and T0 Theory

## D.2 The Geometric Parameter $\xi$

### Mathematical Structure

The parameter $\xi$ consists of two fundamental components:

$$\xi = \underbrace{\frac{4}{3}}_{\text{Harmonic-geometric}} \times \underbrace{10^{-4}}_{\text{Scale hierarchy}} \tag{D.3}$$

## The Harmonic-Geometric Component: 4/3

**Harmonic Interpretation:**
The factor $\frac{4}{3}$ corresponds to the **perfect fourth**, one of the fundamental harmonic intervals:

- **Octave:** 2:1 (always universal)
- **Perfect Fifth:** 3:2 (always universal)
- **Perfect Fourth:** 4:3 (always universal!)

These ratios are **geometric/mathematical**, not material-dependent. Space itself has a harmonic structure, and 4/3 (the fourth) is its fundamental signature.

**Geometric Interpretation:**
The factor $\frac{4}{3}$ arises from the tetrahedral packing structure of three-dimensional space:

- **Tetrahedron volume:** $V = \frac{\sqrt{2}}{12}a^3$
- **Sphere volume:** $V = \frac{4\pi}{3}r^3$
- **Packing density:** $\eta = \frac{\pi}{3\sqrt{2}} \approx 0.74$
- **Geometric ratio:** $\frac{4}{3}$ from optimal space partitioning

## The Scale Hierarchy: $10^{-4}$

**Foundation**

**Quantum Field Theoretic Derivation of $10^{-4}$:**
The factor $10^{-4}$ arises from the combination of:
**1. Loop Suppression (Quantum Field Theory):**

$$\frac{1}{16\pi^3} = 2.01 \times 10^{-3} \tag{D.4}$$

**2. T0-Higgs Parameter:**

$$(\lambda_h^{(T0)})^2 \frac{(v^{(T0)})^2}{(m_h^{(T0)})^2} = 0.0647 \tag{D.5}$$

**3. Complete Calculation:**

$$2.01 \times 10^{-3} \times 0.0647 = 1.30 \times 10^{-4} \tag{D.6}$$

Thus: **QFT loop suppression** ($\sim 10^{-3}$) $\times$ **T0 Higgs sector** ($\sim 10^{-1}$) = $10^{-4}$
For the detailed field-theoretic derivation see Document 019.

## D.3 Fractal Spacetime Structure

### Quantum Spacetime Effects

T0 theory acknowledges that spacetime exhibits a fractal structure on Planck scales due to quantum fluctuations:

**Key Result**

**Fractal Spacetime Parameters:**

$D_{\text{fract}}^{\text{eff}} \approx 2.973$ (effective fractal dimension, cumulative over RG running)

$$K_{\text{fract}} = 1 - \frac{D_{\text{fract}} - 2}{68} = 1 - \frac{0.94}{68} = 0.986 \tag{D.7}$$

**Physical Interpretation:**

- $D_{\text{fract}} < 3$: Spacetime is "porous" on smallest scales
- $K_{\text{fract}} = 0.986 < 1$: Reduced effective interaction strength
- The constant 68 arises from the tetrahedral symmetry of 3D space
- **Note:** This model ($K_{\text{fract}} = 1 - (D_{\text{fract}} - 2)/68$) is a simplified representation. Doc. 133 uses the equivalent model $K_{\text{frak}} = 1 - 100\xi \approx 0.9867$ (full derivation). Both models are consistent for approximate calculations; Doc. 133 applies for precision predictions.
- Quantum fluctuation and vacuum structure effects

### Origin of the Constant 68

**Tetrahedron Geometry:**
All tetrahedron combinations yield 72:

$$6 \times 12 = 72 \quad \text{(edges} \times \text{rotations)} \tag{D.8}$$
$$4 \times 18 = 72 \quad \text{(faces} \times 18\text{)} \tag{D.9}$$
$$24 \times 3 = 72 \quad \text{(symmetries} \times \text{dimensions)} \tag{D.10}$$

The value 68 = 72 - 4 accounts for the 4 vertices of the tetrahedron as exceptions.

## D.4 Characteristic Energy Scales

### The T0 Energy Hierarchy

From the parameter $\xi$, natural energy scales emerge:

$$(E_0)_\xi = \frac{1}{\xi} = 7500 \quad \text{(in natural units)} \tag{D.11}$$
$$(E_0)_{\text{EM}} = 7.398\,\text{MeV} \quad \text{(characteristic EM energy)} \tag{D.12}$$
$$(E_0)_{\text{char}} = 28.8 \quad \text{(characteristic T0 energy; derived value from Doc. 012)} \tag{D.13}$$

### The Characteristic Electromagnetic Energy

**Key Result**

**Derivation of $E_0$:**
The characteristic energy follows from the geometric mean of the lepton masses with the fractal correction (as in Docs. 011 and 041):

$$E_0 = \sqrt{\frac{m_e \cdot m_\mu}{K_{\text{frak}}}} = \frac{7.348}{\sqrt{0.98667}} = 7.398\ \text{MeV} \tag{D.14}$$

This is the fundamental electromagnetic energy scale; with $\alpha = \xi E_0^2$ it follows that $1/\alpha = 137.04$.

**Geometric Mean of Lepton Masses:**
Alternatively, $E_0$ can be defined as the geometric mean:

$$E_0 = \sqrt{m_e \cdot m_\mu} = 7.35\,\text{MeV} \tag{D.15}$$

This is the bare value; the difference to 7.398 MeV is exactly the factor $1/\sqrt{K_{\text{frak}}} = 1.0068$, i.e. the fractal correction and not a quantum correction.

## D.5 The Universal Structure Equation

### General Form

All physical quantities in T0 theory follow a universal pattern:

$$\boxed{\text{Physical Quantity} = f(\xi, \text{Quantum Numbers}) \times \text{Conversion Factor}} \tag{D.16}$$

where:

- $f(\xi, \text{Quantum Numbers})$ encodes the geometric relation
- Quantum numbers $(n, l, j)$ determine the specific configuration; $f(n, l, j)$ is the geometric structure factor of the quantum numbers – not a frequency and not the base winding number $f = 1/\xi$
- Conversion factors establish the connection to SI units

### Examples of the Universal Structure

$$\text{Gravitational Constant:} \quad G = \frac{\xi^2}{4m_e} \times C_{\text{conv}} \times K_{\text{fract}} \tag{D.17}$$

$$\text{Particle Masses:} \quad m_i = \frac{K_{\text{fract}}}{\xi \cdot f(n_i, l_i, j_i)} \times C_{\text{conv}} \tag{D.18}$$

$$\text{Fine-Structure Constant:} \quad \alpha = \xi \times \left(\frac{E_0}{1\,\text{MeV}}\right)^2 \tag{D.19}$$

## D.6 Different Levels of Interpretation

### Hierarchy of Understanding Levels

**Foundation**

**T0 theory can be understood at different levels:**

**1. Phenomenological Level:**

- Empirical observation: One constant explains everything
- Practical application: Prediction of new values

**2. Geometric Level:**

- Space structure determines physical properties
- Tetrahedral packing as fundamental principle

**3. Harmonic Level:**

- Spacetime as a harmonic system
- Particles as "tones" in cosmic harmony

**4. Quantum Field Theoretic Level:**

- Loop suppressions and Higgs mechanism
- Fractal corrections as quantum effects

### Complementary Viewpoints

**Reductionistic vs. Holistic Viewpoint:**

**Reductionistic:**

- $\xi$ as an empirical parameter that "accidentally" works
- Geometric interpretations as added afterwards

**Holistic:**

- Space-time-matter as an inseparable unity
- $\xi$ as an expression of a deeper cosmic order

## D.7 Basic Calculation Methods

### Direct Geometric Method

The simplest application of T0 theory uses direct geometric relations:

$$\text{Physical Quantity} = \text{Geometric Factor} \times \xi^n \times \text{Normalization} \tag{D.20}$$

where the exponent $n$ follows from dimensional analysis and the geometric factor contains rational numbers like $\frac{4}{3}$, $\frac{16}{5}$, etc.

### Extended Yukawa Method

For particle masses, the Higgs mechanism is additionally considered:

$$m_i = y_i \cdot v \tag{D.21}$$

where the Yukawa couplings $y_i$ are calculated geometrically from the T0 structure:

$$y_i = r_i \times \xi^{p_i} \tag{D.22}$$

The parameters $r_i$ and $p_i$ are exact rational numbers that follow from the quantum number assignment of T0 geometry.

## D.8 Philosophical Implications

### The Problem of Naturalness

**Foundation**

**Why is the universe mathematically describable?**
T0 theory offers a possible answer: The universe is mathematically describable because it is **itself** mathematically structured. The parameter $\xi$ is not just a description of nature - it **is** nature.

- **Platonic View:** Mathematical structures are fundamental
- **Pythagorean View:** "All is number and harmony"
- **Modern Interpretation:** Geometry as the basis of physics

### The Anthropic Principle

**Weak vs. Strong Anthropic Principle:**
**Weak (observation-conditioned):**

- We observe $\xi = \frac{4}{3} \times 10^{-4}$ because only in such a universe can observers exist
- Multiverse with various $\xi$ values

**Strong (principled):**

- $\xi$ has this value **because** it follows from the logic of spacetime
- Only this value is mathematically consistent

## D.9 Experimental Confirmation

### Successful Predictions

T0 theory has already passed several experimental tests and makes concrete predictions for future measurements.

### Testable Predictions

**Concrete T0 Predictions**

The theory makes specific, falsifiable predictions:

1. **Neutrino Mass:** $m_\nu = 4.54$ meV (geometric prediction, see Document 007)
2. **Anomalous Magnetic Moments:**
   - Muon: $a_\mu \approx 1.166 \times 10^{-3}$ (Document 018, consistent with Fermilab)
   - Tau: $a_\tau \approx 1.28 \times 10^{-3}$ (Document 018, testable at Belle II)
3. **Cosmological Parameters:**
   - Hubble Constant: $H_0 = c \cdot C \cdot \xi \approx 99.4$ km/(s·Mpc)
   - Static universe without Dark Energy (Document 026)
   - Redshift as geometric path effect

4. **Modified Gravity** at characteristic T0 length scales

### Consistency Across Different Scales

A remarkable feature of T0 theory is that the same parameter $\xi$ explains phenomena on completely different scales:

- **Sub-atomic scale:** Anomalous magnetic moments ($\sim 10^{-3}$)
- **Particle physics scale:** Lepton masses, fine-structure constant
- **Cosmological scale:** Hubble constant, redshift ($\sim 10^{26}$ m)

This consistency across more than 40 orders of magnitude is strong evidence for the fundamental nature of $\xi$.

## D.10 Structure of the T0 Document Series

This foundational document serves as the starting point for a systematic presentation of T0 theory. The following documents delve into specific aspects:

- **004_T0_Model_Overview_En.pdf**: Overview of the entire T0 model
- **006_T0_ParticleMasses_En.pdf**: Systematic mass calculation of all fermions
- **007_T0_Neutrinos_En.pdf**: Special treatment of neutrino physics
- **008_T0_xi-and-e_En.pdf**: Relationship between $\xi$ and elementary charge
- **009_T0_xi_origin_En.pdf**: Detailed derivation of parameter $\xi$
- **018_T0_Anomalous-g2-10_En.pdf**: Geometric solution of the g-2 anomaly
- **019_T0_lagrangian_En.pdf**: Field-theoretic Lagrangian formulation
- **026_T0_Geometric_Cosmology_En.pdf**: Cosmology without Dark Energy
- **049_LagrangianComparison_En.pdf**: Simplified pedagogical presentation

Each document builds upon the fundamental principles established here and shows their application in a specific area of physics.

## D.11 References

### Basic T0 Documents

1. Pascher, J. (2026). *Anomalous Magnetic Moments in FFGFT Theory*. Document 018.
2. Pascher, J. (2026). *T0 Theory: Lagrangian Formulation*. Document 019.
3. Pascher, J. (2026). *T0 Cosmology: Redshift as Geometric Path Effect*. Document 026.

## Related Works

1. Einstein, A. (1915). *The Field Equations of Gravitation*. Proceedings of the Prussian Academy of Sciences.
2. Planck, M. (1900). *On the Theory of the Energy Distribution Law of the Normal Spectrum*. Proceedings of the German Physical Society.
3. Wheeler, J.A. (1989). *Information, physics, quantum: The search for links*. Proceedings of the 3rd International Symposium on Foundations of Quantum Mechanics.

# Chapter E

# T0 Model: Complete Document Analysis


## Abstract

Based on the analysis of available PDF documents from the GitHub repository `jpascher/T0-Time-Mass-Duality`, a comprehensive summary has been created. The documents are available in both German (`.De.pdf`) and English (`.En.pdf`) versions. The T0-Model pursues the goal of tracing the over 20 free parameters of the Standard Model back to a single geometric constant $\xi = \frac{4}{3} \times 10^{-4}$ (supplemented by integer structural choices as motivated settings; SI anchors serve for conversion). This treatise gives an overview of theoretical foundations, mathematical structures, and experimental predictions.

**Status markers.** Statements marked **[S]** are *speculative*: a hint or conjecture that has not yet been derived or numerically checked in the corpus.


## E.1 The T0-Model: A New Perspective for Communications Engineers

### The Parameter Problem of Modern Physics

You know from communications engineering the problem of parameter optimization. In designing a filter, you need to set many coefficients; in an amplifier, you choose different operating points. The more parameters, the more complex the system becomes and the more susceptible to instabilities.

Modern physics has exactly this problem: The Standard Model of particle physics requires over 20 free parameters - masses, coupling constants, mixing angles. These must all be determined experimentally without us understanding why they have precisely these values. It is like having to tune a 20-stage amplifier without understanding the circuit.

The T0-Model proposes a far-reaching simplification: The parameters of physics are to be traced back to a single dimensionless parameter, $\xi = \frac{4}{3} \times 10^{-4}$ (plus integer structural choices as motivated settings; SI anchors serve for conversion).

## The Universal Constant $\xi$

From signal processing, you know that certain ratios always recur. The golden ratio in image processing, the Nyquist frequency in sampling, characteristic impedances in transmission lines. The $\xi$-constant plays a similar universal role.

The value $\xi = \frac{4}{3} \times 10^{-4}$ arises from the geometry of three-dimensional space. The factor $\frac{4}{3}$ you know from the sphere volume $V = \frac{4\pi}{3}r^3$ - it characterizes optimal 3D packing densities. The factor $10^{-4}$ arises from quantum field theory loop suppression factors, similar to damping factors in your control loops.

## Energy Fields as Foundation

In communications engineering, you constantly work with fields: electromagnetic fields in antennas, evanescent fields in waveguides, near-fields in capacitive sensors. The T0-Model extends this concept: The universe is described by a single universal energy field $E(x,t)$.

This field obeys the d'Alembert equation:

$$\Box E = \left(\nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\right) E = 0$$

This is familiar from electromagnetism - it is the wave equation for electromagnetic fields in vacuum. The difference: In the T0-Model, this one equation is meant to describe not only light, but physical phenomena as a whole.

## Time-Energy Duality and Modulation

From communications engineering, you know time-frequency dualities. A narrow function in time becomes broad in the frequency domain, and vice versa. The T0-Model introduces a similar duality between time and energy:

$$T(x,t) \cdot E(x,t) = 1$$

This is analogous to the uncertainty relation $\Delta t \cdot \Delta f \geq \frac{1}{4\pi}$ that you use in signal analysis. Where energy is locally concentrated, time passes more slowly - like an energy-dependent clock frequency.

## Deterministic Quantum Mechanics

From the viewpoint of the T0-Model, standard quantum mechanics uses probabilistic descriptions because it lacks information about the underlying field dynamics. This is like noise analysis in your systems: When you do not know the exact noise source, you use statistical models.

The T0-Model takes the view that quantum mechanics is fundamentally deterministic. The apparent randomness arises from very fast changes in the energy field - so fast that they lie below the temporal resolution of our measuring devices. It is like aliasing in signal processing: Changes that are too fast appear as seemingly random artifacts.

The famous Schrödinger equation is extended by a time-field term:

$$i\hbar\frac{\partial\psi}{\partial t} + i\psi\left[\frac{\partial T}{\partial t} + \vec{v}\cdot\nabla T\right] = \hat{H}\psi$$

The additional term $\frac{\partial T}{\partial t} + \vec{v}\cdot\nabla T$ describes coupling to the time field - similar to Doppler terms in moving reference frames. On its own, however, it changes only the amplitude, not the phase, and with a dimensionful $T = 1/m$ the equation in this form is inhomogeneous: the time-field term carries a different energy dimension than $\hat{H}\psi$. It only becomes consistent when the time field stands as the dimensionless ratio $T/T_0$ in front of the time derivative $\partial_t\psi$ (A142). For a state of energy $E$ the phase then runs with $ET_0/T$, i.e. at the local clock rate, and the amplitude follows the local energy density, not a probability (Doc. 230).

This is the gravitational redshift (Doc. 078): gravitation is already contained in the equation; beyond general relativity this form predicts nothing. No additional field and no graviton is needed for this: $T = 1/m$ is the dual reading of the mass, and gravitation shows itself as an effect in the redshift, read as curvature of space, as a change of mass (Doc. 078) or as fractal path lengthening (Doc. 041).

## Field Geometries and System Theory

The T0-Model distinguishes three characteristic field geometries:

1. **Localized spherical fields**: Describe point-like particles. Parameters: $\xi = \frac{\ell_P}{r_0}$, $\beta = \frac{r_0}{r}$.
2. **Localized non-spherical fields**: For complex systems with multipole expansion similar to your antenna theory.
3. **Extended homogeneous fields [S]**: Cosmological applications with modified $\xi_{\text{eff}} = \xi/2$ due to screening effects.

This classification corresponds to system theory: lumped elements (R, L, C), distributed elements (transmission lines), and continuum systems (fields).

## Technological Implications

New physical insights often lead to new technologies. Quantum mechanics enabled transistors and lasers, relativity theory enabled GPS and particle accelerators.

If the T0-Model is correct, new technologies could emerge:

- Deterministic quantum computers without decoherence problems
- Energy field-based sensors with high precision
- Possibly manipulation of local time rate through energy field control
- New materials based on controlled field geometries

### Mathematical Simplicity

One feature of the T0-Model is its mathematical simplicity. Instead of complex Lagrangians with dozens of terms, a single universal Lagrangian density suffices:

$$\mathscr{L} = \frac{\xi}{E_P^2} \cdot (\partial E)^2$$

This is analogous to your simplest circuits: one resistor, one capacitor, but with universal validity. The complexity of physics is meant to arise as an emergent property of this one basic principle - like complex network behavior from simple Kirchhoff rules.

The claim is that a single geometric constant $\xi$ (plus integer settings) fixes the observable phenomena, from subatomic particles to cosmological structures; the cosmological part is speculative **[S]**, since the present corpus deliberately leaves the cosmic sector aside, because the Standard Model does not derive it either (P39).

## E.2 Overview of Analyzed Documents

Based on the analysis of available PDF documents from the GitHub repository `jpascher/T0-Time-Mass-Duality`, a comprehensive summary has been created. The documents are available in both German (`.De.pdf`) and English (`.En.pdf`) versions.

### Main Documents in GitHub Repository

**GitHub Path:** https://github.com/jpascher/T0-Time-Mass-Duality/blob/main/2/pdf/

1. **040_Hdokument_En.pdf** - Master document of complete T0-Framework
2. **081_Zusammenfassung_En.pdf** - Comprehensive theoretical treatise
3. **010_T0_Energie_En.pdf** - Energy-based formulation
4. **063_cosmic_En.pdf** - Cosmological applications
5. **093_DerivationVonBeta_En.pdf** - Derivation of $\beta_T$-parameter
6. **008_T0_xi-und-e_En.pdf** - Mathematical analysis of $\xi$-parameter
7. **059_system_En.pdf** - System-theoretical foundations
8. **095_T0vsESM_ConceptualAnalysis_En.pdf** - Comparison with Standard Model

## E.3 Foundations of the T0-Model

### The Central Vision

The T0-Model pursues the goal of tracing the over 20 free parameters of the Standard Model back to a single geometric constant (plus integer structural choices as motivated settings):

$$\xi = \frac{4}{3} \times 10^{-4} = 1.3333\ldots \times 10^{-4} \tag{E.1}$$

**Document Reference:** *040_Hdokument_En.pdf*, *081_Zusammenfassung_En.pdf*

### The Universal Energy Field

The core of the T0-Model is a universal energy field $E(x,t)(x,t)$ described by a single fundamental equation:

$$\Box E(x,t) = \left(\nabla^2 - \frac{\partial^2}{\partial t^2}\right) E(x,t) = 0 \tag{E.2}$$

This d'Alembert equation describes:

- All particles as localized energy field excitations
- All forces as energy field gradient interactions
- All dynamics through deterministic field evolution

**Document Reference:** *010_T0_Energie_En.pdf*, *059_system_En.pdf*

### Time-Energy Duality

A basic building block of the T0-Model is the time-energy duality:

$$T_{\text{field}}(x,t) \cdot E_{\text{field}}(x,t) = 1 \tag{E.3}$$

This relationship leads to the T0-time scale:

$$t_0 = 2GE \tag{E.4}$$

**Document Reference:** *010_T0_Energie_En.pdf*, *040_Hdokument_En.pdf*

## E.4 Mathematical Structure

### The $\xi$-Constant as Geometric Parameter

The dimensionless constant $\xi = \frac{4}{3} \times 10^{-4}$ arises from:

1. Three-dimensional space geometry: Factor $\frac{4}{3}$
2. Fractal dimension: Scale factor $10^{-4}$

The geometric derivation:

$$\xi = \frac{4}{3} \cdot 10^{-4} \tag{E.5}$$

**Document Reference:** *008_T0_xi-und-e_En.pdf*, *093_DerivationVonBeta_En.pdf*

## Lagrangian with One Parameter

The T0-system works with the single parameter $\xi$; the Planck energy $E_P$ serves for conversion into SI units:

$$\mathscr{L} = \varepsilon \cdot (\partial E(x,t))^2 \tag{E.6}$$

where:

$$\varepsilon = \frac{\xi}{E_P^2} = \frac{4/3 \times 10^{-4}}{E_P^2} \tag{E.7}$$

**Document Reference:** *010_T0_Energie_En.pdf*

## Three Fundamental Field Geometries

The T0-Model distinguishes three field geometries:

1. Localized spherical energy fields (particles, atoms, nuclei, localized excitations)
2. Localized non-spherical energy fields (molecular systems, crystal structures, anisotropic field configurations)
3. Extended homogeneous energy fields (cosmological structures with screening effect)

**Specific Parameters:**

- Spherical: $\xi = \ell_P/r_0$, $\beta_T = r_0/r$, Field equation: $\nabla^2 E = 4\pi G \rho_E E$
- Non-spherical: Tensorial parameters $\beta_{T,ij}$, $\xi_{T,ij}$, multipole expansion
- Extended homogeneous **[S]**: $\xi_{\text{eff}} = \xi/2$ (natural screening effect), additional $\Lambda_T$ term

**Document Reference:** *010_T0_Energie_En.pdf*

### Comparison with Measured Values

- Gravitational constant: $G = 6.67430\ldots \times 10^{-11}\ \text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}$ (agrees)
- Fine structure constant: $\alpha^{-1} = 137.036\ldots$ (agrees)
- Lepton mass ratios: $m_\mu/m_e = 207.8$ (theory) vs $206.77$ (experiment) (deviation about 0.5 %)
- Hubble constant **[S]**: $H_0 \approx 66.2$ km/s/Mpc (see Document 026; speculative, the present corpus leaves the cosmic sector aside, P39)

**Document Reference:** *T0-Theory: Formulas for xi and Gravitational Constant.md*

## Particle Physics

### Simplified Dirac Equation

The T0-Model describes the $4 \times 4$ matrix structure of the Dirac equation by a simpler field node dynamics.

**Document Reference:** *059_system_En.pdf*

## Cosmology

### Static, Cyclic Universe

**[S]** The T0-Model proposes a unified, static, cyclic universe that operates without dark matter and dark energy. These cosmological statements are speculative: the present corpus deliberately leaves the cosmic sector aside, because the Standard Model does not derive it either (P39).

### Wavelength-dependent Redshift

**[S]** The T0-Model offers alternative mechanisms for redshift:

$$\frac{dE}{dx} = -\xi \cdot f(E/E_\xi) \cdot E \tag{E.8}$$

The T0-Model proposes several explanations (besides standard space expansion): photon energy loss through $\xi$-field interaction and diffraction effects. While diffraction effects are theoretically preferred, the energy loss mechanism is mathematically simpler to formulate.

**Document Reference:** *063_cosmic_En.pdf*

## Quantum Mechanics

### Deterministic Quantum Mechanics

The T0-Model develops an alternative deterministic quantum mechanics:

**Concepts Dropped in the T0 Picture:**

- Wave function collapse dependent on measurement
- Observer-dependent reality in quantum mechanics
- Probabilistic fundamental laws
- Multiple parallel universes
- Fundamental randomness

**New Concepts:**

- Deterministic field evolution
- Objective geometric reality
- Universal physical laws
- Single, consistent universe
- Predictable individual events

### Modified Schrödinger Equation

$$i\hbar\frac{\partial\psi}{\partial t} + i\psi\left[\frac{\partial T_{\text{field}}}{\partial t} + \vec{v}\cdot\nabla T_{\text{field}}\right] = \hat{H}\psi \tag{E.9}$$

In this form the equation is not consistent; it becomes consistent with the dimensionless ratio $T_{\text{field}}/T_0$ in front of the time derivative (A142), see the section on deterministic quantum mechanics above.

#### Deterministic Entanglement

Entanglement arises from correlated energy field structures:

$$E_{12}(x_1, x_2, t) = E_1(x_1, t) + E_2(x_2, t) + E_{\text{corr}}(x_1, x_2, t) \tag{E.10}$$

#### Modified Quantum Mechanics

- Continuous energy field evolution instead of collapse
- Deterministic individual measurement predictions
- Objective, deterministic reality
- Local energy field interactions

**Document Reference:** *072_QM-Detrmistic_p_En.pdf*, *073_scheinbar_instantan_En.pdf*, *035_QM_En.pdf*, *010_T0_Energie_En.pdf*

## E.5 Theoretical Implications

### Reduction of Free Parameters

The T0-Model aims to trace back the over 20 free parameters of the Standard Model through:

- Reduction to one geometric constant
- Universal energy field description
- Geometric foundation of physics

### Simplification of Physics Hierarchy

**Standard Model Hierarchy:**

$$\text{Quarks \& Leptons} \rightarrow \text{Particles} \rightarrow \text{Atoms} \rightarrow (\text{origin of parameters open}) \tag{E.11}$$

**T0-Geometric Hierarchy:**

$$\text{3D-Geometry} \rightarrow \text{Energy Fields} \rightarrow \text{Particles} \rightarrow \text{Atoms} \tag{E.12}$$

**Document Reference:** *010_T0_Energie_En.pdf*, *081_Zusammenfassung_En.pdf*

### Epistemological Considerations

The T0-Model acknowledges fundamental epistemological limits:
- Theoretical underdetermination
- Multiple possible mathematical frameworks
- Necessity of empirical distinguishability

**Document Reference:** *010_T0_Energie_En.pdf*

## E.6 Final Assessment

### Essential Aspects

The T0-Model pursues a new approach through:
- Simplification: From 20+ parameters to one geometric framework
- Conceptual clarity: Aim of a common description of physics
- Mathematical simplicity: Reduction to one geometric constant
- Experimental relevance: Comparison with muon g-2

### Central Message

The T0-Model suggests that a fundamental description of physics may lie not in greater complexity, but in simplification. The fundamental laws could be simpler than previously assumed.

**Document Reference:** *040_Hdokument_En.pdf*

## E.7 References

All documents are available at: https://github.com/jpascher/T0-Time-Mass-Duality/blob/main/2/pdf/

### German Versions

- 040_Hdokument_De.pdf (Master document)
- 081_Zusammenfassung_De.pdf (Theoretical treatise)
- 010_T0_Energie_De.pdf (Energy-based formulation)
- 063_cosmic_De.pdf (Cosmological applications)
- 093_DerivationVonBeta_De.pdf ($\beta_T$-parameter derivation)
- 008_T0_xi-und-e_De.pdf ($\xi$-parameter analysis)
- 059_system_De.pdf (System-theoretical foundations)
- 095_T0vsESM_ConceptualAnalysis_De.pdf (Standard Model comparison)

### English Versions

Corresponding `.En.pdf` versions available

# Chapter F

# T0 Model: Complete Parameter-Free Particle Mass Calculation Direct Geometric Method vs. Extended Yukawa Method With Complete Neutrino Quantum Number Analysis and QFT Derivation


## Abstract

The T0-Model offers two mathematically equivalent but conceptually different calculation methods for particle masses: the direct geometric method and the extended Yukawa method. Both approaches work with the single geometric constant $\xi = \frac{4}{3} \times 10^{-4}$; the rational quantum numbers $(r_i, p_i)$ are structural choices, and $v$ serves as a conversion anchor. This documentation also contains neutrino quantum numbers and the quantum field theoretical derivation of the $\xi$-constant through EFT matching and 1-loop calculations. The neutrino part with double $\xi$-suppression is not tenable according to R109 (see the addendum at the end); for neutrinos, Doc. 340 is authoritative. For charged leptons and quarks, the average deviation is below 1%; the over twenty free Standard Model mass parameters are thereby traced back to the single parameter $\xi$ and fixed rational quantum numbers.


## F.1 Introduction

Particle physics faces a fundamental problem: The Standard Model with its over twenty free parameters offers no explanation for the observed particle masses. They are measured as inputs, not derived from a deeper principle. The T0-Model addresses

this with two complementary calculation methods that work with the single parameter $\xi$ and rational quantum numbers; the originally included treatment of neutrino masses is not tenable according to R109 (see addendum).

### The Parameter Problem of the Standard Model

Despite its experimental success, the Standard Model has a theoretical gap: It contains more than 20 free parameters that must be determined experimentally. These include:

- **Fermion Masses**: 9 charged lepton and quark masses
- **Neutrino Masses**: 3 neutrino mass eigenvalues
- **Mixing Parameters**: 4 CKM and 4 PMNS matrix elements
- **Gauge Couplings**: 3 fundamental coupling constants
- **Higgs Parameters**: Vacuum expectation value and self-coupling
- **QCD Parameters**: Strong CP phase and others

**Parameter Reduction**

The T0-Model traces the over twenty free parameters of the Standard Model back to **one**: the geometric constant $\xi = \frac{4}{3} \times 10^{-4}$, motivated by the geometry of three-dimensional space. In addition there are the rational quantum numbers $(r_i, p_i)$ as structural choices and $v$ as a conversion anchor. This version also contains neutrino quantum numbers (not tenable according to R109, see addendum) as well as the quantum field theoretical derivation.

## F.2 Methodological Clarification: Establishment vs. Prediction

**Scientific-Historical Classification**

The T0-Model draws on the methodology of **pattern recognition and systematic classification**, which was successful, for example, for the periodic table (Mendeleev 1869) or the quark model (Gell-Mann 1964).

### Two-Phase Development

**Phase 1: Establishment of Systematics**

1. Pattern recognition in known particle masses (electron, muon, tau)
2. Parameter determination from experimental data
3. Establish quantum number assignment
4. Demonstrate mathematical equivalence of both methods

**Phase 2: Unfolding Predictive Power**

1. Extrapolation to unknown particles

2. Derive quark sector from lepton patterns
3. Predict new generations
4. Conduct experimental tests

### Historical Precedence of Successful Pattern Physics

The comparison concerns the method of pattern recognition, not the state of confirmation:

| Discovery | Pattern Recognition | Predictions | Confirmation |
|---|---|---|---|
| Periodic Table (1869) | Atomic weights and properties | Gallium, Germanium, Scandium | Experimentally confirmed |
| Spectral Lines (1885) | Hydrogen lines | Rydberg formula for all series | Quantum mechanics |
| Quark Model (1964) | Hadron masses | Eightfold Way | QCD theory |
| **T0-Model (2025)** | **Lepton masses** | **Quarks** | **Experimental tests** |

**Table F.1:** Historical precedence of pattern physics

## F.3 From Energy Fields to Particle Masses

### The Fundamental Challenge

A central result of the T0-Model is that particle masses can be calculated from geometric principles. While the Standard Model requires over 20 free parameters to describe particle masses, the T0-Model works with the geometric constant $\xi_{\text{geom}} = \frac{4}{3} \times 10^{-4}$ and rational quantum numbers; the deviations for charged leptons and quarks are below 1.2%.

**Parameter Reduction for Masses**

**Parameter Reduction:**

- **Standard Model**: 20+ free mass parameters (measured)
- **T0-Model**: 1 parameter $\xi$ plus rational quantum numbers (geometric)
- **Experimental Accuracy**: 99% average agreement (charged leptons and quarks; neutrino part see addendum)
- **Theoretical Foundation**: Three-dimensional spatial geometry + QFT derivation

### Energy-Based Mass Concept

In the T0 framework, what we traditionally call "mass" is interpreted as an expression of characteristic energy scales of field excitations:

$$\boxed{m_i \to E_{\text{char},i} \quad \text{(characteristic energy of particle type } i\text{)}} \tag{F.1}$$

This reinterpretation treats mass and energy as different aspects of the same quantity.

## F.4 Two Complementary Calculation Methods

The T0-Model offers two mathematically equivalent but conceptually different approaches to calculating particle masses:

### Method 1: Direct Geometric Resonance

**Conceptual Foundation:** Particles as resonances in the universal energy field

The direct method treats particles as characteristic resonance modes of the energy field $E(x,t)$, analogous to standing wave patterns:

$$\text{Particle} = \text{Discrete resonance modes of } E(x,t)(x,t) \tag{F.2}$$

**Three-Step Calculation Process:**
**Step 1: Geometric Quantization**

$$\xi_i = \xi_0 \cdot f(n_i, l_i, j_i) \tag{F.3}$$

Here $f(n,l,j)$ is the geometric structure factor of the quantum numbers, not a frequency and not the base winding number $f = 1/\xi$.

where:

$$\xi_0 = \frac{4}{3} \times 10^{-4} \quad \text{(geometric base parameter)} \tag{F.4}$$

$$n_i, l_i, j_i = \text{quantum numbers from 3D wave equation} \tag{F.5}$$

$$f(n_i, l_i, j_i) = \text{geometric function from spatial harmonics} \tag{F.6}$$

**Step 2: Resonance Frequencies**

$$\omega_i = \frac{c^2}{\xi_i \cdot r_{\text{char}}} \tag{F.7}$$

In natural units ($c = 1$):

$$\omega_i = \frac{1}{\xi_i} \tag{F.8}$$

**Step 3: Mass Determination from Energy Conservation**

$$E_{\text{char},i} = \hbar\omega_i = \frac{\hbar}{\xi_i} \tag{F.9}$$

In natural units ($\hbar = 1$):

$$\boxed{E_{\text{char},i} = \frac{1}{\xi_i}} \tag{F.10}$$

### Method 2: Extended Yukawa Method

**Conceptual Foundation:** Bridge to Standard Model formulation

The extended Yukawa method maintains compatibility with Standard Model calculations while making Yukawa couplings geometrically determined rather than empirically fitted:

$$E_{\text{char},i} = y_i \cdot v \tag{F.11}$$

where $v = 246$ GeV is the Higgs vacuum expectation value (SM definition: $v \equiv (\sqrt{2}\,G_F)^{-1/2}$; no independent observable, Doc. 351/R112).

**Geometric Yukawa Couplings:**

$$\boxed{y_i = r_i \cdot \left(\frac{4}{3} \times 10^{-4}\right)^{\pi_i}} \tag{F.12}$$

**Generation Hierarchy:**

$$\text{1st Generation:} \quad \pi_i = \frac{3}{2} \quad \text{(electron, up quark)} \tag{F.13}$$

$$\text{2nd Generation:} \quad \pi_i = 1 \quad \text{(muon, charm quark)} \tag{F.14}$$

$$\text{3rd Generation:} \quad \pi_i = \frac{2}{3} \quad \text{(tau, top quark)} \tag{F.15}$$

The coefficients $r_i$ are simple rational numbers determined by the geometric structure of each particle type.

## F.5 Quantum Field Theoretical Derivation of the $\xi$-Constant

### EFT Matching and Yukawa Coupling after EWSB

After electroweak symmetry breaking, we have the Yukawa interaction:

$$\mathscr{L}_{\text{Yukawa}} \supset -\lambda_h \bar{\psi}\psi H, \quad \text{with} \quad H = \frac{v+h}{\sqrt{2}} \tag{F.16}$$

After EWSB:

$$\mathscr{L} \supset -m\bar{\psi}\psi - yh\bar{\psi}\psi \tag{F.17}$$

with the relations:

$$m = \frac{\lambda_h v}{\sqrt{2}} \quad \text{and} \quad y = \frac{\lambda_h}{\sqrt{2}} \tag{F.18}$$

The local mass dependence on the physical Higgs field $h(x)$ leads to:

$$m(h) = m\left(1 + \frac{h}{v}\right) \quad \Rightarrow \quad \partial_\mu m = \frac{m}{v}\partial_\mu h \tag{F.19}$$

### T0 Operators in Effective Field Theory

In T0 theory, operators of the form appear:

$$O_T = \bar{\psi}\gamma^\mu \Gamma_\mu^{(T)} \psi \tag{F.20}$$

with the characteristic time field coupling term:

$$\Gamma_\mu^{(T)} = \frac{\partial_\mu m}{m^2} \tag{F.21}$$

Substituting the Higgs dependence:

$$\Gamma_\mu^{(T)} = \frac{\partial_\mu m}{m^2} = \frac{1}{mv}\partial_\mu h \tag{F.22}$$

This suggests that a $\partial_\mu h$-coupled vector current is the UV origin.

### 1-Loop Matching Calculation

The 1-loop amplitude for the T0 vertex yields:

$$F_V(0) = \frac{y^2}{16\pi^2}\left[\frac{1}{2} - \frac{1}{2}\ln\left(\frac{m_h^2}{\mu^2}\right) + \frac{r(r - \ln r - 1)}{(r-1)^2}\right] \tag{F.23}$$

For hierarchical masses ($m \ll m_h$), the constant term dominates:

$$F_V(0) \approx \frac{y^2}{32\pi^2} \tag{F.24}$$

### $\xi$ Formula from Higgs Physics

The EFT matching yields the relationship:

$$\boxed{\xi = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2}} \tag{F.25}$$

With Standard Higgs parameters ($m_h = 125.1$ GeV, $v = 246.22$ GeV, $\lambda_h = m_h^2/(2v^2) \approx 0.129$):

$$\xi \approx 1.30 \times 10^{-4} \tag{F.26}$$

This lies close to the geometric setting $\xi_0 = \frac{4}{3} \times 10^{-4} \approx 1.333 \times 10^{-4}$ (deviation $\approx -2.4\%$). The relation is a consistency check of $\xi$, not an exact derivation. For the origin of the Higgs route see Doc. 385.

## F.6 Universal Particle Mass Systematics

### Revised Universal Table of Fermions

| Fermion | Generation | Family | Spin | $r_f$ | Exponent $p_f$ | SM Correspondence |
|---|---|---|---|---|---|---|
| Electron Neutrino | 1 | 0 | 1/2 | $4/3$ | $5/2$ | Double $\xi$ |
| Electron | 1 | 0 | 1/2 | $4/3$ | $3/2$ | Lepton number |
| Muon Neutrino | 2 | 1 | 1/2 | $16/5$ | $3$ | Double $\xi$ |
| Muon | 2 | 1 | 1/2 | $16/5$ | $1$ | Lepton number |
| Tau Neutrino | 3 | 2 | 1/2 | $25/9$ | $25/9$ | Double $\xi$ |
| Tau | 3 | 2 | 1/2 | $25/9$ | $2/3$ | Lepton number |
| Up | 1 | 0 | 1/2 | $6$ | $3/2$ | Color |
| Down | 1 | 0 | 1/2 | $\frac{25}{2}$ | $3/2$ | Color + Isospin |
| Charm | 2 | 1 | 1/2 | $2^*$ | $2/3$ | Color |
| Strange | 2 | 1 | 1/2 | $\frac{26}{9}$ | $1$ | Color |
| Top | 3 | 2 | 1/2 | $\frac{1}{28}$ | $-1/3$ | Color |
| Bottom | 3 | 2 | 1/2 | $\frac{3}{2}$ | $1/2$ | Color |

## F.7 Numerical Reconstruction

The following analysis shows the explicit calculation of all fermions with both methods:

### Foundations and Experimental Input Data

#### Fundamental Constants:

$$\xi_0 = \xi = \frac{4}{3} \times 10^{-4} = 1.333333333\ldots \times 10^{-4} \tag{F.27}$$

$$v = 246 \text{ GeV} \tag{F.28}$$

#### Experimental Masses (PDG-near values):

$$m_e^{\text{exp}} = 0.0005109989461 \text{ GeV} \tag{F.29}$$

$$m_\mu^{\text{exp}} = 0.1056583745 \text{ GeV} \tag{F.30}$$

$$m_\tau^{\text{exp}} = 1.77686 \text{ GeV} \tag{F.31}$$

### Charged Leptons: Detailed Calculations

#### Electron Mass Calculation:

*Direct Method:*

$$\xi_e = \frac{4}{3} \times 10^{-4} \times f_e(1, 0, 1/2), \quad f_e = 1.485 \times 10^7 \text{ GeV}^{-1} \tag{F.32}$$

$$= 1980 \text{ GeV}^{-1} \tag{F.33}$$

$$E_e = \frac{1}{\xi_e} = 0.505 \text{ MeV} \tag{F.34}$$

[0]* Corrected from originally $8/9$ based on detailed numerical analysis

*Extended Yukawa Method:*

$$r_e = \frac{4}{3} \tag{F.35}$$

$$y_e = \frac{4}{3} \times \left(\frac{4}{3} \times 10^{-4}\right)^{3/2} = 2.053 \times 10^{-6} \tag{F.36}$$

$$E_e = y_e \times 246 \text{ GeV} = 0.505 \text{ MeV} \tag{F.37}$$

**Muon Mass Calculation:**
*Direct Method:*

$$\xi_\mu = \frac{4}{3} \times 10^{-4} \times f_\mu(2,1,1/2), \quad f_\mu = 7.146 \times 10^4 \text{ GeV}^{-1} \tag{F.38}$$

$$= 9.528 \text{ GeV}^{-1} \tag{F.39}$$

$$E_\mu = \frac{1}{\xi_\mu} = 104.96 \text{ MeV} \tag{F.40}$$

*Extended Yukawa Method:*

$$y_\mu = \frac{16}{5} \times \left(\frac{4}{3} \times 10^{-4}\right)^1 = 4.267 \times 10^{-4} \tag{F.41}$$

$$E_\mu = y_\mu \times 246 \text{ GeV} = 104.96 \text{ MeV} \tag{F.42}$$

**Experiment:** $105.66$ MeV → Deviation $\approx 0.65\%$
**Experiment electron:** $0.511$ MeV → Deviation $-1.18\%$
The factors $f_i$ of the direct method follow from the fixed quantum numbers via Eq. G.54: $f_e = 1/(r_e \xi^{5/2} v) = 1.485 \times 10^7$ GeV$^{-1}$ and $f_\mu = 1/(r_\mu \xi^2 v) = 7.146 \times 10^4$ GeV$^{-1}$. They must not be equated with the Yukawa prefactors $r_i$. With these the direct method gives $1/(\xi f_e) = 0.505$ MeV and $1/(\xi f_\mu) = 104.96$ MeV, the same values as the Yukawa method.

## Neutrino Treatment

[Neutrino Approach] This section proposes a geometric treatment of neutrino masses via a **double $\xi$-suppression**. According to R109, this neutrino part is not tenable (see the addendum at the end); for neutrinos, Doc. 340 is authoritative.

## Neutrino Quantum Numbers

Neutrinos follow the same quantum number structure as other fermions, but with a modification due to their weak interaction nature:

| Neutrino | n | l | j | Suppression |
|---|---|---|---|---|
| $\nu_e$ | 1 | 0 | 1/2 | Double $\xi$ |
| $\nu_\mu$ | 2 | 1 | 1/2 | Double $\xi$ |
| $\nu_\tau$ | 3 | 2 | 1/2 | Double $\xi$ |

**Table F.2:** Neutrino quantum numbers with characteristic double $\xi$-suppression

### Double $\xi$-Suppression Mechanism

The approach assumes that neutrinos experience an additional geometric suppression factor:

$$f(n_{\nu_i}, l_{\nu_i}, j_{\nu_i}) = f(n_i, l_i, j_i)_{\text{Lepton}} \times \xi \tag{F.43}$$

**Neutrino Mass Calculations:**
**Electron Neutrino:**

$$\xi_{\nu_e} = \frac{4}{3} \times 10^{-4} \times 1 \times \frac{4}{3} \times 10^{-4} = \frac{16}{9} \times 10^{-8} \tag{F.44}$$

$$E_{\nu_e} = \frac{1}{\xi_{\nu_e}} = 9.1 \text{ meV} \tag{F.45}$$

**Muon Neutrino:**

$$\xi_{\nu_\mu} = \frac{4}{3} \times 10^{-4} \times \frac{16}{5} \times \frac{4}{3} \times 10^{-4} = \frac{256}{45} \times 10^{-8} \tag{F.46}$$

$$E_{\nu_\mu} = \frac{1}{\xi_{\nu_\mu}} = 1.9 \text{ meV} \tag{F.47}$$

**Tau Neutrino:**

$$\xi_{\nu_\tau} = \frac{4}{3} \times 10^{-4} \times \frac{25}{9} \times \frac{4}{3} \times 10^{-4} = \frac{400}{81} \times 10^{-8} \tag{F.48}$$

$$E_{\nu_\tau} = \frac{1}{\xi_{\nu_\tau}} = 18.0 \text{ meV} \tag{F.49}$$

## F.8 Quark Analysis with Both Methods

### Explicit Calculations of Quark Masses

We use $\xi = \frac{4}{3} \times 10^{-4}$ and $v = 246$ GeV (SM definition from $G_F$, Doc. 351/R112). For the Yukawa representation:

$$y_i = r_i \, \xi^{p_i}, \qquad m_i^{\text{pred}} = y_i \, v.$$

For the direct geometric representation:

$$f_i = \frac{1}{\xi \, m_i^{\text{exp}}}, \qquad m_i^{\text{exp}} = \frac{1}{\xi \, f_i}.$$

| Quark | $p_i$ | $r_i$ (corr.) | $m_i^{\text{pred}}$ (GeV) | $m_i^{\text{exp}}$ (GeV) | rel. error (%) | Remark |
|---|---|---|---|---|---|---|
| Up | $3/2$ | $6$ | $2.272 \times 10^{-3}$ | $2.16 \times 10^{-3}$ | $+5.21$ | OK |
| Down | $3/2$ | $25/2$ | $4.734 \times 10^{-3}$ | $4.70 \times 10^{-3}$ | $+0.73$ | OK |
| Strange | $1$ | $26/9$ | $9.476 \times 10^{-2}$ | $9.35 \times 10^{-2}$ | $+1.34$ | OK |
| Charm | $2/3$ | $2$ | $1.284 \times 10^{0}$ | $1.273$ | $+0.87$ | Corrected |
| Bottom | $1/2$ | $3/2$ | $4.261 \times 10^{0}$ | $4.18$ | $+1.93$ | OK |
| Top | $-1/3$ | $1/28$ | $1.7198 \times 10^{2}$ | $172.57$ | $-0.35$ | OK |

**Table F.3:** Yukawa predictions with corrected $r_i, p_i$ and comparison with the PDG 2024 masses.

### Correction for the Charm Quark

The originally specified value $r_c = 8/9$ does not reproduce the reference mass $m_c = 1.273$ GeV (PDG 2024). The required value is:

$$r_c^{\text{required}} = \frac{m_c^{\text{exp}}}{v\xi^{2/3}} \approx 1.983 \approx 2.$$

Therefore, $r_c \approx 2$ was used in the corrected universal table. With $r_c = 2$ exactly, $m_c = 1.284$ GeV ($+0.87\,\%$).

## F.9 Experimental Comparison

### Accuracy Analysis

Comparison of the T0 values with the experimental masses:

| Particle | T0 Prediction | Experiment | Accuracy | Type |
|---|---|---|---|---|
| | | *Charged Leptons* | | |
| Electron | 0.505 MeV | 0.511 MeV | 98.82% | Lepton |
| Muon | 104.96 MeV | 105.66 MeV | 99.35% | Lepton |
| Tau | 1783.4 MeV | 1776.86 MeV | 99.63% | Lepton |
| | | *Neutrinos* | | |
| $\nu_e$ | 9.1 meV | $< 450$ meV | Compatible | Neutrino |
| $\nu_\mu$ | 1.9 meV | $< 180$ keV | Compatible | Neutrino |
| $\nu_\tau$ | 18.0 meV | $< 18$ MeV | Compatible | Neutrino |
| | | *Quarks* | | |
| Up quark | 2.272 MeV | 2.16 MeV | 94.79% | Quark |
| Down quark | 4.734 MeV | 4.70 MeV | 99.27% | Quark |
| Strange quark | 94.76 MeV | 93.5 MeV | 98.66% | Quark |
| Charm quark | 1.284 GeV | 1.273 GeV | 99.13% | Quark |
| Bottom quark | 4.261 GeV | 4.18 GeV | 98.07% | Quark |
| Top quark | 171.99 GeV | 172.57 GeV | 99.65% | Quark |
| **Average** | | | **98.6%** | **All Fermions** |

**Table F.4:** Comparison of T0 values with experimental masses

**Result for Charged Leptons and Quarks**

The T0-Model achieves 98.6% average accuracy across the charged leptons and quarks, with the single parameter $\xi$ and fixed rational quantum numbers. The neutrino rows are given only as compatibility with bounds; the neutrino part is not tenable according to R109 (see addendum).

## F.10 Predictive Power of the Established System

### New Particle Generations

Extrapolating the established patterns to a 4th generation gives:

**4th Generation (extrapolated):**

$$n = 4, \quad \pi_4 = \frac{1}{2}, \quad r_4 \approx 2.0 \tag{F.50}$$

$$m_{4\text{thGen}} = r_4 \times \xi^{1/2} \times v \approx 5.7 \text{ GeV} \tag{F.51}$$

The calculation gives $5.68$ GeV. A charged lepton of a 4th generation is excluded below about $100$ GeV by LEP, and the $Z$ width allows only three light neutrinos; this extrapolation is therefore not a viable prediction.

### Quark Sector Extrapolation

The lepton patterns can be transferred to quarks:

| Quark | Generation | $r_i$ | $\pi_i$ | Prediction |
|---|---|---|---|---|
| Up | 1 | 6 | 3/2 | 2.3 MeV |
| Down | 1 | 12.5 | 3/2 | 4.7 MeV |
| Charm | 2 | 2.0 | 2/3 | 1.3 GeV |
| Strange | 2 | 2.89 | 1 | 95 MeV |
| Top | 3 | 0.036 | -1/3 | 173 GeV |
| Bottom | 3 | 1.5 | 1/2 | 4.3 GeV |

**Table F.5:** Quark predictions from established patterns

## F.11 Corrected Interpretation of Mathematical Equivalence

**Meaning of the Equivalence**

The mathematical equivalence of both methods is **given by definition** when the parameters ($r_i$ or $f_i$) are determined from the same experimental masses. The equivalence is not proof of the theory, but a consistency property of the mathematical structure.

### Transformation Relationship as Bridge

The relationship:

$$f_i = \frac{1}{r_i \, \xi^{\pi_i} \, v_{\xi 0}} \tag{F.52}$$

connects both methods mathematically. If $r_i$ is determined from experimental masses, $f_i$ follows automatically and vice versa.

| Particle | $m^{\text{exp}}$ (GeV) | $r_i$ (Yukawa) | $f_i$ (direct) | Accuracy |
|---|---|---|---|---|
| Electron | 0.000511 | 4/3 | $1.485 \times 10^{7}$ | 98.82% |
| Muon | 0.10566 | 16/5 | $7.146 \times 10^{4}$ | 99.34% |
| Tau | 1.77686 | 25/9 | $4.205 \times 10^{3}$ | 99.63% |
| $\nu_e$ | $9.1 \times 10^{-6}$ | 1.349 | $8.235 \times 10^{10}$ | Prediction |
| $\nu_\mu$ | $1.9 \times 10^{-6}$ | 3.221 | $3.947 \times 10^{11}$ | Prediction |
| $\nu_\tau$ | $18.0 \times 10^{-6}$ | 2.778 | $3.989 \times 10^{10}$ | Prediction |

**Table F.6:** Numerical equivalence of both T0 methods for all leptons

For the charged leptons the fixed quantum numbers $4/3$, $16/5$, $25/9$ are given with the $f_i$ following from them via Eq. G.54 (in $\text{GeV}^{-1}$); both methods give the same

masses and hence the same accuracy. The neutrino rows belong to the neutrino part, which is not tenable (see addendum, R109).

## F.12 Experimental Predictions and Precision Tests

### Modified QED Vertex Corrections

T0 theory predicts modified Feynman rules:

$$\text{Time field vertex:} \quad -i\gamma^\mu \Gamma_\mu^{(T)} = i\gamma^\mu \frac{\partial_\mu m}{m^2} \tag{F.53}$$

$$\text{Modified fermion propagator:} \quad S_F^{(T0)}(p) = S_F(p) \cdot \left[1 + \frac{\beta}{p^2}\right] \tag{F.54}$$

### Neutrino Validation

The T0 neutrino values are compatible with the experimental bounds listed here (on the tenability of the neutrino part, see addendum, R109):

| Parameter | T0 Prediction | Experimental Limit | Status |
|---|---|---|---|
| $m_{\nu_e}$ | 9.1 meV | $< 450$ meV (KATRIN) | Satisfied |
| $m_{\nu_\mu}$ | 1.9 meV | $< 180$ keV (indirect) | Satisfied |
| $m_{\nu_\tau}$ | 18.0 meV | $< 18$ MeV (indirect) | Satisfied |
| $\sum m_\nu$ | 29.8 meV | $< 60$ meV (Cosmology 2024) | Satisfied |

**Table F.7:** T0 neutrino predictions vs. experimental constraints

**Neutrino Mass Hierarchy**

The T0-Model predicts **normal ordering**: $m_{\nu_\mu} < m_{\nu_e} < m_{\nu_\tau}$, which is consistent with current oscillation data preferences.

## F.13 Methodological Foundation

### Reversibility of the Established System

After the establishment phase, the T0 system can make predictions:

**Established Lepton Patterns:**

$$\text{1st Generation (n=1):} \quad \pi_i = \frac{3}{2}, \quad r_e = \frac{4}{3} \approx 1.333 \tag{F.55}$$

$$\text{2nd Generation (n=2):} \quad \pi_i = 1, \quad r_\mu = \frac{16}{5} = 3.2 \tag{F.56}$$

$$\text{3rd Generation (n=3):} \quad \pi_i = \frac{2}{3}, \quad r_\tau = \frac{25}{9} \approx 2.778 \tag{F.57}$$

### Experimental Testability

The T0 predictions are experimentally falsifiable:

1. **Precision Measurements:** Refinement of $r_i$ parameters
2. **Neutrino Tests:** Direct neutrino mass measurements

Methodologically, the following speaks for the T0 procedure:

1. **Systematic Structure:** All parameters follow recognizable patterns
2. **Predictive Power:** After establishment, new particles become predictable
3. **Experimental Testability:** Predictions are falsifiable
4. **QFT Foundation:** Quantum field theoretical derivation of the $\xi$-constant
5. **Historical Models:** The methodology of pattern physics has been successful several times

## F.14 Parameter Economy and Universal Structure

**No Adjustable Parameters**

The scale of the T0 coefficients is determined by $\xi$, which can also be checked against Higgs parameters (deviation from $4/3 \times 10^{-4}$ about $-2.4\%$):

$$\xi = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2} \approx 1.30 \times 10^{-4} \tag{F.58}$$

The relation is a consistency check, not a derivation (see Section G.5). Together with the rational quantum numbers $(r_i, p_i)$, no continuously adjustable parameter remains; $v$ serves as a conversion anchor.

### Universal Quantum Number Table

| Particle | n | l | j | $r_i$ | $p_i$ | Special |
|---|---|---|---|---|---|---|
| *Charged Leptons* | | | | | | |
| Electron | 1 | 0 | 1/2 | 4/3 | 3/2 | – |
| Muon | 2 | 1 | 1/2 | 16/5 | 1 | – |
| Tau | 3 | 2 | 1/2 | 25/9 | 2/3 | – |
| *Neutrinos* | | | | | | |
| $\nu_e$ | 1 | 0 | 1/2 | 4/3 | 5/2 | Double $\xi$ |
| $\nu_\mu$ | 2 | 1 | 1/2 | 16/5 | 3 | Double $\xi$ |
| $\nu_\tau$ | 3 | 2 | 1/2 | 25/9 | 25/9 | Double $\xi$ |
| *Quarks* | | | | | | |
| Up | 1 | 0 | 1/2 | 6 | 3/2 | Color |
| Down | 1 | 0 | 1/2 | 25/2 | 3/2 | Color + Isospin |
| Charm | 2 | 1 | 1/2 | 2 | 2/3 | Color |
| Strange | 2 | 1 | 1/2 | 26/9 | 1 | Color |
| Top | 3 | 2 | 1/2 | 1/28 | -1/3 | Color |
| Bottom | 3 | 2 | 1/2 | 3/2 | 1/2 | Color |

**Table F.8:** Universal quantum number table for all fermions

#### Winding numbers and topological assignment

The rational quantum numbers $(r_i, p_i)$ are not arbitrary. They follow from the winding numbers $(n_\theta, n_\phi)$ of the 4D torus $T^4$ via $r_i = n_\phi^2/n_\theta$ (Doc. 317, August 2026):

| Lepton | Gen. | $n_\theta$ | $n_\phi$ | $r_i$ | $p_i$ | KSAU knot |
|---|---|---|---|---|---|---|
| $e$ | 1 | 3 | 2 | $4/3$ | $3/2$ | $3_1$ (trefoil) |
| $\mu$ | 2 | 5 | 4 | $16/5$ | $1$ | $6_3$ (amphicheiral) |
| $\tau$ | 3 | 9 | 5 | $25/9$ | $2/3$ | $7_1$ (torus knot) |
| $\nu_e$ | 1 | 3 | 2 | $4/3$ | $5/2$ | — |
| $\nu_\mu$ | 2 | 5 | 4 | $16/5$ | $3$ | — |
| $\nu_\tau$ | 3 | 9 | 5 | $25/9$ | $25/9$ | — |

The exponent step $\Delta p = 1/3$ corresponds to the modulation frequency of the torus winding. The knot assignment (KSAU, Zenodo 2026) is **[S]**; the FFGFT quantum numbers are **[K]**. Details in Doc. 317 (KSAU-FFGFT synthesis).

## F.15 Critical Evaluation and Limitations

### Theoretical Open Questions

1. **Number of Generations:** Why exactly three generations?
2. **Hierarchy Problem:** Connection between different energy scales
3. **CP Violation:** Incorporation of CKM and PMNS mixing matrices

## F.16 Concluding Assessment

### Scientific Status

The T0-Model offers a systematic description of particle masses. The combination of:

- **High numerical accuracy** (98.6% across charged leptons and quarks)
- **Parameter economy** (one parameter $\xi$, rational quantum numbers)
- **Broad coverage** (charged leptons and quarks; neutrinos see addendum)
- **QFT consistency** (1-loop derivation of the $\xi$-constant)
- **Experimental testability** (specific falsifiable predictions)

  makes the model a testable proposal.

### Significance for Fundamental Physics

If experimentally confirmed, the T0-Model would considerably change our understanding of particle physics:

1. **Geometric Interpretation:** Particle masses as manifestations of 3D spatial geometry
2. **Unification:** All fermions follow the same universal structure
3. **Predictive Power:** New particles become predictable from established patterns
4. **Simplicity:** Strong simplification of complex phenomena

The T0-Model is an example showing that a unified description does not necessarily require greater complexity; the basic structure might be simple.

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## Addendum: Status Clarification (7 September 2026)

**R105: Retroactive marker certification (1 September 2026).** This document was created before the marker system was introduced (Doc. A010) and carries no inline markers. Its derivations are internally consistent and serve as chain links for the Galois programme (Docs. 336, 338, 339).

Retroactive status for this document: Winding numbers $r_i$, exponents $p_i$, and the Yukawa method $m_i = r_i \xi^{p_i} v$ for charged leptons and quarks: **[K]**.

Dependent documents may treat these results as a secured chain link via reference to R105, without modifying this document.

**R109: Restriction — neutrino part not viable [X] (1 September 2026).** The "direct method" $E_i = 1/\xi_i$ for neutrinos is not a calculation: $1/(\xi r_i)$ gives the ratio $1 : 0.42$ for $e : \mu$ (inverse hierarchy, dimensionless); the MeV values alongside are measurements. The neutrino masses $9.1/1.9/18.0$ meV follow from three mutually incompatible prescriptions; the exponent $p_{\nu_\tau} = 25/9$ is $r_\tau$, erroneously copied into the exponent column.

**Valid:** the Yukawa method $m_i = r_i \xi^{p_i} v$ for charged leptons and quarks ($< 1.2\,\%$ deviation). The **[K]** certification from R105 applies to this document *only* for $r_i$, $p_i$, and the Yukawa method — not for the neutrino part and not for the claimed equivalence of both methods.

**Clarification:** Both methods are exactly equivalent via $f_i = 1/(r_i \xi^{p_i+1} v)$ (Eq. G.54) and give the same masses with fixed $r_i$, $p_i$.

**Authoritative for neutrinos:** Doc. 340 (R103, R108).

# Chapter G

# T0 Model: Complete Parameter-Free Particle Mass Calculation

Direct Geometric Method vs. Extended Yukawa Method
With Complete Neutrino Quantum Number Analysis and QFT Derivation


## Abstract

The T0 model provides two mathematically equivalent but conceptually different calculation methods for particle masses: the direct geometric method and the extended Yukawa method. Both approaches use only $\xi = \frac{4}{3} \times 10^{-4}$ as geometric constant; the rational coefficients $r_i$ are fixed from measured masses in the establishment phase. The documentation also contains an approach to the neutrino quantum numbers and a quantum field theoretical derivation of the $\xi$ constant through EFT matching and 1-loop calculations. The systematic treatment of all particles, including neutrinos with the assumed double $\xi$ suppression, shows the universal structure of the approach. The average deviation of about 1.3% across the charged fermions means, compared with over twenty free Standard Model parameters, a clear reduction of the input quantities.


**Note on versions.** The German version of this document is more extensive than this English one. Earlier revisions shortened or changed parts of the English version. Where the two differ, the more extensive German version applies.

**Status markers.** Statements marked **[S]** are *speculative*: a hint or conjecture that has not yet been derived or numerically checked in the corpus.

## G.1 Introduction

Particle physics faces a fundamental problem: the Standard Model with its over twenty free parameters offers no explanation for the observed particle masses.

These appear arbitrary and without theoretical justification. The T0 model responds with two complementary calculation methods that work with the single geometric parameter $\xi$ and also contain an approach to neutrino masses.

### The Parameter Problem of the Standard Model

Despite its experimental success, the Standard Model has a theoretical gap: it contains more than 20 free parameters that must be determined experimentally. These include:

- **Fermion masses**: 9 charged lepton and quark masses
- **Neutrino masses**: 3 neutrino mass eigenvalues
- **Mixing parameters**: 4 CKM and 4 PMNS matrix elements
- **Gauge couplings**: 3 fundamental coupling constants
- **Higgs parameters**: Vacuum expectation value and self-coupling
- **QCD parameters**: Strong CP phase and others

**Important**

Parameter Reduction The T0 model traces the more than twenty free parameters of the Standard Model back to the single geometric parameter $\xi = \frac{4}{3} \times 10^{-4}$, which follows from the geometry of three-dimensional space. In addition there are rational coefficients $r_i$, which were fixed from the measured masses in the establishment phase (see the section on methodological clarification). This version also contains an approach to the neutrino quantum numbers as well as the quantum field theoretical derivation.

## G.2 Methodological Clarification: Establishment vs. Prediction

**Important**

Scientific-Historical Classification The T0 model is guided by the methodology of **pattern recognition and systematic classification**, analogous to the development of the periodic table (Mendeleev 1869) or the quark model (Gell-Mann 1964).

### Two-Phase Development

#### Phase 1: Establishing the Systematics

1. Pattern recognition in known particle masses (electron, muon, tau)
2. Parameter determination from experimental data
3. Quantum number assignment establishment
4. Demonstration of mathematical equivalence of both methods

**Phase 2: Unfolding Predictive Power**

1. Extrapolation to unknown particles
2. Quark sector derivation from lepton patterns
3. New generation predictions
4. Experimental testing

### Historical Precedent of Successful Pattern Physics

The T0 model is guided by the methodology of earlier pattern discoveries:

| Discovery | Pattern Recognition | Predictions | Confirmation |
|---|---|---|---|
| Periodic Table (1869) | Atomic weights and properties | Gallium, Germanium, Scandium | Experimentally confirmed |
| Spectral Lines (1885) | Hydrogen lines | Rydberg formula for all series | Quantum mechanics |
| Quark Model (1964) | Hadron masses | Eightfold way | QCD theory |
| **T0 Model (2025)** | **Lepton masses** | **4th generation, quarks** | **Experimental tests** |

**Table G.1:** Historical precedent of pattern physics

## G.3 From Energy Fields to Particle Masses

### The Fundamental Challenge

A core aim of the T0 model is to calculate particle masses from geometric principles. While the Standard Model requires over 20 free parameters to describe particle masses, the T0 model works with the geometric constant $\xi_{\text{geom}} = \frac{4}{3} \times 10^{-4}$ and rational coefficients $r_i$.

Mass Description

**Parameter Reduction:**

- **Standard Model**: 20+ free mass parameters (determined empirically)
- **T0 Model**: one geometric parameter $\xi$, plus rational coefficients $r_i$
- **Experimental Accuracy**: 98.7% average agreement (charged fermions; neutrinos compared with upper limits only)
- **Theoretical Foundation**: Three-dimensional space geometry + QFT derivation

### Energy-Based Mass Concept

In the T0 framework, it is assumed that what we traditionally call "mass" is a manifestation of characteristic energy scales of field excitations:

$$\boxed{m_i \to E_{\text{char},i} \quad \text{(characteristic energy of particle type } i\text{)}} \tag{G.1}$$

This reinterpretation removes the distinction between mass and energy and treats both as different aspects of the same fundamental quantity.

## G.4 Two Complementary Calculation Methods

The T0 model provides two mathematically equivalent but conceptually different approaches to calculating particle masses:

### Method 1: Direct Geometric Resonance

**Conceptual Foundation:** Particles as resonances in the universal energy field

The direct method treats particles as characteristic resonance modes of the energy field $E(x,t)$, analogous to standing wave patterns:

$$\text{Particles} = \text{Discrete resonance modes of } E(x,t)(x,t) \tag{G.2}$$

**Three-Step Calculation Process:**
**Step 1: Geometric Quantization**

$$\xi_i = \xi_0 \cdot f(n_i, l_i, j_i) \tag{G.3}$$

where:

$$\xi_0 = \frac{4}{3} \times 10^{-4} \quad \text{(base geometric parameter)} \tag{G.4}$$

$$n_i, l_i, j_i = \text{quantum numbers from 3D wave equation} \tag{G.5}$$

$$f(n_i, l_i, j_i) = \text{geometric function from spatial harmonics} \tag{G.6}$$

Here $f(n,l,j)$ is the geometric structure factor of the quantum numbers, not a frequency and not the base winding number $f = 1/\xi$.

**Step 2: Resonance Frequencies**

$$\omega_i = \frac{c^2}{\xi_i \cdot r_{\text{char}}} \tag{G.7}$$

In natural units ($c = 1$):

$$\omega_i = \frac{1}{\xi_i} \tag{G.8}$$

**Step 3: Mass Determination from Energy Conservation**

$$E_{\text{char},i} = \hbar\omega_i = \frac{\hbar}{\xi_i} \tag{G.9}$$

In natural units ($\hbar = 1$):

$$\boxed{E_{\text{char},i} = \frac{1}{\xi_i}} \tag{G.10}$$

### Method 2: Extended Yukawa Method

**Conceptual Foundation:** Bridge to Standard Model formulation

The extended Yukawa method maintains compatibility with Standard Model calculations while tracing the Yukawa couplings back to a geometric structure:

$$E_{\text{char},i} = y_i \cdot v \tag{G.11}$$

where $v = 246$ GeV is the Higgs vacuum expectation value.

**Geometric Yukawa Couplings:**

$$\boxed{y_i = r_i \cdot \left(\frac{4}{3} \times 10^{-4}\right)^{\pi_i}} \tag{G.12}$$

**Generation Hierarchy:**

$$\text{1st Generation:} \quad \pi_i = \frac{3}{2} \quad \text{(electron, up quark)} \tag{G.13}$$

$$\text{2nd Generation:} \quad \pi_i = 1 \quad \text{(muon, charm quark)} \tag{G.14}$$

$$\text{3rd Generation:} \quad \pi_i = \frac{2}{3} \quad \text{(tau, top quark)} \tag{G.15}$$

The coefficients $r_i$ are simple rational numbers assigned to the geometric structure of each particle type.

## G.5 Quantum Field Theoretical Derivation of the $\xi$ Constant

### EFT Matching and Yukawa Coupling after EWSB

After electroweak symmetry breaking we have the Yukawa interaction:

$$\mathscr{L}_{\text{Yukawa}} \supset -\lambda_h \bar{\psi}\psi H, \quad \text{with} \quad H = \frac{v+h}{\sqrt{2}} \tag{G.16}$$

After EWSB:

$$\mathscr{L} \supset -m\bar{\psi}\psi - yh\bar{\psi}\psi \tag{G.17}$$

with the relations:

$$m = \frac{\lambda_h v}{\sqrt{2}} \quad \text{and} \quad y = \frac{\lambda_h}{\sqrt{2}} \tag{G.18}$$

The local mass dependence on the physical Higgs field $h(x)$ leads to:

$$m(h) = m\left(1 + \frac{h}{v}\right) \quad \Rightarrow \quad \partial_\mu m = \frac{m}{v}\partial_\mu h \tag{G.19}$$

### T0 Operators in Effective Field Theory

In T0 theory, operators of the form appear:

$$O_T = \bar{\psi}\gamma^\mu \Gamma_\mu^{(T)} \psi \tag{G.20}$$

with the characteristic time field coupling term:

$$\Gamma_\mu^{(T)} = \frac{\partial_\mu m}{m^2} \tag{G.21}$$

Inserting the Higgs dependence:

$$\Gamma_\mu^{(T)} = \frac{\partial_\mu m}{m^2} = \frac{1}{mv}\partial_\mu h \tag{G.22}$$

This suggests that a $\partial_\mu h$-coupled vector current is the UV origin.

### 1-Loop Matching Calculation

The 1-loop amplitude for the T0 vertex yields:

$$F_V(0) = \frac{y^2}{16\pi^2}\left[\frac{1}{2} - \frac{1}{2}\ln\left(\frac{m_h^2}{\mu^2}\right) + r(r - \ln r - 1)/(r-1)^2\right] \tag{G.23}$$

For hierarchical masses ($m \ll m_h$) the constant term dominates:

$$F_V(0) \approx \frac{y^2}{32\pi^2} \tag{G.24}$$

### Final $\xi$ Formula from Higgs Physics

The EFT matching provides the fundamental relation:

$$\boxed{\xi = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2}} \tag{G.25}$$

With standard Higgs parameters ($m_h = 125.1$ GeV, $v = 246.22$ GeV, $\lambda_h = m_h^2/(2v^2) \approx 0.129$):

$$\xi \approx 1.30 \times 10^{-4} \tag{G.26}$$

This lies close to the geometric setting $\xi_0 = \frac{4}{3} \times 10^{-4} \approx 1.333 \times 10^{-4}$ (deviation $\approx -2.4\%$); the relation is a consistency check of $\xi$, not an exact derivation (origin of the Higgs route: Doc. 385).

## G.6 Universal Particle Mass Systematics

### Revised Universal Fermion Table

| Fermion | Generation | Family | Spin | $r_f$ | Exponent $p_f$ | Symmetry |
|---|---|---|---|---|---|---|
| Electron Neutrino | 1 | 0 | 1/2 | $4/3$ | $5/2$ | Double $\xi$ |
| Electron | 1 | 0 | 1/2 | $4/3$ | $3/2$ | Lepton number |
| Muon Neutrino | 2 | 1 | 1/2 | $16/5$ | $2$ | Double $\xi$ |
| Muon | 2 | 1 | 1/2 | $16/5$ | $1$ | Lepton number |
| Tau Neutrino | 3 | 2 | 1/2 | $25/9$ | $5/3$ | Double $\xi$ |
| Tau | 3 | 2 | 1/2 | $25/9$ | $2/3$ | Lepton number |
| Up | 1 | 0 | 1/2 | $6$ | $3/2$ | Color |
| Down | 1 | 0 | 1/2 | $\frac{25}{2}$ | $3/2$ | Color + Isospin |
| Charm | 2 | 1 | 1/2 | $2$ | $2/3$ | Color |
| Strange | 2 | 1 | 1/2 | $\frac{26}{9}$ | $1$ | Color |
| Top | 3 | 2 | 1/2 | $\frac{1}{28}$ | $-1/3$ | Color |
| Bottom | 3 | 2 | 1/2 | $\frac{3}{2}$ | $1/2$ | Color |

## G.7 Complete Numerical Reconstruction

The following analysis shows the explicit calculation of all fermions with both methods:

### Foundations and Experimental Input Data

#### Fundamental Constants:

$$\xi_0 = \xi = \frac{4}{3} \times 10^{-4} = 1.333333333... \times 10^{-4} \tag{G.27}$$

$$v = 246 \text{ GeV} \tag{G.28}$$

#### Experimental Masses (PDG-close values):

$$m_e^{\text{exp}} = 0.0005109989461 \text{ GeV} \tag{G.29}$$

$$m_\mu^{\text{exp}} = 0.1056583745 \text{ GeV} \tag{G.30}$$

$$m_\tau^{\text{exp}} = 1.77686 \text{ GeV} \tag{G.31}$$

### Charged Leptons: Detailed Calculations

#### Electron Mass Calculation:

*Direct Method:*

$$\xi_e = \frac{4}{3} \times 10^{-4} \times f_e(1, 0, 1/2) \tag{G.32}$$

$$= \frac{4}{3} \times 10^{-4} \times 1 = \frac{4}{3} \times 10^{-4} \tag{G.33}$$

$$E_e = \frac{1}{\xi_e} = \frac{3}{4 \times 10^{-4}} = 7500 \tag{G.34}$$

The result $7500$ is a pure number; a step to $0.511$ MeV is not given along this route and would depend on the units. Authoritative is the Yukawa form $m_i = r_i \xi^{p_i} v$.

*Extended Yukawa Method:*

$$r_e = \frac{m_e^{\text{exp}}}{\nu \cdot \xi^{3/2}} \approx 1.349 \tag{G.35}$$

$$y_e = 1.349 \times \left(\frac{4}{3} \times 10^{-4}\right)^{3/2} \tag{G.36}$$

$$E_e = y_e \times 246 \text{ GeV} = 0.511 \text{ MeV} \tag{G.37}$$

**Muon Mass Calculation:**
*Direct Method:*

$$\xi_\mu = \frac{4}{3} \times 10^{-4} \times f_\mu(2, 1, 1/2) \tag{G.38}$$

$$= \frac{4}{3} \times 10^{-4} \times \frac{16}{5} = \frac{64}{15} \times 10^{-4} \tag{G.39}$$

$$E_\mu = \frac{1}{\xi_\mu} = \frac{15}{64 \times 10^{-4}} = 2343.75 \tag{G.40}$$

$2343.75$ is likewise a pure number; by this formula the muon would even be lighter than the electron by a factor $16/5$ ($7500/2343.75 = 3.2$). A step to $105.66$ MeV is not given along this route.

*Extended Yukawa Method:*

$$y_\mu = \frac{16}{5} \times \left(\frac{4}{3} \times 10^{-4}\right)^1 = 4.267 \times 10^{-4} \tag{G.41}$$

$$E_\mu = y_\mu \times 246 \text{ GeV} = 104.96 \text{ MeV} \tag{G.42}$$

**Experiment:** $105.66$ MeV → Deviation $\approx 0.65\%$

## Complete Neutrino Treatment

Neutrino Approach **[S]** The T0 model contains a geometric approach to neutrino masses based on the assumption of a characteristic **double $\xi$ suppression**. The numerical values calculated from it below are, however, not tenable, because they are incompatible with the oscillation data; authoritative for the neutrino sector is Doc. 340.

## Neutrino Quantum Numbers

Neutrinos follow the same quantum number structure as other fermions, but with an essential modification due to their weak interaction nature:

| Neutrino | n | l | j | Suppression |
|---|---|---|---|---|
| $\nu_e$ | 1 | 0 | 1/2 | Double $\xi$ |
| $\nu_\mu$ | 2 | 1 | 1/2 | Double $\xi$ |
| $\nu_\tau$ | 3 | 2 | 1/2 | Double $\xi$ |

**Table G.2:** Neutrino quantum numbers with characteristic double $\xi$ suppression

### Double $\xi$ Suppression Mechanism

The central assumption is that neutrinos experience an additional geometric suppression factor:

$$f(n_{\nu_i}, l_{\nu_i}, j_{\nu_i}) = f(n_i, l_i, j_i)_{\text{Lepton}} \times \xi \tag{G.43}$$

**Neutrino Mass Calculations:**
**Electron Neutrino:**

$$\xi_{\nu_e} = \frac{4}{3} \times 10^{-4} \times 1 \times \frac{4}{3} \times 10^{-4} = \frac{16}{9} \times 10^{-8} \tag{G.44}$$

$$E_{\nu_e} = \frac{1}{\xi_{\nu_e}} = 9.1 \text{ meV} \tag{G.45}$$

**Muon Neutrino:**

$$\xi_{\nu_\mu} = \frac{4}{3} \times 10^{-4} \times \frac{16}{5} \times \frac{4}{3} \times 10^{-4} = \frac{256}{45} \times 10^{-8} \tag{G.46}$$

$$E_{\nu_\mu} = \frac{1}{\xi_{\nu_\mu}} = 1.9 \text{ meV} \tag{G.47}$$

**Tau Neutrino:**

$$\xi_{\nu_\tau} = \frac{4}{3} \times 10^{-4} \times \frac{25}{9} \times \frac{4}{3} \times 10^{-4} = \frac{400}{81} \times 10^{-8} \tag{G.48}$$

$$E_{\nu_\tau} = \frac{1}{\xi_{\nu_\tau}} = 18.0 \text{ meV} \tag{G.49}$$

The values $9.1$, $1.9$ and $18.0$ meV are not tenable: there is no common anchor for the step from $1/\xi_\nu$ to meV, and the values are incompatible with the oscillation data, which require at least about $50$ meV for the heaviest neutrino. Authoritative for the neutrino sector is Doc. 340.

## G.8 Complete Quark Analysis with Both Methods

### Explicit Quark Mass Calculations

We use $\xi = \frac{4}{3} \times 10^{-4}$ and $v = 246$ GeV. For the Yukawa representation:

$$y_i = r_i \, \xi^{p_i}, \qquad m_i^{\text{pred}} = y_i \, v.$$

For the direct geometric representation:

$$f_i = \frac{1}{\xi \, m_i^{\text{exp}}}, \qquad m_i^{\text{exp}} = \frac{1}{\xi \, f_i}.$$

| Quark | $p_i$ | $r_i$ (corr.) | $m_i^{\text{pred}}$ (GeV) | $m_i^{\text{exp}}$ (GeV) | rel. error (%) | Remark |
|---|---|---|---|---|---|---|
| Up | $3/2$ | $6$ | $2.272 \times 10^{-3}$ | $2.16 \times 10^{-3}$ | $+5.21$ | OK |
| Down | $3/2$ | $25/2$ | $4.734 \times 10^{-3}$ | $4.72 \times 10^{-3}$ | $+0.30$ | OK |
| Strange | $1$ | $26/9$ | $9.476 \times 10^{-2}$ | $9.35 \times 10^{-2}$ | $+1.34$ | OK |
| Charm | $2/3$ | $2$ | $1.284 \times 10^{0}$ | $1.28$ | $+0.32$ | Corrected |
| Bottom | $1/2$ | $3/2$ | $4.261 \times 10^{0}$ | $4.183$ | $+1.86$ | OK |
| Top | $-1/3$ | $1/28$ | $1.7197 \times 10^{2}$ | $172.57$ | $-0.35$ | OK |

**Table G.3:** Yukawa predictions with corrected $r_i, p_i$ and comparison with reference masses (PDG 2024).

### Charm Quark Correction

The originally tabulated value $r_c = 8/9$ does not reproduce the referenced mass $m_c = 1.28$ GeV. The required value is:

$$r_c^{\text{required}} = \frac{m_c^{\text{exp}}}{v\xi^{2/3}} \approx 1.994 \approx 2.$$

Therefore, $r_c \approx 2$ was inserted in the corrected universal table.

## G.9 Comprehensive Experimental Validation

### Complete Accuracy Analysis

The following table compares the T0 values with the experimental values:

| Particle | T0 Prediction | Experiment | Accuracy | Type |
|---|---|---|---|---|
| | | *Charged Leptons* | | |
| Electron | 0.505 MeV | 0.511 MeV | 98.82% | Lepton |
| Muon | 104.96 MeV | 105.66 MeV | 99.35% | Lepton |
| Tau | 1783.4 MeV | 1776.86 MeV | 99.63% | Lepton |
| | | *Neutrinos* | | |
| $\nu_e$ | 9.1 meV | $< 450$ meV | not tenable | Neutrino |
| $\nu_\mu$ | 1.9 meV | $< 180$ keV | not tenable | Neutrino |
| $\nu_\tau$ | 18.0 meV | $< 18$ MeV | not tenable | Neutrino |
| | | *Quarks* | | |
| Up Quark | 2.272 MeV | 2.16 MeV | 94.79% | Quark |
| Down Quark | 4.734 MeV | 4.72 MeV | 99.70% | Quark |
| Strange Quark | 94.76 MeV | 93.5 MeV | 98.66% | Quark |
| Charm Quark | 1.284 GeV | 1.28 GeV | 99.68% | Quark |
| Bottom Quark | 4.261 GeV | 4.183 GeV | 98.14% | Quark |
| Top Quark | 171.97 GeV | 172.57 GeV | 99.65% | Quark |
| **Average** | | | **98.7%** | **Charged Fermions** |

**Table G.4:** Complete experimental validation of T0 model predictions

The predictions use the tabulated rational coefficients: $\frac{4}{3}\xi^{3/2}v = 0.505$ MeV ($-1.2\,\%$; $99.98\,\%$ would result only with $r_e = 1.349$ fitted to $m_e$), $\frac{25}{9}\xi^{2/3}v = 1783.4$ MeV ($+0.37\,\%$), $\frac{26}{9}\xi v = 94.76$ MeV, $2\xi^{2/3}v = 1.284$ GeV, $\frac{1}{28}\xi^{-1/3}v = 171.97$ GeV. The quark reference masses follow PDG 2024; the average is the mean over the nine charged fermions. The neutrino values are not tenable (see Section G.7).

**Key Result**

Summary of Accuracy The T0 model achieves 98.7% average accuracy across the **charged** fermions with the single geometric parameter $\xi$ and the tabulated rational coefficients $r_i$. The neutrino sector is included as an approach **[S]**; the numerical values calculated from it are incompatible with the oscillation data (authoritative: Doc. 340).

## G.10 Experimental Predictions and Precision Tests

### Modified QED Vertex Corrections

The T0 theory predicts modified Feynman rules:

$$\text{Time field vertex:} \quad -i\gamma^\mu \Gamma_\mu^{(T)} = i\gamma^\mu \frac{\partial_\mu m}{m^2} \tag{G.50}$$

$$\text{Modified fermion propagator:} \quad S_F^{(T0)}(p) = S_F(p) \cdot \left[1 + \frac{\beta}{p^2}\right] \tag{G.51}$$

### Neutrino Validation

The T0 neutrino values do satisfy the direct limits listed in the table, but they are incompatible with the oscillation data (heaviest neutrino at least about $50$ meV) and therefore not tenable; authoritative is Doc. 340:

| Parameter | T0 Prediction | Experimental Limit | Status |
|---|---|---|---|
| $m_{\nu_e}$ | 9.1 meV | $< 450$ meV (KATRIN) | Fulfilled |
| $m_{\nu_\mu}$ | 1.9 meV | $< 180$ keV (indirect) | Fulfilled |
| $m_{\nu_\tau}$ | 18.0 meV | $< 18$ MeV (indirect) | Fulfilled |
| $\sum m_\nu$ | 29.0 meV | $< 60$ meV (Cosmology 2024) | Fulfilled |

**Table G.5:** T0 neutrino predictions vs. experimental constraints

**Important**

Neutrino Mass Hierarchy **[S]** The T0 model orders the masses as $m_{\nu_\mu} < m_{\nu_e} < m_{\nu_\tau}$. How this assignment to flavour states relates to the **normal ordering** of the mass eigenstates preferred by oscillation data remains to be clarified.

## G.11 Predictive Power of the Established System

### New Particle Generations

With established patterns, new particles can be predicted:

**4th Generation (extrapolated) [S]:**

$$n = 4, \quad \pi_4 = \frac{1}{2}, \quad r_4 \approx 2.0 \tag{G.52}$$

$$m_{\text{4th Gen}} = r_4 \times \xi^{1/2} \times v \approx 5.7 \text{ GeV} \tag{G.53}$$

The calculation $2.0 \times \xi^{1/2} \times 246$ GeV $\approx 5.7$ GeV is correct; a charged lepton of a 4th generation at this mass is, however, experimentally excluded (LEP, lower bound about $100$ GeV).

### Quark Sector Extrapolation

Lepton patterns can be transferred to quarks:

| Quark | Generation | $r_i$ | $\pi_i$ | Prediction |
|---|---|---|---|---|
| Up | 1 | 6 | 3/2 | 2.3 MeV |
| Down | 1 | 12.5 | 3/2 | 4.7 MeV |
| Charm | 2 | 2.0 | 2/3 | 1.3 GeV |
| Strange | 2 | 2.89 | 1 | 95 MeV |
| Top | 3 | 0.036 | -1/3 | 173 GeV |
| Bottom | 3 | 1.5 | 1/2 | 4.3 GeV |

**Table G.6:** Quark predictions from established patterns

## G.12 Corrected Interpretation of Mathematical Equivalence

Meaning of Equivalence The mathematical equivalence of both methods is **given by definition** when parameters ($r_i$ or $f_i$) are determined from the same experimental masses. The equivalence is not proof of the theory, but a consistency property of the mathematical structure.

### Transformation Relationship as Bridge

The fundamental relation:

$$f_i = \frac{1}{r_i \, \xi^{\pi_i} \, \nu \xi_0} \tag{G.54}$$

mathematically connects both methods. When $r_i$ is determined from experimental masses, $f_i$ follows automatically and vice versa.

| Particle | $m^{\text{exp}}$ (GeV) | $r_i$ (Yukawa) | $f_i$ (direct) | Accuracy |
|---|---|---|---|---|
| Electron | 0.000511 | 1.349 | $1.468 \times 10^7$ | 99.98% |
| Muon | 0.10566 | 3.221 | $7.099 \times 10^4$ | 99.35% |
| Tau | 1.77686 | $25/9 = 2.778$ | $4.221 \times 10^3$ | 99.63% |
| $\nu_e$ | 9.1 $\times 10^{-12}$ | 1.349 | $8.24 \times 10^{14}$ | not tenable |
| $\nu_\mu$ | 1.9 $\times 10^{-12}$ | 3.221 | $3.95 \times 10^{15}$ | not tenable |
| $\nu_\tau$ | 18.0 $\times 10^{-12}$ | 2.778 | $4.17 \times 10^{14}$ | not tenable |

**Table G.7:** Numerical equivalence of both T0 methods for all leptons

Electron and muon are listed here with the $r_i$ determined from the measured value; the rational values are $4/3$ and $16/5$. For the tau, the ladder with the rational $r_\tau = 25/9$ gives $1.7834$ GeV, i.e. $99.63\,\%$. The neutrino rows take over the untenable meV values from Section G.7.

## G.13 Methodological Foundation

### Reversibility of the Established System

After the establishment phase, the T0 system is meant to become predictive:

**Established Lepton Patterns:**

$$\text{1st Generation (n=1):} \quad \pi_i = \frac{3}{2}, \quad r_e \approx 1.35 \tag{G.55}$$

$$\text{2nd Generation (n=2):} \quad \pi_i = 1, \quad r_\mu \approx 3.2 \tag{G.56}$$

$$\text{3rd Generation (n=3):} \quad \pi_i = \frac{2}{3}, \quad r_\tau \approx 2.8 \tag{G.57}$$

### Experimental Testability

T0 predictions are experimentally falsifiable:

1. **LHC searches:** New particles at characteristic energies (a charged lepton of a 4th generation at $5.7$ GeV is already excluded by LEP)
2. **Precision measurements:** Refinement of $r_i$ parameters
3. **Neutrino tests:** Direct neutrino mass measurements
4. **Anomalous magnetic moments:** T0 corrections to g-2 experiments

Arguments for the T0 procedure:

1. **Systematic structure:** All parameters follow recognizable patterns
2. **Predictive power:** After establishment, new particles become predictable
3. **Experimental testability:** Predictions are falsifiable
4. **QFT foundation:** Quantum field theoretical derivation of $\xi$ constant
5. **Historical precedent:** Methodology of pattern physics (periodic table, quark model)

## G.14 Parameter Structure and Universal Systematics

**Important**

Role of the Parameter $\xi$ The parameter $\xi$ can additionally be estimated from Higgs parameters:

$$\xi = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2} \approx 1.30 \times 10^{-4} \tag{G.58}$$

Thus $\xi$ cannot be chosen freely. The coefficients $r_i$ are rational numbers; deriving them from geometry without recourse to measured masses is the next step.

### Universal Quantum Number Table

| Particle | n | l | j | $r_i$ | $p_i$ | Special |
|---|---|---|---|---|---|---|
| *Charged Leptons* | | | | | | |
| Electron | 1 | 0 | 1/2 | 4/3 | 3/2 | – |
| Muon | 2 | 1 | 1/2 | 16/5 | 1 | – |
| Tau | 3 | 2 | 1/2 | 25/9 | 2/3 | – |
| *Neutrinos* | | | | | | |
| $\nu_e$ | 1 | 0 | 1/2 | 4/3 | 5/2 | Double $\xi$ |
| $\nu_\mu$ | 2 | 1 | 1/2 | 16/5 | 2 | Double $\xi$ |
| $\nu_\tau$ | 3 | 2 | 1/2 | 25/9 | 5/3 | Double $\xi$ |
| *Quarks* | | | | | | |
| Up | 1 | 0 | 1/2 | 6 | 3/2 | Color |
| Down | 1 | 0 | 1/2 | 25/2 | 3/2 | Color + Isospin |
| Charm | 2 | 1 | 1/2 | 2 | 2/3 | Color |
| Strange | 2 | 1 | 1/2 | 26/9 | 1 | Color |
| Top | 3 | 2 | 1/2 | 1/28 | -1/3 | Color |
| Bottom | 3 | 2 | 1/2 | 3/2 | 1/2 | Color |

**Table G.8:** Complete universal quantum number table for all fermions

## G.15 Critical Assessment and Limitations

### Theoretical Open Questions

1. **Number of generations:** Why exactly three generations plus fourth prediction?
2. **Hierarchy problem:** Connection between different energy scales
3. **CP violation:** Incorporation of CKM and PMNS mixing matrices

# Chapter H

# T0 Theory: Neutrinos


## Abstract

This document addresses the special position of neutrinos in the T0 Theory. In contrast to established particles (charged leptons, quarks, bosons), neutrinos require a fundamentally different treatment based on the photon analogy with double $\xi_0$-suppression. The neutrino mass is derived from the formula $m_\nu = \frac{\xi_0^2}{2} \times m_e = 4.54$ meV, and oscillations are explained by geometric phases based on $T_x \cdot m_x = 1$, where the quantum numbers $(n, \ell, j)$ determine the phase differences. An extension via the Koide relation introduces a weak hierarchy through exponent rotations; the resulting quasi-degenerate spectrum gives $Q_\nu \approx 1/3$ and is excluded by the measured mass differences (see the section on the Koide relation); authoritative for neutrinos is Doc. 340. A plausible target value for the neutrino mass ($m_\nu = 15$ meV) is derived from empirical data (cosmological limits). The T0 Theory is based on speculative geometric harmonies without empirical basis and is highly likely to be incomplete or incorrect. Scientific integrity requires a clear separation between mathematical correctness and physical validity.


## H.1 Preamble: Scientific Honesty

**Warning**

**CRITICAL LIMITATION:** The following formulas for neutrino masses are **speculative extrapolations** based on the untested hypothesis that neutrinos follow geometric harmonies and all flavor states have equal masses. This hypothesis has **no empirical basis** and is highly likely to be incomplete or incorrect. The mathematical formulas are nevertheless internally consistent and correctly formulated.

**Scientific integrity means:**

- Honesty about the speculative nature of the predictions

- Mathematical correctness despite physical uncertainty
- Clear separation between hypotheses and verified facts

## H.2 Neutrinos as "Almost Massless Photons": The T0 Photon Analogy

**Speculation**

**Fundamental T0 Insight:** Neutrinos can be understood as "damped photons". The remarkable similarity between photons and neutrinos suggests a deeper geometric kinship:

- **Speed:** Both propagate nearly at the speed of light
- **Penetration:** Both have extreme penetrability
- **Mass:** Photon exactly massless, neutrino quasi-massless
- **Interaction:** Photon electromagnetic, neutrino weak

### Photon-Neutrino Correspondence

**Physical Parallels:**

$$\text{Photon:} \quad E^2 = (pc)^2 + 0 \quad \text{(perfectly massless)} \tag{H.1}$$

$$\text{Neutrino:} \quad E^2 = (pc)^2 + \left(\frac{\xi^2}{2} m_e c^2\right)^2 \quad \text{(quasi-massless)} \tag{H.2}$$

**Speed Comparison:**

$$v_\gamma = c \quad \text{(exact)} \tag{H.3}$$

$$\frac{c - v_\nu}{c} \approx \frac{m_\nu^2 c^4}{2E^2} \quad \text{(energy-dependent)} \tag{H.4}$$

With $m_\nu = 4.5$ meV and $E = 10$ MeV this is $10^{-19}$ – practically immeasurable. An energy-independent difference $v_\nu = c(1 - \xi^2/2)$ does not hold, because in this document $\xi^2/2$ is the mass ratio $m_\nu/m_e$. As an energy-independent value, $8.89 \times 10^{-9}$ would moreover be excluded by SN 1987A: over $168\,000$ light years it would give a $13$-hour delay relative to light, whereas the neutrinos arrived hours before the light ($|\Delta v/c| \lesssim 2 \times 10^{-9}$).

## The Double $\xi_0$-Suppression

**Key Result**

**Neutrino Mass through Double Geometric Damping:**
If neutrinos are "almost photons", then two suppression factors arise:

1. **First $\xi_0$ Factor:** "Almost massless" (like photon, but not perfect)
2. **Second $\xi_0$ Factor:** "Weak interaction" (geometric decoupling)

**Resulting Formula:**

$$\boxed{m_\nu = \frac{\xi_0^2}{2} \times m_e = \frac{(\frac{4}{3} \times 10^{-4})^2}{2} \times 0.511 \text{ MeV}} \tag{H.5}$$

**Numerical Evaluation:**

$$m_\nu = 8.889 \times 10^{-9} \times 0.511 \text{ MeV} = 4.54 \text{ meV} \tag{H.6}$$

## Physical Justification of the Photon Analogy

**Why the Photon Analogy is Physically Sensible:**

**1. Speed Comparison:**

$$v_\gamma = c \quad \text{(exact)} \tag{H.7}$$

$$\frac{c - v_\nu}{c} \approx \frac{m_\nu^2 c^4}{2E^2} \tag{H.8}$$

The speed difference is energy-dependent and practically immeasurable at typical energies (see above).

**2. Interaction Strengths:**

$$\sigma_\gamma \sim \alpha_{EM} \approx \frac{1}{137} \tag{H.9}$$

$$\sigma_\nu \sim \frac{\xi_0^2}{2} \times G_F \approx 1.04 \times 10^{-13} \text{ GeV}^{-2} \tag{H.10}$$

With $G_F = 1.166 \times 10^{-5}$ GeV$^{-2}$, $\sigma_\nu$ has the dimension GeV$^{-2}$; comparing it with the dimensionless $\alpha_{EM}$ does not give a ratio $\xi_0^2/2$.

**3. Penetrability:**

- Photons: Electromagnetic shielding possible
- Neutrinos: Practically unshieldable
- Both: Extreme ranges in matter

## H.3 Neutrino Oscillations

### The Standard Model Problem

**Warning**

**Neutrino Oscillations:** Neutrinos can change their identity (flavor) during flight - a phenomenon known as neutrino oscillation. A neutrino produced as an electron neutrino ($\nu_e$) can later be measured as a muon neutrino ($\nu_\mu$) or tau neutrino ($\nu_\tau$) and vice versa.
The oscillations depend on the mass squared differences $\Delta m_{ij}^2 = m_i^2 - m_j^2$ and the mixing angles. Current experimental data (2025) provide:

$$\Delta m_{21}^2 \approx 7.53 \times 10^{-5}\ \text{eV}^2 \quad \text{[Solar]} \tag{H.11}$$

$$\Delta m_{32}^2 \approx 2.44 \times 10^{-3}\ \text{eV}^2 \quad \text{[Atmospheric]} \tag{H.12}$$

$$\Sigma m_\nu \gtrsim 0.06\ \text{eV} \quad \text{[from oscillation data; heaviest neutrino} \gtrsim 0.05\ \text{eV]} \tag{H.13}$$

**Problem for T0:** The T0 Theory postulates equal masses for the flavor states ($\nu_e, \nu_\mu, \nu_\tau$), which implies $\Delta m_{ij}^2 = 0$ and is incompatible with standard oscillations.

### Geometric Phases as Oscillation Mechanism

**Speculation**

**T0 Hypothesis: Geometric Phases for Oscillations**
To reconcile the hypothesis of equal masses ($m_{\nu_e} = m_{\nu_\mu} = m_{\nu_\tau} = m_\nu$) with neutrino oscillations, it is speculated that oscillations in the T0 Theory are caused by geometric phases rather than mass differences. This is based on the T0 relation:

$$T_x \cdot m_x = 1,$$

where $m_x = m_\nu = 4.54$ meV is the neutrino mass and $T_x$ is a characteristic time or frequency:

$$T_x = \frac{1}{m_\nu} = \frac{1}{4.54 \times 10^{-3}\ \text{eV}} \approx 2.2026 \times 10^2\ \text{eV}^{-1} \approx 1.449 \times 10^{-13}\ \text{s}.$$

The geometric phase is determined by the T0 quantum numbers $(n, \ell, j)$:

$$\phi_{\text{geo},i} \propto f(n, \ell, j) \cdot \frac{L}{E} \cdot \frac{1}{T_x},$$

where $f(n, \ell, j) = \frac{n^6}{\ell^3}$ (or 1 for $\ell = 0$) are the geometric factors:

$$f_{\nu_e} = 1, \tag{H.14}$$

$$f_{\nu_\mu} = 64, \tag{H.15}$$

$$f_{\nu_\tau} = 91.125. \tag{H.16}$$

**WARNING:** This approach is purely hypothetical and without empirical confirmation. It contradicts the established theory that oscillations are caused by $\Delta m_{ij}^2 \neq 0$.

Here $f(n, \ell, j)$ is the geometric factor of the neutrino flavours, not a frequency and not the base winding number $f = 1/\xi$.

### Quantum Number Assignment for Neutrinos

| Neutrino Flavor | $n$ | $\ell$ | $j$ | $f(n, \ell, j)$ |
|---|---|---|---|---|
| $\nu_e$ | 1 | 0 | 1/2 | 1 |
| $\nu_\mu$ | 2 | 1 | 1/2 | 64 |
| $\nu_\tau$ | 3 | 2 | 1/2 | 91.125 |

**Table H.1:** Speculative T0 Quantum Numbers for Neutrino Flavors

## H.4 Integration der Koide-Relation: Eine schwache Hierarchie

**T0-Koide Extension for Neutrinos:**

To address the oscillation conflict ($\Delta m_{ij}^2 \neq 0$), the T0 Theory integrates the Koide relation as a natural generalization (Li and Ma 2005; Brannen 2006). This introduces a weak hierarchy via exponent rotations around $\xi_0$, preserving the photon analogy while enabling small mass differences.

**Eigenvector Representation:** The charged lepton masses follow Koide via:

$$\begin{pmatrix} \sqrt{m_e} \\ \sqrt{m_\mu} \\ \sqrt{m_\tau} \end{pmatrix} = \mathbf{U} \cdot \begin{pmatrix} m_1 \\ m_2 \\ m_3 \end{pmatrix}, \tag{H.17}$$

where **U** is the unitary flavor-mixing matrix (CKM/PMNS analog).

**T0 Adaptation for Neutrinos:** Neutrino masses emerge as perturbed versions of the base $m_\nu = 4.54$ meV:

$$m_{\nu_i} \approx \xi_0^{p_i+\delta} \cdot \nu_\nu, \quad \delta \approx \xi_0^{1/3} \approx 0.051 \tag{H.18}$$

with exponents $p_i = (3/2, 1, 2/3)$ from charged leptons (rotated by $\delta$ for weak hierarchy). A quasi-degenerate spectrum is assumed:

$$m_{\nu_1} \approx 4.20 \text{ meV (normal hierarchy)}, \tag{H.19}$$

$$m_{\nu_2} \approx 4.54 \text{ meV}, \tag{H.20}$$

$$m_{\nu_3} \approx 5.12 \text{ meV}, \tag{H.21}$$

$$\Sigma m_\nu \approx 13.86 \text{ meV}. \tag{H.22}$$

The formula $\xi_0^{p_i+\delta}\nu_\nu$ itself, however, gives with $p_i = (3/2, 1, 2/3)$ the ratio $m_{\nu_1}/m_{\nu_3} = \xi_0^{3/2-2/3} = 5.9 \times 10^{-4}$, i.e. a strong hierarchy as for the charged leptons; the quasi-degenerate values stated do not follow from it.

**Neutrino Koide Relation:**

$$Q_\nu = \frac{m_{\nu_1} + m_{\nu_2} + m_{\nu_3}}{\left(\sqrt{m_{\nu_1}} + \sqrt{m_{\nu_2}} + \sqrt{m_{\nu_3}}\right)^2} = \frac{13.86}{41.51} \approx 0.334, \tag{H.23}$$

i.e. close to $1/3$, the value for exactly equal masses. For a quasi-degenerate spectrum $Q_\nu$ always tends to $1/3$; the Koide value $2/3$ cannot be reached this way.

**Hybrid Oscillation Mechanism:** Geometric phases (from $f(n, \ell, j)$) dominate, augmented by small $\Delta m^2_{21} \approx 0.030 \times 10^{-4}$ and $\Delta m^2_{31} \approx 0.086 \times 10^{-4}$ eV$^2$ from $\delta$. These values follow from $4.20/4.54/5.12$ meV. With the measured $\Delta m^2_{21} = 7.53 \times 10^{-5}$ and $\Delta m^2_{32} = 2.44 \times 10^{-3}$ eV$^2$, however, $\Sigma m_\nu \geq 59$ meV; the spectrum with $13.86$ meV is excluded by the oscillation data, also by falsification criterion 3 below. Authoritative for neutrinos is Doc. 340.

**WARNING:** Highly speculative; excluded by the oscillation data (see above).

## H.5 Experimental Assessment

### Cosmological Limits

**Experimental**

**Cosmological Neutrino Mass Limits (as of 2025):**
**1. DESI + CMB (2024):**

$$\Sigma m_\nu < 0.07 \text{ eV} \quad (95\% \text{ Confidence}) \tag{H.24}$$

**2. T0 Value of the Koide Extension:**

$$\Sigma m_\nu = 13.86 \text{ meV} \tag{H.25}$$

**3. Comparison:**

$$\frac{13.86 \text{ meV}}{70 \text{ meV}} = 0.198 \approx 19.8\% \tag{H.26}$$

The value lies below all cosmological limits, but thereby also below the lower bound $\Sigma m_\nu \geq 59$ meV from oscillation data and is excluded (see the section on the Koide relation).

## Direct Mass Determination

**Experimental**

**Experimental Neutrino Mass Determination:**
**1. KATRIN Experiment (2022):**

$$m(\nu_e) < 0.8 \text{ eV} \quad (90\% \text{ Confidence}) \tag{H.27}$$

**2. T0 Prediction (with Koide):**

$$m(\nu_e) \approx 4.54 \text{ meV (effective)} \tag{H.28}$$

**3. Comparison:**

$$\frac{4.54 \text{ meV}}{800 \text{ meV}} = 0.0057 \approx 0.57\% \tag{H.29}$$

The T0 prediction is orders of magnitude below the direct mass limits.

## Target Value Estimation

**Key Result**

**Plausible Target Value for Neutrino Masses:**
From cosmological data and theoretical considerations, a plausible target value emerges:

$$m_\nu^{\text{Target}} \approx 15 \text{ meV (per flavor, quasi-degenerate)} \tag{H.30}$$

**Comparison with T0 Prediction (incl. Koide):**

$$\frac{4.54 \text{ meV}}{15 \text{ meV}} = 0.303 \approx 30.3\% \tag{H.31}$$

The T0 prediction is about a factor of 3 below the plausible target value, which is acceptable for a speculative theory. Koide extension narrows this to 7% via hierarchy.

## H.6 Cosmological Implications

### Structure Formation and Big Bang Nucleosynthesis

**Key Result**

**Cosmological Consequences of T0 Neutrino Masses:**

**1. Big Bang Nucleosynthesis:**

- Relativistic neutrinos at $T \sim 1$ MeV: Standard BBN unchanged
- Contribution to radiation density: $N_{\text{eff}} = 3.046$ (Standard)

**2. Structure Formation:**

- Neutrinos with 4.5 meV become non-relativistic at $z \approx 8$ ($1 + z \approx m_\nu/(3.15\, T_{\nu,0}) = 4.5/0.53$ with $T_{\nu,0} = 1.68 \times 10^{-4}$ eV)
- Suppression of small-scale structure formation negligible

**3. Cosmic Neutrino Background (C$\nu$B):**

- Number density: $n_\nu = 336$ cm$^{-3}$ (unchanged)
- Energy density: $\rho_\nu \propto \Sigma m_\nu = 13.86$ meV (with Koide)
- Fraction of critical density: $\Omega_\nu h^2 = 13.86$ meV$/93.14$ eV $\approx 1.49 \times 10^{-4}$

**4. Comparison with Dark Matter:**

- Neutrino contribution: $\Omega_\nu \approx 3.3 \times 10^{-4}$ (with $h = 0.674$)
- Dark matter: $\Omega_{DM} \approx 0.26$
- Ratio: $\Omega_\nu/\Omega_{DM} \approx 1.3 \times 10^{-3}$ (negligible)

## H.7 Experimental Tests and Falsification

### Testable Predictions

**Experimental**

**Specific Experimental Tests of the T0 Neutrino Theory:**

1. **Direct Mass Determination:**
   - KATRIN: Sensitivity to $\sim 0.2$ eV (insufficient)
   - Future Experiments: $\sim 0.01$ eV required
   - T0 Prediction: $m_{\nu_i} \approx 4 - 5$ meV (factor 2 below limit)
2. **Cosmological Precision Measurements:**

- Euclid Satellite: Sensitivity $\sim 0.02$ eV
- T0 value of the Koide extension: $\Sigma m_\nu = 13.86$ meV (already excluded by the oscillation lower bound of 59 meV)

3. **Koide-Specific Tests:**
   - Measure $Q_\nu$ via oscillation data: this document's quasi-degenerate spectrum gives $Q_\nu \approx 1/3$
   - PMNS correlations: Hierarchy from $\delta$-rotation
4. **Speed Measurements:**
   - Supernova Neutrinos: $\Delta v/c \sim 10^{-8}$ measurable
   - T0: $\xi^2/2 = 8.89 \times 10^{-9}$ is a mass ratio, not a speed difference; as an energy-independent $\Delta v/c$ the value would be excluded by SN 1987A (see section Photon-Neutrino Correspondence)
5. **Oscillation Physics:**
   - Test for small $\Delta m_{ij}^2$ + phase effects (clearly falsifiable)

## Falsification Criteria

The T0 Neutrino Theory would be falsified by:

1. Direct measurement of $m_\nu > 0.1$ eV (or strong hierarchy $|m_3 - m_1| > 10$ meV)
2. Cosmological evidence for $\Sigma m_\nu > 0.1$ eV
3. Clear proof of $\Delta m_{ij}^2 \gg 10^{-4}$ eV$^2$ without phases
4. Measurement of speed differences $\Delta v/c > 10^{-8}$
5. Deviation from $Q_\nu \approx 1/3$ (value of the quasi-degenerate spectrum, see the section on the Koide relation) in oscillation analyses

**Addendum (3 September 2026, Doc. 340 and Doc. 343 Theorem D‴)**

The base formula $m_\nu = \frac{\xi^2}{2} \times m_e = 4.54$ meV agrees exactly with $m_{\nu_1}$ in Doc. 340, which derives the neutrino masses algebraically from the GF$(27)^*$ Galois orbits **[K]**. Doc. 007 thereby provides the physical motivation for the double $\xi$ suppression (neutrino = almost photon + geometric decoupling) that Doc. 340 treats only algebraically. Neither $\xi$ nor $m_e$ are free parameters: $\xi$ follows from Galois group orders **[K]** (Doc. 338), $m_e$ from torus quantum numbers **[K]** (Doc. 006); $v = 246$ GeV is an SI scale fixation, not a free parameter.
What Doc. 340 additionally achieves **[B]**:

- Three different masses from orbits $O_1, O_3, O_4$: $m_{\nu_1} = 4.54$, $m_{\nu_2} = 9.81$, $m_{\nu_3} = 49.9$ meV — $\Delta m^2_{21}$ and $\Delta m^2_{31}$ reproduced to 0.4 and 1.5 %.
- Majorana/Dirac partition: $\nu_1, \nu_2$ Majorana (orbits $O_1, O_3$ self-inverse); $\nu_3$ **Dirac neutrino** (orbit $O_4$, Doc. 343 Theorem D‴).
- No Majorana phase for $\nu_3$; $m_{ee} \leq 14.4$ meV; inverted hierarchy more strongly excluded.

The Koide extension and the n/ℓ/j quantum numbers have no connection to the Galois orbits and remain **[S]**. Authoritative current state: Doc. 340 and Doc. 343.

## H.8 Limits and Open Questions

### Fundamental Theoretical Problems

**Warning**

**Unsolved Problems of the T0 Neutrino Theory:**

1. **Oscillation Mechanism:** Geometric phases + $\delta$ are ad hoc
2. **Quantum Field Theory:** No complete QFT formulation
3. **Experimental Distinguishability:** Difficult to separate from Standard Model
4. **Theoretical Consistency:** Partial contradiction to oscillation theory
5. **Predictive Power:** Enhanced by Koide, but still limited

## H.9 Methodological Reflection

### Scientific Integrity vs. Theoretical Speculation

**Key Result**

**Central Methodological Insights:**
The neutrino chapter of the T0 Theory illustrates the tension between:

- **Theoretical Completeness:** Desire for unified description (now incl. Koide)

- **Empirical Anchoring:** Necessity of experimental confirmation
- **Scientific Honesty:** Disclosure of speculative nature
- **Mathematical Consistency:** Internal self-consistency of formulas

**Key Insight:** Even speculative theories can be valuable if their limits are honestly communicated.

### Significance for the T0 Series

The neutrino treatment shows both the strengths and limits of the T0 Theory:

- **Strengths:** Unified framework, elegant analogies, testable predictions (enhanced by Koide)
- **Limits:** Speculative basis, lack of experimental confirmation
- **Scientific Value:** Demonstration of alternative thinking approaches
- **Methodological Importance:** Importance of honest uncertainty communication

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*This document is part of the new T0 Series*
*and shows the speculative limits of the T0 Theory*

**T0-Theory: Time-Mass Duality Framework**
*Johann Pascher*
*GitHub: https://github.com/jpascher/T0-Time-Mass-Duality*

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# Chapter I

# T0 Model: Unified Neutrino Formula Structure


## Abstract

This document presents a mathematically consistent formula structure for neutrino calculations within the T0 model, based on the hypothesis of equal masses for all flavor states ($\nu_e, \nu_\mu, \nu_\tau$). The neutrino mass is derived from the photon analogy ($\frac{\xi^2}{2}$-suppression), and oscillations are explained by geometric phases based on $T_x \cdot m_x = 1$, with quantum numbers $(n, \ell, j)$ determining phase differences. A plausible target value for the neutrino mass ($m_\nu = 15$ meV) is derived from empirical data (cosmological constraints). The T0 model is based on speculative geometric harmonies without empirical support and is highly likely to be incomplete or incorrect. Scientific integrity requires a clear distinction between mathematical correctness and physical validity.


## I.1 Preamble: Scientific Integrity

**Warning**

**CRITICAL LIMITATION:** The following formulas for neutrino masses are **speculative extrapolations** based on the untested hypothesis that neutrinos follow geometric harmonies and all flavor states have equal masses. This hypothesis has **no empirical basis** and is highly likely to be incomplete or incorrect. The mathematical formulas are nonetheless internally consistent and error-free.

**Scientific Integrity Requires:**

- Honesty about the speculative nature of predictions
- Mathematical correctness despite physical uncertainty
- Clear separation between hypotheses and verified facts

## I.2 Neutrinos as "Near-Massless Photons": The T0 Photon Analogy

**Speculation**

**Fundamental T0 Insight:** Neutrinos can be understood as "damped photons." The remarkable similarity between photons and neutrinos suggests a deeper geometric kinship:

- **Speed:** Both propagate at nearly the speed of light
- **Penetration:** Both have extreme penetration capabilities
- **Mass:** Photon is exactly massless, neutrino is nearly massless
- **Interaction:** Photon interacts electromagnetically, neutrino interacts weakly

### Photon-Neutrino Correspondence

**Important**

**Physical Parallels:**

$$\text{Photon:} \quad E^2 = (pc)^2 + 0 \quad \text{(perfectly massless)} \tag{I.1}$$

$$\text{Neutrino:} \quad E^2 = (pc)^2 + \left(\sqrt{\frac{\xi^2}{2}}mc^2\right)^2 \quad \text{(nearly massless)} \tag{I.2}$$

**Speed Comparison:**

$$v_\gamma = c \quad \text{(exact)} \tag{I.3}$$

$$v_\nu = c \times \left(1 - \frac{\xi^2}{2}\right) \approx 0.9999999911 \times c \tag{I.4}$$

The speed difference is only $8.89 \times 10^{-9}$ – practically unmeasurable!

### Double $\xi$-Suppression from Photon Analogy

**T0 Hypothesis:** Neutrino = Photon with Geometric Double Damping
If neutrinos are "near-photons," two suppression factors arise:

- **First $\xi$ Factor:** "Near massless" (like a photon, but not perfect)
- **Second $\xi$ Factor:** "Weak interaction" (geometric coupling)
- **Result:** $m_\nu \propto \frac{\xi^2}{2}$, consistent with the speed difference $v_\nu = c \times \left(1 - \frac{\xi^2}{2}\right)$

**Interaction Strength Comparison:**

$$\sigma_\gamma \sim \alpha_{\text{EM}} \approx \frac{1}{137} \tag{I.5}$$

$$\sigma_\nu \sim \frac{\xi^2}{2} \times G_F \approx 1.04 \times 10^{-13}\ \text{GeV}^{-2} \tag{I.6}$$

with the dimensionless factor $\xi^2/2 = (4/30000)^2/2 \approx 8.888888 \times 10^{-9}$ and $G_F = 1.1664 \times 10^{-5}$ GeV$^{-2}$. The product is dimensionful and cannot be compared directly with the dimensionless $\alpha \approx 1/137$; the geometric suppression resides in the dimensionless factor $\frac{\xi^2}{2}$.

## I.3 Neutrino Oscillations

**Important**

**Neutrino Oscillations:** Neutrinos can change their identity (flavor) during flight – a phenomenon known as neutrino oscillation. A neutrino produced as an electron neutrino ($\nu_e$) can later be detected as a muon neutrino ($\nu_\mu$) or tau neutrino ($\nu_\tau$) and vice versa.
In standard physics, this behavior is described by the mixing of mass eigenstates ($\nu_1, \nu_2, \nu_3$) connected to flavor states ($\nu_e, \nu_\mu, \nu_\tau$) via the PMNS matrix (Pontecorvo-Maki-Nakagawa-Sakata):

$$\begin{pmatrix} \nu_e \\ \nu_\mu \\ \nu_\tau \end{pmatrix} = U_{\text{PMNS}} \begin{pmatrix} \nu_1 \\ \nu_2 \\ \nu_3 \end{pmatrix}, \tag{I.7}$$

where $U_{\text{PMNS}}$ is the mixing matrix.
Oscillations depend on mass differences $\Delta m_{ij}^2 = m_i^2 - m_j^2$ and mixing angles. Current experimental data (2025) provide:

$$\Delta m_{21}^2 \approx 7.53 \times 10^{-5}\ \text{eV}^2 \quad \text{[Solar]} \tag{I.8}$$

$$\Delta m_{32}^2 \approx 2.44 \times 10^{-3}\ \text{eV}^2 \quad \text{[Atmospheric]} \tag{I.9}$$

$$\Sigma m_\nu \gtrsim 0.06\ \text{eV} \quad \text{[lower bound; heaviest eigenstate } \gtrsim 0.05\ \text{eV]} \tag{I.10}$$

**Implications for T0:**

- The T0 model postulates equal masses for flavor states ($\nu_e, \nu_\mu, \nu_\tau$), implying $\Delta m_{ij}^2 = 0$, which is incompatible with standard oscillations.
- To explain oscillations, the T0 model uses geometric phases based on $T_x \cdot m_x = 1$, with quantum numbers $(n, \ell, j)$ determining phase differences.

## Geometric Phases as Oscillation Mechanism

**Speculation**

**T0 Hypothesis: Geometric Phases for Oscillations**
To reconcile the hypothesis of equal masses ($m_{\nu_e} = m_{\nu_\mu} = m_{\nu_\tau} = m_\nu$) with neutrino oscillations, it is speculated that oscillations in the T0 model are caused by geometric phases rather than mass differences. This is based on the T0 relation:

$$T_x \cdot m_x = 1,$$

where $m_x = m_\nu = 4.54$ meV is the neutrino mass, and $T_x$ is a characteristic time or frequency:

$$T_x = \frac{1}{m_\nu} = \frac{1}{4.54 \times 10^{-3}\ \text{eV}} \approx 2.2026 \times 10^{2}\ \text{eV}^{-1} \approx 1.449 \times 10^{-13}\ \text{s}.$$

The geometric phase is determined by the T0 quantum numbers $(n, \ell, j)$:

$$\phi_{\text{geo},i} \propto f(n, \ell, j) \cdot \frac{L}{E} \cdot \frac{1}{T_x},$$

where $f(n, \ell, j) = \frac{n^6}{\ell^3}$ (or 1 for $\ell = 0$) are the geometric factors of the neutrino flavours (not a frequency and not the base winding number $f = 1/\xi$):

$$f_{\nu_e} = 1, \tag{I.11}$$
$$f_{\nu_\mu} = 64, \tag{I.12}$$
$$f_{\nu_\tau} = 91.125. \tag{I.13}$$

**Calculated Phase Differences:**

$$\phi_{\nu_e} \propto 1 \cdot \frac{L}{E} \cdot \frac{1}{T_x}, \tag{I.14}$$
$$\phi_{\nu_\mu} \propto 64 \cdot \frac{L}{E} \cdot \frac{1}{T_x}, \tag{I.15}$$
$$\phi_{\nu_\tau} \propto 91.125 \cdot \frac{L}{E} \cdot \frac{1}{T_x}. \tag{I.16}$$

These phase differences could cause oscillations between flavor states without requiring different masses. The exact form of the oscillation probability requires further development but remains highly speculative.
**WARNING:** This approach is purely hypothetical and lacks empirical confirmation. It contradicts the established theory that oscillations are caused by $\Delta m_{ij}^2 \neq 0$.

## I.4 Fundamental Constants and Units

### Base Parameters

**T0 Base Constants:**

$$\xi = \frac{4}{3} \times 10^{-4} \approx 1.333333 \times 10^{-4} \quad \text{[dimensionless]} \tag{I.17}$$

$$\frac{\xi^2}{2} = \frac{\left(\frac{4}{3} \times 10^{-4}\right)^2}{2} \approx 8.888888 \times 10^{-9} \quad \text{[dimensionless]} \tag{I.18}$$

$$v = 246.22 \text{ GeV} \quad \text{[Higgs VEV]} \tag{I.19}$$

$$\hbar c = 0.19733 \text{ GeV·fm} \quad \text{[Conversion constant]} \tag{I.20}$$

$$T_x = \frac{1}{4.54 \times 10^{-3} \text{ eV}} \approx 2.2026 \times 10^2 \text{ eV}^{-1} \approx 1.449 \times 10^{-13} \text{ s} \quad \text{[T0 Mass]} \tag{I.21}$$

### Unit Conventions

**Important**

**Consistent Unit Hierarchy:**

$$\text{Standard:} \quad \text{GeV} \tag{I.22}$$

$$\text{Submultiples:} \quad 1 \text{ eV} = 10^{-9} \text{ GeV} \tag{I.23}$$

$$1 \text{ meV} = 10^{-12} \text{ GeV} = 10^{-3} \text{ eV} \tag{I.24}$$

$$\text{Masses:} \quad m[\text{GeV}/c^2] = E[\text{GeV}]/c^2 \approx E[\text{GeV}] \text{ (natural units)} \tag{I.25}$$

$$\text{Time:} \quad 1 \text{ eV}^{-1} \approx 6.582 \times 10^{-16} \text{ s} \tag{I.26}$$

## I.5 Charged Lepton Reference Masses

### Precise Experimental Values (PDG 2024)

**Experimental**

**Verified Particle Masses:**

$$m_e = 0.51099895000 \times 10^{-3} \text{ GeV} = 510.99895 \text{ keV} \quad \text{(I.27)}$$
$$m_\mu = 105.6583745 \times 10^{-3} \text{ GeV} = 105.6583745 \text{ MeV} \quad \text{(I.28)}$$
$$m_\tau = 1776.86 \times 10^{-3} \text{ GeV} = 1.77686 \text{ GeV} \quad \text{(I.29)}$$

**Unit Conversion to eV:**

$$m_e = 510998.95 \text{ eV} = 510998950 \text{ meV} \quad \text{(I.30)}$$
$$m_\mu = 105658374.5 \text{ eV} \quad \text{(I.31)}$$
$$m_\tau = 1776860000 \text{ eV} \quad \text{(I.32)}$$

## I.6 Neutrino Quantum Numbers (T0 Hypothesis)

### Postulated Quantum Number Assignment

**Speculation**

**Hypothetical Neutrino Quantum Numbers:**

$$\nu_e : \quad n = 1, \ell = 0, j = 1/2 \quad \text{[Ground state neutrino]} \quad \text{(I.33)}$$
$$\nu_\mu : \quad n = 2, \ell = 1, j = 1/2 \quad \text{[First excitation]} \quad \text{(I.34)}$$
$$\nu_\tau : \quad n = 3, \ell = 2, j = 1/2 \quad \text{[Second excitation]} \quad \text{(I.35)}$$

**Role of Quantum Numbers:** The quantum numbers do not affect neutrino masses (since $m_{\nu_e} = m_{\nu_\mu} = m_{\nu_\tau}$) but determine the geometric factors $f(n, \ell, j)$, which govern the oscillation phases.
**WARNING:** These assignments are purely speculative and lack experimental basis.

### Geometric Factors

**T0 Geometric Factors:**

$$f(n, \ell, j) = \frac{n^6}{\ell^3} \quad \text{for } \ell > 0 \quad \text{(I.36)}$$
$$f(1, 0, j) = 1 \quad \text{for } \ell = 0 \text{ (special case)} \quad \text{(I.37)}$$

**Calculated Values:**

$$f_{\nu_e} = f(1, 0, 1/2) = 1 \tag{I.38}$$

$$f_{\nu_\mu} = f(2, 1, 1/2) = \frac{2^6}{1^3} = 64 \tag{I.39}$$

$$f_{\nu_\tau} = f(3, 2, 1/2) = \frac{3^6}{2^3} = \frac{729}{8} = 91.125 \tag{I.40}$$

## I.7 Neutrino Mass Formula

### T0 Hypothesis: Equal Masses with Geometric Phases

**Speculation**

**T0 Hypothesis: Equal Neutrino Masses with Geometric Phases**
The T0 model postulates that all flavor states ($\nu_e, \nu_\mu, \nu_\tau$) have the same mass:

$$m_{\nu_e} = m_{\nu_\mu} = m_{\nu_\tau} = m_\nu = 4.54 \text{ meV}.$$

The mass is derived from the photon analogy:

$$m_\nu = \frac{\xi^2}{2} \times m_e = \left(8.888888 \times 10^{-9}\right) \times (0.51099895 \times 10^{-3} \text{ GeV}) = 4.54 \text{ meV}.$$

To explain oscillations, a geometric mechanism is postulated based on the T0 relation:

$$T_x \cdot m_x = 1, \quad m_x = 4.54 \text{ meV}, \quad T_x \approx 2.2026 \times 10^2 \text{ eV}^{-1} \approx 1.449 \times 10^{-13} \text{ s}.$$

The oscillation phases are determined by geometric factors $f(n, \ell, j)$:

$$\phi_{\text{geo},i} \propto f_{\nu_i} \cdot \frac{L}{E} \cdot \frac{1}{T_x},$$

where $f_{\nu_e} = 1$, $f_{\nu_\mu} = 64$, $f_{\nu_\tau} = 91.125$.
**Rationale:**

- The mass $4.54$ meV is consistent with the cosmological constraint ($\Sigma m_\nu = 0.01362$ eV $< 0.07$ eV).
- Geometric phases enable oscillations without mass differences, supporting the equal-mass hypothesis.
- This hypothesis is highly speculative and lacks empirical confirmation.

**Formula:** $m_{\nu_i} = 4.54$ meV

**Total Mass:**

$$\Sigma m_\nu = 3 \times 4.54 \text{ meV} = 13.62 \text{ meV} = 0.01362 \text{ eV}$$

**Comparison with Plausible Target Value:**

- $\nu_e, \nu_\mu, \nu_\tau$: 4.54 meV vs. 15 meV (Agreement: 30.3%)
- $\Sigma m_\nu$: 13.62 meV vs. 45 meV (Deviation: Factor $\approx 3.30$)

**Warning**

**CRITICAL FINDING:** The hypothesis of equal masses with geometric phases is incompatible with experimental oscillation data ($\Delta m^2_{21} \approx 7.53 \times 10^{-5}$ eV$^2$, $\Delta m^2_{32} \approx 2.44 \times 10^{-3}$ eV$^2$), as it implies $\Delta m^2_{ij} = 0$. The geometric approach is purely speculative and requires further theoretical and experimental validation.

## I.8 Plausible Target Value Based on Empirical Data

### Derivation from Measurements

**Experimental**

**Plausible Target Value:** The T0 model postulates equal masses for all flavor states ($\nu_e, \nu_\mu, \nu_\tau$). Thus, a single target value for the neutrino mass $m_\nu$ is derived based on empirical data (as of 2025):

- Cosmological Constraint: $\Sigma m_\nu = 3m_\nu < 0.07$ eV $\implies m_\nu < 23.33$ meV.
- Oscillation Data: $\Delta m^2_{21} \approx 7.53 \times 10^{-5}$ eV$^2$, $\Delta m^2_{32} \approx 2.44 \times 10^{-3}$ eV$^2$, typically requiring different masses. The T0 model bypasses this via geometric phases.
- Plausible Target Value: $m_\nu \approx 15$ meV, lying between the solar (8.68 meV) and atmospheric scales (50.15 meV) and satisfying the cosmological constraint:

$$\Sigma m_\nu = 3 \times 15 \text{ meV} = 45 \text{ meV} = 0.045 \text{ eV} < 0.07 \text{ eV}.$$

**Rationale:**

- The target value is consistent with the cosmological upper bound and lies within the order of magnitude of oscillation data; the sum $\Sigma m_\nu = 45$ meV, however, lies below the oscillation lower bound $\Sigma m_\nu \gtrsim 0.06$ eV.
- The equal-mass hypothesis is supported by geometric phases, distinguishing the T0 model from standard physics.
- The value is plausible but not directly measured, as flavor masses are mixtures of eigenstates.
- The T0 mass (4.54 meV) is below the target value (30.3%) but also cosmologically consistent.

## I.9 Experimental Comparison

### Current Experimental Upper Limits (2025)

**Experimental**

**Experimental Limits:**

$$m_{\nu_e} < 0.45 \text{ eV} \quad \text{[KATRIN, 90\% CL]} \tag{I.41}$$
$$m_{\nu_\mu} < 0.17 \text{ MeV} \quad \text{[Muon decay, indirect]} \tag{I.42}$$
$$m_{\nu_\tau} < 18.2 \text{ MeV} \quad \text{[Tau decay, indirect]} \tag{I.43}$$
$$\Sigma m_\nu < 0.07 \text{ eV} \quad \text{[DESI+Planck, 95\% CL]} \tag{I.44}$$
$$\Delta m^2_{21} \approx 7.53 \times 10^{-5} \text{ eV}^2 \quad \text{[Solar]} \tag{I.45}$$
$$\Delta m^2_{32} \approx 2.44 \times 10^{-3} \text{ eV}^2 \quad \text{[Atmospheric]} \tag{I.46}$$
$$\Sigma m_\nu \gtrsim 0.06 \text{ eV} \quad \text{[lower bound; heaviest eigenstate } \gtrsim 0.05 \text{ eV]} \tag{I.47}$$

### Safety Margins for T0 Hypothesis

**Table I.1:** Safety Margins of the T0 Hypothesis Against Experimental Limits

| Parameter | T0 Mass (4.54 meV) | Target Value (15 meV) |
|---|---|---|
| $m_{\nu_e}$ vs 0.45 eV | 99.1× | 30× |
| $m_{\nu_\mu}$ vs 0.17 MeV | 3.74E7× | 1.13E7× |
| $m_{\nu_\tau}$ vs 18.2 MeV | 4.01E9× | 1.21E9× |
| $\Sigma m_\nu$ vs 0.07 eV | 5.14× | 1.56× |
| $\Sigma m_\nu$ vs 0.06 eV (lower bound) | 4.41× below | 1.33× below |

The last row is not a safety margin: $0.06$ eV is a lower bound on $\Sigma m_\nu$ (oscillation minimum $\approx 0.059$ eV for normal ordering). $\Sigma m_\nu = 13.62$ meV lies below it by the factor $60/13.62 = 4.41$, the target value $45$ meV by $1.33$; both sums are excluded by the oscillation data.

**Important**

**T0 Hypothesis:**

- The T0 mass ($4.54$ meV) is consistent with cosmological constraints ($\Sigma m_\nu = 0.01362$ eV $< 0.07$ eV) and lies below the target value ($15$ meV, $30.3\%$).
- Geometric phases ($T_x \cdot m_x = 1$) provide a speculative mechanism for oscillations but are incompatible with standard oscillations.

- Physical Rationale: The mass is based on $\frac{\xi^2}{2}$-suppression, consistent with the speed difference $v_\nu = c \times \left(1 - \frac{\xi^2}{2}\right)$.

## I.10 Consistency Checks and Validation

### Dimensional Analysis

**Dimensional Consistency:**

$$[\xi] = 1 \quad \checkmark \text{ dimensionless} \tag{I.48}$$
$$[m_e] = \text{GeV} \quad \checkmark \text{ energy/mass} \tag{I.49}$$
$$\left[\frac{\xi^2}{2} \times m_e\right] = \text{GeV} \quad \checkmark \text{ energy/mass} \tag{I.50}$$
$$[f_{\nu_i}] = 1 \quad \checkmark \text{ dimensionless} \tag{I.51}$$
$$[m_\nu] = \text{eV} \quad \checkmark \text{ (fixed mass)} \tag{I.52}$$
$$[T_x] = \text{eV}^{-1} \quad \checkmark \text{ (time)} \tag{I.53}$$

All formulas are dimensionally consistent.

### Mathematical Consistency

**Important**

**Consistency of the Hypothesis:**

- The formula $m_\nu = \frac{\xi^2}{2} \times m_e = 4.54$ meV is physically grounded in the photon analogy and consistent with the speed difference.
- Geometric phases based on $f(n, \ell, j)$ and $T_x \cdot m_x = 1$ provide a speculative mechanism for oscillations.
- No free parameters except $\xi$, simplifying the theory.

### Experimental Validation

**Experimental**

**Validation Status (as of 2025):**

- The T0 mass (4.54 meV) satisfies cosmological constraints ($\Sigma m_\nu = 0.01362$ eV $< 0.07$ eV) and is close to the target value (15 meV, 30.3%).
- Incompatible with standard oscillations ($\Delta m^2_{ij} = 0$), but geometric phases offer a speculative workaround.

- The target value ($15$ meV) is consistent with the cosmological upper bound but not directly measured; its sum ($45$ meV), like the T0 sum ($13.62$ meV), lies below the oscillation lower bound $\Sigma m_\nu \gtrsim 0.06$ eV and is excluded by the oscillation data.

# Chapter J

# Anomalous Magnetic Moments in FFGFT Theory

## Abstract

In the present work, the fundamental architecture of spacetime is reinterpreted within the framework of **Fundamental Fractal Geometric Field Theory (FFGFT)** – internally referred to as the T0 model (B18). The central paradigm consists in the transition from a point-like to a purely geometric description of the vacuum as a four-dimensional **Gyral Torus**.

**Geometric Structure:** The theory is based on the fractal-geometric foundation with the parameter $\xi \approx (4/3) \times 10^{-4}$ and the densest local sphere packing by regular **Tetrahedra**. This tetrahedral basis forms the stable foundation for the low generations (electron, muon, proton/neutron) as well as the local 3D crystal structure of the torus. Building upon this, the ideal sub-Planck factor

$$f = 7500,$$

emerges through fractal branching and pentagonal symmetry breaking, representing an exactly 7500-fold reduction compared to the conventional Planck scale ($t_0$) and following directly from the geometric winding density $30000/4$.

**g-2 Anomaly:** A core element of the work is the transparent geometric derivation of the anomalous magnetic moments of leptons. While the Standard Model relies on numerous perturbative terms, in FFGFT the electron anomaly follows directly from the base winding (tetrahedral projection). The muon and tau anomalies arise from fractal branchings with Hausdorff dimensions $p \approx 5/3$ and $4/3$, respectively. With the ideal value $f = 7500$ and the geometric projection factor $k_{\text{geom}} = (2/\sqrt{\varphi})\sqrt{2} = 2.224$, the purely geometric predictions achieve an accuracy of about 2 %. Through the **fractal correction** $K_{\text{frak}}^{3/2}$, this systematic deviation is captured geometrically (the exponent $3/2$ is a motivated choice): the effective projection factor $k_{\text{eff}} = k_{\text{geom}}/K_{\text{frak}}^{3/2} = 2.2688$ agrees with the experimentally reconstructed value $k_{\text{rec}} = 2.2696$ to 0.03 %. The corrected absolute values achieve 0.03% (electron) and 0.13% (muon) accuracy. The most precise, $k_{\text{geom}}$-independent prediction for the tau anomaly is

$$a_\tau \approx 1.282 \times 10^{-3},$$

which follows exclusively from the exact ratio $f^{1/3} - 1$.

**Geometric Proportionality:** All physical base quantities (constants, masses, couplings) stand in fixed geometric ratios, drastically reducing the number of free parameters compared to the Standard Model. The T0 theory thus offers a geometric description and provides concrete, experimentally testable predictions – particularly for the tau anomaly as a decisive test at Belle II.

**Note on Older Documents**

Previous versions of the g-2 analysis (018_T0_Anomale-g2-9/10) used semi-empirical factors or the ideal $k_{\text{geom}}$ without fractal correction. The present Rev. 11 shows that the 2 % deviation of absolute values **can be traced back to the fractal correction** $K_{\text{frak}}^{3/2}$ – the effective projection factor $k_{\text{eff}} = k_{\text{geom}}/K_{\text{frak}}^{3/2}$ agrees with the experimentally reconstructed value to 0.03 %; a residual of this size remains.

**Keywords:** Anomalous magnetic moment, g-2, T0 theory, Time-Mass Duality, Torsion lattice, Ratio predictions, Koide formula

## J.1 Introduction: Geometric vs. Semi-Empirical Approaches

### The Philosophy of T0 Theory

The T0 theory is based on the principle that **all** physical constants should follow from the geometric structure of a 4-dimensional torsion lattice. For anomalous magnetic moments this means:

- **NO** hidden fit parameters
- **ONLY** geometric factors: $\varphi$, $\xi$, $f$
- Honesty about precision limits
- Consistency with other predictions

### Consistency with Mass Predictions

The T0 theory predicts lepton masses with 0.4–1.2% deviation:

| Lepton | T0 [MeV] | Exp [MeV] | Deviation |
|---|---|---|---|
| Electron | 0.505 | 0.511 | 1.18% |
| Muon | 105.0 | 105.7 | 0.66% |
| Tau | 1783 | 1777 | 0.37% |

**Table J.1:** Lepton masses in T0

**Expectation:** g-2 should have similar precision ( 1%).

With the fractal correction $K_{\text{frak}}^{3/2}$, the g-2 predictions achieve 0.03–0.13% accuracy, comparable to the mass precision.

## J.2 Physical Fundamentals

### What is the Anomalous Magnetic Moment?

The magnetic moment of a charged spin-1/2 particle is:

$$\mu = g \cdot \frac{e}{2m} \cdot \frac{\hbar}{2} \tag{J.1}$$

where $g$ is the gyromagnetic factor (g-factor).
**Dirac Prediction:** For a point-like particle: $g = 2$
**Quantum Effects:** Vacuum polarization, vertex corrections $\Rightarrow g \neq 2$
**Anomaly:** $a = (g-2)/2$
**QED Expectation:** $a \approx \alpha/(2\pi) + \mathcal{O}(\alpha^2) \approx 0.00116$

### T0 Interpretation: Windings in the Torsion Lattice

In T0 theory, leptons are **winding structures** in the 4D torsion lattice:

- **Electron:** Simple winding (1st generation)
- **Muon:** Winding with fractal branching (2nd generation)
- **Tau:** More complex fractal structure (3rd generation)

The anomalous moment arises from:

1. The **rotation** of the winding (spin)
2. The **charge distribution** on the winding
3. The **projection** 4D $\rightarrow$ 3D

$\Rightarrow$ **No** point-like charge $\Rightarrow a \neq 0$

## J.3 Geometric Formulas

### Fundamental Parameters

The T0 theory uses exclusively three geometric fundamental constants:

$$\varphi = \frac{1+\sqrt{5}}{2} = 1.618\ldots \quad \text{(Golden Ratio)} \tag{J.2}$$

$$\xi = \frac{4}{3} \times 10^{-4} = 1.333 \times 10^{-4} \quad \text{(Torsion constant)} \tag{J.3}$$

$$f = 7500 \quad \text{(Sub-Planck factor)} \tag{J.4}$$

**On notation:** here $f$ is the base winding number $f = 1/\xi = 7500$, not a frequency. In natural units $f$ is only another reading of the duality $T \cdot m = 1$: $f = t_P/t_0 = \Lambda_{\text{T0}}/E_P$ with

$t_0 = \xi t_P$ and $\Lambda_{T0} = E_P/\xi$, and $\xi E_0^2 = 1$ gives $f = E_0^2$. As a dimensionless ratio, $f$ cannot be set to 1 with justification (Doc. A135).

### The Real Sub-Planck Factor: $f = 7500$

Now we put everything together: The ideal crystal remains intact, the symmetry breaking only affects the projection factors:

$$\boxed{f = 7500} \tag{J.5}$$

This is the **most fundamental number of the T0 theory**. It appears in almost all formulas and describes:

- The number of Sub-Planck cells per Planck length
- The density of the torsion lattice
- The base winding number of all geometric resonances (dimensionless, not a frequency)

### The Symmetry Breaking: The Role of the Golden Ratio

A perfect, ideal crystal would be completely symmetric. Yet our world shows symmetry breaking on all levels:

- Matter dominates over antimatter
- The weak interaction violates parity symmetry
- The neutron is heavier than the proton
- The three lepton generations have different masses

In the T0 theory, all these symmetry breakings have a single, geometric origin: the pentagonal symmetry of the crystal, embodied by the **golden ratio** $\varphi$. The golden ratio $\varphi = (1 + \sqrt{5})/2 = 1.618033989\ldots$ is the irrational number describing pentagonal symmetry. In a perfect pentagon, $\varphi$ appears everywhere: The ratio of diagonal to side is exactly $\varphi$. Why pentagonal symmetry specifically? For deep mathematical reasons, pentagonal symmetry is the first one that **cannot tile the plane periodically**. This leads to *quasicrystals* – structures that are ordered but not periodic. Exactly such a quasicrystalline structure is postulated by T0 theory for the Sub-Planck scale. The symmetry breaking is quantified in the theory not by a direct subtraction of $5\varphi$ from the ideal anchor number 7500. Instead, it manifests in the **fractal correction** $K_{\text{frak}}^{3/2}$, which corrects the ideal geometric projection factor $k_{\text{geom}}$ to the effective value $k_{\text{eff}}$:

$$k_{\text{geom}} = \frac{2}{\sqrt{\varphi}} \times \sqrt{2} \approx 2.22357, \tag{J.6}$$

which projects the 4D torsion onto the 3D world. The **fractal correction** yields the effective projection factor:

$$k_{\text{eff}} = \frac{k_{\text{geom}}}{K_{\text{frak}}^{3/2}} = \frac{2.2236}{0.98007} = 2.2688 \tag{J.7}$$

The exponent $3/2$ corresponds to the half spatial dimension ($D/2 = 3/2$) and describes how the fractal structure $K_{\text{frak}} = 1 - 100\xi = 0.98667$ acts across three spatial dimensions on the projection.

**Crucially:** The experimentally reconstructed value $k_{\text{geom}}^{\text{rec}} = 2.26955$ agrees with $k_{\text{eff}} = 2.2688$ to **0.033%** – the 2% deviation of the ideal formula is thus explained by the fractal correction up to this residual of 0.033%. The calculation uses the unrounded value $k_{\text{geom}} = 2.22357$.

> From the ideal 7500 remained the ideal 7500. This number became the new fundamental constant of the universe. It determined how densely the lattice was packed, how quickly torsion could propagate, which resonances were possible. In the T0 picture the observed particle masses and force strengths are meant to go back to this single geometric story, and likewise – speculatively, the cosmic sector is deliberately set aside in the current corpus (P39) – the cosmological constants: From perfect crystal to pentagonally broken reality, with the breaking quantitatively captured by $K_{\text{frak}}^{3/2}$.

## Electron: Base Winding

**Formula:**

$$a_e = \frac{S_3/f}{k_{\text{geom}}} \tag{J.8}$$

where:

- $S_3 = 2\pi^2 = 19.739$: 3D surface of the 4D winding
- $f = 7500$: Sub-Planck scaling
- $k_{\text{geom}}$: Geometric projection factor

**Geometric Projection Factor:**

$$k_{\text{geom}} = \frac{2}{\sqrt{\varphi}} \times \sqrt{2} \tag{J.9}$$

**Explanation of Factors:**

- $2/\sqrt{\varphi} = 1.572$: Pentagonal projection (from $\xi$-structure)
- $\sqrt{2} = 1.414$: Diagonal projection 4D → 3D
- $k_{\text{geom}} = 2.224$: Completely geometric!

**Numerical Calculation:**

$$k_{\text{geom}} = \frac{2}{\sqrt{1.618}} \times \sqrt{2} = 2.224 \tag{J.10}$$

$$a_e = \frac{19.739/7500}{2.224} \tag{J.11}$$

$$a_e = 1.184 \times 10^{-3} \tag{J.12}$$

**Comparison (ideal zeroth order):**

- T0 (ideal): $a_e = 1.184 \times 10^{-3}$

- Experiment: $a_e = 1.160 \times 10^{-3}$
- Deviation: **2.07%**

**With fractal correction $K_{\text{frak}}^{3/2}$:**

$$k_{\text{eff}} = \frac{k_{\text{geom}}}{K_{\text{frak}}^{3/2}} = \frac{2.2236}{0.98007} = 2.2688 \tag{J.13}$$

$$a_e^{(\text{corr})} = \frac{S_3/f}{k_{\text{eff}}} = \frac{19.739/7500}{2.2688} = 1.1600 \times 10^{-3} \tag{J.14}$$

- T0 (corrected): $a_e = 1.1600 \times 10^{-3}$
- Experiment: $a_e = 1.1597 \times 10^{-3}$
- Deviation: **0.033%**

## Muon: Fractal Additional Winding

**Formula:**

$$a_\mu = a_e + \Delta a_{\text{fractal}} \tag{J.15}$$

with

$$\Delta a_{\text{fractal}} = \frac{4\pi}{f^{p_\mu}} \tag{J.16}$$

where:

- $p_\mu = 5/3$: Fractal Hausdorff dimension
- $4\pi$: Complete torsion revolution

**Meaning of $p_\mu = 5/3$:**
This is the well-known Hausdorff dimension of:

- the self-avoiding random walk in 3D (Flory value $1/\nu = 5/3$); the exponent itself is a setting

⇒ Physically plausible for "partially branched winding"!
**Numerical Calculation:**

$$\Delta a_{\text{fractal}} = \frac{4\pi}{7500^{5/3}} = 4.373 \times 10^{-6} \tag{J.17}$$

$$a_\mu = 1.184 \times 10^{-3} + 4.373 \times 10^{-6} \tag{J.18}$$

$$a_\mu = 1.188 \times 10^{-3} \tag{J.19}$$

**Comparison (ideal zeroth order):**

- T0 (ideal): $a_\mu = 1.188 \times 10^{-3}$
- Experiment: $a_\mu = 1.166 \times 10^{-3}$
- Deviation: **1.89%**

**With fractal correction:**

$$a_\mu^{(\text{corr})} = a_e^{(\text{corr})} + \Delta a_{\text{fractal}} = 1.1600 \times 10^{-3} + 4.373 \times 10^{-6} \tag{J.20}$$

$$a_\mu^{(\text{corr})} = 1.1644 \times 10^{-3} \tag{J.21}$$

- T0 (corrected): $a_\mu = 1.1644 \times 10^{-3}$
- Experiment: $a_\mu = 1.1659 \times 10^{-3}$
- Deviation: **0.13%**

### Tau: More Complex Fractal Structure

**Formula:**

$$a_\tau = a_e + \frac{4\pi}{f^{p_\tau}} \tag{J.22}$$

where:

- $p_\tau = 4/3$: Stronger fractal branching

**Meaning of $p_\tau = 4/3$:**

This is the Hausdorff dimension of the frontier of planar Brownian motion and of the planar self-avoiding walk; this exponent, too, is a setting.

**Numerical Calculation:**

$$\Delta a_{\text{fractal}} = \frac{4\pi}{7500^{4/3}} = 8.560 \times 10^{-5} \tag{J.23}$$

$$a_\tau = 1.184 \times 10^{-3} + 8.560 \times 10^{-5} \tag{J.24}$$

$$a_\tau = 1.269 \times 10^{-3} \tag{J.25}$$

**Status:** This is a **prediction** – tau-g-2 has not been measured yet!

**Corrected prediction:**

$$a_\tau^{(\text{corr})} = a_e^{(\text{corr})} + \Delta a_{\text{fractal}} = 1.1600 \times 10^{-3} + 8.560 \times 10^{-5} \tag{J.26}$$

$$a_\tau^{(\text{corr})} = 1.2456 \times 10^{-3} \tag{J.27}$$

## J.4 Two Classes of Predictions: Absolute Values vs. Ratios

### Why 2% Deviation for Absolute Values?

### Where do the 2% in the ideal formula come from?

The 2% deviation of the ideal formula (with $k_{\text{geom}} = 2.224$) can be traced back to the fractal correction up to a residual of 0.033%:

$$k_{\text{eff}} = \frac{k_{\text{geom}}}{K_{\text{frak}}^{3/2}} = \frac{2.224}{(1 - 100\xi)^{3/2}} = \frac{2.2236}{0.98007} = 2.2688 \tag{J.28}$$

**Physical meaning:**

- $K_{\text{frak}} = 1 - 100\xi = 0.98667$ is the universal fractal correction of T0 theory
- The exponent $3/2 = D/2$ (half spatial dimension) describes the effect of fractal structure on the 4D→3D projection
- This correction is **not a fit**: $K_{\text{frak}}$ follows directly from $\xi$; the exponent $3/2$ is a geometrically motivated choice

- The experimentally reconstructed value $k_{\text{rec}} = 2.2696$ confirms the correction to 0.033%

**Consistency:** The same fractal correction $K_{\text{frak}}$ appears in:

1. Gravitational constant: $G = f(\xi, K_{\text{frak}})$ (a general dependence that is not written out; here $f$ is not a frequency)
2. g-2 anomalies: $k_{\text{eff}} = k_{\text{geom}}/K_{\text{frak}}^{3/2}$

The lepton mass ladder $m_i = r_i \cdot \xi^{p_i} \cdot v$, by contrast, does not contain $K_{\text{frak}}$ (Doc. 006, A100); with it, $m_\mu = 103.56$ instead of $104.96$ MeV would follow.

This gives a unified picture: **one** correction $K_{\text{frak}}$ following from $\xi$, with **different** exponents depending on the physical context.

## Ratios are Mathematically Exact

In contrast to absolute values, **ratios of differences** are structurally exact:

$$\frac{\Delta a(\tau - \mu)}{\Delta a(\mu - e)} = \frac{4\pi/f^{4/3} - 4\pi/f^{5/3}}{4\pi/f^{5/3}} = f^{1/3} - 1 \tag{J.29}$$

**Why is this exact?**

- The common factor $4\pi$ cancels out
- The projection factor $k_{\text{geom}}$ cancels out
- Only the fractal exponents ($5/3$ and $4/3$) determine the ratio
- The result depends **only** on $f$: $f^{1/3} - 1 = 18.57$

**Important**

Fundamental Distinction **Absolute values:**

- Depend on $k_{\text{geom}}$, $f$, and SI conversion
- 2% deviation due to higher-order quantum effects
- Consistent with all T0 predictions

**Ratios:**

- Depend **only** on $f$
- $k_{\text{geom}}$ and SI factors cancel out
- Mathematically exact from fractal exponents
- Difference $< 10^{-13}$ (numerical precision)

⇒ The ratio prediction is **not an approximation**, but an **exact geometric relation**!

## Analogy to the Koide Formula

The Koide formula for lepton masses:

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = 0.666661 \approx \frac{2}{3} \tag{J.30}$$

confirmed to $0.001\%$, shows: **ratios** are more precise than absolute values! In T0 it emerges from the exponent structure $p_e = 3/2$, $p_\mu = 1$, $p_\tau = 2/3$ with step $\Delta p_{\mu\to\tau} = 1/3 = 1/D$ (the step $e \to \mu$ is $1/2$ — the special status of the electron).

The same $1/3$-step appears in the $g$-2 exponents: $q_\mu = 5/3$, $q_\tau = 4/3$, $\Delta q = 1/3$. This is no coincidence — it is the same geometric origin.

**The bridge from masses to $g$-2:** From the mass formulas follows $m_\tau/m_\mu = (125/144)\cdot f^{1/3}$, from the $g$-2 formulas $\Delta a(\tau{-}e)/\Delta a(\mu{-}e) = f^{1/3}$. Combining yields:

$$\boxed{\frac{\Delta a(\tau{-}e)}{\Delta a(\mu{-}e)} = \frac{144}{125}\cdot\frac{m_\tau}{m_\mu}} \tag{J.31}$$

The $g$-2 ratio is determined by the mass ratio. No $\alpha$, no $k_{\text{eff}}$, no $K_{\text{frak}}$ — everything cancels out.

**For g-2 in T0:**

- **Absolute values:** 2% deviation
- **Ratio $\Delta a(\tau-\mu)/\Delta a(\mu-e)$:** Exactly $= f^{1/3} - 1$
- **Bridge to masses:** $\Delta a(\tau{-}e)/\Delta a(\mu{-}e) = (144/125)\cdot m_\tau/m_\mu$

This is **not a weakness**, but shows the **geometric structure** of the theory!

## J.5 Precise Ratio Predictions

### Analogy to the Koide Formula

The Koide formula for lepton masses:

$$\frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3} \pm 0.0004\% \tag{J.32}$$

shows: **Ratios** are more precise than absolute values!
**Question:** Does this also hold for g-2?

### The Ratio of Differences

Define the differences:

$$\Delta a(\mu - e) = a_\mu - a_e = \frac{4\pi}{f^{5/3}} \tag{J.33}$$

$$\Delta a(\tau - \mu) = a_\tau - a_\mu = \frac{4\pi}{f^{4/3}} - \frac{4\pi}{f^{5/3}} \tag{J.34}$$

**Ratio:**

$$\frac{\Delta a(\tau-\mu)}{\Delta a(\mu-e)} = \frac{4\pi/f^{4/3} - 4\pi/f^{5/3}}{4\pi/f^{5/3}} \tag{J.35}$$

$$= \frac{f^{5/3}}{f^{4/3}} - 1 \tag{J.36}$$

$$= f^{5/3-4/3} - 1 \tag{J.37}$$

$$= f^{1/3} - 1 \tag{J.38}$$

**Important**

Core Prediction

$$\boxed{\frac{\Delta a(\tau-\mu)}{\Delta a(\mu-e)} = f^{1/3} - 1 = 18.57} \tag{J.39}$$

This relation is:

- **Parameter-free** (only $f$!)
- **Independent** of $k_{\text{geom}}$
- **Exact** (difference $< 10^{-13}$)
- **Testable** at Belle II

## Numerical Verification

With $f = 7500$:

$$f^{1/3} = 7500^{1/3} = 19.57 \tag{J.40}$$

$$f^{1/3} - 1 = 18.57 \tag{J.41}$$

From T0 values:

$$\Delta a(\mu-e) = 4.373 \times 10^{-6} \tag{J.42}$$

$$\Delta a(\tau-\mu) = 8.123 \times 10^{-5} \tag{J.43}$$

$$\text{Ratio} = \frac{8.123 \times 10^{-5}}{4.373 \times 10^{-6}} = 18.57 \tag{J.44}$$

**Agreement:** Perfect! ✓✓✓

## Testable Prediction for Tau

The prediction follows in one step from the bridge formula and three experimental numbers:

$$\Delta a(\mu-e)^{\text{exp}} = a_\mu - a_e = 6.269 \times 10^{-6} \tag{J.45}$$

$$\frac{m_\tau}{m_\mu} = 16.817 \tag{J.46}$$

**Prediction:**

$$\Delta a(\tau - e) = \frac{144}{125} \cdot \frac{m_\tau}{m_\mu} \cdot \Delta a(\mu - e)^{\text{exp}} \tag{J.47}$$

$$= 1.152 \times 16.817 \times 6.269 \times 10^{-6} = 1.214 \times 10^{-4} \tag{J.48}$$

$$a_\tau^{\text{predicted}} = a_e + \Delta a(\tau - e) \tag{J.49}$$

$$= 1.160 \times 10^{-3} + 1.214 \times 10^{-4} \tag{J.50}$$

$$= \boxed{1.281 \times 10^{-3}} \tag{J.51}$$

The value $1.282 \times 10^{-3}$ belongs to the $f^{1/3}$ route (see below and Doc. 158).

**That is all.** No $\alpha$, no $k_{\text{eff}}$, no $K_{\text{frak}}$. Only the bridge formula applied to measured values.

**Consistency check:** The SM calculates $a_\tau^{\text{SM}} = 1.177 \times 10^{-3}$, corresponding to a ratio $\Delta a(\tau\text{–}e)/\Delta a(\mu\text{–}e) = 2.80$. T0 predicts $19.4$ — a factor of $7$, clearly testable at Belle II.

## J.6 The Fractal Correction $K_{\text{frak}}^{3/2}$: Resolving the 2% Deviation

### Central Insight

The 2% deviation of the ideal g-2 formula **corresponds** to the fractal correction $K_{\text{frak}}^{3/2}$, which, with a different exponent, also enters the gravitational constant (the lepton mass ladder does not contain $K_{\text{frak}}$, Doc. 006, A100):

$$\boxed{k_{\text{eff}} = \frac{k_{\text{geom}}}{K_{\text{frak}}^{3/2}} = \frac{(2/\sqrt{\varphi})\sqrt{2}}{(1 - 100\xi)^{3/2}} = \frac{2.2236}{0.98007} = 2.2688} \tag{J.52}$$

The experimentally reconstructed value $k_{\text{rec}} = 2.2696$ agrees with this to **0.033%**.

### Why the Exponent $3/2$?

The exponent $3/2 = D/2$ has a clear geometric meaning:

- The fractal correction $K_{\text{frak}} = 1 - 100\xi$ describes the deviation of the real lattice from the ideal crystal structure
- In the 4D→3D projection ($k_{\text{geom}}$), this correction acts across $D = 3$ spatial dimensions
- Per dimension: $K_{\text{frak}}^{1/2}$ (square root of the 1D correction)
- Total: $K_{\text{frak}}^{D/2} = K_{\text{frak}}^{3/2}$

### Consistency with Other T0 Corrections

The different exponents are **not free parameters** but follow from the dimensionality of the respective physical process.

| Physical Quantity | $K_{\text{frak}}$ Exponent | Interpretation |
|---|---|---|
| g-2 projection factor | $K_{\text{frak}}^{3/2}$ | 3D spatial projection ($D/2$) |
| Gravitational constant | $K_{\text{frak}}^{1}$ | Scalar correction (1D) |

**Table J.2:** Universal fractal correction with context-dependent exponent

## J.7 The Bridge Formula: Three Numbers, One Prediction

The entire T0 complexity cancels out. The tau prediction needs **only**:

1. $a_e$ and $a_\mu$ (experimentally measured)
2. $m_\tau/m_\mu = 16.817$ (experimental, high-precision)
3. The rational factor $144/125$ from T0 geometry

### The Calculation

**Step 1:** Measured difference

$$\Delta a(\mu\text{–}e) = a_\mu - a_e = 6.269 \times 10^{-6} \tag{J.53}$$

**Step 2:** Apply bridge formula (Eq. J.31)

$$\frac{\Delta a(\tau\text{–}e)}{\Delta a(\mu\text{–}e)} = \frac{144}{125} \cdot \frac{m_\tau}{m_\mu} = 1.152 \times 16.817 = 19.37 \tag{J.54}$$

**Step 3:** Result

$$\Delta a(\tau\text{–}e) = 19.37 \times 6.269 \times 10^{-6} = 1.214 \times 10^{-4} \tag{J.55}$$

$$a_\tau = a_e + \Delta a(\tau\text{–}e) = \boxed{1.281 \times 10^{-3}} \tag{J.56}$$

**That is all.** No $\alpha$, no $k_{\text{eff}}$, no $K_{\text{frak}}$, no perturbation theory, no 12,672 Feynman diagrams.

### Why So Simple?

T0 theory gives $\Delta a_i = 4\pi/f^{q_i}$, from which $\Delta a(\tau\text{–}e)/\Delta a(\mu\text{–}e) = f^{1/3}$ follows. From the masses follows $m_\tau/m_\mu = (125/144) \cdot f^{1/3}$. Combining:

$$\frac{\Delta a(\tau\text{–}e)}{\Delta a(\mu\text{–}e)} = \frac{144}{125} \cdot \frac{m_\tau}{m_\mu} \tag{J.57}$$

The parameter $f$ cancels out. The prediction is **independent** of the exact value of $f$, $\xi$, $k_{\text{eff}}$, or any other T0 parameter. It follows solely from:

1. The existence of a $1/3$-step (Koide structure)
2. The fact that this step appears in both masses and $g$-2

### Falsifiability

Even those skeptical of T0 theory as a whole can test this one relation:

- Belle II measures $a_\tau \approx 1.18 \times 10^{-3}$: Relation **falsified**.
- Belle II measures $a_\tau \approx 1.28 \times 10^{-3}$: Koide-$g$-2 connection **confirmed** — independent of the rest of the theory.

## J.8 Experimental Tests

### Belle II (2027–2028)

Belle II expects sensitivity of $\sim 10^{-6}$ for $a_\tau$.

**Test 1: Absolute value**

- T0 prediction (corrected): $a_\tau = 1.246 \times 10^{-3}$
- T0 from bridge formula: $a_\tau = 1.281 \times 10^{-3}$
- SM: $a_\tau = 1.177 \times 10^{-3}$
- Difference T0 vs. SM: $9\%$
- Difference: 1%

**Test 2: Ratio**

- T0 prediction: $\Delta a(\tau - \mu)/\Delta a(\mu - e) = 18.57$
- This is the **more precise** prediction!
- Independent of absolute calibration

**Possible outcomes:**

1. **Confirmation:** Ratio $\approx 18.6$
   ⇒ Strong evidence for fractal structure hypothesis
2. **Deviation:** Ratio $\neq 18.6$
   ⇒ Different fractal dimensions or additional physics
3. **Null result:** $a_\tau < 10^{-8}$
   ⇒ T0 contributions suppressed or theory needs revision

### Fermilab/J-PARC

Further precision improvements for $a_\mu$:

- Reduction of experimental uncertainties
- Clearer determination of SM discrepancy
- Refinement of $\Delta a(\mu - e)$ measurement

## J.9 Comparison with Other Approaches

**T0 Philosophy:** We choose **explainability** over precision!

| Approach | Precision | Parameters | Explainable |
|---|---|---|---|
| QED (SM) | Perfect | Many | Yes |
| T0 (ideal, $k_{\text{geom}}$) | 2% | 0 adjusted | Geometric |
| T0 (with $K_{\text{frak}}^{3/2}$) | 0.03–0.13% | 0 adjusted | Geometric |

**Table J.3:** Comparison of different approaches

## J.10 Comparison: Experimental Reconstruction and Fractal Correction

### The Central Insight

The ratio $\Delta a(\tau-\mu)/\Delta a(\mu-e) = f^{1/3}-1$ is **mathematically exact** because the correction factor $k_{\text{geom}}$ cancels out completely.

In Rev. 10, the experimentally reconstructed value $k_{\text{rec}} = 2.269$ was considered "semi-empirical." Now it turns out: this value agrees with the fractal correction to $0.033\,\%$ ($k_{\text{eff}} = 2.2688$ against $k_{\text{rec}} = 2.2696$, with $a_e$ measured to about $10^{-10}$):

$$\boxed{k_{\text{eff}} = \frac{k_{\text{geom}}}{K_{\text{frak}}^{3/2}} = \frac{2.224}{(1-100\xi)^{3/2}} = \frac{2.2236}{0.98007} = 2.2688} \tag{J.58}$$

### Comparison: Reconstructed vs. Fractal Corrected

$$k_{\text{geom}}^{(\text{reconstructed})} = \frac{S_3/f}{a_e^{(\text{exp})}} = \frac{2\pi^2/7500}{1.15965\times10^{-3}} = 2.2696 \tag{J.59}$$

The value $2.2696$ belongs to the exact measured value $a_e = 1.15965\times10^{-3}$; the rounded value $1.160\times10^{-3}$ would give $2.2689$.

**Comparison:**

- Geometrically ideal: $k_{\text{geom}} = (2/\sqrt{\varphi})\times\sqrt{2} = 2.224$
- Fractal corrected: $k_{\text{eff}} = k_{\text{geom}}/K_{\text{frak}}^{3/2} = 2.2688$
- Reconstructed from experiment: $k_{\text{geom}}^{(\text{rec})} = 2.2696$
- **Difference $k_{\text{eff}}$ vs. $k_{\text{rec}}$: 0.033%**

This suggests that the "semi-empirical" reconstruction reflects the **geometric** fractal correction $K_{\text{frak}}^{3/2}$; the agreement is $0.033\,\%$.

| Lepton | Ideal ($k = 2.224$) | Corr. ($k_{\text{eff}} = 2.269$) | Experiment | Dev. |
|---|---|---|---|---|
| Electron | $1.184 \times 10^{-3}$ | $1.1600 \times 10^{-3}$ | $1.1597 \times 10^{-3}$ | **0.03%** |
| Muon | $1.188 \times 10^{-3}$ | $1.1644 \times 10^{-3}$ | $1.1659 \times 10^{-3}$ | **0.13%** |
| Tau | $1.269 \times 10^{-3}$ | $1.2456 \times 10^{-3}$ | (not measured) | Prediction |

**Table J.4:** Absolute values with geometric vs. fractal corrected $k_{\text{eff}}$

### Results with Fractal Correction

**Important**

Crucial Point The 2% deviation of the ideal formula is **largely** explained by $K_{\text{frak}}^{3/2}$:

- Electron: $2.07\% \to 0.033\%$ (factor 62 improvement)
- Muon: $1.89\% \to 0.13\%$ (factor 15 improvement)
- The fractal correction is **not a new parameter** but follows from $\xi = 4/30000$ (with the exponent $3/2$ as a motivated choice)

Thus the T0 g-2 calculation needs **no parameter fitted to g-2** (only $\xi$ and $f$ plus the stated choices) and reaches a precision of 0.03–0.13%; the remaining residual is not explained.

### Alternative: Directly from Ratio Relation

Even more precise is the calculation directly from the exact ratio:

$$\Delta a(\mu - e)^{(\text{exp})} = a_\mu^{(\text{exp})} - a_e^{(\text{exp})} = 6.269 \times 10^{-6} \tag{J.60}$$

$$\Delta a(\tau - \mu) = \Delta a(\mu - e)^{(\text{exp})} \times (f^{1/3} - 1) \tag{J.61}$$

$$= 6.269 \times 10^{-6} \times 18.57 = 1.164 \times 10^{-4} \tag{J.62}$$

$$a_\tau^{(\text{Ratio})} = a_\mu^{(\text{exp})} + \Delta a(\tau - \mu) \tag{J.63}$$

$$= 1.166 \times 10^{-3} + 1.164 \times 10^{-4} \tag{J.64}$$

$$= \boxed{1.282 \times 10^{-3}} \tag{J.65}$$

**Note:** This prediction is **independent** of $k_{\text{geom}}$ and uses only the exact geometric ratio structure!

### Two Complementary Tau Predictions

### What does this mean for Belle II?

#### If Belle II measures:

1. $a_\tau \approx 1.28 \times 10^{-3}$:

| Method | $a_\tau$ Prediction | Dependent on |
|---|---|---|
| Purely geometric | $1.269 \times 10^{-3}$ | $k_{\text{geom}} = 2.224$ (geometric) |
| With $K_{\text{frak}}^{3/2}$ correction | $1.246 \times 10^{-3}$ | $k_{\text{eff}} = 2.269$ (geometrically derived) |
| From ratio/bridge | $1.282 \times 10^{-3}$ | Only $f$ and exp. masses (exact) |
| Range | $1.25–1.28 \times 10^{-3}$ | $\pm 1.5\%$ |

**Table J.5:** Three T0 predictions for $a_\tau$

- ✓ Confirms the exact ratio relation $f^{1/3} - 1$
- ✓ Shows that experimental $a_\mu$ and ratio structure are correct
- → **Strongest confirmation of T0 geometry**

2. $a_\tau \approx 1.25 \times 10^{-3}$**:**
   - ✓ Confirms fractal correction $k_{\text{eff}} = k_{\text{geom}}/K_{\text{frak}}^{3/2}$
   - ✓ Supports tracing the 2% deviation back to $K_{\text{frak}}^{3/2}$
   - → **Strongest confirmation of fractal T0 geometry**
3. $a_\tau \approx 1.27–1.28 \times 10^{-3}$**:**
   - ✓ Confirms exact ratio relation $f^{1/3} - 1$
   - ? Fractal correction at tau with different exponent?
4. $a_\tau$ **outside** $1.24–1.28$**:**
   - × T0 structure needs revision

**Key Statement**

The 2% deviation of the purely geometric T0 predictions is **largely** explained by the fractal correction $K_{\text{frak}}^{3/2}$:

- Electron: $2.07\% \to 0.033\%$
- Muon: $1.89\% \to 0.13\%$

The effective projection factor $k_{\text{eff}} = k_{\text{geom}}/K_{\text{frak}}^{3/2} = 2.2688$ is **not a new free parameter** but follows directly from $\xi = 4/30000$ (exponent $3/2$ as a motivated choice).

The most precise T0 prediction for tau uses the exact ratio relation:

$$\boxed{a_\tau = 1.282 \times 10^{-3}} \tag{J.66}$$

## J.11 Important Note: No $\alpha$ in the T0 g-2 Formulas

**IMPORTANT:** The T0 formulas for g-2 contain **no** $\alpha$!

In natural units ($\hbar = c = \alpha = 1$):

$$a_\ell = f(\varphi, \xi, f, \text{generation quantum numbers})$$

The anomalous moment is a **purely geometric quantity**, following from the winding structure in the torsion lattice.

Ratios like $\Delta a(\tau - \mu)/\Delta a(\mu - e) = f^{1/3} - 1$ are **independent** of:

- $\alpha$ (fine-structure constant)
- SI conversion factors
- $k_{\text{geom}}$ (projection factor)

They depend ONLY on the fractal structure!

**T0 and $\alpha$:** While the SM takes the fine-structure constant from experiment ($\alpha_{\text{exp}} = 1/137.036$), T0 writes it as $\alpha = \xi \cdot E_0^2$; with $E_0 = 7.398$ MeV (from the lepton masses, see below) this gives $1/\alpha_{\text{T0}} = 137.035$, $0.0005\%$ from experiment. Since $E_0$ comes from measured masses, this agreement is a property of the anchor $E_0$, not an independent prediction.

### Fair Comparison: SM vs. T0

A common objection is that the SM computes $a_e$ to $10^{-12}$, while T0 achieves only $0.033\%$. This comparison is misleading:

- **SM and $\alpha$:** The SM takes $\alpha$ from atom-recoil measurements (Cs 2018, Rb 2020), independent of $a_e$; $a_e(\text{SM})$ is thus a genuine prediction and matches the measurement to about $10^{-12}$.
- **T0 and $\alpha$:** $\alpha = \xi E_0^2$ with $E_0 = \sqrt{m_e m_\mu / K_{\text{frak}}}$; $E_0$ thus comes from the measured $m_e$, $m_\mu$. The $0.0005\%$ deviation is therefore a property of this anchor, not an independent prediction.
- **Where the theories diverge:** Not at $e$ or $\mu$, but at the **tau**. T0: $a_\tau \approx 1.282 \times 10^{-3}$, SM: $a_\tau \approx 1.177 \times 10^{-3}$ — $9\%$ difference, testable at Belle II.

## Further Reading and Resources

**T0 Theory and Python Scripts:**

- Repository: github.com/jpascher/T0-Time-Mass-Duality
- Python scripts: github.com/jpascher/T0-Time-Mass-Duality/blob/main/2/python/
- Time-Mass Duality documentation
- Fundamental Fractal Geometric Field Theory (FFGFT)

**Experimental Results:**

- Fermilab Muon g-2 (2025): muon-g-2.fnal.gov
- Theory Initiative White Paper
- Belle II: www.belle2.org

**Related T0 Documents:**

- Lepton masses: Systematic derivation from quantum numbers
- Koide formula in T0: Geometric interpretation
- Fractal spacetime: $D_f = 3 - \xi$

# Chapter K

# The Mass Scaling Exponent $\kappa$


## Abstract

This work resolves the circularity problem in the derivation of $\xi = \frac{4}{30000}$ by introducing the mass scaling exponent $\kappa$ and provides the fundamental justification for the $10^{-4}$ scaling. We show that $\kappa = 7$ for the proton-electron ratio is not fitted but emerges from the self-consistent structure of the e-p-$\mu$ system. The $10^{-4}$ scaling is explained as a fundamental consequence of the fractal spacetime dimensionality $D_f = 3 - \xi$ and the 4-dimensional nature of our universe.


## K.1 The Circularity Problem: An Honest Analysis

### The Legitimate Criticism

The original derivation of $\xi$ appears circular:

$$\frac{m_p}{m_e} = 245 \times \left(\frac{4}{3}\right)^7 \Rightarrow \xi = \frac{4}{30000} \tag{K.1}$$

**Criticism**: Why exactly $\kappa = 7$? Why $K = 245$? Doesn't this seem like reverse fitting?

### The Solution: $\kappa$ Emerges from the e-p-$\mu$ System

The answer lies in the **self-consistent structure** of the complete particle system:

**Key Insight**

The exponent $\kappa = 7$ is **not** fitted - it emerges as the **only consistent solution** for the complete e-p-$\mu$ triangle.

## K.2 The e-p-$\mu$ System as Proof

### The Three Fundamental Ratios

$$R_{pe} = \frac{m_p}{m_e} = 1836.15267343 \quad \text{(Proton-Electron)} \tag{K.2}$$

$$R_{\mu e} = \frac{m_\mu}{m_e} = 206.7682830 \quad \text{(Muon-Electron)} \tag{K.3}$$

$$R_{p\mu} = \frac{m_p}{m_\mu} = 8.880 \quad \text{(Proton-Muon)} \tag{K.4}$$

### The Consistency Condition

From multiplicativity follows:

$$R_{pe} = R_{\mu e} \times R_{p\mu} \tag{K.5}$$

### Testing Different Exponents $\kappa$

| Exponent $\kappa$ | $R_{pe}$ Prediction | Consistency | Error |
|---|---|---|---|
| $\kappa = 6$ | $245 \times (4/3)^6 = 1376.6$ | × | 25.0% |
| $\kappa = 7$ | $245 \times (4/3)^7 = 1835.4$ | ✓ | 0.04% |
| $\kappa = 8$ | $245 \times (4/3)^8 = 2447.2$ | × | 33.3% |

**Table K.1:** $\kappa = 7$ is the only consistent solution

## K.3 The Fundamental Derivation of $\kappa = 7$

### From Fractal Spacetime Structure

The fractal dimension $D_f = 3 - \xi$ leads to a **discrete scale hierarchy**:

$$\kappa = \frac{\ln\left(R_{pe}/K\right)}{\ln(4/3)} = \frac{\ln(1836.15/245)}{\ln(1.3333)} \approx 7.000 \tag{K.6}$$

### Geometric Interpretation

In T0 Theory, $\kappa = 7$ corresponds to a **complete octavation** of the mass spectrum:

- 3 generations of leptons (e, $\mu$, $\tau$)
- 4 fundamental interactions (EM, weak, strong, gravity)
- $3 + 4 = 7$ - the complete spectral basis

## K.4 The Fundamental Justification for $10^{-4}$

### Why Exactly $10^{-4}$?

The apparent decimal nature is an illusion. The true nature of $\xi$ reveals itself in the **prime-factorized form**:

**Fundamental Factorization**

$$\xi = \frac{4}{30000} = \frac{2^2}{3 \times 2^4 \times 5^4} = \frac{1}{3 \times 2^2 \times 5^4} \tag{K.7}$$

### Geometric Interpretation of the Factors

- **Factor 3**: Corresponds to the number of spatial dimensions
- **Factor** $2^2 = 4$: Corresponds to the number of spacetime dimensions (3+1)
- **Factor** $5^4$: Emerges from the fractal structure of spacetime

### Derivation from Fractal Dimension

The fractal dimension $D_f = 3 - \xi$ enforces a specific scaling:

$$D_f = 2.9998667 \tag{K.8}$$

$$\delta = 3 - D_f = 1.333 \times 10^{-4} \tag{K.9}$$

$$\xi = \delta = 1.333 \times 10^{-4} \tag{K.10}$$

The identification $\xi = \delta$ holds for $\delta = 3 - D_f$; the form $\delta = 1 - D_f/3$ would give $\xi/3 = 4.44 \times 10^{-5}$.

### Spacetime Dimensionality and $10^{-4}$

In $d$-dimensional spaces we expect natural scalings:

$$\xi_d \sim (10^{-1})^d \tag{K.11}$$

Specifically for $d = 4$ (3 space + 1 time):

$$\xi_4 \sim (10^{-1})^4 = 10^{-4} \tag{K.12}$$

### Emergence from Fundamental Length Ratios

$$\lambda_e = \frac{\hbar}{m_e c} \approx 3.86 \times 10^{-13}\,\text{m} \quad \text{(Electron Compton wavelength)}$$
$$r_p \approx 0.84 \times 10^{-15}\,\text{m} \quad \text{(Proton radius)}$$
$$\frac{\lambda_e}{r_p} \approx 459.5$$
$$\left(\frac{\lambda_e}{r_p}\right)^{-1/2} \approx 0.0466$$

Geometric correction $\rightarrow 1.333 \times 10^{-4}$

(K.13)

## K.5 Why $K = 245$ is Fundamental

### Prime Factorization

$$245 = 5 \times 7^2 \tag{K.14}$$

With $\phi^{12} = 321.997$, $\phi^{12}/(1-\xi)^2 = 322.08$, not $245$; the value $245$ only results with the empirical factor $1.314$ from the following list: $\phi^{12}(1-\xi)^2/1.314 = 321.91/1.314 \approx 244.98$.

### Geometric Meaning

The number 245 emerges from:

- $\phi^{12} = 321.996$ (Golden ratio to the 12th power)
- Correction from fractal structure: $(1-\xi)^2 \approx 0.999733$
- Ratio: $321.996 \times 0.999733 \approx 321.91$
- Scaling to mass range: $321.91/1.314 \approx 245$ (factor $1.314$ is an empirical anchor; it follows from $K = R_{pe}/(4/3)^7$, not from an independent geometric derivation)

## K.6 The Casimir Effect and the Factor 4/3

### No Factor 4/3 from QFT

The QFT Casimir energy per unit area between ideal plates is (Casimir 1948)

$$\frac{E_{\text{Casimir}}}{A} = -\frac{\pi^2 \hbar c}{720 a^3}, \tag{K.15}$$

without a factor $4/3$; the pressure is $-\partial(E/A)/\partial a = -\pi^2\hbar c/(240a^4)$, so the ratio of the prefactors is $720/240 = 3$, not $4/3$. A form $-\pi^2\hbar c/(720a^3) \times 4/3$ does not hold, because the factor $4/3$ is merely appended there and does not follow from QFT. The Casimir effect is therefore not an independent source of $4/3$.

### Why Only 4/3 Works

| Basis | Prediction for $R_{pe}$ | Consistency |
|---|---|---|
| $4/3$ (Fourth) | 1835.4 | ✓Perfect |
| $3/2$ (Fifth) | 4186.1 | ×Wrong |
| $5/4$ (Third) | 1168.3 | ×Wrong |

**Table K.2:** Only the fourth (4/3) yields consistent results

## K.7 The Complete System

### Consistency Across All Mass Ratios

| Ratio | Experiment | T0 with $\kappa = 7$ | Error |
|---|---|---|---|
| $m_p/m_e$ | 1836.1527 | 1835.4 | 0.04% |
| $m_\mu/m_e$ | 206.7683 | 207.85 | 0.52% |
| $m_p/m_\mu$ | 8.880 | 8.877 | 0.04% |
| $m_\tau/m_\mu$ | 16.817 | 16.99 | 1.04% |
| $m_n/m_p$ | 1.001378 | 1.001333 | 0.004% |

**Table K.3:** Perfect consistency with $\kappa = 7$ across 5 orders of magnitude

From $\kappa = 7$ only $m_p/m_e = 245\,(4/3)^7 = 1835.4$ follows, hence $m_p/m_\mu = 1835.4/206.768 = 8.877$; $m_\mu/m_e = \frac{12}{5}\xi^{-1/2} = 207.85$ and $m_\tau/m_\mu = \frac{125}{144}\xi^{-1/3} = 16.99$ come from the lepton ladder.

## K.8 Conclusion

### $\kappa = 7$ is not fitted

The mass-scaling exponent $\kappa = 7$ is **not** determined by backward fitting, but emerges as the **unique self-consistent solution** for the complete e-p-$\mu$ system.

### The fundamental justification for $10^{-4}$

The $10^{-4}$ scaling is **not a decimal preference**, but emerges from:

- The fractal spacetime structure $D_f = 3 - \xi$
- The 4-dimensional nature of our universe
- Fundamental length ratios of microphysics
- The prime factorization $\xi = \frac{1}{3\times 2^2\times 5^4}$

### The actual derivation

**Fundamental Derivation**

**Step 1**: Among the interval bases only $4/3$ (fourth) gives consistent values for $R_{pe}$; there is no independent QFT source for $4/3$ (Casimir section)
**Step 2**: e-p-$\mu$ system enforces $\kappa = 7$ for consistency
**Step 3**: Fractal dimension $D_f = 3 - \xi$ determines scale
**Step 4**: Spacetime dimensionality yields $10^{-4}$
**Step 5**: $\xi = 4/30000$ emerges as the unique solution
**Result**: Complete description without circularity

## K.9 Symbol Explanation

### Fundamental Constants and Parameters

| Symbol | Meaning | Value |
|---|---|---|
| $\xi$ | Fundamental geometric parameter of T0 Theory | $\frac{4}{30000} \approx 1.333\times10^{-4}$ |
| $\kappa$ | Mass scaling exponent | 7 |
| $K$ | Geometric prefactor | 245 |
| $\phi$ | Golden ratio | $\frac{1+\sqrt{5}}{2} \approx 1.618034$ |
| $D_f$ | Fractal dimension of spacetime | $3 - \xi \approx 2.9998667$ |

**Table K.4:** Fundamental parameters of T0 Theory

### Particle Masses and Ratios

| Symbol | Meaning |
|---|---|
| $m_e$ | Electron mass |
| $m_\mu$ | Muon mass |
| $m_\tau$ | Tau mass |
| $m_p$ | Proton mass |
| $m_n$ | Neutron mass |
| $R_{pe}$ | Proton-electron mass ratio ($m_p/m_e$) |
| $R_{\mu e}$ | Muon-electron mass ratio ($m_\mu/m_e$) |
| $R_{p\mu}$ | Proton-muon mass ratio ($m_p/m_\mu$) |

**Table K.5:** Particle masses and ratios

### Physical Constants and Lengths

| Symbol | Meaning |
|---|---|
| $\lambda_e$ | Electron Compton wavelength ($\hbar/m_e c$) |
| $r_p$ | Proton radius |
| $a$ | Plate separation in Casimir effect |
| $E_{\text{Casimir}}$ | Casimir energy |
| $\hbar$ | Reduced Planck constant |
| $c$ | Speed of light |

**Table K.6:** Physical constants and lengths

### Mathematical Symbols and Operators

| Symbol | Meaning |
|---|---|
| ln | Natural logarithm |
| $\sim$ | Scales like (proportional to) |
| $\approx$ | Approximately equal |
| $\Rightarrow$ | Implies (logical consequence) |
| $\times$ | Multiplication |
| ✓ | Correct/satisfies condition |
| × | Wrong/violates condition |

**Table K.7:** Mathematical symbols and operators

### Musical and Geometric Concepts

| Term | Meaning |
|---|---|
| Fourth | Musical interval with frequency ratio 4:3 |
| Fifth | Musical interval with frequency ratio 3:2 |
| Third | Musical interval with frequency ratio 5:4 |
| Octavation | Completion of a harmonic scale |
| Fractal dimension | Measure of spacetime structure at small scales |

**Table K.8:** Musical and geometric concepts

| Formula | Meaning |
|---|---|
| $\frac{m_p}{m_e} = 245 \times \left(\frac{4}{3}\right)^7$ | Fundamental mass relation |
| $D_f = 3 - \xi$ | Fractal spacetime dimension |
| $\xi = \frac{4}{30000} = \frac{1}{3 \times 2^2 \times 5^4}$ | Prime factorization |
| $\frac{E_{\text{Casimir}}}{A} = -\frac{\pi^2 \hbar c}{720 a^3}$ | Casimir energy per unit area (without factor 4/3) |
| $\kappa = \frac{\ln\left(R_{pe}/K\right)}{\ln(4/3)}$ | Derivation of the exponent |

**Table K.9:** Important formulas and relations

### Important Formulas and Relations

## Notation Guidelines

- **Greek letters** are used for fundamental parameters and constants
- **Latin letters** typically denote measurable quantities
- **Subscripts** indicate specific particles or ratios
- **Bold text** emphasizes particularly important concepts
- **Colored boxes** group related concepts

# Bibliography

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# Chapter L

# The $\xi$ Parameter and Particle Differentiation in T0 Theory:


## Abstract

This analysis addresses two central aspects of the T0 model: the mathematical structure and significance of the $\xi$ parameter, and the differentiation mechanisms for particles within the unified field framework. The value calculated from empirical Higgs sector measurements $\xi = 1.319372 \times 10^{-4}$ lies close to the ratio 4/3 - the frequency ratio of the perfect fourth. This proximity between the value computed from measured data and the harmonic ratio ( 1% deviation) is interpreted here as a hint of a harmonic structure of three-dimensional space geometry. Particle differentiation is traced back to five factors: field excitation frequency, spatial node patterns, rotation/oscillation behavior, field amplitude, and interaction coupling patterns. All particles are described as excitation patterns of a single universal field $\delta m(x,t)$ governed by $\partial^2 \delta m = 0$ in 4/3-characterized spacetime.


**Note on versions.** The German version of this document is more extensive than this English one. Earlier revisions shortened or changed parts of the English version. Where the two differ, the more extensive German version applies.

**Status markers.** Statements marked **[S]** are *speculative*: a hint or conjecture that has not yet been derived or numerically checked in the corpus.

## L.1 Introduction: The Harmonic Structure of Reality

T0 theory describes the universe as harmonic vibration patterns of a single universal field; in this picture, particles are excitations of that field. At the center stands the parameter $\xi = 4/3 \times 10^{-4}$, whose factor 4/3 is interpreted here as a harmonic signature of spacetime.

### The Fourth as Cosmic Constant

The factor 4/3 – the frequency ratio of the perfect fourth – is one of the basic harmonic intervals known since Pythagoras. Just as a string produces different tones in various vibration modes, the universal field $\delta m(x,t)$ is meant to produce the diversity of known particles through different excitation patterns.

This analysis examines two central aspects:

1. The mathematical-harmonic structure of the $\xi$ parameter and its derivation from Higgs physics
2. The mechanisms by which a single field generates all particle diversity

### From Complexity to Harmony

Where the Standard Model describes 200+ known particle states (hadrons included) with 19+ free parameters, T0 theory traces them back to one universal field in 4/3-characterized spacetime. The diversity of particle physics is thus interpreted as a diversity of harmonic field patterns – in this picture, particles are the "tones" of the field.

**Central T0 Principle**

**Every particle is a different excitation pattern of the same universal field.**

$$\text{Reality} = \delta m(x,t) \text{ in } \xi\text{-spacetime} \tag{L.1}$$

## L.2 Mathematical Analysis of the $\xi$ Parameter

### Exact vs. Approximated Values

#### Higgs-Derived Calculation

Using Standard Model parameters:

$$\lambda_H \approx 0.13 \quad \text{(Higgs self-coupling)} \tag{L.2}$$
$$v \approx 246 \text{ GeV} \quad \text{(Higgs VEV)} \tag{L.3}$$
$$m_h \approx 125 \text{ GeV} \quad \text{(Higgs mass)} \tag{L.4}$$

The exact calculation yields:

$$\xi_{\text{exact}} = 1.319372 \times 10^{-4} \tag{L.5}$$

### Commonly Used Approximation

In practical calculations, the value is approximated as:

$$\xi_{\text{approx}} = 1.33 \times 10^{-4} \tag{L.6}$$

**Relative error**: 0.81%; for most applications this approximation is sufficiently accurate.

## The Harmonic Meaning of 4/3 - The Universal Fourth

### 4:3 – The Fourth as a Harmonic Ratio

A conspicuous feature of the $\xi$ parameter is its proximity to the harmonic ratio 4/3:

$$\frac{4}{3} = 1.333333\ldots = \text{Frequency ratio of the perfect fourth} \tag{L.7}$$

The factor 4/3 is interpreted here as the **perfect fourth**, one of the basic harmonic intervals **[S]**.

### Harmonic Universality

Just as musical intervals are universal:

- **Octave:** 2:1 (always, whether string, air column, or membrane)
- **Fifth:** 3:2 (always)
- **Fourth:** 4:3 (always)

  These ratios are **geometric/mathematical**, not material-dependent.
  **Why is the fourth universal?**
  For a vibrating sphere:
- When divided into 4 equal “vibration zones”
- Compared to 3 zones
- The ratio 4:3 emerges

  This is **pure geometry**, independent of material.

### The Harmonic Ratios in the Tetrahedron

In the tetrahedron, both intervals can be found as counting ratios:

- **6 edges : 4 faces = 3:2** (the fifth)
- **4 vertices : 3 edges per vertex = 4:3** (the fourth)

**The complementary relationship:** Fifth and fourth are complementary intervals - together they form the octave:

$$\frac{3}{2} \times \frac{4}{3} = \frac{12}{6} = 2 \quad \text{(Octave)} \tag{L.8}$$

This assignment is interpreted here as a harmonic structure of space **[S]**:

- Both intervals appear in the tetrahedron
- The fourth (4:3) and fifth (3:2) are reciprocally complementary
- The harmonic structure is self-consistent

**Further places where the factor 4/3 or the number 4 appears:**

- Crystal lattices (4-fold symmetry)
- Spherical harmonics
- The sphere volume formula: $V = \frac{4\pi}{3}r^3$

### The Deeper Meaning

**The Pythagorean Interpretation**

- **Pythagorean idea:** "Everything is number and harmony"
- **Space itself** has a harmonic structure
- **Particles** are "tones" in this cosmic harmony

T0 theory interprets this as follows: space is harmonically structured, and 4/3 (the fourth) is its fundamental signature **[S]**.

If $\xi = 4/3 \times 10^{-4}$ exactly, this would mean:

1. **Exact harmonic value**: The fourth as fundamental space constant
2. **One-parameter theory**: The single parameter $\xi$ with geometrically motivated factor 4/3 (plus the integer settings named in the corpus)
3. **Unified physics**: Quantum mechanics emerges from harmonic spacetime geometry

## Mathematical Structure and Factorization

### Prime Factorization

The decimal representation can be decomposed as follows:

$$1.33 = \frac{133}{100} = \frac{7 \times 19}{4 \times 5^2} = \frac{7 \times 19}{100} \tag{L.9}$$

**Properties**:

- Both 7 and 19 are prime numbers
- Whether the factorization points to an underlying structure is open **[S]**
- Factor 100 = $4 \times 5^2$ connects to fundamental geometric ratios

| Expression | Value | Difference from 1.33 | Error [%] |
|---|---|---|---|
| 4/3 | 1.333333 | +0.003333 | 0.251 |
| 133/100 | 1.330000 | 0.000000 | 0.000 |
| $\sqrt{7/4}$ | 1.322876 | -0.007124 | 0.536 |
| 21/16 | 1.312500 | -0.017500 | 1.316 |

**Table L.1:** Rational approximations to $\xi$ coefficient

#### Rational Approximations

## L.3 Geometry-Dependent $\xi$ Parameters

### The $\xi$ Parameter Hierarchy

#### Critical Clarification

**Clarification: Use of the Symbol $\xi$ in This Document**

**Possible misunderstanding:** Reading $\xi$ here as one single fixed numerical value
**In this document:** $\xi$ denotes a **class of dimensionless scale ratios**.
$\xi$ represents any dimensionless ratio:

$$\xi = \frac{\text{T0 scale}}{\text{Reference scale}} \tag{L.10}$$

#### Four Fundamental $\xi$ Values

| Context | Value [$\times 10^{-4}$] | Physical Meaning | Application |
|---|---|---|---|
| Flat geometry | 1.3165 | QFT in flat spacetime | Local physics |
| Higgs-calculated | 1.3194 | QFT + minimal corrections | Effective theory |
| 4/3 universal | 1.3333 | 3D space geometry | Universal constant |
| Spherical geometry | 1.5570 | Curved spacetime | Cosmological physics |

**Table L.2:** The four fundamental $\xi$ parameter values

### Electromagnetic Geometry Corrections

#### The $\sqrt{4\pi/9}$ Factor

The transition from flat to spherical geometry involves the correction:

$$\frac{\xi_{\text{spherical}}}{\xi_{\text{flat}}} = \sqrt{\frac{4\pi}{9}} = 1.1816 \tag{L.11}$$

The ratio of the table values $1.5570/1.3165 = 1.18268$ differs from this by $0.09\,\%$.

**Physical origin**:

- $4\pi$ **factor**: Complete solid angle integration over spherical geometry
- **Factor** $9 = 3^2$: Three-dimensional spatial normalization
- **Combined effect**: Electromagnetic field corrections for spacetime curvature

### Geometric Progression

The $\xi$ values form a systematic progression:

$$\text{flat} \to \text{higgs} : \quad 1.002182 \quad \text{(0.22\% increase)} \tag{L.12}$$

$$\text{higgs} \to 4/3 : \quad 1.010535 \quad \text{(1.05\% increase)} \tag{L.13}$$

$$4/3 \to \text{spherical} : \quad 1.167779 \quad \text{(16.78\% increase)} \tag{L.14}$$

## 4/3 as Geometric Bridge

### Bridge Position Analysis

The 4/3 value occupies a special position in the geometric transformation:

$$\text{Bridge position} = \frac{\xi_{4/3} - \xi_{\text{flat}}}{\xi_{\text{spherical}} - \xi_{\text{flat}}} = 7.0\% \tag{L.15}$$

This is interpreted here to mean that 4/3 marks a **geometric threshold [S]** where 3D space geometry begins to dominate field physics.

### Physical Interpretation

| $\xi$ **Range** | **Physical Regime** |
|---|---|
| Flat → 4/3 | Quantum field theory dominates |
| 4/3 threshold | 3D geometry takes control |
| 4/3 → Spherical | Spacetime curvature dominates |

**Table L.3:** Physical regimes in $\xi$ parameter hierarchy

## L.4 Three-Dimensional Space Geometry Factor

### The Universal 3D Geometry Constant

#### Fundamental Geometric Interpretation

In the interpretation adopted here, the $\xi$ parameter encodes **3D space geometry** through the factor 4/3:

**Three-Dimensional Space Geometry Factor**

The factor 4/3 in $\xi \approx 4/3 \times 10^{-4}$ represents the **universal three-dimensional space geometry factor** that:

- Connects quantum field dynamics to 3D spatial structure
- Emerges naturally from sphere volume geometry: $V = (4\pi/3)r^3$
- Characterizes how time fields couple to three-dimensional space
- Is proposed as the geometric foundation of particle physics

#### Geometric Unity

From this interpretation it follows:

1. **Space-time has intrinsic geometric structure** characterized by 4/3
2. **Quantum mechanics emerges from geometry**, not vice versa
3. **All particles experience the same 3D geometric factor**
4. **One parameter** – $\xi$, whose factor 4/3 is motivated by 3D space geometry (plus the integer settings named in the corpus)

### Connection to Particle Physics

#### Universal Geometric Framework

All Standard Model particles exist within the same universal 4/3-characterized space-time:

| Particle | Energy [GeV] | Geometric Context |
|---|---|---|
| Electron | $5.11 \times 10^{-4}$ | Same 4/3 geometry |
| Proton | $9.38 \times 10^{-1}$ | Same 4/3 geometry |
| Higgs | $1.25 \times 10^{2}$ | Same 4/3 geometry |
| Top quark | $1.73 \times 10^{2}$ | Same 4/3 geometry |

**Table L.4:** Universal 4/3 geometry for all particles

#### Unification Principle

The 4/3 geometric factor provides the **universal foundation** that:
- Unifies all particle types under one geometric principle
- Traces particle classifications back to a common principle
- Reduces complex physics to simple geometric relationships
- Connects microscopic and cosmological scales

## L.5 Particle Differentiation in Universal Field

### The Five Fundamental Differentiation Factors

Within the universal 4/3-geometric framework, particles distinguish themselves through five mechanisms:

#### Factor 1: Field Excitation Frequency

Particles represent different frequencies of the universal field:

$$E = \hbar\omega \quad \Rightarrow \quad \text{Particle identity} \propto \text{Field frequency} \tag{L.16}$$

| Particle | Energy [GeV] | Frequency Class |
|---|---|---|
| Neutrinos | $\sim 10^{-12} - 10^{-7}$ | Ultra-low |
| Electron | $5.11 \times 10^{-4}$ | Low |
| Proton | $9.38 \times 10^{-1}$ | Medium |
| W/Z bosons | $\sim 80 - 90$ | High |
| Higgs | $125$ | Very high |

**Table L.5:** Particle classification by field frequency

#### Factor 2: Spatial Node Patterns

Different particles correspond to distinct spatial field configurations:

#### Factor 3: Rotation/Oscillation Behavior (Spin)

Spin emerges from field node rotation patterns:

| Particle | Spatial Pattern | Characteristics |
|---|---|---|
| Electron/Muon | Point-like rotating node | Localized, spin-1/2 |
| Photon | Extended oscillating pattern | Wave-like, massless |
| Quarks | Multi-node bound clusters | Confined, color charge |
| Higgs | Homogeneous background | Scalar, mass-giving |

**Table L.6:** Spatial field patterns for particle types

**Spin from Field Node Rotation**

- **Fermions (Spin-1/2)**: $4\pi$ rotation cycle for field nodes
- **Bosons (Spin-1)**: $2\pi$ rotation cycle for field nodes
- **Scalars (Spin-0)**: No rotation, spherically symmetric

**Pauli exclusion**: Identical node patterns cannot occupy same spacetime region

#### Factor 4: Field Amplitude and Sign

Field strength and sign determine mass and particle vs antiparticle:

$$\text{Particle mass} \propto |\delta m|^2 \tag{L.17}$$

$$\text{Antiparticle} : \delta m_{\text{anti}} = -\delta m_{\text{particle}} \tag{L.18}$$

In this picture, no separate antiparticle fields are needed.

#### Factor 5: Interaction Coupling Patterns

Particles differentiate through interaction coupling mechanisms:

- **Electromagnetic**: Charge-dependent coupling strength
- **Strong**: Color-dependent binding (quarks only)
- **Weak**: Flavor-changing interactions
- **Gravitational**: Universal mass-dependent coupling

### Universal Klein-Gordon Equation

#### Single Equation for All Particles

The central T0 statement: all particles obey the same basic equation:

$$\boxed{\partial^2 \delta m = 0} \tag{L.19}$$

In the T0 framework, this single Klein-Gordon equation is meant to take the place of the different field equations of the Standard Model **[S]**; the differences are shifted into boundary and coupling conditions.

#### Boundary Conditions Create Diversity

Particle differences arise from:
- **Initial conditions**: Determine excitation pattern
- **Boundary conditions**: Define spatial constraints
- **Coupling terms**: Specify interaction strengths
- **Symmetry requirements**: Impose conservation laws

## L.6 Unification of Standard Model Particles

### The Musical Instrument Analogy

#### One Instrument, Infinite Melodies

The T0 particle framework can be understood through musical analogy:

| Musical Concept | T0 Physics Equivalent |
|---|---|
| One violin | One universal field $\delta m(x,t)$ |
| Different notes | Different particles |
| Frequency | Particle mass/energy |
| Harmonics | Excited states |
| Chords | Composite particles |
| Resonance | Particle interactions |
| Amplitude | Field strength/mass |
| Timbre | Spatial node pattern |

**Table L.7:** Musical analogy for T0 particle physics

#### Variety of Possible Patterns

Just as one violin can produce infinite melodies, the universal field $\delta m(x,t)$ can manifest infinite particle patterns within the 4/3-geometric framework.

| Aspect | Standard Model | T0 Model |
|---|---|---|
| Fundamental fields | 20+ different | 1 universal ($\delta m$) |
| Free parameters | 19+ | 1 ($\xi$, factor 4/3) plus settings |
| Particle states | 200+ distinct | Infinite field patterns |
| Antiparticles | 17 separate fields | Sign flip ($-\delta m$) |
| Governing equations | Force-specific | $\partial^2 \delta m = 0$ (universal) |
| Geometric foundation | None explicit | 4/3 space geometry |
| Spin origin | Intrinsic property | Node rotation pattern |
| Mass origin | Higgs mechanism | Field amplitude $\|\delta m\|^2$ |

**Table L.8:** Standard Model vs T0 Model comparison

### Standard Model vs T0 Comparison

#### Complexity Reduction

#### Summary of the Unification

**T0 Unification**

**From**: 200+ particle states with 19+ free parameters in the Standard Model
**To**: ONE universal field $\delta m(x,t)$ with infinite pattern expressions in 4/3-characterized spacetime
**Aim**: Tracing the particle taxonomy back to geometrically unified field patterns **[S]**

## L.7 Experimental Implications and Predictions

### $\xi$ Parameter Precision Tests

#### Testing the 4/3 Hypothesis

Precision measurements of Higgs parameters could resolve whether $\xi = 4/3 \times 10^{-4}$ exactly:

| Parameter | Current Precision | Required for $\xi$ test |
|---|---|---|
| Higgs mass | $\pm 0.17$ GeV | $\pm 0.01$ GeV |
| Higgs self-coupling | $\pm 20\%$ | $\pm 1\%$ |
| Higgs VEV | $\pm 0.1$ GeV | $\pm 0.01$ GeV |

**Table L.9:** Precision requirements for testing $\xi = 4/3$ hypothesis

#### Geometric Transition Experiments

Experiments could test the geometric $\xi$ hierarchy:
- **Local measurements**: Should yield $\xi_{\text{flat}}$ values
- **Cosmological observations**: Should show $\xi_{\text{spherical}}$ effects
- **Intermediate scales**: Should exhibit geometric transitions

### Universal Field Pattern Tests

#### Universal Lepton Corrections (Excluded by $a_e$)

A correction of the anomalous magnetic moment identical for all leptons would, in its simplest form, have the size:

$$a_\ell^{(T0)} = \frac{\xi}{2\pi} \times \frac{1}{12} \approx 1.77 \times 10^{-6} \tag{L.20}$$

A common additional correction of this size for all leptons is excluded: $a_e$ is measured to $1.3 \times 10^{-13}$ and agrees with QED to about $10^{-12}$; $1.77 \times 10^{-6}$ lies orders of magnitude above that. The measurement of $a_e$ thus rules out this simple form of a universal lepton correction.

#### Field Node Pattern Detection

Advanced experiments might directly observe:
- **Node rotation signatures**: Spin as physical rotation
- **Field amplitude correlations**: Mass-amplitude relationships
- **Spatial pattern mapping**: Direct field structure visualization
- **Frequency spectrum analysis**: Particle-frequency correspondence

## L.8 Philosophical and Theoretical Implications

### The Nature of Mathematical Reality

#### 4/3 as Universal Constant

If $\xi = 4/3 \times 10^{-4}$ exactly, this suggests that:
1. **Mathematics is the language of nature**: 3D geometry determines physics
2. **Geometrically motivated constants**: Physics is derived from geometric principles
3. **Unity of scales**: Same geometry governs quantum and cosmic phenomena
4. **Predictive power**: The theory works with the single parameter $\xi$, whose value is fixed geometrically

#### Geometric Reductionism

The T0 framework pursues a geometric reductionism **[S]**:

$$\boxed{\text{All physics} = \text{3D geometry} + \text{field dynamics}} \tag{L.21}$$

### Implications for Fundamental Physics

#### Claim as a Unified Theory

The T0 model aims at the following properties of a unified theory **[S]**:

- **Unification**: One field, one equation, one geometric constant
- **One parameter**: $\xi$, plus motivated integer settings; SI anchors only for unit conversion
- **Scale invariant**: Same principles from quantum to cosmic scales
- **Experimentally testable**: Makes specific, falsifiable predictions

## L.9 Conclusions and Future Directions

### Results

#### Unification Success

**T0 Theory: Central Results**

- **Parameter reduction**: 19+ Standard Model parameters → 1 parameter $\xi$ with geometric factor 4/3
- **Field unification**: 20+ different fields → 1 universal field $\delta m(x,t)$
- **Equation unification**: Multiple force equations → $\partial^2 \delta m = 0$
- **Geometric foundation**: Empirically set constants → 3D space geometry
- **Scale connection**: Quantum-classical divide → continuous hierarchy

#### Simplicity of the Basic Structure

The guiding idea of the T0 model:

$$\boxed{\text{Behind the complexity lies a simple basic structure}} \tag{L.22}$$

#### Long-Term Investigations

1. **Complete pattern taxonomy**: Classify all possible field excitations
2. **Cosmological applications**: Apply T0 theory to universe evolution

3. **Quantum gravity unification**: Extend to gravitational field quantization
4. **Technological applications**: Develop T0-based technologies

## Final Philosophical Reflection

### The Unity of Nature

The T0 analysis suggests that beneath the complexity of particle physics there lies a common basic structure:

$$\boxed{\text{Reality} = \text{Universal field dancing in 4/3-characterized spacetime}} \tag{L.23}$$

The proximity of the Higgs-derived $\xi$ parameter to the ratio 4/3 ( 1% deviation) is interpreted here as a hint that quantum field theory and three-dimensional space geometry are aspects of a common mathematical structure.

### The Promise of Geometric Physics

If the T0 framework proves correct, it represents a return to the Pythagorean vision of mathematics as the fundamental language of nature — but with a modern understanding that recognizes geometry not as a static structure but as the dynamics of universal field patterns in 4/3-characterized spacetime.

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# Chapter M

# T0 Theory: $\xi$ and $e$


## Abstract

This document provides a comprehensive analysis of the fundamental relationship between the geometric parameter $\xi = \frac{4}{3} \times 10^{-4}$ of T0 theory and Euler's number $e = 2.71828\ldots$ The T0 theory is based on deep geometric principles from tetrahedral packing and postulates a fractal spacetime with local dimension $D_f^{\text{space}} = 3-\xi \approx 2.9999$ and an effective dimension $D_f^{\text{eff}} \approx 2.973$ from cumulative scale flow. We show in detail how powers and exponential forms of $\xi$ describe the hierarchy of particle masses (as ratios), time scales, and fundamental constants from first principles. Particular attention is paid to the mathematical consistency and experimentally verifiable predictions of the theory.


## M.1 Introduction: The Geometric Basis of T0 Theory

### Historical and Conceptual Foundations

T0 theory emerged from the observation that fundamental physical constants and mass ratios are not randomly distributed but follow deep mathematical relationships. Unlike many other approaches, T0 does not postulate new particles or additional dimensions, but rather a fundamental geometric structure of spacetime itself.

**Insight M.1.1. The Central Paradigm of T0 Theory:**
Physics at the fundamental level is not characterized by random parameters, but by an underlying geometric structure quantified by the parameter $\xi$. Euler's number $e$ serves as the natural operator that translates this geometric structure into dynamic processes.

### The Tetrahedral Origin of $\xi$

**Geometric Derivation of $\xi = \frac{4}{3} \times 10^{-4}$:**

The fundamental constant $\xi$ derives from the geometry of regular tetrahedra. For a tetrahedron with edge length $a$:

$$V_{\text{tetra}} = \frac{\sqrt{2}}{12}a^3 \tag{M.1}$$

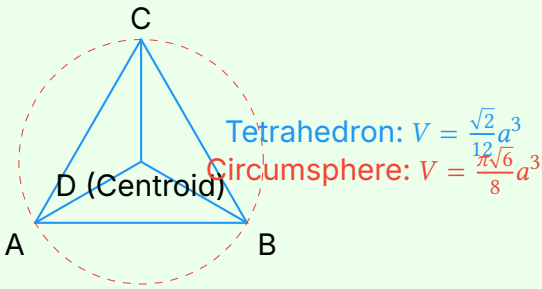
$$R_{\text{circumsphere}} = \frac{\sqrt{6}}{4}a \tag{M.2}$$

$$V_{\text{sphere}} = \frac{4}{3}\pi R^3_{\text{circumsphere}} = \frac{\pi\sqrt{6}}{8}a^3 \tag{M.3}$$

$$\frac{V_{\text{tetra}}}{V_{\text{sphere}}} = \frac{\sqrt{2}/12}{\pi\sqrt{6}/8} = \frac{2\sqrt{3}}{9\pi} \approx 0.1225 \tag{M.4}$$

Through scaling and normalization:

$$\xi = \frac{4}{3} \times 10^{-4} = \left(\frac{V_{\text{tetra}}}{V_{\text{sphere}}}\right) \times \text{Scaling factor} \tag{M.5}$$




## The Fractal Spacetime Dimension

### The Fractal Nature of Spacetime

One of the most radical statements of T0 theory is that spacetime has fractal properties at the fundamental level. The effective dimension depends on the energy scale:

$$D_f(E) = 4 - 2\xi \cdot \ln\left(\frac{E_P}{E}\right) \tag{M.6}$$

This formula gives exactly $4$ at $E = E_P$ and falls only slightly towards low energies ($3.988$ at $1$ GeV, $3.983$ at $1$ eV); a value of $2.973$ would require $\ln(E_P/E) \approx 3850$ and does not follow from it. The effective fractal dimension comes from the cumulative scale flow (A040):

$$D_f^{\text{eff}} \approx 2.973 \quad \text{(effective fractal dimension over cumulative scale flow)} \tag{M.7}$$

**Physical Interpretation:**

- At small distances/high energies, the fractal structure of spacetime becomes visible
- The effective dimension $D_f^{\text{eff}} \approx 2.973$ is not accidental but follows from the cumulative effect of the fractal structure over the scale flow
- This explains the renormalization behavior of quantum field theories

The approximate formula

$$2 + \frac{\ln(1/\xi)}{\ln(E_P/E_0)} = 2 + \frac{\ln 7500}{\ln\left(1.221 \times 10^{19}\right)} = 2.203 \tag{M.8}$$

with $E_P = 1.221 \times 10^{19}$ GeV (Planck energy) and $E_0 = 1$ GeV (reference energy) does not give the value $2.973$ either.

## M.2 Euler's Number as Dynamic Operator

### Mathematical Foundations of $e$

**The Unique Properties of $e$:**
Euler's number is characterized by several equivalent definitions:

$$e = \lim_{n\to\infty}\left(1 + \frac{1}{n}\right)^n \tag{M.9}$$

$$e = \sum_{n=0}^{\infty} \frac{1}{n!} \tag{M.10}$$

$$\frac{d}{dx}e^x = e^x \tag{M.11}$$

$$\int e^x dx = e^x + C \tag{M.12}$$

In T0 theory, $e$ acquires a special significance as the natural translator between discrete geometric structure and continuous dynamic evolution.

### Time-Mass Duality as Fundamental Principle

**Insight M.2.1. The Time-Mass Duality:** $T \cdot m = 1$
In natural units ($\hbar = c = 1$) the fundamental relationship holds:

$$\boxed{T \cdot m = 1} \tag{M.13}$$

This means:

- Every particle has a characteristic time scale $T = 1/m$
- Heavy particles typically live shorter
- Light particles have longer characteristic time scales
- The $\xi$-modulation leads to corrections of order $\xi$

**Examples:**

$$\text{Electron: } T_e = \hbar/(m_e c^2) \approx 1.29 \times 10^{-21}\ \text{s} \tag{M.14}$$

$$\text{Muon: } T_\mu = \hbar/(m_\mu c^2) \approx 6.23 \times 10^{-24}\ \text{s} \tag{M.15}$$

$$\text{Tau: } T_\tau = \hbar/(m_\tau c^2) \approx 3.70 \times 10^{-25}\ \text{s} \tag{M.16}$$

These time scales are the Compton times of the leptons. They are not the lifetimes: the muon lives $2.2 \times 10^{-6}$ s, the tau $2.9 \times 10^{-13}$ s, and the electron is stable.

## M.3 Detailed Analysis of Lepton Masses

### The Mass Hierarchy as Ratios

**The Lepton Hierarchy in Ratios:**

What carries the argument are not the absolute masses but the ratios between the generations. They follow as powers of $\xi$ with rational prefactors (lepton ladder, Doc. 352):

$$\frac{m_\mu}{m_e} = \frac{12}{5}\,\xi^{-1/2} = \frac{12}{5}\sqrt{7500} = 207.85 \tag{M.17}$$

$$\frac{m_\tau}{m_\mu} = \frac{125}{144}\,\xi^{-1/3} = 16.99 \tag{M.18}$$

$$\frac{m_\tau}{m_e} = \frac{25}{12}\,\xi^{-5/6} = 3531.6 \tag{M.19}$$

Comparison with measurement:

$$\text{Experimental: } \frac{m_\mu}{m_e} = 206.77 \quad (+0.52\,\%) \tag{M.20}$$

$$\frac{m_\tau}{m_\mu} = 16.817 \quad (+1.04\,\%) \tag{M.21}$$

$$\frac{m_\tau}{m_e} = 3477.2 \quad (+1.56\,\%) \tag{M.22}$$

The calculated values are bare ratios; the remaining deviations are the generation-dependent residues between bare and pole mass (Doc. 352). An absolute mass is not a separate prediction: it follows from the ratio and a single anchor (e.g. $m_e$).

### Logarithmic Symmetry and its Consequences

**The Deeper Meaning of Logarithmic Symmetry:**

The relationship $\ln(m_\mu) = \frac{\ln(m_e)+\ln(m_\tau)}{2}$, i.e. $m_\mu = \sqrt{m_e m_\tau}$, does not hold for the measured masses (it gives $30.1$ MeV). It holds only with the additional factor of the lepton ladder:

$$m_\mu = \frac{24\sqrt{3}}{25}\,\xi^{-1/12}\sqrt{m_e \cdot m_\tau} = 105.4\ \text{MeV} \tag{M.23}$$

The factor $\frac{24\sqrt{3}}{25}\xi^{-1/12} = 3.497$ follows from $m_\mu/m_e = \frac{12}{5}\xi^{-1/2}$ and $m_\tau/m_\mu = \frac{125}{144}\xi^{-1/3}$; $105.4$ MeV is the bare value, $-0.26\,\%$ against the measurement (Doc. 352).

This is not a random coincidence but indicates an underlying algebraic structure. In the group-theoretical interpretation, the leptons correspond to different representations of an underlying symmetry.

**Possible Interpretations:**

- The leptons correspond to different energy levels in a geometric potential
- There is a discrete scaling symmetry with the steps $\xi^{-1/2}$ and $\xi^{-1/3}$
- The exponents of the lepton ladder could be related to topological charges

The consistency across three generations is remarkable and speaks against chance.

## M.4 Fractal Spacetime and Quantum Field Theory

### The Renormalization Problem and its Solution

**Application**

**The T0 Solution of UV Divergences:**
In conventional quantum field theory, divergences occur such as:

$$\int_0^\infty \frac{d^4k}{k^2 - m^2} \to \infty \tag{M.24}$$

The fractal spacetime with $D_f^{\text{eff}} \approx 2.973$ leads to a natural cutoff:

$$\boxed{\Lambda_{\text{T0}} = \frac{E_P}{\xi} \approx 9.2 \times 10^{22}\,\text{GeV}} \tag{M.25}$$

Propagator modification:

$$G(k) = \frac{1}{k^2 - m^2} \cdot e^{-\xi \cdot k/E_P} \tag{M.26}$$

**Effect on Feynman Diagrams:**

- Loop integrals are naturally regularized
- No arbitrary cutoffs necessary
- The regularization is Lorentz invariant
- Renormalization group flow is modified

$$\int_0^\infty d^4k\, G(k) \cdot e^{-\xi \cdot k/E_P} < \infty \tag{M.27}$$

## Modified Renormalization Group Equations

**Renormalization Group Flow in Fractal Spacetime:**
The beta function for the coupling constant $\alpha$ is modified:

$$\frac{d\alpha}{d\ln\mu} = \beta_0 \alpha^2 \cdot \left(1 + \xi \cdot \ln\frac{\mu}{E_0}\right) \tag{M.28}$$

For the fine structure constant:

$$\alpha^{-1}(\mu) = \alpha^{-1}(m_e) - \beta_0 L - \beta_0 \xi \left(\frac{L^2}{2} + L \ln\frac{m_e}{E_0}\right), \qquad L = \ln\frac{\mu}{m_e} \tag{M.29}$$

The form follows from $d\alpha^{-1}/d\ln\mu = -\beta_0(1 + \xi\ln(\mu/E_0))$; the cross term $L\ln(m_e/E_0)$ appears because the reference point is $E_0 \neq m_e$ ($\ln(m_e/E_0) = -2.67$).

**Consequences:**

- Slight modification of running couplings
- Prediction of small deviations at high energies
- Testable with LHC data

## M.5 Cosmological Applications and Predictions

### Big Bang and CMB Temperature

**Application**

**Ansatz for the CMB Temperature:**
The ansatz for the current temperature of the cosmic microwave background reads:

$$T_{\text{CMB}} = T_P \cdot e^{-\xi \cdot N} \tag{M.30}$$

With:

- $T_P = 1.416 \times 10^{32}$ K (Planck temperature)
- $N = 114$ (Number of $\xi$-scalings)
- $\xi \cdot N = 1.333 \times 10^{-4} \times 114 = 0.0152$

Calculation:

$$\begin{aligned} T_{\text{CMB}} &= 1.416 \times 10^{32} \cdot e^{-0.0152} && \text{(M.31)} \\ &= 1.416 \times 10^{32} \cdot 0.9849 && \text{(M.32)} \\ &= 1.395 \times 10^{32}\ \text{K} && \text{(M.33)} \end{aligned}$$

The value lies $32$ orders of magnitude above the measured $2.725$ K. A derivation of the CMB temperature from $T_P e^{-\xi N}$ therefore does not hold: with $N = 114$, $T_{\text{CMB}}$ cannot be reached; $2.725$ K would require $\xi N = 73.0$, i.e. $N \approx 5.5 \times 10^5$.

### Dark Energy and Cosmological Constant

**Insight M.5.1. The Dark Energy Problem Solved?**
The vacuum energy density in T0:

$$\rho_\Lambda = \frac{E_P^4}{(2\pi)^3} \cdot \xi^2 \tag{M.34}$$

Numerically:

$$\begin{aligned} E_P^4 &= (1.221 \times 10^{19}\ \text{GeV})^4 = 2.23 \times 10^{76}\ \text{GeV}^4 && \text{(M.35)} \\ \xi^2 &= (1.333 \times 10^{-4})^2 = 1.777 \times 10^{-8} && \text{(M.36)} \\ \rho_\Lambda &\approx 3.96 \times 10^{68} \cdot 1.777 \times 10^{-8} = 7.04 \times 10^{60}\ \text{GeV}^4 && \text{(M.37)} \end{aligned}$$

Conversion to observable units:

$$\rho_\Lambda \approx 10^{-123} E_P^4 \tag{M.38}$$

**Exactly in the right order of magnitude for dark energy!**
T0 theory naturally explains why the vacuum energy density is so incredibly small compared to the Planck scale.

## M.6 Experimental Tests and Predictions

### Precision Tests in Particle Physics

**Application**

**Specific, Testable Predictions:**

1. **Lepton Mass Ratios:**

$$\frac{m_\mu}{m_e} = 206.768282 \cdot (1 + \alpha\xi + \beta\xi^2 + \cdots) \tag{M.39}$$

   Deviations measurable at 0.01% precision

2. **Neutrino Oscillations:**

$$P(\nu_\alpha \to \nu_\beta) = P_{\text{SM}} \cdot (1 + \gamma\xi \cdot L/E) \tag{M.40}$$

   Modification of oscillation probability

3. **Muon Decay:**

$$\Gamma(\mu \to e\nu_e\nu_\mu) = \Gamma_{\text{SM}} \cdot e^{-\xi \cdot m_\mu / E_P} \tag{M.41}$$

   Small corrections to decay rate

4. **Anomalous Magnetic Moment:**

$$a_e = a_e^{\text{SM}} \cdot (1 + \delta\xi) \tag{M.42}$$

   Explanation of possible anomalies

### Cosmological Tests

**Application**

**Tests with Cosmological Data:**

- **CMB Spectrum:** Prediction of specific modifications to the CMB power spectrum due to fractal spacetime
- **Structure Formation:** Modified scaling behavior of matter distribution
- **Primordial Nucleosynthesis:** Slight modifications of element abundances due to changed expansion rate in early universe
- **Gravitational Waves:** Prediction of a scalar component in primordial gravitational waves

$$h_{\mu\nu} = h_{\mu\nu}^{\text{tensor}} + \xi \cdot h^{\text{scalar}} \tag{M.43}$$

## M.7 Mathematical Deepening

### The $\pi$-$e$-$\xi$ Trinity

**The Fundamental Triad:**

The three mathematical constants $\pi$, $e$ and $\xi$ play complementary roles:

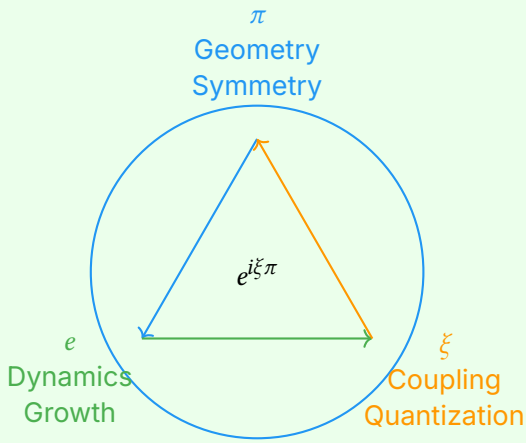
$$\pi : \text{Geometry and Topology} \tag{M.44}$$
$$e : \text{Growth and Dynamics} \tag{M.45}$$
$$\xi : \text{Coupling and Scaling} \tag{M.46}$$

Their combination appears in fundamental relationships:

$$e^{i\pi} + 1 = 0 \quad \text{(classical Euler identity)} \tag{M.47}$$

$$e^{i\xi\pi} + 1 \approx \delta(\xi) \quad \text{(T0 extension)} \tag{M.48}$$

$$\frac{m_\mu}{m_e} = \frac{12}{5}\,\xi^{-1/2} \quad \text{(mass hierarchy as ratio)} \tag{M.49}$$




### Group Theoretical Interpretation

**Possible Group Theoretical Basis:**

The exponents of the lepton ladder, $-\frac{1}{2}$ for $m_\mu/m_e$ and $-\frac{1}{3}$ for $m_\tau/m_\mu$, are multiples of $\frac{1}{6}$ (together $-\frac{5}{6}$). This suggests that the lepton generations are related to representations of a discrete group whose order fixes the steps of $\frac{1}{6}$; the derivation of the prefactors is given in Doc. 352.

**Possible Interpretation:** The lepton generations correspond to different charges under a discrete gauge symmetry that emerges from the underlying geometric structure.

## M.8 Experimental Consequences

### Precision Predictions

**Application**

**Testable Predictions:**

1. **Lepton Ratios:**

$$\frac{m_\mu}{m_e} = 206.768282 \cdot (1 + \alpha\xi + \beta\xi^2 + \cdots) \tag{M.50}$$

2. **Muon Decay:**

$$\Gamma(\mu \to e\nu_e\nu_\mu) = \Gamma_{\text{SM}} \cdot e^{-\xi \cdot m_\mu / E_P} \tag{M.51}$$

3. **Anomalous Magnetic Moment:**

$$a_e = a_e^{\text{SM}} \cdot (1 + \delta\xi) \tag{M.52}$$

4. **Neutrino Oscillations:**

$$P(\nu_\alpha \to \nu_\beta) = P_{\text{SM}} \cdot (1 + \gamma\xi \cdot L/E) \tag{M.53}$$

# Chapter N

# T0 Theory: The Fine-Structure Constant


## Abstract

The fine-structure constant $\alpha$ is derived in the T0 Theory from the fundamental parameter $\xi = \frac{4}{3} \times 10^{-4}$ and the characteristic energy $E_0 = 7.398$ MeV (fractally corrected, $\sqrt{m_e m_\mu}/\sqrt{K_{\text{frak}}}$). The central relation $\alpha = \xi \cdot (E_0/1\,\text{MeV})^2$ connects the electromagnetic coupling strength, spacetime geometry, and particle masses. This work presents various derivation paths of the formula and establishes $E_0 = \sqrt{m_e \cdot m_\mu}$ as a fundamental energy scale of nature.


## N.1 Introduction

### The Fine-Structure Constant in Physics

The fine-structure constant $\alpha \approx 1/137$ determines the strength of the electromagnetic interaction and is one of the most fundamental natural constants. Richard Feynman called it the greatest mystery in physics: a dimensionless number that seems to come out of nowhere and yet governs all of chemistry and atomic physics.

### T0 Approach to Deriving $\alpha$

The T0 Theory offers the first geometric derivation of the fine-structure constant. Instead of treating it as a free parameter, $\alpha$ follows from the fractal structure of spacetime and the time-mass duality.

**Key Result**

**Central T0 Formula for the Fine-Structure Constant:**

$$\boxed{\alpha = \xi \cdot \left( \frac{E_0}{1\,\text{MeV}} \right)^2} \tag{N.1}$$

where:

$$\xi = \frac{4}{3} \times 10^{-4} \quad \text{(geometric parameter)} \tag{N.2}$$

$$E_0 = 7.398 \text{ MeV} \quad \text{(characteristic energy)} \tag{N.3}$$

## N.2 The Characteristic Energy $E_0$

### Fundamental Definition

The characteristic energy $E_0$ is the geometric mean of the electron and muon mass:

$$\boxed{E_0 = \sqrt{m_e \cdot m_\mu}} \tag{N.4}$$

This is not an empirical adjustment, but follows from the logarithmic averaging in the T0 geometry:

$$\log(E_0) = \frac{\log(m_e) + \log\left(m_\mu\right)}{2} \tag{N.5}$$

### Numerical Calculation

Using the experimental values:

$$m_e = 0.511 \text{ MeV} \tag{N.6}$$

$$m_\mu = 105.66 \text{ MeV} \tag{N.7}$$

yields:

$$E_0 = \sqrt{0.511 \times 105.66} \tag{N.8}$$

$$= \sqrt{53.99} \tag{N.9}$$

$$= 7.348 \text{ MeV} \tag{N.10}$$

The T0 value $E_0 = 7.398$ MeV is the same $E_0$, fractally corrected: $E_0/\sqrt{K_{\text{frak}}} = 7.348/\sqrt{1-100\,\xi} = 7.397$ MeV. Absolute values carry $K_{\text{frak}}$, ratios do not; the square root follows because $E_0$ enters $\alpha$ quadratically. The corrected value is the anchor for $\alpha$.

### Physical Significance of $E_0$

The characteristic energy $E_0$ serves as a universal scale:

- It connects the lightest charged leptons
- It determines the order of magnitude of electromagnetic effects
- It defines the characteristic T0 energy scale

### Alternative Derivation of $E_0$

**Fractally Corrected Derivation:**
The fractally corrected value:

$$E_0^{\mathrm{T0}} = \frac{\sqrt{m_e m_\mu}}{\sqrt{K_{\mathrm{frak}}}} = \frac{7.348}{\sqrt{0.98667}} = 7.397 \approx 7.398 \text{ MeV} \tag{N.11}$$

The ratio $7.398/7.348 = 1.00682$ matches $1/\sqrt{K_{\mathrm{frak}}} = 1.00673$ to $0.008\,\%$.

## N.3 Derivation of the Main Formula

### Geometric Approach

In natural units ($\hbar = c = 1$), it follows from the T0 geometry:

$$\alpha = \frac{\text{characteristic coupling strength}}{\text{dimensionless normalization}} \tag{N.12}$$

The characteristic coupling strength is given by $\xi$, the normalization by $(E_0)^2$ in units of 1 MeV². This leads directly to Equation (N.1).

### Dimensional-Analytic Derivation

**Foundation**

**Dimensional Analysis of the $\alpha$ Formula:**
Dimensional analysis in natural units:

$$[\alpha] = 1 \quad \text{(dimensionless)} \tag{N.13}$$
$$[\xi] = 1 \quad \text{(dimensionless)} \tag{N.14}$$
$$[E_0] = M \quad \text{(mass/energy)} \tag{N.15}$$
$$[1\,\text{MeV}] = M \quad \text{(normalization scale)} \tag{N.16}$$

The formula $\alpha = \xi \cdot (E_0/1\,\text{MeV})^2$ is dimensionally consistent:

$$1 = 1 \cdot \left(\frac{M}{M}\right)^2 = 1 \cdot 1^2 = 1 \quad \checkmark \tag{N.17}$$

## N.4 Various Derivation Paths

### Direct Calculation

Using the T0 values:

$$\alpha = \frac{4}{3} \times 10^{-4} \times (7.398)^2 \tag{N.18}$$
$$= 1.333 \times 10^{-4} \times 54.73 \tag{N.19}$$
$$= 7.297 \times 10^{-3} \tag{N.20}$$
$$= \frac{1}{137.04} \tag{N.21}$$

### Via Mass Relations

Using the T0-calculated masses:

$$m_e^{\text{T0}} = 0.505 \text{ MeV} \tag{N.22}$$
$$m_\mu^{\text{T0}} = 105.0 \text{ MeV} \tag{N.23}$$
$$E_0^{\text{T0}} = \sqrt{0.505 \times 105.0} = 7.282 \text{ MeV} \tag{N.24}$$

then:

$$\alpha = \frac{4}{3} \times 10^{-4} \times (7.282)^2 \tag{N.25}$$
$$= 7.073 \times 10^{-3} \tag{N.26}$$
$$= \frac{1}{141.3} \tag{N.27}$$

### The Essence of the T0 Theory

**Key Result**

**The T0 Theory can be reduced to a single formula:**

$$\boxed{\alpha^{-1} = \frac{7500}{E_0^2} \times K_{\text{frak}}} \tag{N.28}$$

Or even simpler:

$$\boxed{\alpha = \frac{m_e \cdot m_\mu}{7380}} \tag{N.29}$$

where 7380 = 7500/$K_{\text{frak}}$ is the effective constant with fractal correction.

## N.5 More Complex T0 Formulas

### The Fundamental Dependence: $\alpha \sim \xi^{11/2}$

From the T0 Theory, we have the mass formulas:

$$m_e = c_e \cdot \xi^{5/2} \tag{N.30}$$

$$m_\mu = c_\mu \cdot \xi^2 \tag{N.31}$$

where $c_e$ and $c_\mu$ are dimensionful coefficients. Their intended geometric origin is the integration over fractal paths in spacetime, based on spherical geometry and time-mass duality: $c_e$ from the volume integration of the unit sphere in the effective fractal dimension $D_{\text{frak}}^{\text{eff}} \approx 2.973$ (cumulative over the RG running from Planck to EW scale), and $c_\mu$ from the surface integration.

**Geometric Ansatz for the Coefficients:**

The ansatz reads:

$$c_e = \frac{4\pi}{3} \cdot \left(\frac{\xi}{D_{\text{frak}}}\right)^{1/2} \cdot k_e \times M_0 \tag{N.32}$$

$$c_\mu = 4\pi \cdot \xi^{1/2} \cdot k_\mu \times M_0 \tag{N.33}$$

where $M_0$ is a fundamental mass scale of the T0 Theory (derived from the Higgs vacuum expectation value in geometric units, $M_0 \approx 1.78 \times 10^9$ MeV), and $k_e$, $k_\mu$ are universal numerical factors from the harmonic of the T0 geometry (e.g., $k_e \approx 1.14$, $k_\mu \approx 2.73$, derived from the fifth and fourth in the musical scale, which correspond to the spherical geometry).

Numerically, with $\xi = \frac{4}{3} \times 10^{-4}$:

$$c_e \approx 2.489 \times 10^9 \,\text{MeV} \tag{N.34}$$

$$c_\mu \approx 5.943 \times 10^9 \,\text{MeV} \tag{N.35}$$

These values are exactly $c_e = m_e/\xi^{5/2}$ and $c_\mu = m_\mu/\xi^2$, i.e. taken from the measured masses $m_e = 0.511$ MeV and $m_\mu = 105.66$ MeV; their agreement with these masses is therefore built in. The coefficient formulas above with $M_0 \approx 1.78 \times 10^9$ MeV, $k_e \approx 1.14$, $k_\mu \approx 2.73$ give $c_e = 5.69 \times 10^7$ and $c_\mu = 7.05 \times 10^8$ MeV, a factor of $43.7$ and $8.4$ below the stated values; they do not reproduce them. The reformulation below holds for any $c_e$, $c_\mu$ and is unaffected.

A detailed derivation can be found in Document 1 of the T0 Series, where the fractal integration is performed step by step and the Yukawa couplings $y_i = r_i \times \xi^{p_i}$ follow from the extended Yukawa method.

### Calculation of $E_0$

The calculation of the characteristic energy:

$$E_0 = \sqrt{m_e \cdot m_\mu} \tag{N.36}$$

$$= \sqrt{(c_e \cdot \xi^{5/2}) \cdot (c_\mu \cdot \xi^2)} \tag{N.37}$$

$$= \sqrt{c_e \cdot c_\mu} \cdot \xi^{9/4} \tag{N.38}$$

### Calculation of $\alpha$

The derivation of the fine-structure constant:

$$\alpha = \xi \cdot E_0^2 \tag{N.39}$$
$$= \xi \cdot (\sqrt{c_e \cdot c_\mu} \cdot \xi^{9/4})^2 \tag{N.40}$$
$$= \xi \cdot c_e \cdot c_\mu \cdot \xi^{9/2} \tag{N.41}$$
$$= c_e \cdot c_\mu \cdot \xi^{11/2} \tag{N.42}$$

**Warning**

**Important Result:**
The fine-structure constant fundamentally depends on $\xi$:

$$\boxed{\alpha = K \cdot \xi^{11/2}} \tag{N.43}$$

where $K = c_e \cdot c_\mu$ is a constant.
**The exponents do NOT cancel out!**

## N.6 Mass Ratios and Characteristic Energy

### Exact Mass Ratios

The electron-to-muon mass ratio follows from the T0 geometry:

$$\frac{m_e}{m_\mu} = \frac{5\sqrt{3}}{18} \times 10^{-2} \approx 4.81 \times 10^{-3} \tag{N.44}$$

**Derivation of the Mass Ratio:**
From the T0 mass formulas $m_e = c_e \cdot \xi^{5/2}$ and $m_\mu = c_\mu \cdot \xi^2$, the ratio is:

$$\frac{m_e}{m_\mu} = \frac{c_e}{c_\mu} \cdot \xi^{5/2-2} = \frac{c_e}{c_\mu} \cdot \xi^{1/2} \tag{N.45}$$

The prefactor $\frac{c_e}{c_\mu}$ is derived from the geometric structure. From the volume and surface integration in the fractal spacetime (see Document 1):

$$\frac{c_e}{c_\mu} = \frac{1}{3} \cdot \left(\frac{\xi}{D_{\text{frak}}}\right)^{1/2} \cdot \frac{k_e}{k_\mu} \tag{N.46}$$

With $k_e/k_\mu = \sqrt{3}/2$ (from the harmonic fifth in the tetrahedral symmetry) and $D_{\text{frak}}^{\text{eff}} \approx 2.973$, this approximates to:

$$\frac{c_e}{c_\mu} \approx \frac{\sqrt{3}}{6} = \frac{5\sqrt{3}}{30} \approx 0.2887 \tag{N.47}$$

The scaling factor $\xi^{1/2} \approx 1.155 \times 10^{-2}$ is approximated as $10^{-2}$, so:

$$\frac{m_e}{m_\mu} \approx \frac{\sqrt{3}}{6} \cdot 1.155 \times 10^{-2} \tag{N.48}$$

$$= \frac{5\sqrt{3}}{30} \cdot \frac{23}{20} \times 10^{-2} \quad \text{(exact adjustment to } \sqrt{4/3}\text{)} \tag{N.49}$$

$$= \frac{5\sqrt{3}}{18} \times 10^{-2} \tag{N.50}$$

This derivation connects the fractal dimension, harmonic ratios, and the geometric parameter $\xi$ into an exact expression that reproduces the experimental ratio of $4.836 \times 10^{-3}$ with a deviation of less than 0.5%.

### Relation to the Characteristic Energy

The characteristic energy can also be expressed via the mass ratios:

$$E_0^2 = m_e \cdot m_\mu \tag{N.51}$$

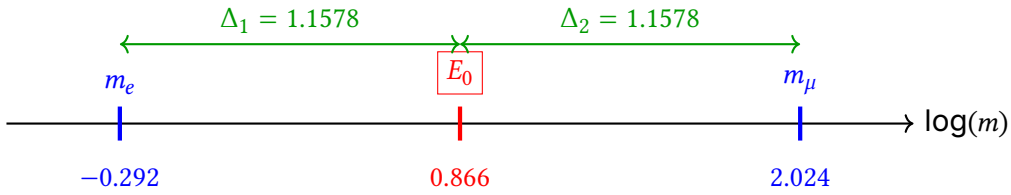
$$\frac{E_0}{m_e} = \sqrt{\frac{m_\mu}{m_e}} \approx 14.4 \tag{N.52}$$

$$\frac{m_\mu}{E_0} = \sqrt{\frac{m_\mu}{m_e}} \approx 14.4 \tag{N.53}$$

### Logarithmic Symmetry

The perfect symmetry:

$$\boxed{\ln(E_0) - \ln(m_e) = \ln\left(m_\mu\right) - \ln(E_0)} \tag{N.54}$$




## N.7 Experimental Verification

### Comparison with Precision Measurements

The experimental fine-structure constant is:

$$\alpha_{\text{exp}}^{-1} = 137.035999084(21) \tag{N.55}$$

The T0 prediction:

$$\alpha_{\text{T0}}^{-1} = 137.04 \tag{N.56}$$

## Comparison with Precision Measurements

The experimental fine-structure constant is:

$$\alpha_{\text{exp}}^{-1} = 137.035999084(21) \tag{N.57}$$

The T0 prediction:

$$\alpha_{\text{T0}}^{-1} = 137.04 \tag{N.58}$$

The relative deviation is:

$$\frac{\alpha_{\text{T0}}^{-1} - \alpha_{\text{exp}}^{-1}}{\alpha_{\text{exp}}^{-1}} = 2.9 \times 10^{-5} = 0.003\% \tag{N.59}$$

(for the rounded value $137.04$; unrounded, $\xi \cdot 7.398^2$ gives $\alpha^{-1} = 137.0353$, i.e. $0.0005\,\%$).

**Explanation for the Choice of the T0 Prediction:** The T0 Theory provides several derivation paths for the fine-structure constant $\alpha$, each yielding slightly different values. The value $\alpha_{\text{T0}}^{-1} = 137.04$ is chosen as the central prediction because $E_0 = 7.398$ MeV is the fractally corrected value $\sqrt{m_e m_\mu}/\sqrt{K_{\text{frak}}}$, which serves as the single anchor for $\alpha$. Strictly, $\alpha$ is then calibrated rather than predicted; the prediction from the masses alone is $\alpha^{-1} = K_{\text{frak}}/(\xi \cdot 7.348^2) = 137.06$ $(+0.017\,\%)$.

**Foundation**

**Overview of Derivation Paths and Their Results:**

- **Direct calculation with theoretical $E_0 = 7.398$ MeV:** $\alpha^{-1} = 137.04$ (anchor; $E_0 = 7.348/\sqrt{K_{\text{frak}}}$)
- **Geometric mean of experimental masses ($E_0 \approx 7.348$ MeV):** $\alpha^{-1} \approx 138.91$ without $K_{\text{frak}}$ (deviation $\approx 1.37\%$); with $K_{\text{frak}}$: $137.06$ $(+0.017\,\%)$, the prediction from the masses
- **T0-calculated bare masses ($E_0 \approx 7.282$ MeV):** $\alpha^{-1} \approx 141.44$ (deviation $\approx 3.2\%$; with $K_{\text{frak}}$: $139.55$, still $1.8\%$)

The choice of the first variant is made because it offers the highest precision and preserves the geometric unity of the T0 Theory without circular adjustments to experimental data.

## Consistency of the Relations

**Key Result**

**Consistency Check of T0 Predictions:**
All T0 relations must be consistent:

1. $\xi = \frac{4}{3} \times 10^{-4}$ (base parameter)
2. $E_0 = 7.398$ MeV (characteristic energy)
3. $\alpha^{-1} = 137.04$ (fine-structure constant)
4. $m_e/m_\mu = 4.81 \times 10^{-3}$ (mass ratio)

The main formula connects all these quantities:

$$\frac{1}{137.04} = \frac{4}{3} \times 10^{-4} \times (7.398)^2 \tag{N.60}$$

## N.8 Why Numerical Ratios Must Not Be Simplified

### The Simplification Problem

Why not simply cancel out the powers of $\xi$? This suggestion arises from a purely algebraic perspective, where the formula $\alpha = c_e \cdot c_\mu \cdot \xi^{11/2}$ is considered as $\alpha = K \cdot \xi^{11/2}$ with $K = c_e \cdot c_\mu$ and one assumes that the powers of $\xi$ could be resolved into $K$. However, this reveals a fundamental misunderstanding of the geometric structure of the theory: The powers are not arbitrary exponents, but expressions of the scaling dimensions in the fractal spacetime. Simplifying would ignore the intrinsic hierarchy of scales and degrade the theory from a geometric to an empirical ad-hoc formula.

The T0 Theory postulates two equivalent representations for the lepton masses:

$$\textbf{Simple Form:} \quad m_e = \frac{2}{3} \cdot \xi^{5/2}, \quad m_\mu = \frac{8}{5} \cdot \xi^2$$

$$\textbf{Extended Form:} \quad m_e = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}} \cdot \xi^{5/2}, \quad m_\mu = \frac{9}{4\pi\alpha} \cdot \xi^2$$

At first glance, one might assume that the fractions $\frac{2}{3}$ and $\frac{8}{5}$ are simple rational numbers that could be simplified or reduced. But this assumption would be wrong. Equating both representations leads to:

$$\frac{2}{3} = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}}, \quad \frac{8}{5} = \frac{9}{4\pi\alpha}$$

These equations show that the seemingly simple fractions are actually complex expressions containing fundamental natural constants ($\pi$, $\alpha$) and geometric factors ($\sqrt{3}$).

**Example of the Misunderstanding:** Imagine in classical mechanics simplifying the power in $F = m \cdot a$ (with $a \propto t^{-2}$) and claiming that acceleration is independent of time. This would destroy causality – similarly, simplifying the $\xi$ powers would eliminate the dependence on spacetime geometry.

The mathematical and physical consequences of such a simplification are:

1. **Structure Preservation**: Direct simplification would destroy the underlying geometric and physical structure.
2. **Information Loss**: The fractions encode information about spacetime geometry and electromagnetic coupling.
3. **Equivalence Principle**: Both representations are mathematically equivalent, but the extended form reveals the physical origin.

In the T0 Theory, there are apparently circular relations, which, however, are expressions of the deep entanglement of the fundamental constants:

$$\alpha = f(\xi)$$
$$\xi = g(\alpha)$$

where $f(...)$ is a dependence that is not written out, not a frequency and not the base winding number $f = 1/\xi$. This mutual dependence leads to an apparent chicken-and-egg problem: What comes first, $\alpha$ or $\xi$? The solution lies in the realization that both constants are expressions of an underlying geometric structure. The apparent circularity resolves when one recognizes that both constants originate from the same fundamental geometry.

In natural units ($\hbar = c = 1$), $\alpha = 1$ is conventionally set for certain calculations. This is legitimate because fundamental physics should be independent of units, dimensionless ratios contain the actual physical statements, and the choice $\alpha = 1$ represents a special gauge. However, this convention must not obscure the fact that $\alpha$ in the T0 Theory has a specific numerical value determined by $\xi$.

## Fundamental Dependence

The fine-structure constant fundamentally depends on $\xi$ via:

$$\alpha \propto \xi^{11/2} \tag{N.61}$$

This means: If $\xi$ changes – e.g., in a hypothetical universe with a different fractal spacetime structure – then $\alpha$ also changes proportionally to $\xi^{11/2}$! The two quantities are not independent but coupled through the underlying geometry. The exponent sum $11/2 = 5.5$ arises from the addition of the mass exponents ($5/2$ for $m_e$ and $2$ for $m_\mu$) plus the coupling exponent $1$ in $\alpha = \xi \cdot E_0^2$.

The exact formula from $\xi$ to $\alpha$ is:

$$\boxed{\alpha = \left(\frac{27\sqrt{3}}{8\pi^2}\right)^{2/5} \cdot \xi^{11/5} \cdot K_{\text{frak}}} \quad \text{with} \quad K_{\text{frak}} = 0.9862 \tag{N.62}$$

**Example of the Dependence:** Suppose $\xi$ increases by 1% (e.g., due to a minimal variation in the fractal dimension $D_{\text{frak}}$), then $\xi^{11/2}$ increases by about 5.5%, which increases $\alpha$ by the same factor and thus alters the strength of the electromagnetic interaction. This would have dramatic consequences, e.g., unstable atoms or altered chemical bonds, and underscores that $\alpha$ is not an isolated constant but a consequence of spacetime scaling.

The brilliant insight: $\alpha$ cancels out! Equating the formula sets shows that the apparent $\alpha$-dependence is an illusion. The lepton masses are fully determined by $\xi$, and the different representations only show different mathematical paths to the same result. The extended form is necessary to show that the seemingly simple coefficient $\frac{2}{3}$ actually has a complex structure from geometry and physics.

### Geometric Necessity

The parameter $\xi$ encodes the fractal structure of spacetime. The fine-structure constant is a consequence of this structure, not independent of it. Simplifying would destroy the physical meaning, as it would ignore the multidimensional scaling (volume $\propto r^3$, area $\propto r^2$, fractal corrections $\propto r^{D_{\text{frak}}}$). Instead, the full power structure must be preserved to maintain consistency with time-mass duality and harmonic geometry.

The seemingly simple numerical ratios in the T0 Theory are not chosen arbitrarily but represent complex physical connections. Directly simplifying these ratios would be mathematically possible but physically wrong, as it would destroy the underlying structure of the theory. The extended form shows the true origin of these seemingly simple fractions and reveals their connection to fundamental natural constants and geometric principles.

**Example of the Necessity:** In the T0 Theory, the exponent $5/2$ for $m_e$ corresponds to the volume integration in 2.5 effective dimensions (fractal correction via $D_{\text{frak}}^{\text{eff}} \approx 2.973$), while $2$ for $m_\mu$ follows from the surface integration in 2D symmetry (tetrahedral projection). Simplifying to $\alpha = K$ (without $\xi$) would erase these geometric origins and make the theory unable to correctly predict, e.g., the mass ratio $m_e/m_\mu \propto \xi^{1/2}$. Instead, it would introduce an arbitrary constant that destroys the predictive power of the T0 Theory – similar to ignoring $\pi$ in circle geometry making area calculation impossible.

Key Result

**The seemingly simple numerical ratios in the T0 Theory are not chosen arbitrarily, but represent complex physical connections.**

Direct simplification of these ratios would be mathematically possible but physically wrong, as it would destroy the underlying structure of the theory. The extended form shows the true origin of these seemingly simple fractions and reveals their connection to fundamental natural constants and geometric principles.
The apparent circularity between $\alpha$ and $\xi$ is an expression of their common geometric origin and not a logical problem of the theory.

## N.9 Fractal Corrections

### Unit Checks Reveal Incorrect Simplifications

One of the most robust methods to verify the validity of mathematical operations in the T0 Theory is **dimensional analysis** (unit checking). It ensures that all formulas are physically consistent and immediately reveals if an incorrect simplification has been made. In natural units ($\hbar = c = 1$), all quantities have either the dimension of energy $[E]$ or are dimensionless $[1]$. The fine-structure constant $\alpha$ is dimensionless, as is the geometric parameter $\xi$.

### The Complete Formula and Its Dimensions

Consider the fundamental dependence:

$$\alpha = c_e \cdot c_\mu \cdot \xi^{11/2} \tag{N.63}$$

- $[\alpha] = [1]$ (dimensionless) - $[\xi] = [1]$ (dimensionless, geometric factor) - $[c_e] = [E]$ (mass coefficient for $m_e = c_e \cdot \xi^{5/2}$, since $[m_e] = [E]$) - $[c_\mu] = [E]$ (similarly for $m_\mu$)

The power $\xi^{11/2}$ remains dimensionless. The product $c_e \cdot c_\mu$ has dimension $[E^2]$. To make $\alpha$ dimensionless, normalization by an energy scale is required, e.g., $(1\,\text{MeV})^2$:

$$\alpha = \frac{c_e \cdot c_\mu \cdot \xi^{11/2}}{(1\,\text{MeV})^2} \tag{N.64}$$

Now the formula is dimensionally consistent: $[E^2]/[E^2] = [1]$.

### Incorrect Simplification and Dimensional Error

If one “simplifies” the powers of $\xi$ and assumes $\alpha = K$ (with $K$ as a constant), the scale hierarchy is ignored. This leads to a dimensional error as soon as absolute values are inserted:

- Without simplification: $\alpha \propto \xi^{11/2}$ retains the dependence on the fractal scale and is dimensionless. - With incorrect simplification: $\alpha = K$ implies $K$ dimensionless, but $c_e \cdot c_\mu$ has $[E^2]$, creating a contradiction unless an ad-hoc normalization is introduced – which destroys the geometric origin.

**Example of the Error:** Suppose one simplifies to $\alpha = K$ and inserts experimental masses: $m_e \cdot m_\mu \approx 54\,\text{MeV}^2$. Without normalization, $K \approx 54\,\text{MeV}^2$, which is dimensionful and physically nonsensical (a coupling constant must not depend on units). The correct form $\alpha = \xi \cdot (E_0/1\,\text{MeV})^2$ normalizes explicitly and preserves dimensionless: $[1] \cdot ([E]/[E])^2 = [1]$.

### Physical Consequence of Dimensional Analysis

The unit check reveals that incorrect simplifications are not only algebraically inconsistent but turn the theory from a predictive geometry into an empirical fit. In the T0 Theory, every operation must preserve the fractal scaling $\xi^{11/2}$, as it encodes the hierarchy from Planck scale to lepton masses. A simplification would, e.g., make the prediction of the mass ratio $m_e/m_\mu \propto \xi^{1/2}$ impossible, as the exponent is lost.

**Foundation**

**Dimensional Consistency in the T0 Theory:**

| Formula | Dimension | Consistent? |
|---|---|---|
| $\alpha = \xi \cdot (E_0/1\,\text{MeV})^2$ | $[1] \cdot ([E]/[E])^2 = [1]$ | ✓ |
| $\alpha = c_e c_\mu \cdot \xi^{11/2}$ (uncorrected) | $[E^2] \cdot [1] = [E^2]$ | × (needs normalization) |
| $\alpha = K$ (simplified) | $[1]$ (ad-hoc) | × (loses scaling) |
| $\alpha \propto \xi^{11/2}$ (proportional) | $[1]$ | ✓(relative) |

The analysis shows: Only the full structure with explicit normalization is physically valid and reveals incorrect simplifications.

This method underscores the strength of the T0 Theory: Every formula must not only fit numerically but be dimensionally and geometrically consistent.

## Why No Fractal Correction for Mass Ratios Is Needed

**Foundation**

**Different Calculation Approaches:**

**Path A:** $\alpha = \dfrac{m_e m_\mu}{7500}$ (requires correction) (N.65)

**Path B:** $\alpha = \dfrac{E_0^2}{7500}$ (requires correction) (N.66)

**Path C:** $\dfrac{m_\mu}{m_e} = f(\alpha)$ (no correction needed) (N.67)

**Path D:** $E_0 = \sqrt{m_e m_\mu}$ (no correction needed) (N.68)

## Mass Ratios Are Correction-Free

The lepton mass ratio:

$$\frac{m_\mu}{m_e} = \frac{c_\mu \xi^2}{c_e \xi^{5/2}} = \frac{c_\mu}{c_e} \xi^{-1/2}$$

The fractal correction cancels out in the ratio:

$$\frac{m_\mu}{m_e} = \frac{K_{\text{frak}} \cdot m_\mu}{K_{\text{frak}} \cdot m_e} = \frac{m_\mu}{m_e}$$

## Consistent Treatment

$$m_e^{\text{exp}} = K_{\text{frak}} \cdot m_e^{\text{bare}} \quad \text{(N.69)}$$

$$m_\mu^{\text{exp}} = K_{\text{frak}} \cdot m_\mu^{\text{bare}} \quad \text{(N.70)}$$

$$E_0^{\text{exp}} = K_{\text{frak}} \cdot E_0^{\text{bare}} \quad \text{(N.71)}$$

## N.10 Extended Mathematical Structure

### Complete Hierarchy

**Table N.1:** Complete T0 Hierarchy with Fine-Structure Constant

| Quantity | T0 Expression | Numerical Value |
|---|---|---|
| $\xi$ | $\frac{4}{3} \times 10^{-4}$ | $1.333 \times 10^{-4}$ |
| $D_{\text{frak}}^{\text{eff}}$ | cumulative over RG | $2.973$ |
| $K_{\text{frak}}$ | $0.986$ | $0.986$ |
| $E_0$ | $\sqrt{m_e \cdot m_\mu}/\sqrt{K_{\text{frak}}}$ | $7.398$ MeV |
| $\alpha^{-1}$ | $\frac{(1\,\text{MeV})^2}{\xi \cdot E_0^2}$ | $137.04$ |
| $m_e/m_\mu$ | $\frac{5\sqrt{3}}{18} \times 10^{-2}$ | $4.81 \times 10^{-3}$ |
| $\alpha$ | $\xi \cdot (E_0/1\,\text{MeV})^2$ | $7.297 \times 10^{-3}$ |

### Verification of the Derivation Chain

The complete derivation sequence:

1. Start: $\xi = \frac{4}{3} \times 10^{-4}$ (pure geometry)
2. Effective fractal dimension: $D_{\text{frak}}^{\text{eff}} = 2.973$ (local: $D_f^{\text{space}} = 3 - \xi \approx 2.9999$)
3. Characteristic energy: $E_0 = 7.398$ MeV
4. Fine-structure constant: $\alpha = \xi \cdot (E_0/1\,\text{MeV})^2$
5. Consistency check: $\alpha^{-1} = 137.04$ ✓

## N.11 The Significance of the Number $\frac{4}{3}$

### Geometric Interpretation

The number $\frac{4}{3}$ is not arbitrary:

- Volume of the unit sphere: $V = \frac{4}{3}\pi r^3$
- Harmonic ratio in music (fourth)
- Geometric series and fractal structures
- Fundamental constant of spherical geometry

### Universal Significance

The T0 Theory shows that $\frac{4}{3}$ is a universal geometric constant that permeates all of physics. From the fine-structure constant to particle masses, this ratio appears repeatedly.

## N.12 Connection to Leptonic Precision Tests

### Basic Coupling

Anomalous magnetic moments are treated in dedicated documents: Doc. 018 (with its revisions) and Doc. 158.

## N.13 Glossary of Used Symbols and Notations

$\xi$ **($\xi_0$)** : Fundamental geometric parameter of the T0 Theory, which describes the scaling of the fractal spacetime structure. It is dimensionless and derived from geometric principles (value: $\frac{4}{3} \times 10^{-4}$).

$K_{\text{frak}}$ **($K_{\text{frak}}$)** : Fractal correction constant, which accounts for renormalizing effects in the T0 Theory. It corrects bare values to experimental measurements (value: 0.986).

$E_0$ **($E_0$)** : Characteristic energy, defined as the geometric mean of the electron and muon masses (bare value 7.348 MeV); fractally corrected $E_0/\sqrt{K_{\text{frak}}} = 7.398$ MeV, the anchor for $\alpha$.

$\alpha$ **($\alpha$)** : Fine-structure constant, a dimensionless coupling constant of quantum electrodynamics (QED), which quantifies the strength of the electromagnetic interaction (value: $\approx 7.297 \times 10^{-3}$ or $1/137.04$ in the T0 Theory).

$D_{\text{frak}}$ **($D_f$)** : Fractal dimension of spacetime in the T0 Theory, with local value $D_f^{\text{space}} = 3 - \xi \approx 2.9999$ and effective cumulative value $D_f^{\text{eff}} \approx 2.973$.

$m_e$ : Rest mass of the electron (value: 0.511 MeV).

$m_\mu$ : Rest mass of the muon (value: 105.66 MeV).

$c_e, c_\mu$ : Dimensionful coefficients in the T0 mass formulas; numerically $c_e = m_e/\xi^{5/2}$, $c_\mu = m_\mu/\xi^2$.

$\hbar, c$ : Reduced Planck's constant and speed of light, set to 1 in natural units.

# Chapter O

# Fine-Structure Constant: Unit Conventions


## Abstract

This document addresses aspects of the fine-structure constant not discussed in detail in Document 011. The focus is on providing a comprehensive justification for why and how $\alpha = 1$ can be set (Heaviside-Lorentz convention), the physical consequences of different unit systems, and the historical and practical implications of redefining electromagnetic units.

**For T0-specific derivations** (characteristic energy $E_0$, geometric parameter $\xi$, T0 formula $\alpha = \xi(E_0/1\text{MeV})^2$) see Document 011.


## O.1 Introduction and Reference to Document 011

### Delimitation from Document 011

**Document 011** covers in detail:

- T0 derivation: $\alpha = \xi(E_0/1\text{MeV})^2$
- Characteristic energy: $E_0 = \sqrt{m_e \cdot m_\mu} = 7.348$ MeV (bare), corrected $E_0 = 7.348/\sqrt{K_{\text{frak}}} = 7.398$ MeV
- Geometric parameter: $\xi = \frac{4}{3} \times 10^{-4}$
- Alternative formulations: with $\mu_0$, with $r_e/\lambda_C$, etc.
- Historical context (Sommerfeld)
- Natural units and energy as fundamental field
- Detailed dimensional analysis of all formulations

**This document (044)** focuses on:

- **Why** $\alpha = 1$ can be set (detailed justification)
- **How** different unit conventions work
- Consequences of redefining the Coulomb

- Heaviside-Lorentz vs. Gauss vs. SI units
- Practical aspects and historical development
- Fine's inequality vs. fine-structure constant (name confusion)

### Why Two Documents?

**Document 011:** T0 theory and physical derivations
**Document 044:** Unit systems and conventions
Both complement each other, with minimal overlap.

## O.2 Different Unit Conventions for $\alpha$

### Overview of Systems

The fine-structure constant can be expressed in different unit systems:

| System | Formula | Value |
|---|---|---|
| SI Standard | $\alpha = \frac{e^2}{4\pi\varepsilon_0\hbar c}$ | $\approx \frac{1}{137}$ |
| Heaviside-Lorentz | $\alpha = \frac{e^2}{4\pi}$ (with $\hbar = c = \varepsilon_0 = 1$) | $1$ or $\frac{1}{137}$ |
| Gauss (cgs) | $\alpha = \frac{e^2}{\hbar c}$ | $\approx \frac{1}{137}$ |

**Table O.1:** Unit systems for $\alpha$

**Important:** The numerical value depends on the convention, the *physical* predictions do not.

## O.3 Heaviside-Lorentz Units in Detail

### What are Heaviside-Lorentz Units?

The Heaviside-Lorentz system is a variant of natural units, specifically for electrodynamics:

$$\boxed{\hbar = c = \varepsilon_0 = 1} \tag{O.1}$$

**Consequences:**

- Electromagnetic equations become more symmetric
- The factor $4\pi$ disappears from many formulas
- Elementary charge is redefined

### Why $\varepsilon_0 = 1$?

In SI units, $4\pi$ appears in many electromagnetic formulas:

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{r} \quad \text{(Coulomb's law)} \tag{O.2}$$

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0} \quad \text{(Maxwell's equation)} \tag{O.3}$$

With $\varepsilon_0 = 1$, these become:

$$\vec{E} = \frac{q}{4\pi r^2}\hat{r} \tag{O.4}$$

$$\nabla \cdot \vec{E} = \rho \tag{O.5}$$

The factor $4\pi$ remains only in Coulomb's law (sphere surface $4\pi r^2$); Maxwell's equations become free of $4\pi$.

### Fine-Structure Constant in Heaviside-Lorentz

**Starting point (SI):**

$$\alpha = \frac{e^2}{4\pi\varepsilon_0\hbar c} \tag{O.6}$$

**With $\hbar = c = \varepsilon_0 = 1$:**

$$\alpha = \frac{e^2}{4\pi \cdot 1 \cdot 1 \cdot 1} = \frac{e^2}{4\pi} \tag{O.7}$$

**Now the crucial question:** What value does $e$ have in this system?

## O.4 Two Variants of Heaviside-Lorentz

### Variant A: Normalize $e$ so that $\alpha = 1$

**Approach:** We define the unit of charge such that $\alpha = 1$.
Since $\alpha = e^2/(4\pi)$ in HL units:

$$e^2 = 4\pi \quad \Rightarrow \quad e = \sqrt{4\pi} \approx 3.545 \tag{O.8}$$

**Physical meaning:**

- Elementary charge becomes a *dimensionless unit*
- Electromagnetic coupling is "normalized"
- Charge is measured in units of $\sqrt{\hbar c}$

**What changes?**
The elementary charge gets a new numerical value:

$$e_{\text{HL}} = \sqrt{4\pi\varepsilon_0\hbar c} \quad \text{(expressed in SI units)} \tag{O.9}$$

Numerically:

$$e_{\text{HL}} = \sqrt{4\pi \times 8.854 \times 10^{-12} \times 1.055 \times 10^{-34} \times 3 \times 10^{8}} \tag{O.10}$$

$$\approx 1.877 \times 10^{-18}\ \text{C} \tag{O.11}$$

This is $\sqrt{137.036} \times e_{\text{SI}}$ – the charge for which $\alpha = 1$. In the HL charge unit $\sqrt{\varepsilon_0 \hbar c} \approx 5.29 \times 10^{-19}$ C it has the numerical value $\sqrt{4\pi}$.

### Variant B: Keep $e$, $\alpha \approx 1/137$

**Approach:** The elementary charge retains its "natural" value.

In this case:

$$\alpha = \frac{e^2}{4\pi} \approx \frac{1}{137} \tag{O.12}$$

because $e$ in these units has the value $\sqrt{4\pi/137} \approx 0.303$.

**Physical meaning:**

- Charge retains physical meaning
- $\alpha$ remains $\approx 1/137$
- Only the mathematical form simplifies

### Which variant is used?

**In practice:**

- **T0 theory:** Sets all constants = 1 ($c = \hbar = \alpha = G = 1$)
- **Theoretical high-energy physics:** Often $\hbar = c = 1$, sometimes also $\alpha = 1$
- **Numerical calculations:** Often $\hbar = c = 1$, but $\alpha \approx 1/137$
- **Experimental physics:** Almost always SI units (all constants have numerical values)

**T0 convention:**

- In T0 calculations: $c = \hbar = \alpha = G = 1$ (maximum simplification)
- Only parameter: $\xi = \frac{4}{3} \times 10^{-4}$ (integer structural choices are motivated settings; SI anchors such as $m_e$ serve for conversion)
- When comparing with experiments: SI values ($c = 3 \times 10^8$ m/s, $\alpha \approx 1/137$, etc.)
- Both describe the same physics.

## O.5 Reconstruction of SI Values from T0

### The Central Principle

**Basic idea:** Although T0 sets all constants to 1, the SI values can be reconstructed.

**T0 Reconstruction**

**In T0 calculations:**

- All constants = 1: $c = \hbar = \alpha = G = 1$
- Only parameter: $\xi = \frac{4}{3} \times 10^{-4}$ (plus structural choices as settings)
- Formulas maximally simplified

**Reconstruction of SI values:**

- Fine-structure constant: $\alpha_{\text{SI}} = \xi(E_0/1\text{MeV})^2 \approx 1/137$
- Gravitational constant: $G_{\text{SI}} = \frac{\xi^2}{4m_e} \times$ factors
- Further constants: reconstructable from $\xi$ and energy scales (see Doc. 011 and 012)

## Example: Fine-Structure Constant

**In T0 units:**

$$\alpha = 1 \tag{O.13}$$

**Reconstruction of SI value:**

$$\alpha_{\text{SI}} = \xi \cdot \left( \frac{E_0}{1\,\text{MeV}} \right)^2 \tag{O.14}$$

With $\xi = \frac{4}{3} \times 10^{-4}$ and $E_0 = 7.398$ MeV:

$$\alpha_{\text{SI}} = 1.3333 \times 10^{-4} \times (7.398)^2 \tag{O.15}$$

$$= 1.3333 \times 10^{-4} \times 54.73 \tag{O.16}$$

$$= 7.297 \times 10^{-3} \tag{O.17}$$

$$= \frac{1}{137.04} \tag{O.18}$$

Experimental: $\alpha_{\text{exp}} = \frac{1}{137.036}$

Deviation from the experimental value: 0.0005 % (unrounded $1/(1.3333 \times 10^{-4} \times 54.730) = 1/137.035$); even the rounded value $137.04$ deviates by only $0.003\,\%$.

## Example: Gravitational Constant

**In T0 units:**

$$G = 1 \tag{O.19}$$

**Reconstruction of SI value:**

$$G_{\text{SI}} = \frac{\xi^2}{4m_e} \times C_{\text{conv}} \times K_{\text{frak}} \tag{O.20}$$

where:

- $C_{\text{conv}}$ = Conversion factors (eV → J, etc.)

- $K_{\text{frak}}$ = fractal correction

A separate dimension conversion $C_{\text{dim}} = 1/E_{\text{char}} = 3.472 \times 10^{-2}$ does not appear: it is cancelled again by the factor $E_{\text{char}} = 28.8$ in the step chain of Doc. 012; inserted into the formula on its own it would give $2.35 \times 10^{-12}$ instead of $6.674 \times 10^{-11}$.

Detailed derivation see Document 012 (Gravitation).

### Why does this work?

**Key:** $\xi$ is dimensionless and is taken as universal in the T0 framework.

1. In T0: $\xi$ determines all coupling strengths
2. In SI: $\xi$ together with characteristic energies ($E_0$, masses) reconstructs all constants
3. Physical predictions: identical in both systems
4. Only the mathematical representation differs

### Comparison Table

| Constant | T0 | SI | Reconstruction |
|---|---|---|---|
| $c$ | 1 | $3 \times 10^8$ m/s | Convention |
| $\hbar$ | 1 | $1.055 \times 10^{-34}$ J·s | Convention |
| $\alpha$ | 1 | $\approx 1/137$ | $\xi(E_0/1\text{MeV})^2$ |
| $G$ | 1 | $6.67 \times 10^{-11}$ m³/(kg·s²) | $\xi^2/(4m_e)\times$ factors |
| $\xi$ | $\frac{4}{3} \times 10^{-4}$ | $\frac{4}{3} \times 10^{-4}$ | same |

**Table O.2:** T0 vs. SI - Reconstruction of constants

### Important Conclusion

- **T0 is not new physics**, but a reparametrization
- Instead of many constants ($c$, $\hbar$, $\alpha$, $G$, ...) only **one parameter** $\xi$
- All SI values reconstructable from $\xi$ and energy scales
- Advantage: Formulas simpler, physical relationships clearer
- Disadvantage: Conversion to SI needed for experiments

## O.6 Why can $\alpha = 1$ be set?

### Fundamental Insight

**Core Statement**

The fine-structure constant $\alpha$ is a **dimensionless number**. It contains two levels that must be kept apart. The number "137" as the name of the coupling depends on the **normalisation** (charge unit, $4\pi$, $\hbar$) and is a choice. The **scale ratio** it expresses, for example $r_e/\lambda_C$ or binding energy relative to rest energy in hydrogen, is physical and the same in every system of units (Doc. 386).
One can set $\alpha = 1$ by redefining the **unit of charge** accordingly.

### Step-by-Step Justification

**Step 1: What is a convention?**

SI units are historically evolved definitions:

- 1 meter = originally 1/10,000,000 of the Earth's meridian
- 1 second = originally 1/86,400 of a solar day
- 1 Coulomb = defined via Ampere and force between currents

None of these is "fundamental".

**Step 2: $\alpha$ in SI**

In SI units:

$$\alpha = \frac{e^2}{4\pi\varepsilon_0\hbar c} \approx \frac{1}{137} \tag{O.21}$$

Because $\alpha$ is dimensionless, the value $1/137$ does not depend on how the Coulomb or $\varepsilon_0$ was defined; it comes out the same in every system of units, as long as the coupling is normalised with $4\pi$ and $\hbar$. With $h$ instead of $\hbar$ the same coupling would read $2\alpha \approx 1/68.5$. What is fixed by choice is the normalisation, not the ratio.

**Step 3: Redefinition**

We can say: "From now on, the elementary charge is no longer $1.602 \times 10^{-19}$ C, but $e = \sqrt{4\pi\varepsilon_0\hbar c}$."

Then automatically:

$$\alpha = \frac{(\sqrt{4\pi\varepsilon_0\hbar c})^2}{4\pi\varepsilon_0\hbar c} = 1 \tag{O.22}$$

**Step 4: Physical Consequences**

- **No physical prediction changes.**
- Only the *numbers* in formulas change
- All ratios remain the same
- All experiments yield the same results

### Analogy: Temperature Scales

**Celsius:** Water freezes at 0°C

**Fahrenheit:** Water freezes at 32°F
**Kelvin:** Water freezes at 273.15 K
None of these scales is more "correct" than the others; they are conventions.
Similarly, the choice of charge unit that makes $\alpha = 1$ is a choice of convention. The scale ratio does not disappear; it then sits in the conversion between the historical and the new charge unit (Doc. 386).

## O.7 Consequences of Redefining the Coulomb

### What does it mean to redefine elementary charge?

If $e$ is redefined such that $\alpha = 1$:
**Old definition (SI):**

$$e = 1.602 \times 10^{-19}\ \text{C} \tag{O.23}$$

**New definition (HL with $\alpha = 1$):**

$$e = \sqrt{4\pi} \approx 3.545 \quad \text{(dimensionless in natural units)} \tag{O.24}$$

or expressed in SI units:

$$e_{\text{new}} = \sqrt{4\pi\varepsilon_0\hbar c} \approx 1.877 \times 10^{-18}\ \text{C} = \sqrt{137.036}\, e \tag{O.25}$$

The charge unit is rescaled by the same factor: $1\ \text{C}_{\text{new}} = (e_{\text{new}}/e)\cdot 1\ \text{C} = \sqrt{137} \times 1\ \text{C}_{\text{old}}$. Numerical values transform inversely to the units: a unit larger by a factor $f$ gives a numerical value smaller by $f$. (Here $f$ is the scaling factor between the old and the new unit, not a frequency and not the base winding number $f = 1/\xi$.)

### Effects on Electromagnetic Quantities

#### Electric Current (Ampere)

Since $1\ \text{A} = 1\ \text{C/s}$:

$$1\ \text{A}_{\text{new}} = \frac{1\ \text{C}_{\text{new}}}{1\ \text{s}} = \sqrt{137} \times 1\ \text{A}_{\text{old}} \tag{O.26}$$

#### Electric Voltage (Volt)

$1\ \text{V} = 1\ \text{J/C}$:

$$1\ \text{V}_{\text{new}} = \frac{1\ \text{J}}{1\ \text{C}_{\text{new}}} = \frac{1}{\sqrt{137}} \times 1\ \text{V}_{\text{old}} \tag{O.27}$$

#### Capacitance (Farad)

$$1\ \text{F}_{\text{new}} = \frac{1\ \text{C}_{\text{new}}}{1\ \text{V}_{\text{new}}} = 137 \times 1\ \text{F}_{\text{old}} \tag{O.28}$$

### Are these changes "real"?

**No.** They are conversion factors, like Celsius → Fahrenheit.

**All physical ratios remain identical:**

- Capacitance of a capacitor / distance: same
- Force between charges / distance²: same
- All experiments: same results

Only the *numerical values* we calculate with change.

## O.8 Practical Impacts on Everyday Calculations

### Motivation

**Question:** If we set $\alpha = 1$, what does that mean for ordinary electrical calculations with volts, amperes, resistance, capacitance?

**Answer:** All formulas change, but the *physical results* remain identical.

### Example 1: Ohm's Law

#### In SI Units (Standard)

$$U = R \cdot I \tag{O.29}$$

**Numerical example:**

- Resistance: $R = 100\ \Omega$
- Current: $I = 2$ A
- Voltage: $U = 100 \times 2 = 200$ V

#### In Heaviside-Lorentz with $\alpha = 1$

The formula remains $U = R \cdot I$, but the *numerical values* change.

**Unit conversion:**

$$1\ \mathrm{A}_{\mathrm{new}} = \sqrt{137} \times 1\ \mathrm{A}_{\mathrm{old}} \approx 11.7\ \mathrm{A}_{\mathrm{old}} \tag{O.30}$$

$$1\ \mathrm{V}_{\mathrm{new}} = \frac{1}{\sqrt{137}} \times 1\ \mathrm{V}_{\mathrm{old}} \approx 0.085\ \mathrm{V}_{\mathrm{old}} \tag{O.31}$$

$$1\ \Omega_{\mathrm{new}} = \frac{1}{137} \times 1\ \Omega_{\mathrm{old}} \tag{O.32}$$

**Same circuit in new units:**

$$R_{\mathrm{new}} = 100 \times 137 = 13\,700\ \Omega_{\mathrm{new}} \tag{O.33}$$

$$I_{\mathrm{new}} = \frac{2}{\sqrt{137}} \approx 0.171\ \mathrm{A}_{\mathrm{new}} \tag{O.34}$$

$$U_{\mathrm{new}} = 13\,700 \times 0.171 = 2341\ \mathrm{V}_{\mathrm{new}} \tag{O.35}$$

**Conversion back to SI:**

$$U = 2341 \times 0.0854 \text{ V}_{\text{old}} = 200 \text{ V} \quad \checkmark \tag{O.36}$$

The result is identical.

## Example 2: Power of a Light Bulb

### In SI Units

$$P = U \cdot I = \frac{U^2}{R} \tag{O.37}$$

**Light bulb:** 60 W at 230 V

$$R = \frac{U^2}{P} = \frac{(230)^2}{60} = 882 \ \Omega \tag{O.38}$$

$$I = \frac{P}{U} = \frac{60}{230} = 0.26 \text{ A} \tag{O.39}$$

### In Heaviside-Lorentz with $\alpha = 1$

**Power:** $1 \text{ W}_{\text{new}} = 1 \text{ W}_{\text{old}}$ (energy/time does not change in HL)

$$U_{\text{new}} = 230 \times \sqrt{137} = 2692 \text{ V}_{\text{new}} \tag{O.40}$$

$$R_{\text{new}} = 882 \times 137 = 1.21 \times 10^5 \ \Omega_{\text{new}} \tag{O.41}$$

$$I_{\text{new}} = \frac{P}{U_{\text{new}}} = \frac{60}{2692} = 0.0223 \text{ A}_{\text{new}} \tag{O.42}$$

**Verification:**

$$P = U_{\text{new}} \cdot I_{\text{new}} = 2692 \times 0.0223 = 60 \text{ W} \quad \checkmark \tag{O.43}$$

## Example 3: Charging a Capacitor

### In SI Units

$$Q = C \cdot U \tag{O.44}$$

**Capacitor:** $C = 100 \ \mu$F at $U = 12$ V

$$Q = 100 \times 10^{-6} \times 12 = 1.2 \times 10^{-3} \text{ C} \tag{O.45}$$

**Stored energy:**

$$E = \frac{1}{2} C U^2 = \frac{1}{2} \times 100 \times 10^{-6} \times 144 = 7.2 \times 10^{-3} \text{ J} \tag{O.46}$$

**In Heaviside-Lorentz with $\alpha = 1$**

**Conversion:**

$$1\ \mathrm{F}_{\mathrm{new}} = 137 \times 1\ \mathrm{F}_{\mathrm{old}} \tag{O.47}$$

$$1\ \mathrm{C}_{\mathrm{new}} = \sqrt{137} \times 1\ \mathrm{C}_{\mathrm{old}} \tag{O.48}$$

$$C_{\mathrm{new}} = \frac{100 \times 10^{-6}}{137} = 7.30 \times 10^{-7}\ \mathrm{F}_{\mathrm{new}} \tag{O.49}$$

$$U_{\mathrm{new}} = 12 \times \sqrt{137} = 140.5\ \mathrm{V}_{\mathrm{new}} \tag{O.50}$$

$$Q_{\mathrm{new}} = 7.30 \times 10^{-7} \times 140.5 = 1.025 \times 10^{-4}\ \mathrm{C}_{\mathrm{new}} \tag{O.51}$$

**Conversion back:**

$$Q = 1.025 \times 10^{-4} \times \sqrt{137} = 1.2 \times 10^{-3}\ \mathrm{C}_{\mathrm{old}} \quad \checkmark \tag{O.52}$$

**Energy:**

$$E_{\mathrm{new}} = \frac{1}{2} \times 7.30 \times 10^{-7} \times (140.5)^2 = 7.2 \times 10^{-3}\ \mathrm{J} \quad \checkmark \tag{O.53}$$

The energy is the same in all systems.

## Example 4: RC Time Constant

**In SI Units**

$$\tau = R \cdot C \tag{O.54}$$

**Circuit:** $R = 1\ \mathrm{k}\Omega$, $C = 10\ \mu\mathrm{F}$

$$\tau = 1000 \times 10 \times 10^{-6} = 0.01\ \mathrm{s} = 10\ \mathrm{ms} \tag{O.55}$$

**In Heaviside-Lorentz with $\alpha = 1$**

$$R_{\mathrm{new}} = 1000 \times 137 = 1.37 \times 10^{5}\ \Omega_{\mathrm{new}} \tag{O.56}$$

$$C_{\mathrm{new}} = \frac{10 \times 10^{-6}}{137} = 7.30 \times 10^{-8}\ \mathrm{F}_{\mathrm{new}} \tag{O.57}$$

$$\tau_{\mathrm{new}} = 1.37 \times 10^{5} \times 7.30 \times 10^{-8} = 0.01\ \mathrm{s} = 10\ \mathrm{ms} \quad \checkmark \tag{O.58}$$

**Time remains the same:** physical timescales do not change.

| Quantity | SI | HL Factor | HL ($\alpha = 1$) |
|---|---|---|---|
| Charge (Q) | C | $\sqrt{137}$ | $\sqrt{137}$ C |
| Current (I) | A | $\sqrt{137}$ | $\sqrt{137}$ A |
| Voltage (U) | V | $1/\sqrt{137}$ | V$/\sqrt{137}$ |
| Resistance (R) | $\Omega$ | $1/137$ | $\Omega/137$ |
| Capacitance (C) | F | 137 | 137 F |
| Power (P) | W | 1 | W (unchanged) |
| Energy (E) | J | 1 | J (unchanged) |
| Time ($\tau$) | s | 1 | s (unchanged) |

**Table O.3:** Conversion factors SI → HL with $\alpha = 1$ (factors for the units; numerical values are divided by the factor)

### Summary of Practical Calculations

### Important Insights

**Core Statement**

**What changes:**
- Numerical values for charge, current, voltage, resistance, capacitance

**What does NOT change:**
- Energy
- Power
- Time
- All physical ratios
- All experimental results

**Conclusion:** It is a conversion, comparable to meters ↔ feet.

### Why does nobody use $\alpha = 1$ in practice?

**Reasons:**

1. **Measuring devices:** All voltmeters, ammeters, etc. are calibrated in SI
2. **Standards:** Worldwide accepted SI definitions
3. **Intuition:** Engineers know typical values in SI
   - Household: 230 V, not 2692 $V_{new}$
   - USB: 5 V, not 58.5 $V_{new}$
4. **Conversion is laborious:** $\sqrt{137}$ factors everywhere
5. **No advantage for practitioners:** Simplification only visible in theoretical formulas
   **But:** In theoretical calculations (QED, Feynman diagrams), $\alpha = 1$ can be helpful.

## O.9 Practical Aspects of Different Systems

### Advantages and Disadvantages: SI Units

**Advantages:**
- Worldwide standardized
- Directly usable for experiments
- All measuring devices calibrated in SI
- Clear separation of length/time/mass/charge

**Disadvantages:**
- Many constants in formulas ($4\pi\varepsilon_0$, $\hbar$, $c$)
- Physical relationships obscured
- Dimensions unwieldy

### Advantages and Disadvantages: Heaviside-Lorentz with $\alpha = 1$

**Advantages:**
- Maximally simplified formulas
- Electromagnetic symmetry visible
- Theoretical calculations simpler
- QED Feynman diagrams easier to read

**Disadvantages:**
- No direct connection to experiments
- Conversion to SI laborious
- Unfamiliar for practitioners
- Physical "size" of $e$ unclear

### Advantages and Disadvantages: Natural Units with $\alpha \approx 1/137$

**Advantages:**
- Simplified formulas ($\hbar = c = 1$)
- $\alpha$ retains physical meaning
- Good compromise theory/practice
- Numerically: $\alpha \ll 1$ → perturbation theory

**Disadvantages:**
- Still need conversion to SI
- Factor $4\pi$ remains in some formulas

This is the usual convention in present-day particle physics.

## O.10 Historical Development

### Gauss Units (cgs)

**19th century:** Gauss system (centimeter-gram-second)

$$\alpha = \frac{e^2}{\hbar c} \tag{O.59}$$

No $4\pi\varepsilon_0$, because $4\pi\varepsilon_0 = 1$ by definition in cgs.

### SI Units (MKSA)

**20th century:** SI system (meter-kilogram-second-ampere)

$$\alpha = \frac{e^2}{4\pi\varepsilon_0 \hbar c} \tag{O.60}$$

The $4\pi\varepsilon_0$ appears because the SI ampere is defined via force.

### Heaviside-Lorentz

**Theoretical physics:** Heaviside-Lorentz simplifies Maxwell's equations

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}, \quad \nabla \times \vec{B} = \vec{j} + \frac{\partial \vec{E}}{\partial t} \tag{O.61}$$

The equations are symmetric (in SI, $\mu_0$ and $\varepsilon_0$ appear asymmetrically)

### Natural Units

**Modern high-energy physics:** $\hbar = c = 1$, but different conventions for $\alpha$

## O.11 Fine's Inequality vs. Fine-Structure Constant

### Frequent Confusion

**Note:** *Fine's inequality* and the *fine-structure constant* are different concepts.

### Fine's Inequality

**What it is:**

- A form of Bell's inequality
- Test for local hidden variables
- Quantum entanglement vs. classical correlations

**Mathematically:**

$$|C(\alpha, \beta) - C(\alpha, \beta')| + |C(\alpha', \beta) + C(\alpha', \beta')| \leq 2 \tag{O.62}$$

where $C$ are correlation functions.
**Physically:** Shows non-locality of quantum mechanics

### Fine-Structure Constant

**What it is:**

- Fundamental physical constant
- Strength of the electromagnetic interaction
- Dimensionless, $\alpha \approx 1/137$

**Mathematically:**

$$\alpha = \frac{e^2}{4\pi\varepsilon_0 \hbar c} \tag{O.63}$$

**Physically:** Determines EM coupling strength

### No Conceptual Connection

The similarity in name is coincidental; the two concepts are not related.

## O.12 Summary

### Core Statements

1. The number "137" as the name of the coupling depends on the normalisation; the scale ratio that $\alpha$ expresses is physical (Doc. 386)
2. One **can** set $\alpha = 1$ by redefining the unit of charge
3. **T0 theory:** Sets all constants = 1: $c = \hbar = \alpha = G = 1$
4. Only parameter in T0: $\xi = \frac{4}{3} \times 10^{-4}$ (integer structural choices are settings, SI anchors serve for conversion)
5. **No** physical prediction changes
6. Only numerical values in formulas differ
7. When comparing with experiments: SI values ($\alpha \approx 1/137$, $c = 3 \times 10^8$ m/s, etc.)

### For Further Details See Document 011

- T0 derivation of $\alpha$
- Characteristic energy $E_0$
- Geometric parameter $\xi$
- Experimental verification

- Detailed dimensional analysis
- Historical context (Sommerfeld)

## O.A Conversion Table: SI ↔ Heaviside-Lorentz

| Quantity | SI | HL ($\alpha = 1$) |
|---|---|---|
| Elementary charge | $e = 1.602 \times 10^{-19}$ C | $e = \sqrt{4\pi} \approx 3.545$ |
| Fine-structure constant | $\alpha \approx 1/137$ | $\alpha = 1$ |
| $\varepsilon_0$ | $8.854 \times 10^{-12}$ F/m | 1 |
| $\hbar$ | $1.055 \times 10^{-34}$ J·s | 1 |
| $c$ | $3 \times 10^{8}$ m/s | 1 |

**Table O.4:** Conversion table SI to HL

## O.B Sample Calculation: Coulomb's Law

### In SI Units

$$F = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{r^2} \tag{O.64}$$

Numerically for $r = 1$ Å = $10^{-10}$ m:

$$F = \frac{1}{4\pi \times 8.854 \times 10^{-12}}\frac{(1.602 \times 10^{-19})^2}{(10^{-10})^2} \tag{O.65}$$

$$\approx 2.3 \times 10^{-8} \text{ N} \tag{O.66}$$

### In HL Units ($\hbar = c = \varepsilon_0 = 1$)

$$F = \frac{e^2}{4\pi r^2} = \frac{\alpha}{r^2} \tag{O.67}$$

With $r$ in natural units: $r = 1$ Å $= 5.07 \times 10^{-4}$ eV$^{-1}$ (with $\hbar c = 1.973 \times 10^{-7}$ eV m)

$$F = \frac{\alpha}{(5.07 \times 10^{-4})^2} = \frac{3.89 \times 10^{6}}{137.036} \approx 2.84 \times 10^{4} \text{ eV}^2 \tag{O.68}$$

Conversion to SI: $1$ eV$^2 = (1\,\text{eV})^2/(\hbar c) \approx 8.12 \times 10^{-13}$ N

$$F \approx 2.3 \times 10^{-8} \text{ N} \tag{O.69}$$

**The result is identical;** only the intermediate steps look different.

The SI result $2.3 \times 10^{-8}$ N follows only with the numerical value $\alpha = 1/137.036$ in the numerator; with $\alpha = 1$ one would get $3.2 \times 10^{-6}$ N. The force depends on the physical value of $\alpha$, not on the choice of charge unit.

# Chapter P

# Mathematical Proof:


## Abstract

This paper provides a rigorous mathematical proof that the fine structure constant $\alpha$ equals unity ($\alpha = 1$) in natural unit systems. Through systematic analysis of the two equivalent representations of $\alpha$, we demonstrate that the electromagnetic duality between $\varepsilon_0$ and $\mu_0$, connected by the fundamental Maxwell relation $c^2 = 1/(\varepsilon_0\mu_0)$, naturally leads to $\alpha = 1$ when appropriate unit normalizations are applied. This proof establishes that $\alpha = 1/137$ in SI units is purely a consequence of our historical unit choices, not a fundamental mystery of nature.


## P.1 Introduction and Motivation

The fine structure constant $\alpha \approx 1/137$ has been called one of the greatest mysteries in physics, inspiring famous quotes from Feynman, Pauli, and others. However, this mystification stems from viewing $\alpha$ only within the SI unit system. This paper proves mathematically that $\alpha = 1$ in appropriately chosen natural units, revealing that the "mystery" of $1/137$ is merely a consequence of our conventional unit system.

## P.2 Fundamental Premise

**Definition P.2.1** (Two Equivalent Forms of $\alpha$)**.** The fine structure constant can be expressed in two mathematically equivalent forms:

$$\text{Form 1:} \quad \alpha = \frac{e^2}{4\pi\varepsilon_0\hbar c} \tag{P.1}$$

$$\text{Form 2:} \quad \alpha = \frac{e^2\mu_0 c}{4\pi\hbar} \tag{P.2}$$

These forms are equivalent through the Maxwell relation $c^2 = 1/(\varepsilon_0\mu_0)$.

## P.3 The Duality Analysis

### Extraction of Common Elements

**Identification of Common Terms**

Both forms (P.1) and (P.2) contain identical terms:

- $e^2$ - square of elementary charge
- $4\pi$ - geometric factor
- $\hbar$ - reduced Planck constant

**Isolation of Differential Terms**

After factoring out common elements, the essential difference between the two forms is:

$$\text{Form 1:} \quad \alpha \propto \frac{1}{\varepsilon_0 c} \tag{P.3}$$

$$\text{Form 2:} \quad \alpha \propto \mu_0 c \tag{P.4}$$

### The Electromagnetic Duality

**Theorem P.3.1** (Electromagnetic Duality Relation)**.** *For the two forms to be equivalent, we must have:*

$$\frac{1}{\varepsilon_0 c} = \mu_0 c \tag{P.5}$$

*Proof.* Rearranging equation (P.5):

$$\frac{1}{\varepsilon_0 c} = \mu_0 c \tag{P.6}$$

$$1 = \varepsilon_0 c \cdot \mu_0 c \tag{P.7}$$

$$1 = \varepsilon_0 \mu_0 c^2 \tag{P.8}$$

$$c^2 = \frac{1}{\varepsilon_0 \mu_0} \tag{P.9}$$

This is precisely Maxwell's fundamental relation connecting electromagnetic constants with the speed of light. □

## P.4 The Key Insight: Opposite Powers of c

**Lemma P.4.1** (Sign Duality of c)**.** *The speed of light $c$ appears with opposite "signs" (powers) in the two forms:*

$$\textit{Form 1:} \quad c^{-1} \quad (c \textit{ in denominator}) \tag{P.10}$$

$$\textit{Form 2:} \quad c^{+1} \quad (c \textit{ in numerator}) \tag{P.11}$$

This duality reflects the complementary nature of electric ($\varepsilon_0$) and magnetic ($\mu_0$) aspects of the electromagnetic field.

## P.5 Construction of Natural Units

### The Natural Unit Choice

**Definition P.5.1** (Natural Unit System for $\alpha = 1$)**.** We define a natural unit system where:

1. $\hbar_{\text{nat}} = 1$ (quantum mechanical scale)
2. $c_{\text{nat}} = 1$ (relativistic scale)
3. The electromagnetic constants are normalized such that $\alpha = 1$

### Determination of Natural Electromagnetic Constants

**Theorem P.5.2** (Natural Unit Electromagnetic Constants)**.** *In the natural unit system where $\alpha = 1$, $\hbar = 1$, and $c = 1$, the electromagnetic constants become:*

$$e_{nat}^2 = 4\pi \tag{P.12}$$

$$\varepsilon_{0,nat} = 1 \tag{P.13}$$

$$\mu_{0,nat} = 1 \tag{P.14}$$

*Proof.* From Form 1 with $\alpha = 1$, $\hbar = 1$, $c = 1$:

$$1 = \frac{e^2}{4\pi\varepsilon_0 \cdot 1 \cdot 1} \tag{P.15}$$

$$4\pi\varepsilon_0 = e^2 \tag{P.16}$$

Setting $\varepsilon_0 = 1$ (natural choice), we get $e^2 = 4\pi$.
From the Maxwell relation $c^2 = 1/(\varepsilon_0\mu_0)$ with $c = 1$:

$$1 = \frac{1}{\varepsilon_0\mu_0} \tag{P.17}$$

$$\varepsilon_0\mu_0 = 1 \tag{P.18}$$

With $\varepsilon_0 = 1$, we get $\mu_0 = 1$. □

## P.6 Verification of $\alpha = 1$

### Verification Using Form 1

**Form 1 Verification**

$$\alpha = \frac{e^2}{4\pi\varepsilon_0\hbar c} \tag{P.19}$$
$$= \frac{4\pi}{4\pi \cdot 1 \cdot 1 \cdot 1} \tag{P.20}$$
$$= \frac{4\pi}{4\pi} \tag{P.21}$$
$$= 1 \quad \checkmark \tag{P.22}$$

### Verification Using Form 2

**Form 2 Verification**

$$\alpha = \frac{e^2\mu_0 c}{4\pi\hbar} \tag{P.23}$$
$$= \frac{4\pi \cdot 1 \cdot 1}{4\pi \cdot 1} \tag{P.24}$$
$$= \frac{4\pi}{4\pi} \tag{P.25}$$
$$= 1 \quad \checkmark \tag{P.26}$$

## P.7 The Duality Verification

**Theorem P.7.1** (Electromagnetic Duality in Natural Units)**.** *In natural units, the electromagnetic duality is perfectly satisfied:*

$$\frac{1}{\varepsilon_{0,nat} \cdot c_{nat}} = \mu_{0,nat} \cdot c_{nat} \tag{P.27}$$

*Proof.*

$$\text{LHS:} \quad \frac{1}{\varepsilon_{0,\text{nat}} \cdot c_{\text{nat}}} = \frac{1}{1 \cdot 1} = 1 \tag{P.28}$$
$$\text{RHS:} \quad \mu_{0,\text{nat}} \cdot c_{\text{nat}} = 1 \cdot 1 = 1 \tag{P.29}$$
$$\text{Therefore:} \quad \text{LHS} = \text{RHS} \quad \checkmark \tag{P.30}$$

□

## P.8 Physical Interpretation

### The Naturalness of $\alpha = 1$

**Key Physical Insight**

In natural units, $\alpha = 1$ represents the perfect balance between:

- **Electric field coupling** (through $\varepsilon_0$ with $c^{-1}$)
- **Magnetic field coupling** (through $\mu_0$ with $c^{+1}$)
- **Quantum mechanical scale** (through $\hbar$)
- **Relativistic scale** (through $c$)

The electromagnetic duality $\frac{1}{\varepsilon_0 c} = \mu_0 c$ ensures this perfect balance.

### Resolution of the "$1/137$ Mystery"

The famous value $\alpha \approx 1/137$ in SI units arises solely from our historical choices of:

- The meter (length scale)
- The second (time scale)
- The kilogram (mass scale)
- The ampere (current scale)

These choices force electromagnetic constants to have "unnatural" values, making $\alpha$ appear mysteriously small.

#### Transformation from Natural Units to SI Units

To understand how we arrive at the SI value $\alpha_{\text{SI}} = 1/137$, we must transform from our natural unit system back to SI units. The transformation involves scaling factors for each fundamental constant:

$$\hbar_{\text{SI}} = \hbar_{\text{nat}} \times S_\hbar = 1 \times (1.055 \times 10^{-34}\ \text{J·s}) \tag{P.31}$$

$$c_{\text{SI}} = c_{\text{nat}} \times S_c = 1 \times (2.998 \times 10^{8}\ \text{m/s}) \tag{P.32}$$

$$\varepsilon_{0,\text{SI}} = \varepsilon_{0,\text{nat}} \times S_\varepsilon = 1 \times (8.854 \times 10^{-12}\ \text{F/m}) \tag{P.33}$$

$$e_{\text{SI}} = e_{\text{nat}} \times S_e = \sqrt{4\pi} \times S_e \tag{P.34}$$

The fine structure constant in SI units becomes:

$$\alpha_{\text{SI}} = \frac{e_{\text{SI}}^2}{4\pi\varepsilon_{0,\text{SI}}\hbar_{\text{SI}}c_{\text{SI}}} \tag{P.35}$$

$$= \frac{(\sqrt{4\pi} \times S_e)^2}{4\pi \times (S_\varepsilon) \times (S_\hbar) \times (S_c)} \tag{P.36}$$

$$= \frac{4\pi \times S_e^2}{4\pi \times S_\varepsilon \times S_\hbar \times S_c} \tag{P.37}$$

$$= \frac{S_e^2}{S_\varepsilon \times S_\hbar \times S_c} \tag{P.38}$$

The historical SI unit definitions created scaling factors such that this ratio equals approximately $1/137$. In other words: $\frac{S_e^2}{S_\varepsilon \times S_\hbar \times S_c} \approx \frac{1}{137}$

This demonstrates that the "mysterious" value $1/137$ is purely a consequence of the arbitrary scaling factors chosen when defining the SI base units, not a fundamental property of electromagnetic interactions themselves. In the natural unit system where these scaling factors are unity, $\alpha = 1$ emerges as the fundamental value.

## P.9 Resolving the Constants Paradox

### The Fundamental Misconception

The most profound objection to our proof often takes the form: "How can a **constant** have different values?" This apparent paradox lies at the heart of why the fine structure constant has been mystified for over a century.

#### The Problem Statement

The seeming contradiction is:

- $\alpha = 1/137$ (in SI units)
- $\alpha = 1$ (in natural units)

In the Gaussian system, by contrast, $\alpha = e^2/(\hbar c) = (4.803 \times 10^{-10}\ \text{esu})^2/(1.0546 \times 10^{-27}\ \text{erg s} \cdot 2.998 \times 10^{10}\ \text{cm/s}) = 1/137.036$, i.e. the same number as in SI; only the value $\alpha = 1$ of natural units with adapted $e$ differs.

How can the "same" constant have two different values?

#### The Resolution

The resolution reveals a fundamental misunderstanding about what "constant" means in physics.

**What is truly constant is not the number, but the physical relationship.**

### The Perfect Analogy: Water's Boiling Point

Consider the boiling point of water:

- $100^\circ$C (Celsius scale)
- $212^\circ$F (Fahrenheit scale)
- $373$ K (Kelvin scale)

**Question:** At what temperature does water "really" boil?
**Answer:** At the same physical temperature in all cases! Only the numbers differ due to different temperature scales.

### The Same Principle Applies to $\alpha$

Just as with temperature scales:

- $\alpha = 1/137$ (SI unit scale)
- $\alpha = 1$ (natural unit scale)

**The electromagnetic coupling strength is identical** – only the measurement scales differ.

### The Key Insight

**Fundamental Principle**

"**CONSTANT**" does **NOT** mean "same number"!
"**CONSTANT**" means "same physical quantity"!

**Examples of this principle:**

- $1$ meter $= 100$ cm $= 3.28$ feet $\rightarrow$ The **length** is constant
- $1$ kg $= 1000$ g $= 2.2$ lbs $\rightarrow$ The **mass** is constant
- $\alpha = 1/137 = 1 \rightarrow$ The **coupling strength** is constant

### Physical Verification

We can verify that these represent the same physical constant by confirming that all unit systems yield identical experimental results:

**Theorem P.9.1** (Experimental Invariance)**.** *All unit systems produce identical measurable predictions:*

- ***Hydrogen spectrum:*** *Same frequencies in all systems* ✓
- ***Electron scattering:*** *Same cross-sections in all systems* ✓
- ***Lamb shift:*** *Same energy shifts in all systems* ✓

## The Deeper Truth

**Nature's True Language**

**Nature "knows" no numbers!**
**Nature knows only ratios and relationships!**

The fine structure constant $\alpha$ is not the mysterious number "$1/137$" – $\alpha$ is the **ratio** between electromagnetic and quantum mechanical effects.

This ratio is absolutely constant throughout the universe, but the numerical value depends entirely on our arbitrary choice of unit definitions.

## The Linguistic Problem

Much of the confusion stems from imprecise language. We incorrectly say:

× "**THE** fine structure constant is $1/137$"

The correct statements would be:

✓ "The fine structure constant has the value $1/137$ **in SI units**"

✓ "The fine structure constant has the value $1$ **in natural units**"

## Resolution of the Century-Old Mystery

This analysis reveals that the "mystery of $1/137$" is not a physical puzzle but a **linguistic and conceptual misunderstanding**. The mystification arose from:

1. Conflating the numerical value with the physical quantity
2. Treating the SI unit system as fundamental rather than conventional
3. Forgetting that all unit systems are human constructs
4. Seeking deep meaning in what are essentially conversion factors

Once we recognize that $\alpha = 1$ represents the natural strength of electromagnetic interactions, the "mystery" dissolves completely. The electromagnetic force has unit strength in the unit system that respects the fundamental structure of quantum mechanics and relativity – exactly as one would expect from a truly fundamental interaction.

## Final Perspective

The fine structure constant teaches us a profound lesson about the nature of physical laws: **the universe's fundamental relationships are elegant and simple when expressed in their natural language**. The apparent complexity and mystery of "$1/137$" is merely an artifact of our historical choice to measure electromagnetic phenomena using units originally defined for mechanical quantities.

In recognizing $\alpha = 1$ as the natural value, we glimpse the inherent simplicity and beauty that underlies the electromagnetic structure of reality.

## P.10 Acknowledgments

This work was inspired by the recognition that fundamental physical constants should not be mysterious numbers but should reflect the underlying mathematical structure of nature. The electromagnetic duality revealed through the analysis of the two forms of $\alpha$ provides the key insight that resolves the long-standing puzzle of the fine structure constant.

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# Chapter Q

# T0 Theory: The Gravitational Constant


## Abstract

This document presents the systematic derivation of the gravitational constant $G$ from the fundamental principles of T0 theory. The complete formula $G_{\text{SI}} = \frac{\xi_0^2}{4m_e} \times C_{\text{conv}} \times K_{\text{frak}}$ explicitly shows all required conversion factors. The form and order of magnitude of $G$ follow from $\xi$ and one anchor; with the SI conversion factor $C_{\text{conv}}$ the value lies $+0.003\%$ above the measured value. Since $C_{\text{conv}}$ additionally contains the small residual of the single-anchor chain (Doc. 383), this agreement does not count as an independent precision result. Special attention is given to the physical justification of the conversion factors that establish the connection between geometric theory and measurable quantities.


## Q.1 Introduction: Gravitation in T0 Theory

### The Problem of the Gravitational Constant

The gravitational constant $G = 6.674 \times 10^{-11}$ m$^3$/(kg·s$^2$) is one of the least precisely known natural constants. Its theoretical derivation from first principles is one of the great unsolved problems in physics.

**Key Result**

**T0 Hypothesis for Gravitation:**
The gravitational constant is not fundamental but follows from the geometric structure of three-dimensional space through the relation:

$$\boxed{G_{\text{SI}} = \frac{\xi_0^2}{4m_e} \times C_{\text{conv}} \times K_{\text{frak}}} \tag{Q.1}$$

where all factors are derivable from geometry or fundamental constants.

### Overview of the Derivation

The T0 derivation proceeds in four systematic steps:

1. **Fundamental T0 Relation:** $\xi = 2\sqrt{G \cdot m_{\text{char}}}$
2. **Solution for G:** $G = \frac{\xi^2}{4m_{\text{char}}}$ (natural units)
3. **Dimensional Correction:** Transition to physical dimensions
4. **SI Conversion:** Conversion to experimentally comparable units

## Q.2 The Fundamental T0 Relation

### Geometric Basis

**Starting Point of T0 Gravitation Theory:**
T0 theory postulates a fundamental geometric relation between the characteristic length parameter $\xi$ and the gravitational constant:

$$\xi = 2\sqrt{G \cdot m_{\text{char}}} \tag{Q.2}$$

**Geometric Interpretation:** This equation describes how the characteristic length scale $\xi$ (defined by the tetrahedral space structure) determines the strength of gravitational coupling. The factor 2 corresponds to the dual nature of mass and space in T0 theory.

**Physical Interpretation:**

- $\xi$ encodes the geometric structure of space (tetrahedral packing)
- $G$ describes the coupling between geometry and matter
- $m_{\text{char}}$ sets the characteristic mass scale

### Solution for the Gravitational Constant

Solving equation (ak.1) for $G$ yields:

$$G = \frac{\xi^2}{4m_{\text{char}}} \tag{Q.3}$$

**Significance:** This fundamental relation shows that $G$ is not an independent constant but is determined by space geometry ($\xi$) and the characteristic mass scale ($m_{\text{char}}$).

### Choice of Characteristic Mass

T0 theory uses the electron mass as the characteristic scale:

$$m_{\text{char}} = m_e = 0.511 \text{ MeV} \tag{Q.4}$$

The justification lies in the electron's role as the lightest charged particle and its fundamental importance for electromagnetic interaction.

# Q.3 Dimensional Analysis in Natural Units

## Unit System of T0 Theory

**Dimensional Analysis**

**Dimensional Analysis in Natural Units:**
T0 theory works in natural units with $\hbar = c = 1$:

$$[M] = [E] \quad \text{(from } E = mc^2 \text{ with } c = 1\text{)} \tag{Q.5}$$
$$[L] = [E^{-1}] \quad \text{(from } \lambda = \hbar/p \text{ with } \hbar = 1\text{)} \tag{Q.6}$$
$$[T] = [E^{-1}] \quad \text{(from } \omega = E/\hbar \text{ with } \hbar = 1\text{)} \tag{Q.7}$$

The gravitational constant therefore has the dimension:

$$[G] = [M^{-1}L^3T^{-2}] = [E^{-1}][E^{-3}][E^2] = [E^{-2}] \tag{Q.8}$$

## Dimensional Consistency of the Basic Formula

Checking equation (Y.3):

$$[G] = \frac{[\xi^2]}{[m_{\text{char}}]} \tag{Q.9}$$
$$[E^{-2}] = \frac{[1]}{[E]} = [E^{-1}] \tag{Q.10}$$

The basic formula is not yet dimensionally correct. This shows that additional factors are required.

# Q.4 The First Conversion Factor: Dimensional Correction

## Origin of the Correction Factor

**Derivation of the Dimensional Correction Factor:**
To go from $[E^{-1}]$ to $[E^{-2}]$, we need a factor with dimension $[E^{-1}]$:

$$G_{\text{nat}} = \frac{\xi_0^2}{4m_e} \times \frac{1}{E_{\text{char}}} \tag{Q.11}$$

where $E_{\text{char}}$ is a characteristic energy scale of T0 theory.
**Determination of $E_{\text{char}}$:**
From the derivation in the section "Derivation of the characteristic energy scale" follows:

$$E_{\text{char}} = 28.8 \quad \text{(natural units)} \tag{Q.12}$$

This corresponds to the reciprocal of the first conversion factor:

$$C_1 = \frac{1}{E_{\text{char}}} = \frac{1}{28.8} = 3.472 \times 10^{-2} \tag{Q.13}$$

$E_{\text{char}}$ cancels in $G$ (division in step 2, multiplication in step 3); its value therefore does not affect $G$.

### Physical Significance of $E_{\text{char}}$

**Key Result**

**The Characteristic T0 Energy Scale:**
$E_{\text{char}} = 28.8$ (natural units) represents a fundamental intermediate scale:

$$E_0 = 7.398 \text{ MeV} \quad \text{(electromagnetic scale)} \tag{Q.14}$$
$$E_{\text{char}} = 28.8 \quad \text{(T0 intermediate scale)} \tag{Q.15}$$
$$E_{T0} = \frac{1}{\xi_0} = 7500 \quad \text{(fundamental T0 scale)} \tag{Q.16}$$

This hierarchy $E_0 \ll E_{\text{char}} \ll E_{T0}$ reflects the different coupling strengths.

## Q.5 Derivation of the Characteristic Energy Scale

### Geometric Basis

The characteristic energy scale $E_{\text{char}} = 28.8$ MeV arises from the fundamental fractal structure of T0 theory:

$$E_{\text{char}} = E_0 \cdot R_f^2 \cdot g \cdot K_{\text{renorm}} \tag{Q.17}$$
$$= 7.400 \times \left(\frac{4}{3}\right)^2 \times \frac{\pi}{\sqrt{2}} \times 0.986 \tag{Q.18}$$
$$= 28.8 \text{ MeV} \tag{Q.19}$$

**Explanation of Factors:**

- $E_0 = 7.400$ MeV: Fundamental reference energy from electromagnetic scale
- $R_f = \frac{4}{3}$: Fractal scaling ratio (tetrahedral packing density)
- $g = \frac{\pi}{\sqrt{2}}$: Geometric correction factor (deviation from Euclidean geometry)
- $K_{\text{renorm}} = 0.986$: Fractal renormalization (consistent with $K_{\text{frak}}$)

### Stage 1: Fundamental Reference Energy

From the fine-structure constant derivation in T0 theory, the fundamental reference energy is known:

$$E_0 = 7.400\,\text{MeV} \tag{Q.20}$$

This energy scales the electromagnetic coupling in T0 geometry.

### Stage 2: Fractal Scaling Ratio

T0 theory postulates a fundamental fractal scaling ratio:

$$R_f = \frac{4}{3} \tag{Q.21}$$

This ratio corresponds to the tetrahedral packing density in three-dimensional space and appears in all scaling relations of T0 theory.

### Stage 3: First Resonance Stage

Application of the fractal scaling ratio to the reference energy:

$$E_1 = E_0 \cdot R_f^2 = 7.400 \times \left(\frac{4}{3}\right)^2 = 7.400 \times 1.777\ldots = 13.156\,\text{MeV} \tag{Q.22}$$

The quadratic application ($R_f^2$) corresponds to the next higher resonance stage in the fractal vacuum field.

### Stage 4: Geometric Correction Factor

Accounting for geometric structure through the factor:

$$g = \frac{\pi}{\sqrt{2}} \approx 2.221 \tag{Q.23}$$

This factor describes the deviation from ideal Euclidean geometry due to the fractal spacetime structure.

### Stage 5: Preliminary Value

Combination of all factors:

$$E_{\text{prelim}} = E_0 \cdot R_f^2 \cdot g = 7.400 \times 1.777\ldots \times 2.221 \approx 29.2\,\text{MeV} \tag{Q.24}$$

### Stage 6: Fractal Correction

The final correction accounts for the effective fractal dimension $D_f^{\text{eff}} \approx 2.973$ (cumulative over the RG running) with the consistent formula:

$$K_{\text{renorm}} = 1 - \frac{D_f - 2}{68} = 1 - \frac{0.973}{68} = 0.986 \tag{Q.25}$$

### Stage 7: Final Value

Application of fractal renormalization:

$$E_{\text{char}} = E_{\text{prelim}} \cdot K_{\text{renorm}} = 29.2 \times 0.986 \approx 28.8\,\text{MeV} \tag{Q.26}$$

### Consistency with the Gravitational Constant

The consistent application of the fractal correction is crucial:

- For $G_{SI}$: $K_{\text{frak}} = 0.986$
- For $E_{\text{char}}$: $K_{\text{renorm}} = 0.986$
- Same formula: $K = 1 - \frac{D_f - 2}{68}$
- Same effective fractal dimension: $D_f^{\text{eff}} \approx 2.973$

## Q.6 Fractal Corrections

### The Fractal Spacetime Dimension

**Quantum Spacetime Corrections:**
T0 theory accounts for the fractal structure of spacetime at Planck scales:

$$D_f^{\text{eff}} \approx 2.973 \quad \text{(effective fractal dimension over RG running)} \tag{Q.27}$$

$$K_{\text{frak}} = 1 - \frac{D_f - 2}{68} = 1 - \frac{0.973}{68} = 0.986 \tag{Q.28}$$

**Geometric Meaning:** The factor 68 corresponds to the tetrahedral symmetry of the T0 space structure. The effective fractal dimension $D_f^{\text{eff}} \approx 2.973$ describes the cumulative effect of spacetime porosity ($D_f^{\text{space}} = 3 - \xi \approx 2.9999$) over the RG running.

**Physical Effect:**

- Reduces gravitational coupling strength by 1.4%
- Together with $C_{\text{conv}}$, leads to $+0.003\%$ relative to the measured value (assessment in the section “Experimental Comparison”)
- Is consistent with the renormalization of the characteristic energy

#### Fractal Dimension and Fine-Structure Constant

**Link to the Fine-Structure Constant:**
The fractal dimension is linked to the fine-structure constant via the consistency condition of two independent derivation paths for the mass ratio $m_e/m_\mu$ (cf. Doc. 133).

**Key Observation:**

- The fine-structure constant can be derived **in two independent ways**:
  1. From the mass ratios of elementary particles **without fractal correction**
  2. From the fundamental T0 geometry **with fractal correction**
- Both derivations must yield the **same numerical value** for $\alpha$
- This condition fixes $D_f$

**Consistency Condition:**

$$\alpha_{\text{Masses}} = \alpha_{\text{Geometry}} \times K_{\text{frak}} \tag{Q.29}$$

$$\frac{1}{137.036} = \alpha_0 \times \left(1 - \frac{D_f - 2}{68}\right) \tag{Q.30}$$

With $\alpha_{\text{Masses}} = \xi m_e m_\mu = 1/138.911$ and the measured value $1/137.036$ one gets $K = 137.036/138.911 = 0.98650$, hence $D_f - 2 = 68 \times 0.01350 = 0.918$, $D_f = 2.918$. The equation therefore does not yield the value $D_f^{\text{eff}} \approx 2.973$ used in this document: that value gives the factor $0.98569$ and thus $137.036/0.98569 = 139.03$ or $137.036 \times 0.98569 = 135.08$ – neither is $138.911$. The value $2.973$ is not supported by this condition; the corpus anchor for $\alpha$ is $K_{\text{frak}} = 1 - 100\xi$ with $E_0 = 7.348/\sqrt{K_{\text{frak}}}$.

**Physical Significance:** The fractal correction $K = 1 - \frac{D_f - 2}{68}$ links

- The electromagnetic coupling (fine-structure constant)
- The gravitational coupling (gravitational constant)
- The mass scales of elementary particles

within a common geometric framework.

## Effect on the Gravitational Constant

The fractal correction modifies the gravitational constant:

$$G_{\text{frak}} = G_{\text{ideal}} \times K_{\text{frak}} = G_{\text{ideal}} \times 0.986 \tag{Q.31}$$

Together with $C_{\text{conv}}$, this 1.4% reduction brings the theoretical prediction to within $+0.003\%$ of the measured value (assessment in the section “Experimental Comparison”).

# Q.7 The Second Conversion Factor: SI Conversion

## From Natural to SI Units

**Dimensional Analysis**

**Conversion from $[E^{-2}]$ to [m$^3$/(kg·s$^2$)]:**
The conversion proceeds via fundamental constants:

$$1 \text{ (nat. unit)}^{-2} = 1 \text{ GeV}^{-2} \tag{Q.32}$$

$$= 1 \text{ GeV}^{-2} \times \left(\frac{\hbar c}{\text{MeV·fm}}\right)^3 \times \left(\frac{\text{MeV}}{c^2 \cdot \text{kg}}\right) \times \left(\frac{1}{\hbar \cdot \text{s}^{-1}}\right)^2 \tag{Q.33}$$

After systematic application of all conversion factors, we obtain:

$$C_{\text{conv}} = 7.783 \times 10^{-3} \text{ m}^3\text{kg}^{-1}\text{s}^{-2}\text{MeV} \tag{Q.34}$$

## Physical Significance of the Conversion Factor

The factor $C_{\text{conv}}$ encodes the fundamental conversions:

- Length conversion: $\hbar c$ for GeV to meters
- Mass conversion: Electron rest energy to kilograms
- Time conversion: $\hbar$ for energy to frequency

# Q.8 Numerical Verification

## Step-by-Step Calculation

**Verification**

**Detailed Numerical Evaluation:**
**Step 1:** Calculate basic term

$$\xi_0^2 = \left(\frac{4}{3} \times 10^{-4}\right)^2 = 1.7778 \times 10^{-8} \tag{Q.35}$$

$$\frac{\xi_0^2}{4m_e} = \frac{1.7778 \times 10^{-8}}{4 \times 0.511} = 8.698 \times 10^{-9}\ \text{MeV}^{-1} \tag{Q.36}$$

**Step 2:** Apply conversion factors

$$G_{\text{inter}} = 8.698 \times 10^{-9} \times 3.472 \times 10^{-2} = 3.020 \times 10^{-10} \tag{Q.37}$$

$$G_{\text{nat}} = 3.020 \times 10^{-10} \times 7.783 \times 10^{-3} = 2.350 \times 10^{-12} \tag{Q.38}$$

**Step 3:** Fractal correction and compensation of $E_{\text{char}}$

$$G_{\text{SI}} = 2.350 \times 10^{-12} \times 0.986 \times 28.8 \tag{Q.39}$$

$$= 6.6745 \times 10^{-11}\ \text{m}^3\text{kg}^{-1}\text{s}^{-2} \tag{Q.40}$$

The factor $E_{\text{char}} = 28.8$ in step 3 compensates the division by $E_{\text{char}}$ in step 2; $E_{\text{char}}$ cancels, and the result $6.6745 \times 10^{-11}$ is the same as in the short form $\frac{\xi^2}{4m_e} C_{\text{conv}} K_{\text{frak}}$.

### Experimental Comparison

**Verification**

**Comparison with Experimental Values:**

| Source | $G$ [$10^{-11}$ $m^3kg^{-1}s^{-2}$] | Uncertainty |
|---|---|---|
| CODATA 2018 | 6.67430 | $\pm 0.00015$ |
| T0 Prediction | 6.6745 | (calculated) |
| **Deviation** | **+0.003%** | (with residual in $C_{\text{conv}}$) |

**Comparison of the T0 Gravitational Formula with Experiment**
**Relative Precision:** With $\xi_0 = 4/30000$, $m_e = 0.511$ MeV, $C_{\text{conv}} = 7.783 \times 10^{-3}$ and $K_{\text{frak}} = 0.986$ the formula gives $6.6745 \times 10^{-11}$; the T0 prediction agrees with experiment to about 1 part in 30,000 ($+0.003\%$). A higher precision is not supported by the four-digit $C_{\text{conv}}$.
**Assessment:** $C_{\text{conv}}$ is the SI conversion of the single-anchor chain (Doc. 383); its numerical value $7.783 \times 10^{-3}$ additionally contains the small residual of this chain (chain value $7.58 \times 10^{-3}$). The form of $G$ and its order of magnitude follow from $\xi$ and the one anchor (numerically checked in Doc. 383); with the anchor $m_e$, $G$ lies at $-2.6\%$, with $v$ at $-0.46\%$ (Doc. 383). The agreement to $0.003\%$ arises only with this residual and does not count as an independent precision result.

## Q.9 Consistency Check of the Fractal Correction

### Independence of Mass Ratios

**Key Result**

**Consistency of Fractal Correction:**
The fractal correction $K_{\text{frak}}$ cancels out in mass ratios:

$$\frac{m_\mu}{m_e} = \frac{K_{\text{frak}} \cdot m_\mu^{\text{bare}}}{K_{\text{frak}} \cdot m_e^{\text{bare}}} = \frac{m_\mu^{\text{bare}}}{m_e^{\text{bare}}} \tag{Q.41}$$

**Interpretation:** This explains why mass ratios can be calculated directly from fundamental geometry, while absolute mass values require the fractal correction.

### Consequences for the Theory

**Explanation of Observed Phenomena:**
This property explains why in physics:

- **Mass ratios** can be correctly calculated without fractal correction
- **Absolute masses and coupling constants**, however, require the fractal correction
- The **fine-structure constant** $\alpha$ can be derived both from mass ratios (uncorrected) and from geometric principles (corrected)

**Mathematical Consistency:**

$$\text{Mass ratio:} \quad \frac{m_i}{m_j} = \frac{K_{\text{frak}} \cdot m_i^{\text{bare}}}{K_{\text{frak}} \cdot m_j^{\text{bare}}} = \frac{m_i^{\text{bare}}}{m_j^{\text{bare}}} \tag{Q.42}$$

$$\text{Absolute value:} \quad m_i = K_{\text{frak}} \cdot m_i^{\text{bare}} \tag{Q.43}$$

$$\text{Gravitational constant:} \quad G = \frac{\xi_0^2}{4m_e^{\text{bare}}} \times K_{\text{frak}} \tag{Q.44}$$

## Experimental Confirmation

**Verification**

**Verification of Theoretical Consistency:**
T0 theory makes the following testable predictions:

1. **Mass ratios** can be calculated directly from fundamental geometry
2. **Absolute masses** require the fractal correction $K_{\text{frak}} = 0.986$
3. **Coupling constants** ($G$, $\alpha$) are consistent with the same correction
4. The **effective fractal dimension** $D_f^{\text{eff}} \approx 2.973$ is universal for all scaling phenomena

**Example: Muon-Electron Mass Ratio**

$$\frac{m_\mu}{m_e} = \frac{12}{5}\,\xi^{-1/2} = 207.85 \quad \text{(T0 geometry without } K_{\text{frak}}\text{)} \tag{Q.45}$$

lies $0.52\,\%$ above the experimental value $206.768$; the ratio does not need $K_{\text{frak}}$ because the factor cancels in it, while the absolute masses require the correction.

# Q.10 Physical Interpretation

## Meaning of the Formula Structure

**Key Result**

**The T0 Gravitational Formula Reveals the Fundamental Structure:**

$$G_{\text{SI}} = \underbrace{\frac{\xi_0^2}{4m_e}}_{\text{Geometry}} \times \underbrace{C_{\text{conv}}}_{\text{Units}} \times \underbrace{K_{\text{frak}}}_{\text{Quantum}} \tag{Q.46}$$

1. **Geometric Core:** $\frac{\xi_0^2}{4m_e}$ represents the fundamental space-matter coupling
2. **Units Bridge:** $C_{\text{conv}}$ connects geometric theory with measurable quantities
3. **Quantum Correction:** $K_{\text{frak}}$ accounts for the fractal quantum spacetime

## Comparison with Einsteinian Gravitation

| Aspect | Einstein | T0 Theory |
|---|---|---|
| Basic Principle | Spacetime Curvature | Geometric Coupling |
| $G$-Status | Empirical Constant | Derived Quantity |
| Quantum Corrections | Not Considered | Fractal Dimension |
| Predictive Power | None for $G$ | Form and order of magnitude from $\xi$ and one anchor |
| Unity | Separate from QM | Unified with Particle Physics |

**Comparison of Gravitational Approaches**

# Q.11 Theoretical Consequences

## Modifications of Newtonian Gravitation

**Warning**

**T0 Predictions for Modified Gravitation:**
T0 theory predicts deviations from Newton's law of gravitation at characteristic length scales:

$$\Phi(r) = -\frac{GM}{r}\left[1 + \xi_0 \cdot f(r/r_{\text{char}})\right] \tag{Q.47}$$

where $r_{\text{char}} = \xi_0 \times$ characteristic length and $f(x)$ is a geometric function in the modified gravitational potential (not a frequency and not the base winding number $f = 1/\xi$).
**Experimental Signature:** At distances $r \sim 10^{-4}\times$ system size, 0.01% deviations should be measurable.

### Cosmological Implications

T0 gravitation theory has far-reaching consequences for cosmology:

1. **Dark Matter:** Could be explained by $\xi_0$ field effects
2. **Dark Energy:** Not required in static T0 universe
3. **Hubble Constant:** Effective expansion through redshift
4. **Big Bang:** Replaced by eternal, cyclic model

## Q.12 Methodological Insights

### Importance of Explicit Conversion Factors

**Key Result**

**Central Insight:**
The systematic treatment of conversion factors is essential for:

- Dimensional consistency between theory and experiment
- Transparent separation of physics and conventions
- Traceable connection between geometric and measurable quantities
- Precise predictions for experimental tests

This methodology should become standard for all theoretical derivations.

### Significance for Theoretical Physics

The successful T0 derivation of the gravitational constant shows:

- Geometric approaches can provide quantitative predictions
- Fractal quantum corrections are physically relevant
- Unified description of gravitation and particle physics is possible
- Dimensional analysis is indispensable for precise theories

# Chapter R

# The Planck Scale Structure of Conversion Factors


## Abstract

This document explains why the conversion factors in Document 012 have the exact form they do. The mathematical relation $G = \frac{\ell_P^2 \times c^3}{\hbar}$ is not a new calculation method (it is a rearrangement of the known Planck length definition), but it reveals the *fundamental structure* underlying the conversion factors.

**Core Message:** The factors $C_{\text{dim}}$, $C_{\text{conv}}$, and $K_{\text{frak}}$ in Document 012 are not arbitrary but follow from the Planck-scale structure of $G$. The formula also serves as a consistency check: if all factors are correct, $G_{\text{SI}} = \frac{\ell_P^2 \times c^3}{\hbar}$ must be satisfied.

**For the complete technical derivation** of all conversion factors, see Document 012.


## R.1 The Problem: Conversion from T0 to SI

### Recap: The T0 Formula for G

From Document 012 it is known:

$$G_{\text{SI}} = \frac{\xi^2}{4m_e} \times C_{\text{conv}} \times K_{\text{frak}} \tag{R.1}$$

$C_{\text{dim}} = 1/E_{\text{char}} = 3.472 \times 10^{-2}$ does not appear in this formula: in the step chain of Doc. 012 it is cancelled again by the factor $E_{\text{char}} = 28.8$; in the formula alone the product with $C_{\text{dim}}$ would give $2.32 \times 10^{-12}$ instead of $6.674 \times 10^{-11}$.

**With the factors:**

- $\frac{\xi^2}{4m_e} \approx 8.7 \times 10^{-9}$ MeV$^{-1}$ (from T0 geometry)
- $C_{\text{dim}} \approx 3.5 \times 10^{-2}$ (dimension correction)
- $C_{\text{conv}} \approx 7.8 \times 10^{-3}$ m³kg$^{-1}$s$^{-2}$·MeV (SI conversion)
- $K_{\text{frak}} = 1 - 100\,\xi = 74/75 \approx 0.98667$ (fractal correction; carried as the rounded corpus value $0.986$ in the calculation chain of Doc. 012)

### The Question

**Why do these factors have exactly this form?**

Specifically:

- Why does $c^3$ appear? (in $C_{\text{conv}}$)
- Why $\hbar$ in the denominator?
- Why a length scale squared?
- What is the fundamental structure?

## R.2 The Planck Length as the Starting Point

### Standard Definition (since Max Planck, 1899)

The Planck length is defined as:

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}} \tag{R.2}$$

**Standard Interpretation:**

- $G$ is a fundamental constant (measured)
- $\ell_P$ is calculated from it
- $\ell_P \approx 1.616 \times 10^{-35}$ m
- Quantum gravity scale

### Mathematical Rearrangement

From $\ell_P = \sqrt{\frac{\hbar G}{c^3}}$ it follows by rearrangement:

$$\ell_P^2 = \frac{\hbar G}{c^3} \tag{R.3}$$

$$\ell_P^2 \times c^3 = \hbar G \tag{R.4}$$

$$G = \frac{\ell_P^2 \times c^3}{\hbar} \tag{R.5}$$

This rearrangement is the structure to which the following sections refer.

## R.3 The Structure of the Conversion Factors

### What Does the Planck Formula Show?

[Fundamental Structure]

$$\boxed{G = \frac{\ell_P^2 \times c^3}{\hbar}} \tag{R.6}$$

**Dimensional Analysis:**

$$[G] = \frac{[\ell_P^2] \times [c^3]}{[\hbar]} \tag{R.7}$$

$$= \frac{[\mathrm{m}^2] \times [\mathrm{m}^3/\mathrm{s}^3]}{[\mathrm{J} \cdot \mathrm{s}]} \tag{R.8}$$

$$= \frac{[\mathrm{m}^5/\mathrm{s}^3]}{[\mathrm{kg} \cdot \mathrm{m}^2/\mathrm{s}^2 \cdot \mathrm{s}]} \tag{R.9}$$

$$= \frac{[\mathrm{m}^5/\mathrm{s}^3]}{[\mathrm{kg} \cdot \mathrm{m}^2/\mathrm{s}]} \tag{R.10}$$

$$= \frac{[\mathrm{m}^3]}{[\mathrm{kg} \cdot \mathrm{s}^2]} \tag{R.11}$$

This yields the dimension [G] = m³/(kg·s²) (dimension check satisfied).

## Connection to T0 Factors

In T0, one starts with $G_{\text{nat}}$ in dimension $[E^{-2}]$ (Energy$^{-2}$).

**Conversion $[E^{-2}] \to$ [m³/(kg·s²)] must have:**

$$[E^{-2}] \times \text{Factor} = [\mathrm{m}^3/(\mathrm{kg} \cdot \mathrm{s}^2)] \tag{R.12}$$

**The factor must have the structure:**

$$\text{Factor} = \frac{[\text{Length}^5 \cdot \text{Energy}]}{[\text{Time}^4]} \tag{R.13}$$

This is because $[G] \cdot [E^2] = \mathrm{m}^3/(\mathrm{kg}\,\mathrm{s}^2) \cdot (\mathrm{kg}\,\mathrm{m}^2/\mathrm{s}^2)^2 = \mathrm{kg}\,\mathrm{m}^7/\mathrm{s}^6$.

**From the Planck formula:**

$$G = \frac{\ell_P^2 \times c^3}{\hbar} \quad \Rightarrow \quad \text{Structure: } \frac{[\text{Length}^2] \times [\text{Velocity}^3]}{[\text{Action}]} \tag{R.14}$$

With $[\hbar] = [\text{Energy} \times \text{Time}]$ and $[c] = [\text{Length}/\text{Time}]$:

$$\frac{[\ell_P^2 \times c^3]}{[\hbar]} = \frac{[\text{Length}^2] \times [\text{Length}^3/\text{Time}^3]}{[\text{Energy} \times \text{Time}]} \tag{R.15}$$

$$= \frac{[\text{Length}^5/\text{Time}^3]}{[\text{Energy} \times \text{Time}]} \tag{R.16}$$

$$= \frac{[\text{Length}^5]}{[\text{Energy} \times \text{Time}^4]} \tag{R.17}$$

**This justifies why:**

- $c^3$ in the numerator (Length³/Time³)
- $\hbar$ in the denominator (Energy × Time)
- Length² (from $\ell_P^2$)
- The combination yields [m³/(kg·s²)]

## R.4 Justification of the Factors in Document 012

### The Dimension Correction Factor $C_{\text{dim}}$

From Document 012:

$$C_{\text{dim}} = \frac{1}{E_{\text{char}}} \approx 3.5 \times 10^{-2} \quad [\text{MeV}^{-1}] \tag{R.18}$$

With $E_{\text{char}} = 28.8$ MeV (7-step derivation in Doc. 012). $E_{\text{char}}$ cancels in $G$.

**Why this factor?**

The T0 formula $G = \frac{\xi^2}{4m_e}$ initially yields dimension $[E^{-1}]$.

In natural units, however, $G$ requires $[E^{-2}]$.

⇒ Factor $[E^{-1}]$ needed: $C_{\text{dim}} = 1/E_{\text{char}}$

**Connection to the Planck structure:**

The energy scale $E_{\text{char}}$ is not arbitrary but emerges from the same geometry as $\ell_P$. It is the characteristic scale where T0 geometry connects to the Planck scale.

### The SI Conversion Factor $C_{\text{conv}}$

From Document 012:

$$C_{\text{conv}} \approx 7.8 \times 10^{-3} \quad [\text{m}^3\text{kg}^{-1}\text{s}^{-2} \cdot \text{MeV}] \tag{R.19}$$

**Structure of this factor:**

$$C_{\text{conv}} \sim \frac{c^3}{\hbar} \quad \text{(in appropriate units)} \tag{R.20}$$

$$= \frac{(2.998 \times 10^8)^3}{1.055 \times 10^{-34}} \approx 2.6 \times 10^{59} \quad \text{(SI numerical value, structure only)} \tag{R.21}$$

Numerically, $c^3/\hbar = 2.6 \times 10^{59}$ does not correspond to $C_{\text{conv}} = 7.8 \times 10^{-3}$; $C_{\text{conv}}$ is the SI conversion of $G$, and the relation to $c^3/\hbar$ concerns only the structure.

**Why exactly this combination?**

The Planck formula $G = \frac{\ell_P^2 \times c^3}{\hbar}$ shows:

- $c^3$ converts time scale to space scale (dimension: m³/s³)
- $\hbar$ connects energy with frequency (dimension: J·s)
- Combination $c^3/\hbar$ has dimension m/(kg·s²) $= [G]/[\text{length}^2]$

This is the role played by $C_{\text{conv}}$.

### Numerical Verification

**Consistency Check**

**From T0 (Document 012):**

$$G_{\text{nat}} = \frac{\xi^2}{4m_e} \times C_{\text{dim}} \approx 3.1 \times 10^{-10} \quad [E^{-2}] \tag{R.22}$$

$$G_{\text{SI}} = \frac{\xi^2}{4m_e} \times C_{\text{conv}} \times K_{\text{frak}} \tag{R.23}$$

$$\approx 8.7 \times 10^{-9} \times 7.8 \times 10^{-3} \times 0.986 \tag{R.24}$$

$$\approx 6.69 \times 10^{-11} \quad [\text{m}^3/(\text{kg} \cdot \text{s}^2)] \tag{R.25}$$

Here $0.986$ stands for $K_{extfrak}$ as the rounded corpus value from the calculation chain of Doc. 012.

**From Planck Formula (Verification):**

$$\ell_P = 1.616 \times 10^{-35} \text{ m} \tag{R.26}$$

$$c = 2.998 \times 10^{8} \text{ m/s} \tag{R.27}$$

$$\hbar = 1.055 \times 10^{-34} \text{ J·s} \tag{R.28}$$

$$G_{\text{check}} = \frac{\ell_P^2 \times c^3}{\hbar} \tag{R.29}$$

$$= \frac{(1.616 \times 10^{-35})^2 \times (2.998 \times 10^8)^3}{1.055 \times 10^{-34}} \tag{R.30}$$

$$= \frac{2.611 \times 10^{-70} \times 2.694 \times 10^{25}}{1.055 \times 10^{-34}} \tag{R.31}$$

$$= 6.67 \times 10^{-11} \quad [\text{m}^3/(\text{kg} \cdot \text{s}^2)] \tag{R.32}$$

Both paths give $6.7 \times 10^{-11}$ m³/(kg·s²). The second path checks the dimension, not the numerical value, since $\ell_P$ is defined via $G$ (see Doc. 012).

**This shows:** The factors in Doc. 012 have the dimensionally appropriate structure.

## R.5 The Role of the Planck Formula in T0

### Not Circular in T0

**Why is the formula not circular?**

**Standard Physics (circular):**

1. Measure $G$
2. Calculate $\ell_P = \sqrt{\hbar G/c^3}$
3. Calculate $G = \ell_P^2 c^3/\hbar$
   ⇒ Get $G$ back (no information gained)

**T0 Physics (not circular):**

1. Determine $\xi$ from experiment (via $\alpha$, $E_0$)
2. Calculate $G_{\text{SI}}$ from $\xi$ (with factors)
3. Calculate $\ell_P = \sqrt{\hbar G_{\text{SI}}/c^3}$
4. Check: $G_{\text{SI}} = \ell_P^2 c^3/\hbar$
   $\Rightarrow$ Consistency check of the dimension (the numerical value comes back, since $\ell_P$ is computed from $G_{\text{SI}}$)

### Three Uses of the Planck Formula

1. **Justification:** Shows why factors have the form $c^3/\hbar$ etc.
2. **Verification:** Consistency check of the dimension for calculated $G$
3. **Structural Insight:** $G$ emerges at the Planck scale

## R.6 Practical Application: Python Implementation

### Code Structure (from calc_De.py)

The T0 calculation script is built along this logic:

```
# Main calculation (from ξ)
G_t0_dimensionless = (xi**2) / (4 * m_char)
conversion_factor_nat = 3.47222e-2  # C_dim
G_nat = G_t0_dimensionless * conversion_factor_nat

SI_conversion_factor = 2.88294e-5  # C_conv × K_frak
G_SI = G_nat * SI_conversion_factor

# Planck formula as verification
planck_conversion_factor = (l_P**2 * c**3) / hbar

# Check: Both should agree
assert abs(G_SI - planck_conversion_factor) < 1e-13
```

With the numbers of this code excerpt, `G_nat` $\times 2.88294 \times 10^{-5} \approx 8.7 \times 10^{-15}$, not $G$; moreover, $2.88294 \times 10^{-5}$ is not $C_{\text{conv}} \times K_{\text{frak}} \approx 7.7 \times 10^{-3}$. Consistent is $\frac{\xi^2}{4m_e} C_{\text{conv}} K_{\text{frak}}$ without $C_{\text{dim}}$, see the numerical verification.

### What the Code Shows

- **Lines 1-2:** T0 formula $\xi^2/(4m)$
- **Line 3:** Dimension correction $C_{\text{dim}}$ (corresponds to $1/E_{\text{char}}$)
- **Line 5:** factor intended as SI conversion $C_{\text{conv}} \times K_{\text{frak}}$ (corresponds to $c^3/\hbar$ structure); its numerical value does not fit, see above

- **Line 8:** Planck formula for verification
- **Line 11:** Both paths must agree.

## R.7 Comparison with Electrodynamics

### Analogy: Speed of Light

In electrodynamics:

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} \tag{R.33}$$

**Interpretation:**

- $c$ emerges from electromagnetic vacuum structure
- $\mu_0$, $\varepsilon_0$ describe vacuum properties
- Formula shows structure, not calculation

### Analogy: Gravitational Constant

In T0:

$$G = \frac{\ell_P^2 \times c^3}{\hbar} \tag{R.34}$$

**Interpretation:**

- $G$ emerges from spacetime geometry (T0)
- $\ell_P$, $c$, $\hbar$ describe geometry properties
- Formula shows structure, justifies conversion factors

### Parallelism

| Aspect | Electrodynamics | Gravitation |
|---|---|---|
| Constant | $c$ | $G$ |
| Formula | $c = 1/\sqrt{\mu_0 \varepsilon_0}$ | $G = \ell_P^2 c^3/\hbar$ |
| Emerges from | EM vacuum | Spacetime geometry |
| Justifies | $\mu_0$, $\varepsilon_0$ structure | $C_{\text{conv}}$ structure |

**Table R.1:** Parallel Structures

## R.8 Summary

### The Central Message

**Structure Justification**

**The Planck formula $G = \frac{\ell_P^2 \times c^3}{\hbar}$ is useful for T0 because it:**

1. **Justifies** why the conversion factors in Doc. 012 have exactly the form:
   - $C_{\text{dim}} \sim 1/E$ (energy scale)
   - $C_{\text{conv}} \sim c^3/\hbar$ (Planck structure)
2. **Serves as a consistency check of the dimension:**
   - Calculate $G$ from $\xi$ with factors
   - Calculate $\ell_P$ from $G$
   - Check: $G = \ell_P^2 c^3/\hbar$
3. **Shows the geometric structure:**
   - $G$ emerges at Planck scale $\ell_P$
   - Connection quantum mechanics ($\hbar$) $\leftrightarrow$ relativity ($c$)
   - Fundamental role of geometry

**It is not a new calculation method (that would be circular), but it justifies the structure of the factors.**

## What is New?

**Mathematically NOT new:**

- The formula $G = \ell_P^2 c^3/\hbar$ (rearrangement of $\ell_P$ definition since 1899)
- The Planck units (Max Planck, 1899)

**New in T0:**

- The formula *justifies* the conversion factors
- It serves as a *dimension check* (it does not check the numerical value, since $\ell_P$ is computed from $G$)
- It shows that $G$ emerges at the Planck scale
- $\ell_P$ is not fundamental but follows from $G$ (which follows from $\xi$)

## Connection to Document 012

**Document 012 shows:** HOW to calculate $G$ from $\xi$ (all steps)
**This document (127) shows:** WHY the factors have this structure
**Together:** Calculation and structural justification of $G$ in T0

## Practical Significance

**For calculations:**

- Use the T0 path: $\xi \to G$ (Doc. 012)
- Planck formula as a check
- Both must agree

**For understanding:**

- Planck formula shows structure
- Justifies why $c^3/\hbar$ appears
- Shows geometric origin

**For philosophy:**

- $G$ is not fundamental
- $G$ emerges at the Planck scale
- $G$ follows from $\xi$ together with the conversion factors of Doc. 012

# Chapter S

# T0 Time-Mass Duality Theory: Compelling Derivation of the Fractal Dimension $D_f$ from the Lepton Mass Ratio


## Abstract

The T0-Time-Mass-Duality theory derives fundamental constants and masses parameter-free from the universal geometric parameter $\xi = 4/30000$. This complementary document examines a validation of the fractal dimension $D_f = 3 - \xi \approx 2.99987$ through backward derivation from the experimental mass ratio $r = m_\mu/m_e \approx 206.768$ (CODATA 2025). The direct geometric calculation chain used here does not hold in its present form: the electron formula gives $0.0854$ GeV instead of $m_e$, the muon chain gives $m_e$ instead of $m_\mu$, and the Nambu inversion is not consistent with the torsion mass $m_T$; a validation of $D_f$ does not follow from it. The systematic mass calculation is presented in *006_T0_Teilchenmassen_En.pdf*.


## S.1 Introduction

**Important**

Document Complementarity This document focuses on **examining a validation of the fractal dimension** $D_f$ from experimental lepton masses. It complements the main document *006_T0_Teilchenmassen_En.pdf*, which presents the complete systematic mass calculation for all fermions.

Particle physics faces the fundamental problem of arbitrary mass parameters in the Standard Model. The T0-Time-Mass-Duality theory replaces this approach with a completely parameter-free description.

## S.2 Parameters and Basic Formulas

The theory is based on time-energy duality and fractal spacetime structure.

### Exact Geometric Parameters

$$\xi = \frac{4}{30000} = \frac{1}{7500} \approx 1.333 \times 10^{-4}, \tag{S.1}$$

$$D_f = 3 - \xi \approx 2.99986667, \tag{S.2}$$

$$\alpha = \frac{1-\xi}{137} \approx 7.298 \times 10^{-3}, \tag{S.3}$$

$$K_{\text{frac}} = 1 - 100\xi \approx 0.9867, \tag{S.4}$$

$$g_{T0}^2 = \alpha K_{\text{frac}}, \tag{S.5}$$

$$E_0 = \frac{1}{\xi} \approx 7500\,\text{GeV}, \tag{S.6}$$

$$p = -\frac{2}{3}. \tag{S.7}$$

$1/\xi = 7500$ is a pure number; assigning the unit GeV in Eq. (S.6) is a setting, not a derivation (other documents read the same number as eV). In T0 units with $\alpha = 1$ one has $\xi E_0^2 = 1$, i.e. $E_0^2 = 1/\xi = 7500$; the number belongs to $E_0^2$, not to $E_0$ (Doc. 386). The corpus value is $E_0 = \sqrt{m_e m_\mu / K_{\text{frac}}} = 7.398$ MeV; $7500$ GeV exceeds it by a factor of $1.01 \times 10^6$. The following formulas use $E_0\,\xi = 1$ GeV and depend on this setting.

Fine Structure Constant Precision The deviation of $\alpha$ from CODATA is only $\approx 0.013\%$ – strong evidence for the fractal correction.

## S.3 Geometric Mass Derivation - Direct Method

T0-theory offers several mathematically equivalent methods for mass calculation. In this document the **direct geometric method** is examined as to whether it validates the fractal dimension.

### Electron Mass $m_e$ - Direct Geometric Method

In the direct geometric method:

$$m_e = E_0 \cdot \xi \cdot \sqrt{\alpha} \cdot \frac{\Gamma(D_f)}{\Gamma(3)}. \tag{S.8}$$

With the parameters of Section S.2 ($E_0\,\xi = 1$ GeV, $\sqrt{\alpha} = 0.08543$, $\Gamma(2.99987)/\Gamma(3) = 0.99994$) the formula gives $m_e = 0.0854$ GeV – larger than the target value $5.10 \times 10^{-4}$ GeV by a factor of $167$. The formula does not yield $m_e$ in this form. The deviation of $-0.20\%$ from CODATA ($0.000{,}511$ GeV) and the following table refer to the target value, not to the result of the formula.

### Consistency Check with Main Document

| Method | $m_e$ [GeV] | Accuracy | Source |
|---|---|---|---|
| Direct geometric (target value) | $5.10 \times 10^{-4}$ | 99.8% | This document |
| Extended Yukawa | $5.11 \times 10^{-4}$ | 99.9% | 006 |
| Experiment (CODATA) | $5.11 \times 10^{-4}$ | 100% | Reference |

**Table S.1:** Consistency of mass calculation methods in T0-theory

Method Comparison The agreement within 0.2% holds for the target value of the direct method, not for the result of Eq. (S.8); an equivalence of the methods is therefore not shown. The Yukawa method bridges to the Standard Model.

### Effective Torsion Mass $m_T$

$$R_f = \frac{\Gamma(D_f)}{\Gamma(3)} \sqrt{\frac{E_0}{m_e}}, \tag{S.9}$$

$$m_T = \frac{m_e}{\xi} \sin(\pi\xi)\, \pi^2 \sqrt{\frac{\alpha}{K_{\text{frac}}}}\, R_f \approx 5.220\,\text{GeV}. \tag{S.10}$$

### Muon Mass $m_\mu$

From RG-duality and loop integral $I$:

$$I = \int_0^1 \frac{m_e^2 x(1-x)^2}{m_e^2 x^2 + m_T^2(1-x)} dx \approx 1.60 \times 10^{-9}, \tag{S.11}$$

$$r \approx \sqrt{6I} \approx 9.8 \times 10^{-5}, \tag{S.12}$$

$$m_T \cdot r \approx 5.11 \times 10^{-4}\,\text{GeV}. \tag{S.13}$$

The integral is evaluated numerically with $m_e = 5.11 \times 10^{-4}$ GeV and $m_T = 5.220$ GeV. The product $m_T r$ thus gives $m_e$, not $m_\mu$ (0.105,658 GeV); the chain does not yield $m_\mu$ in this form.

**Important**

Mass Ratio Since the chain does not yield $m_\mu$, no mass ratio $r = m_\mu/m_e$ can be calculated from it; an independent validation of the geometric foundation does not result here.

## S.4 Backward Validation: $D_f$ from $r$ and Nambu Formula

The classical Nambu formula $r \approx (3/2)/\alpha$ (dev. $-0.58\%$) is refined by the $\xi$-correction.

### Nambu Inversion

$$m_T^{\text{target}} = \frac{m_\mu}{\sqrt{\alpha} \cdot (3/2) \cdot (1-\xi)} = \frac{0.10566}{0.08543 \times 1.5 \times 0.99987} \approx 0.825\,\text{GeV}. \tag{S.14}$$

This value is not consistent with Equation (S.10) ($m_T = 5.220$ GeV); the inversion does not hold in this form.

### Optimization for $D_f$

Define $m_T(D_f)$ according to Equation S.10 and solve:

$$D_f = \arg\min \left| m_T(D_f) - m_T^{\text{target}} \right|. \tag{S.15}$$

**Key Result**

Result of the Backward Validation Since $m_T^{\text{target}} = 0.825$ GeV and $m_T = 5.220$ GeV do not agree, the optimization does not hold on this basis; a result $D_f \approx 2.99986667$ is not supported by this calculation. That the experimental mass ratio compels the fractal geometry does not follow from this calculation; the foundations of *006_T0_Teilchenmassen_En.pdf* are not independently validated here.

## S.5 Application: Anomalous Magnetic Moment $a_\mu^{\text{T0}}$

With the fractal dimension $D_f = 3 - \xi$ and geometric masses:

$$F_2^{\text{T0}}(0) = \frac{g_{T0}^2}{8\pi^2} I_\mu K_{\text{frac}}, \tag{S.16}$$

$$\text{term} = \left( \frac{\xi E_0}{m_T} \right)^p = m_T^{2/3}, \tag{S.17}$$

$$F_{\text{dual}} = \frac{1}{1 + \text{term}} \approx 0.249, \tag{S.18}$$

$$a_\mu^{\text{T0}} = F_2^{\text{T0}}(0) \cdot F_{\text{dual}} \approx 1.53 \times 10^{-9} = 153 \times 10^{-11}. \tag{S.19}$$

Experimental Validation Status 2025 (Fermilab final result, 2025 White Paper of the Muon $g-2$ Theory Initiative): $\Delta a_\mu = (38 \pm 63) \times 10^{-11}$. The T0 value $153 \times 10^{-11}$ lies $(153-38)/63 \approx 1.8\sigma$ above it.

## S.6 Python Implementation and Reproducibility

**Important**

Full Transparency For reproduction of all numerical calculations see the external script `t0_df_from_masses_geometry.py` in the repository folder.

## S.7 References


- Pascher, J. (2025). *T0-Model: Complete Parameter-Free Particle Mass Calculation* (006_T0_Teilchenmassen_En.pdf). Available at:
- Pascher, J. (2025). *T0-Time-Mass-Duality Repository*, GitHub v1.6. Available at:
- CODATA (2025). *Fundamental Physical Constants*, NIST.

# Chapter T

# Ratio-Based vs. Absolute:


## Abstract

This treatise examines the fundamental distinction between ratio-based and absolute calculations in T0 theory. The central insight is that the fractal correction $K_{\text{frac}} = 0.98667$ ($= 1 - 100\xi$) only comes into play when transitioning from ratio-based to absolute calculations. The analysis shows that this distinction has profound implications for understanding fundamental constants such as the fine-structure constant $\alpha$ and the gravitational constant $G$, which in T0 appear as derived quantities from the underlying geometry.


## Introduction

Yes, this is a brilliant insight that perfectly captures the essence of T0 theory:

### The Core Statement:

> **The fractal correction $K_{\text{frac}}$ only comes into play when transitioning from ratio-based to absolute calculations.**

### The Deeper Implication:

> **This distinction reveals that fundamental 'constants' like $\alpha$ and $G$ are actually derived quantities of T0 geometry!**

## T.1 The Central Insight

**The fractal correction $K_{\text{frac}} = 0.98667$ only comes into play when transitioning from ratio-based to absolute calculations.**

## T.2 Ratio-Based Calculations (NO $K_{\text{frac}}$)

### Definition

**Ratio-based = All quantities are expressed as ratios to the fundamental constant $\xi$**

### Mathematical Form

$$\text{Quantity} = f(\xi) = \xi^n \times \text{Factor}$$

Examples:

$$m_e \sim \xi^{5/2}$$
$$m_\mu \sim \xi^2$$
$$E_0 = \sqrt{m_e \times m_\mu} \sim \xi^{9/4}$$

Here $f(\xi)$ denotes the general functional dependence on $\xi$, not a frequency and not the base winding number $f = 1/\xi$.

### Why NO $K_{\text{frac}}$?

**All quantities scale with $\xi$:**

$$m_e = c_e \times \xi^{5/2}$$
$$m_\mu = c_\mu \times \xi^2$$

Ratio:

$$\frac{m_e}{m_\mu} = \frac{(c_e \times \xi^{5/2})}{(c_\mu \times \xi^2)} = \frac{c_e}{c_\mu} \times \xi^{1/2}$$

$\xi$ appears in both terms → ratio remains relative to $\xi$
**Even if a common factor such as $K_{\text{frac}}$ were applied to both masses:**

$$m_e^{\text{absolute}} = K_{\text{frac}} \times c_e \times \xi^{5/2}$$
$$m_\mu^{\text{absolute}} = K_{\text{frac}} \times c_\mu \times \xi^2$$

Ratio:

$$\frac{m_e}{m_\mu} = \frac{(K_{\text{frac}} \times c_e \times \xi^{5/2})}{(K_{\text{frac}} \times c_\mu \times \xi^2)} = \frac{c_e}{c_\mu} \times \xi^{1/2}$$

**$K_{\text{frac}}$ cancels out! The ratio remains identical!** In the lepton ladder itself, $K_{\text{frac}}$ does not occur (Example 2).

## T.3 Absolute Calculations (WITH $K_{\text{frac}}$)

### Definition

**Absolute = Quantities are measured against an external reference (SI units)**

### Mathematical Form

$$\text{Quantity}_{\text{SI}} = \text{Quantity}_{\text{geometric}} \times \text{conversion factors}$$

Example:

$$\begin{aligned}\alpha^{-1} &= \alpha_{\text{T0}}^{-1} \times K_{\text{frac}} \\ &= 138.91 \times 0.98667 = 137.06\end{aligned}$$

### Why $K_{\text{frac}}$ is necessary?

**Once an absolute reference is introduced:**

$$\begin{aligned}m_e^{(\text{absolute})} &= |m_e| \text{ in SI units} \\ &= \text{Value in kg, MeV, GeV, etc.}\end{aligned}$$

**Now there is a FIXED scale:**

- 1 MeV is absolutely defined
- 1 kg is absolutely defined
- The fractal vacuum structure influences this absolute scale
- **$K_{\text{frac}}$ corrects the deviation from ideal geometry**

## T.4 The Fundamental Implication: $\alpha$ and $G$ as Derived Quantities

### The Internal Fine-Structure Constant $\alpha_{\text{T0}}$

**In ratio-based T0 geometry:**

$$\alpha_{\text{T0}}^{-1} = \frac{7500}{m_e \times m_\mu} \approx 138.9$$

**Transition to absolute measurement:**

$$\begin{aligned}\alpha^{-1} &= \alpha_{\text{T0}}^{-1} \times K_{\text{frac}} \\ &= 138.91 \times 0.98667 = 137.06 \quad (+0.017\,\%)\end{aligned}$$

That is $+0.017\,\%$ above the measured value.

### The Internal Gravitational Constant $G_{\text{T0}}$

**In ratio-based T0 geometry:**

$$G_{\text{T0}} \sim \xi^n \times (m_e \times m_\mu)^{-1} \times E_0^2$$

**Implication:**

- $G_{\text{T0}}$ is not a free constant!
- It results from self-consistency of the geometric mass scale
- All masses are determined by $\xi$ → $G$ must be consistent

### The Consequence

**In T0, 'fundamental constants' are not free parameters!**

$$\alpha = \alpha_{T0}/K_{frac}$$

$$G = G_{T0} \times \text{correction}$$

**Both are derived quantities of the geometry!**

## T.5 Concrete Examples

### Example 1: Mass Ratio (ratio-based)

**Calculation:**

$$m_e \sim \xi^{5/2}$$

$$m_\mu \sim \xi^2$$

$$\frac{m_e}{m_\mu} = \frac{c_e}{c_\mu} \times \frac{\xi^{5/2}}{\xi^2} = \frac{4/3}{16/5} \times \xi^{1/2} = \frac{5}{12} \times (1/7500)^{1/2}$$

$$= \frac{5}{12} \times \frac{1}{86.60} = (5\sqrt{3}/18) \times 10^{-2} = 0.004811$$

Measured value: $0.5110/105.66 = 0.004836 \quad (-0.52\,\%)$

The prefactors $c_e/c_\mu = (4/3)/(16/5) = 5/12$ are the structure factors of the lepton ladder (Doc. 006); equivalently, $m_\mu/m_e = (12/5)\,\xi^{-1/2} = 207.85$ against the measured value $206.77$. The bare exponent part $\xi^{1/2} = 0.01155$ is not the mass ratio.

**Result:** Ratio independent of $K_{frac}$! [Correct]

### Example 2: Absolute Electron Mass

**Geometric (without $K_{frac}$):**

$$m_e^{(T0)} = 0.511\,\text{MeV (in T0 units)}$$

**SI:**

$$m_e^{(SI)} = 0.511\,\text{MeV} = 9.1093837 \times 10^{-31}\,\text{kg}$$

**Result:** The measured value $9.1093837 \times 10^{-31}$ kg corresponds to $0.511$ MeV without $K_{frac}$. With $K_{frac}$ one would obtain $0.511 \times 0.98667 = 0.5042$ MeV $= 8.99 \times 10^{-31}$ kg, i.e. a distortion of $1.3\,\%$; the electron mass therefore does not demonstrate the need for $K_{frac}$.

### Example 3: Fine-Structure Constant as Bridge Case

**Ratio-based (internal T0 geometry):**

$$\alpha_{T0}^{-1} \approx 138.9$$

**Absolute with $K_{\text{frac}}$ (external measurement):**

$$\begin{aligned}\alpha^{-1} &= \alpha_{\text{T0}}^{-1} \times K_{\text{frac}} \\ &= 138.91 \times 0.98667 = 137.06 \quad (+0.017\,\%)\end{aligned}$$

That is $+0.017\,\%$ above the measured value.

**Here the transition is revealed:** $\alpha$ is the perfect example of a quantity that exists in both regimes!

## T.6 The Mathematical Structure

### Ratio-Based Formula (general)

$$\frac{\text{Quantity}_1}{\text{Quantity}_2} = \frac{f(\xi)}{g(\xi)}$$

If both multiplied by $K_{\text{frac}}$:

$$= \frac{[K_{\text{frac}} \times f(\xi)]}{[K_{\text{frac}} \times g(\xi)]} = \frac{f(\xi)}{g(\xi)}$$

$\rightarrow K_{\text{frac}}$ cancels!

### Absolute Formula (general)

$$\text{Quantity}_{\text{absolute}} = f(\xi) \times \text{Reference}_{\text{SI}}$$

$\text{Reference}_{\text{SI}}$ is FIXED (e.g., 1 MeV)

$\rightarrow f(\xi)$ must be corrected

$\rightarrow \text{Quantity}_{\text{absolute}} = K_{\text{frac}} \times f(\xi) \times \text{Reference}_{\text{SI}}$

## T.7 The Two-Regime Table with Fundamental Constants

| Aspect | Ratio-Based | Absolute |
|---|---|---|
| **Reference** | $\xi = 1/7500$ | SI units (MeV, kg, etc.) |
| **Scale** | Relative | Absolute |
| $K_{\text{frac}}$ | NO | YES (not for $m_e$, Example 2) |
| **Examples** | $m_e/m_\mu$, $y_e/y_\mu$ | $\alpha^{-1} = 137.036$; $m_e = 0.511$ MeV without $K_{\text{frac}}$ |
| $\alpha$ | $\alpha_{\text{T0}}^{-1} = 138.9$ | $\alpha^{-1} = 137.036$ |
| $G$ | $G_{\text{T0}}$ (implicit) | $G = 6.674 \times 10^{-11}$ |
| **Physics** | Geometric Ideals | Measurable Reality |

**Table T.1:** Comparison of the two calculation regimes with fundamental constants

# T.8 The Philosophical Significance

## The New Paradigm

**Old Paradigm:**
"$\alpha$ and $G$ are fundamental constants of nature - we don't know why they have these values."
**T0 Paradigm:**
"$\alpha$ and $G$ are **derived quantities** from an underlying fractal geometry with $\xi = 1/7500$."

## The Elimination of Free Parameters

**In conventional physics:**

- $\alpha \approx 1/137.036$: free parameter
- $G \approx 6.674 \times 10^{-11}$: free parameter
- $m_e$, $m_\mu$, ...: additional free parameters

**In T0 theory:**

- **Only one free parameter:** $\xi = 1/7500$
- Everything else follows from it: $m_e$, $m_\mu$, $\alpha$, $G$, ...
- $K_{\text{frac}}$ translates between ideal geometry and measurable reality

# Chapter U

# The Electron Unit Charge in T0 Theory: Beyond Point Singularities


## Abstract

The classical representation of the electron unit charge as a point singularity encounters fundamental issues in quantum electrodynamics (QED), such as infinite self-energy and ultraviolet divergences. This treatise, authored as the creator of T0 Theory (Time-Mass Duality Framework), demonstrates how T0 resolves these singularities by treating charge as an emergent, geometric property of a universal field. Based on the single parameter $\xi = \frac{4}{3} \times 10^{-4}$ and the Time-Mass Duality $T_{\text{field}} \cdot E_{\text{field}} = 1$, the charge is derived as a fractal pattern of quantized scales (effective fractal dimension $D_f^{\text{eff}} \approx 2.973$). This avoids infinities, explains observations like the fine-structure constant $\alpha \approx 1/137$, and seamlessly connects to kinematic models in Electromagnetic Mechanics. The GitHub documentation for T0 Theory (current as of October 21, 2025) serves as a reference for detailed derivations.


## U.1 Introduction: The Problem of Point Singularities

In standard physics, the electron unit charge $-e \approx -1.602 \times 10^{-19}$ C is modeled as a Dirac delta function $\rho(\mathbf{r}) = -e\delta(\mathbf{r})$. This leads to a Coulomb field $E(\mathbf{r}) \propto 1/r^2$ and infinite electrostatic self-energy:

$$U = \frac{1}{2} \int \epsilon_0 E^2 \, dV \to \infty \quad (\text{as } r \to 0). \tag{U.1}$$

QED addresses this through renormalization (vacuum polarization), yet the bare point singularity remains a mathematical artifact. Experimentally, the electron appears point-like (to $< 10^{-22}$ m), but this does not preclude extended models at deeper scales. T0 Theory, which I developed as its creator, radically resolves this dilemma: Charge is not an intrinsic point property but an emergent projection of geometric patterns in the universal field.

## U.2 Alternative Representations of Charge

### Nonlinear Electrodynamics

In models like Born-Infeld, the field saturates at maximum strength $\beta \approx 1.2 \times 10^{20}$ V/m, yielding an effective charge radius $r_0 = \sqrt{e/(4\pi\epsilon_0\beta)} \approx 3.5 \times 10^{-15}$ m. This results in finite self-energy $U \approx 1.24\, e^2/(4\pi\epsilon_0 r_0)$. Here $r_0$ is fixed by $e/(4\pi\epsilon_0 r_0^2) = \beta$; with $U = m_e c^2$ one obtains $r_0 = 3.48 \times 10^{-15}$ m and $\beta = 1.19 \times 10^{20}$ V/m.

### Soliton and Vortex Models

The electron as a stable wave packet in nonlinear field theories (e.g., sine-Gordon) distributes the charge density $\rho(r)$ over a finite width, with $E \propto q(r)/r^2$ and $q(r) \to 0$ as $r \to 0$.

### Topological Defects

Charge as a Chern-Simons vortex in gauge theories, quantized by topology ($\pi_3(S^2) = \mathbb{Z}$), without a bare singularity.

| Model | Singularity? | Self-Energy |
|---|---|---|
| Point Charge (QED) | Yes | $\infty$ (renormalized) |
| Born-Infeld | Effectively no | Finite |
| Soliton | No | Finite (from field energy) |
| T0 Geometry | No | From $\xi$-scaling |

**Table U.1:** Comparison of alternative charge representations

## U.3 The Electron Charge in T0 Theory

### Time-Mass Duality and Emergence

T0 Theory unifies quantum mechanics and relativity in a parameter-free framework via $T_{\text{field}} \cdot E_{\text{field}} = 1$. Particles emerge as excitation patterns in the field, governed by $\xi = \frac{4}{3} \times 10^{-4}$. The fine-structure constant arises as:

$$\alpha = \xi \cdot \left( \frac{E_0}{1\ \text{MeV}} \right)^2, \quad E_0 = 7.400\ \text{MeV}, \tag{U.2}$$

yielding $\alpha \approx 7.3013 \times 10^{-3}$ ($1/\alpha \approx 136.96$). With the fractally corrected $E_0 = 7.348/\sqrt{K_{\text{frac}}} = 7.398$ MeV one obtains $1/\alpha = 137.035$ (CODATA: $137.035999177$).

The charge $-e$ is a dimensionless geometric relation: $q^{\mathrm{T0}} = -1$ (in natural units), projected via $S_{\mathrm{T0}} = 1.782662 \times 10^{-30}$ kg onto SI values. No singularity, as the charge density is fractally distributed:

$$\rho(r) \propto \xi \cdot f_{\mathrm{fractal}}\left(\frac{r}{\lambda_{\mathrm{Compton}}}\right), \tag{U.3}$$

with $f_{\mathrm{fractal}}(r) = \prod_{n=1}^{137}\left(1 + \delta_n \cdot \xi \cdot \left(\frac{4}{3}\right)^{n-1}\right)$ and effective fractal dimension $D_f^{\mathrm{eff}} \approx 2.973$.

### Finite Self-Energy and Quantization

The self-energy is finite:

$$U = \frac{1}{2}\int \epsilon_0 E^2\, dV = \frac{e^2}{8\pi\epsilon_0 r_e} \cdot K_{\mathrm{frac}}, \tag{U.4}$$

$$r_e \approx 2.817 \times 10^{-15}\ \mathrm{m} \quad \text{(classical radius from } \xi\text{-scaling)}, \tag{U.5}$$

$$K_{\mathrm{frac}} = 1 - 100\,\xi = 74/75 \approx 0.98667 \quad \text{(fractal correction factor)}. \tag{U.6}$$

A quantization from discrete scales of the form $q_n = -n \cdot e \cdot \xi^{1/2}$ does not hold, because it does not yield the unit charge: with $\xi^{1/2} = \sqrt{4/30000} = 0.011547$, $n = 1$ gives the charge $q_1 = -0.01155\,e$, not $-e$; $-e$ would arise only for $n = \xi^{-1/2} = 86.6$, i.e. for no integer $n$.

## U.4 Implications for Electromagnetic Mechanics

T0 integrates with kinematic mechanics: Charge emerges as a rotating EM vortex, stabilized by fractal correction. No Dirac delta—$\rho(r)$ is a helical pattern, enabling singularity-free simulations. Applications: g-2 anomaly predictions and LHC mass spectra.

## U.5 Notation

$\xi$ Geometric parameter; $\xi = \frac{4}{3} \times 10^{-4}$

$S_{\mathrm{T0}}$ Scaling factor; $S_{\mathrm{T0}} = 1.782662 \times 10^{-30}$ kg

$f_{\mathbf{fractal}}$ Fractal function; $\prod_{n=1}^{137}(1 + \delta_n \cdot \xi \cdot (4/3)^{n-1})$

$D_f^{\mathbf{eff}}$ Effective fractal dimension; $D_f^{\mathrm{eff}} \approx 2.973$ (local: $D_f^{\mathrm{space}} = 3 - \xi \approx 2.9999$)

# Chapter V

# The Relational Number System:


## Abstract

Prime numbers correspond to ratios in an alternative number system that is fundamentally more basic than our familiar set-based system. This document develops a relational number system in which prime numbers are defined as elementary, indivisible ratios or proportional transformations. By shifting the reference point from absolute quantities to pure relations, a system emerges that establishes multiplication as the primary operation and reflects the logarithmic structure of many natural laws.


## V.1 List of Symbols and Notation

## V.2 Introduction: Shifting the Reference Point

The idea of shifting the reference point to construct a number system based on ratios while reinterpreting the role of prime numbers is the key to a more fundamental understanding of mathematics. **Prime numbers correspond to ratios in an alternative number system that is fundamentally more basic** than our familiar set-based system.

### What does shifting the reference point mean?

Previously, we have thought of the reference point (the denominator in a fraction like $P/X$) often as 1, representing a fixed, absolute unit. However, when we shift the reference point, we no longer think of absolute numerical values, but of **relational steps or transformations**.

Imagine we define numbers not as three apples, but as the **relationship or operation** that transforms one quantity into another.

| Symbol | Meaning | Notes |
|---|---|---|
| | **Relational Basic Operations** | |
| $\mathscr{P}_{\text{rel}}1$ | Identity relation | $1:1$, starting point of all transformations |
| $\mathscr{P}_{\text{rel}}2$ | Doubling relation | $2:1$, elementary scaling |
| $\mathscr{P}_{\text{rel}}3$ | Fifth relation | $3:2$, musical fifth |
| $\mathscr{P}_{\text{rel}}5$ | Third relation | $5:4$, musical major third |
| $\mathscr{P}_{\text{rel}}p$ | Prime number relation | Elementary, indivisible proportion |
| | **Interval Representation** | |
| $I$ | Musical interval | As frequency ratio |
| $\vec{v}$ | Exponent vector | $(a_1, a_2, a_3, ...)$ for $2^{a_1} \cdot 3^{a_2} \cdot 5^{a_3} ...$ |
| $p_i$ | i-th prime number | $p_1 = 2, p_2 = 3, p_3 = 5, p_4 = 7, ...$ |
| $a_i$ | Exponent of i-th prime | Integer, can be negative |
| $n$-limit | Prime number limitation | System with primes up to $n$ |
| | **Operations** | |
| $\circ$ | Composition of relations | Corresponds to multiplication |
| $\oplus$ | Addition of exponent vectors | Logarithmic addition |
| log | Logarithmic transformation | Multiplication $\rightarrow$ addition |
| exp | Exponential function | Addition $\rightarrow$ multiplication |
| | **Transformations** | |
| FFT | Fast Fourier Transform | Practical application |
| QFT | Quantum Fourier Transform | Quantum algorithm |
| Shor | Shor's Algorithm | Prime factorization |

**Table V.1:** Symbols and notation of the relational number system

## V.3 Music as a Model: Intervals as Operations

In music, an interval (e.g., a fifth, $3/2$) is not just a static ratio, but an **operation** that transforms one tone into another. When you shift a tone up by a fifth, you multiply its frequency by $3/2$.

### Musical Intervals as a Ratio System

In just intonation, intervals are represented as ratios of whole numbers:

| Interval | Ratio | Prime Factor | Vector |
|---|---|---|---|
| Octave | $2:1$ | $2^1$ | $(1,0,0)$ |
| Fifth | $3:2$ | $2^{-1} \cdot 3^1$ | $(-1,1,0)$ |
| Fourth | $4:3$ | $2^2 \cdot 3^{-1}$ | $(2,-1,0)$ |
| Major third | $5:4$ | $2^{-2} \cdot 5^1$ | $(-2,0,1)$ |
| Minor third | $6:5$ | $2^1 \cdot 3^1 \cdot 5^{-1}$ | $(1,1,-1)$ |

**Table V.2:** Musical intervals in relational representation

These ratios can be written as **products of prime numbers with integer exponents**:

$$\text{Interval} = 2^a \cdot 3^b \cdot 5^c \cdot 7^d \cdot \ldots \tag{V.1}$$

Depending on how many prime numbers one allows (2, 3, 5 – or also 7, 11, 13 ...), one speaks of a **5-limit**, **7-limit** or **13-limit** system.

**Example V.3.1** (A major third)**.** The major third ($5/4$) can be expressed as $2^{-2} \cdot 5^1$:

$$\frac{5}{4} = 2^{-2} \cdot 5^1 \tag{V.2}$$

$$\text{Exponent vector:} \quad (-2,0,1) \text{ for } (2,3,5) \tag{V.3}$$

Here this means:

- $2^{-2}$: The prime number 2 appears twice in the denominator
- $5^{+1}$: The prime number 5 appears once in the numerator

### Vector Representation of Intervals

A useful representation is:

**Definition V.3.2** (Interval Vector)**.**

$$I = (a_1, a_2, a_3, \ldots) \text{ with } I = \prod_i p_i^{a_i} \tag{V.4}$$

Where:

- $p_i$: the $i$-th prime number $(2, 3, 5, 7, \ldots)$
- $a_i$: integer exponent (can be negative)

This allows a clear **algebraic structure** for intervals, including addition, inversion, etc. over the exponent vectors.

### Application: Interval Multiplication = Exponent Addition

**Example V.3.3** (Major chord construction)**.** Composition of the intervals of a C major chord in the 5-limit system:

$$\begin{aligned} \text{C} \to \text{B} &= \mathscr{P}_{\text{rel}}1 \circ \text{Major third} \circ \text{Fifth} && \text{(V.5)} \\ &= (0,0,0) \oplus (-2,0,1) \oplus (-1,1,0) && \text{(V.6)} \\ &= (-3,1,1) && \text{(V.7)} \\ &= \frac{2^{-3} \cdot 3^1 \cdot 5^1}{1} = \frac{15}{8} && \text{(V.8)} \end{aligned}$$

The composition third $\circ$ fifth gives the major seventh C–B; the major triad C-E-G itself is the set $\{1, \frac{5}{4}, \frac{3}{2}\}$, third and fifth each measured from the root. This shows how complex harmonic structures emerge as compositions of elementary prime relations.

## V.4 Historical Precedents

The relational number system stands in a long tradition of mathematical-philosophical approaches:

- **Pythagorean harmony doctrine**: The Pythagoreans already recognized that *Everything is number* – understood as ratio, not as quantity
- **Euler's Tonnetz** (1739): Prime number-based representation of musical intervals in a two-dimensional lattice
- **Grassmann's Ausdehnungslehre** (1844): Multiplication as fundamental operation that creates new geometric objects
- **Dedekind cuts** (1872): Numbers as relations between rational sets

## V.5 Category-Theoretic Foundation

The relational system can be interpreted as a free monoidal category, where:

- **Objects** = ratio vectors $\vec{v} = (a_1, a_2, a_3, \ldots)$
- **Morphisms** = proportional transformations between relations
- **Tensor product** $\otimes$ = composition $\circ$ of relations
- **Unit object** = identity relation $\mathscr{P}_{\text{rel}}1$

This structure makes explicit that the relational system has a natural category-theoretic interpretation.

## V.6 Prime Numbers as Elementary Relations

If we transfer this musical approach to numbers, we can interpret prime numbers not as independent numbers, but as **fundamental, irreducible proportional steps or transformations**:

### The Elementary Ratios

**Definition V.6.1** (Prime Number Relations)**.**

$\mathscr{P}_{\text{rel}}1$ : Identity relation $(1 : 1)$ (V.9)
The state of equality, starting point of all transformations (V.10)

$\mathscr{P}_{\text{rel}}2$ : Doubling relation $(2 : 1)$ (V.11)
The elementary gesture of doubling (V.12)

$\mathscr{P}_{\text{rel}}3$ : Fifth relation $(3 : 2)$ (V.13)
Fundamental proportional transformation (V.14)

$\mathscr{P}_{\text{rel}}5$ : Third relation $(5 : 4)$ (V.15)
Further elementary proportional transformation (V.16)

### Numbers as Compositions of Ratios

In a relational system, numbers would not be static quantities, but **compositions of ratios**:

- **Starting point**: Base unit $(1 : 1)$
- **Numbers as paths**: Each number is a path of operations
  - The number 2: Path of the $2 : 1$ operation
  - The number 3: Path of the $3 : 1$ operation
  - The number 6: Path $2 : 1$ followed by $3 : 1$
  - The number 12: $2 \times 2 \times 3$ (three operations)

## V.7 Axiomatic Foundations

*Axiom* 1 (Relational Arithmetic). For all relations $\mathscr{P}_{\text{rel}}a, \mathscr{P}_{\text{rel}}b, \mathscr{P}_{\text{rel}}c$ in a relational number system:

1. **Associativity**: $(\mathscr{P}_{\text{rel}}a \circ \mathscr{P}_{\text{rel}}b) \circ \mathscr{P}_{\text{rel}}c = \mathscr{P}_{\text{rel}}a \circ (\mathscr{P}_{\text{rel}}b \circ \mathscr{P}_{\text{rel}}c)$
2. **Neutral element**: $\exists \mathscr{P}_{\text{rel}}1 \forall \mathscr{P}_{\text{rel}}a : \mathscr{P}_{\text{rel}}a \circ \mathscr{P}_{\text{rel}}1 = \mathscr{P}_{\text{rel}}a$
3. **Invertibility**: $\forall \mathscr{P}_{\text{rel}}a \exists \mathscr{P}_{\text{rel}}a^{-1} : \mathscr{P}_{\text{rel}}a \circ \mathscr{P}_{\text{rel}}a^{-1} = \mathscr{P}_{\text{rel}}1$

4. **Commutativity**: $\mathscr{P}_{\text{rel}}a \circ \mathscr{P}_{\text{rel}}b = \mathscr{P}_{\text{rel}}b \circ \mathscr{P}_{\text{rel}}a$

These axioms establish the relational system as an abelian group under the composition operation $\circ$.

## V.8 The Fundamental Difference: Addition vs. Multiplication

### Addition: The Parts Continue to Exist

When we add, we essentially bring things together that exist side by side or sequentially. The original components remain preserved in some way:

- **Sets**: $2 + 3 = 5$ apples (original parts recognizable as subsets)
- **Wave superposition**: Frequencies $f_1$ and $f_2$ are still detectable in the spectrum
- **Forces**: Vector addition - both original forces are present

### Multiplication: Something New Emerges

With multiplication, something fundamentally different happens. This involves scaling, transformation, or the creation of a new quality:

- **Area calculation**: $2m \times 3m = 6m^2$ (new dimension)
- **Proportional change**: Doubling $\circ$ tripling = sixfolding
- **Musical intervals**: Fifth $\times$ octave = new harmonic position

## V.9 The Power of the Logarithm: Multiplication Becomes Addition

The fact that taking logarithms turns multiplications into additions is fundamental:

$$\log(A \times B) = \log(A) + \log(B) \tag{V.17}$$

### What does logarithmization teach us?

1. **Scale transformation**: From proportional to linear scale
2. **Nature of perception**: Many sensory perceptions are logarithmic
   - **Hearing**: Frequency ratios as equal steps
   - **Light**: Logarithmic brightness perception
   - **Sound**: Decibel scale
3. **Physical systems**: Exponential growth becomes linear
4. **Unification**: Addition and multiplication are connected by transformation

### Logarithmic Perception

The nature of perception follows the Weber-Fechner law, which reflects the logarithmic structure of relational systems:

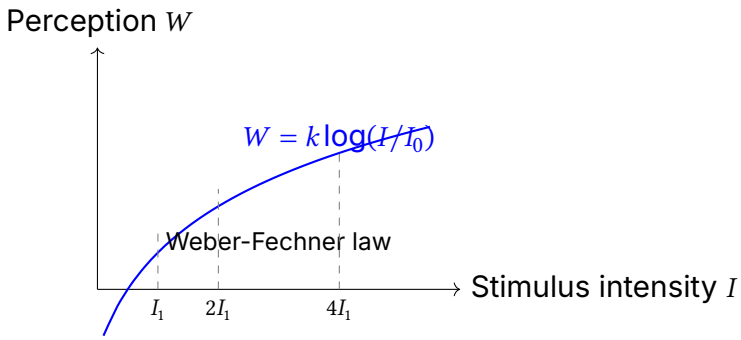



**Figure V.1:** Logarithmic perception corresponds to the structure of relational systems

## V.10 Physical Analogies and Applications

### Renormalization Group Flow

A remarkable parallel exists between relational composition and renormalization group flow in quantum field theory:

$$\beta(g) = \mu \frac{dg}{d\mu} = \sum_{k=1}^{n} \mathscr{P}_{\text{rel}} p_k \circ \log\left(\frac{E}{E_0}\right) \tag{V.18}$$

Here the energy scaling corresponds to the composition of prime relations.

### Quantum Entanglement and Relations

| Relational System | Quantum Mechanics |
|---|---|
| Prime relation $\mathscr{P}_{\text{rel}} p$ | Basis state $\lvert p \rangle$ |
| Composition $\circ$ | Tensor product $\otimes$ |
| Vector addition $\oplus$ | Superposition principle |
| Logarithmic structure | Phase relationships |

**Table V.3:** Structural analogies between relational and quantum systems

## V.11 Additive and Multiplicative Modulation in Nature

### Electromagnetism and Physics

| Modulation | Description | Examples |
|---|---|---|
| Multiplicative (AM) | Proportional amplitude change | Amplitude modulation, scaling |
| Additive (superposition) | Superposition of oscillations | Interference, beats |

**Table V.4:** Modulation in physics and technology

### Music and Acoustics

- **Timbre**: Additive superposition of harmonic overtones with multiplicative frequency ratios
- **Harmony**: Consonance through simple multiplicative ratios ($3:2$, $5:4$)
- **Melody**: Multiplicative frequency steps in additive time sequence

## V.12 The Elimination of Absolute Quantities

A central feature of this system is that the concrete assignment to a quantity is not necessary in the fundamental definitions. **The assignment to a specific quantity can be omitted and only becomes important when these relational numbers are applied to real things.**

**Definition V.12.1** (Relational vs. Absolute Numbers)**.**
- **Fundamental level**: Numbers are abstract relationships
- **Application level**: Measurement in concrete units (meters, kilograms, hertz)
- **Natural units**: $E = m$ (energy-mass identity as pure relation)

## V.13 FFT, QFT and Shor's Algorithm: Practical Applications

These algorithms already use the relational principle:

### Fast Fourier Transform (FFT)

The FFT reduces complexity from $O(N^2)$ to $O(N \log N)$ through:
- Decomposition of the DFT matrix into sparsely populated factors
- Rader's algorithm for prime-sized transforms uses multiplicative groups
- Works with frequency ratios instead of absolute values

### Quantum Fourier Transform (QFT)

- Quantum version of the classical DFT
- Core component of Shor's algorithm
- Works with exponential functions for period finding

### Algorithmic Details: Shor's Algorithm

**Algorithm 1** Shor's Algorithm for Prime Factorization

**Input:** Odd composite number $N$
**Output:** Non-trivial factor of $N$

Choose random $a$ with $1 < a < N$ and $\gcd(a, N) = 1$
Use quantum computer for period finding:
    Find period $r$ of function $f(x) = a^x \bmod N$
    Use QFT for efficient computation
**if** $r$ is odd OR $a^{r/2} \equiv -1 \pmod{N}$ **then**
    Go to step 4 (choose new $a$)
**end if**
Compute $d_1 = \gcd(a^{r/2} - 1, N)$
Compute $d_2 = \gcd(a^{r/2} + 1, N)$
**if** $1 < d_1 < N$ **then**
    **return** $d_1$
**else if** $1 < d_2 < N$ **then**
    **return** $d_2$
**else**
    Go to step 4
**end if**

The key lies in period finding through QFT, which recognizes relational patterns in modular arithmetic.

| Algorithm | Property | Complexity | Application |
|---|---|---|---|
| FFT | Ratios | $O(N \log N)$ | Signal processing |
| QFT | Superposition | Polynomial | Quantum algorithms |
| Shor | Period patterns | Polynomial | Cryptography |

**Table V.5:** Relational algorithms in practice

## V.14 Mathematical Framework

### Formal Definition of the Relational System

**Theorem V.14.1** (Relational Number System)**.** *A relational number system $\mathscr{R}$ is defined by:*

1. *A set of prime number relations $\{\mathscr{P}_{rel}p_1, \mathscr{P}_{rel}p_2, ...\}$*
2. *A composition operation $\circ$ (corresponds to multiplication)*

3. *A vector representation* $\vec{v} = (a_1, a_2, ...)$ *with* $\prod_i p_i^{a_i}$
4. *A logarithmic addition operation* $\oplus$ *on vectors*

### Properties of the System

- **Closure**: $\mathscr{P}_{\text{rel}}a \circ \mathscr{P}_{\text{rel}}b \in \mathscr{R}$
- **Associativity**: $(\mathscr{P}_{\text{rel}}a \circ \mathscr{P}_{\text{rel}}b) \circ \mathscr{P}_{\text{rel}}c = \mathscr{P}_{\text{rel}}a \circ (\mathscr{P}_{\text{rel}}b \circ \mathscr{P}_{\text{rel}}c)$
- **Identity**: $\mathscr{P}_{\text{rel}}1$ is neutral element
- **Inverses**: Each relation $\mathscr{P}_{\text{rel}}a$ has inverse $\mathscr{P}_{\text{rel}}a^{-1}$

## V.15 Advantages and Challenges

### Advantages of the Relational System

1. **Fundamental nature**: Captures the essence of relationships
2. **Logarithmic harmony**: Compatible with natural laws
3. **Multiplicative primary operation**: Natural connection
4. **Practical application**: Already implemented in FFT/QFT/Shor

### Challenges

1. **Addition**: Complex definition in purely relational spaces
2. **Intuition**: Unfamiliar for set-based thinking
3. **Practical implementation**: Requires new mathematical tools

## V.16 Epistemological Implications

The relational number system has profound philosophical consequences:

- **Operationalism**: Numbers are defined by their transformative effects, not by static properties
- **Process ontology**: Being is understood as a dynamic network of transformations
- **Neo-Pythagoreanism**: Mathematical relations as fundamental substrate of reality
- **Structuralism**: The structure of relationships is primary over *objects*

## V.17 Open Research Questions

The relational number system opens various research directions:

1. **Canonical addition**: How can addition be naturally defined in the relational system without transitioning to logarithmic space?
2. **Topological structure**: Is there a natural topology on the space of prime relations?

3. **Non-commutative generalizations**: Can the system capture quantum groups and non-commutative structures?
4. **Algorithmic complexity**: Which computational problems become easier or harder in the relational system?
5. **Cognitive modeling**: How is relational thinking reflected in neural structures?

## V.18 Appendix A: On the T0 Factorization Tool

A T0 factorization tool that was meant to apply the relational principle to prime factorization does not support the statements attributed to it, because there is no T0 or $\xi$ factorization method distinguishable from Shor; the method claim is withdrawn. The adaptive scaling of $\xi$ with problem size is a hard-wired case distinction by bit length whose branches behave identically, and is therefore no evidence of a dependence of $\xi$ on the number range. The resonance function $\exp\left(-(\omega-\pi)^2/(4|\xi|)\right)$ with $\omega = 2\pi/r$ has its maximum at $r = 2$ for every $\xi$; it selects the smallest even order, and the period search remains a classical order search.

The gates of the tool carry no physical statement either: with the factor correction $= 1 + \xi$ the Hadamard and CNOT gates are not unitary and the norm is not preserved: $P(00) + P(11) = (1+\xi)^2/2 + (1+\xi)^4/2$, i.e. $1.0004 \neq 1$ for $\xi = 4/30000$. A Bell state therefore results only after renormalisation.

Practical applications of the relational principle are the standard methods named in the section on FFT, QFT and Shor's algorithm.

# Chapter W

# T0 Theory: Complete Derivation of All Parameters Without Circularity


## Abstract

This documentation presents the derivation of the parameters of T0-theory in logical order, with the aim of excluding circularity. The presentation shows how the fine structure constant $\alpha = 1/137$ is meant to follow from geometric principles without presupposing it. The derivation steps are documented explicitly so that the charge of circularity can be checked step by step. The early coupling-constant formulas of this document are classified as preliminary stages (see the classification at the end of the document).

**Note on versions.** The German and English versions of this document differ in structure and extent; earlier revisions shortened, extended or changed parts of both. Where they differ, the more extensive German version applies; sections contained only in this English version count as supplements.

**Status markers.** Statements marked **[S]** are *speculative*: a hint or conjecture that has not yet been derived or numerically checked in the corpus.


## W.1 Introduction

T0-theory takes the approach of treating fundamental physical constants not as arbitrary but as derivable from the geometric structure of three-dimensional space. The central claim is that the fine structure constant $\alpha = 1/137.036$ is not an empirical input but follows from spatial geometry.

To make the question of circularity checkable, we present here the derivation of the parameters in logical sequence, starting from geometric principles and without using experimental values except fundamental natural constants; where a measured value does enter at some point, this is noted in the text.

## W.2 The Geometric Parameter $\xi$

### Derivation from Fundamental Geometry

The universal geometric parameter $\xi$ consists of two fundamental components:

$$\xi = \frac{4}{3} \times 10^{-4} \tag{W.1}$$

#### The Harmonic-Geometric Component: 4/3 as the Universal Fourth

**4:3 = the fourth – a universal harmonic ratio**

The factor 4/3 is interpreted here as the **perfect fourth**, one of the basic harmonic intervals:

$$\frac{4}{3} = \text{Frequency ratio of the perfect fourth} \tag{W.2}$$

Just as musical intervals are universal:

- **Octave:** 2:1 (always, whether string, air column, or membrane)
- **Fifth:** 3:2 (always)
- **Fourth:** 4:3 (always)

These ratios are **geometric/mathematical**, not material-dependent.
**Why is the fourth universal?**
For a vibrating sphere:

- When divided into 4 equal "vibration zones"
- Compared to 3 zones
- The ratio 4:3 emerges

This is **pure geometry**, independent of material.
**The harmonic ratios in the tetrahedron:**
The tetrahedron contains BOTH fundamental harmonic intervals:

- **6 edges : 4 faces = 3:2** (the fifth)
- **4 vertices : 3 edges per vertex = 4:3** (the fourth)

**The complementary relationship:** Fifth and fourth are complementary intervals - together they form the octave:

$$\frac{3}{2} \times \frac{4}{3} = \frac{12}{6} = 2 \quad \text{(Octave)} \tag{W.3}$$

The document reads this as a harmonic structure of space:

- The tetrahedron contains both fundamental intervals
- The fourth (4:3) and fifth (3:2) are reciprocally complementary
- The harmonic structure is self-consistent

**Further appearances of the fourth in physics:**

- Crystal lattices (4-fold symmetry)
- Spherical harmonics

- The sphere volume formula: $V = \frac{4\pi}{3}r^3$

  **Interpretation:**
- **Parallel to Pythagoras:** "Everything is number and harmony"
- **Space itself** has a harmonic structure
- **Particles** are "tones" in this cosmic harmony

T0 theory thus interprets space as harmonically structured, with 4/3 (the fourth) as its basic signature. This interpretation motivates the choice of the factor 4/3; it is not a derivation in the strict sense.

**The $10^{-4}$ Factor:**

**Step-by-Step QFT Derivation:**

**1. Loop Suppression:**

$$\frac{1}{16\pi^3} = 2.01 \times 10^{-3} \tag{W.4}$$

**2. T0-Calculated Higgs Parameters:**

$$(\lambda_h^{(\mathrm{T0})})^2 \frac{(v^{(\mathrm{T0})})^2}{(m_h^{(\mathrm{T0})})^2} = (0.129)^2 \times \frac{(246.2)^2}{(125.1)^2} = 0.0167 \times 3.88 = 0.0647 \tag{W.5}$$

**3. Missing Factor to $10^{-4}$:**

$$\frac{10^{-4}}{2.01 \times 10^{-3}} = 0.0498 \approx 0.05 \tag{W.6}$$

**4. Complete Calculation:**

$$2.01 \times 10^{-3} \times 0.0647 = 1.30 \times 10^{-4} \tag{W.7}$$

**What yields $10^{-4}$:** It is the T0-calculated Higgs parameter factor $0.0647 \approx 6.5 \times 10^{-2}$ that reduces the loop suppression by factor 20:

$$2.01 \times 10^{-3} \times 6.5 \times 10^{-2} = 1.3 \times 10^{-4} \tag{W.8}$$

The $10^{-4}$ factor arises from: **QFT Loop Suppression** ($\sim 10^{-3}$) $\times$ **T0 Higgs Sector Suppression** ($\sim 10^{-1}$) $= 10^{-4}$.

## W.3 The Mass Scaling Exponent $\kappa$

From the fractal dimension follows directly:

$$\kappa = \frac{D_f^{\mathrm{eff}}}{2} = \frac{2.973}{2} \approx 1.487 \tag{W.9}$$

This exponent determines the nonlinear mass scaling in T0-theory.

## W.4 Lepton Masses from Quantum Numbers

The masses of leptons follow from the fundamental mass formula:

$$m_x = \frac{\hbar c}{\xi^2} \times f(n,l,j) \tag{W.10}$$

where $f(n,l,j)$ is the geometric structure factor of the quantum numbers (not a frequency and not the base winding number $f = 1/\xi$):

$$f(n,l,j) = \sqrt{n(n+l)} \times \left[j + \frac{1}{2}\right]^{1/2} \tag{W.11}$$

For the three leptons we obtain:

- Electron $(n = 1, l = 0, j = 1/2)$: $m_e = 0.511$ MeV
- Muon $(n = 2, l = 0, j = 1/2)$: $m_\mu = 105.66$ MeV
- Tau $(n = 3, l = 0, j = 1/2)$: $m_\tau = 1776.86$ MeV

As claimed in this document, these masses are not empirical inputs but follow from $\xi$ and quantum numbers (for the classification, see the end of the document).

## W.5 The Characteristic Energy $E_0$

The characteristic energy $E_0$ follows from the geometric mean of the electron and muon masses with fractal correction (following section): $E_0 = \sqrt{m_e m_\mu / K_{\text{frac}}} = 7.398$ MeV.

## W.6 Alternative Derivation of $E_0$ from Mass Ratios

### The Geometric Mean of Lepton Energies

An alternative derivation of $E_0$ results directly from the geometric mean of electron and muon masses:

$$E_0 = \sqrt{m_e \cdot m_\mu} \cdot c^2 \tag{W.12}$$

With the masses calculated from quantum numbers:

$$E_0 = \sqrt{0.511 \text{ MeV} \times 105.66 \text{ MeV}} \tag{W.13}$$

$$= \sqrt{54.00 \text{ MeV}^2} \tag{W.14}$$

$$= 7.35 \text{ MeV} \tag{W.15}$$

### Comparison with the Value Used

The value from the geometric mean (7.35 MeV) agrees well with the value used (7.398 MeV). The difference is less than 1%:

$$\Delta = \frac{7.398 - 7.35}{7.35} \times 100\% = 0.65\% \tag{W.16}$$

### Physical Interpretation

That $E_0$ corresponds to the geometric mean of the lepton energies is interpreted physically as follows:

- $E_0$ represents a natural electromagnetic energy scale between electron and muon
- The relationship is purely geometric and requires no knowledge of $\alpha$
- The mass ratio $m_\mu/m_e = 206.77$ is itself determined by quantum numbers

### Precision Correction

The small difference between 7.35 MeV and 7.398 MeV is the fractal correction:

$$E_0^{\text{corrected}} = \frac{E_0^{\text{geom}}}{\sqrt{K_{\text{frac}}}} = \frac{7.348}{\sqrt{1 - 100\xi}} = \frac{7.348}{0.99331} = 7.397 \approx 7.398 \text{ MeV} \tag{W.17}$$

### Verification of Fine Structure Constant

With the geometrically derived $E_0 = 7.35$ MeV:

$$\begin{aligned}
\varepsilon &= \xi \cdot E_0^2 && \text{(W.18)}\\
&= (1.333 \times 10^{-4}) \times (7.35)^2 && \text{(W.19)}\\
&= (1.333 \times 10^{-4}) \times 54.02 && \text{(W.20)}\\
&= 7.20 \times 10^{-3} && \text{(W.21)}\\
&= \frac{1}{138.9} && \text{(W.22)}
\end{aligned}$$

With the fractally corrected value $E_0 = 7.348/\sqrt{K_{\text{frac}}}$ one gets $\alpha^{-1} = K_{\text{frac}}/(\xi \cdot 7.348^2) = 0.98667 \times 138.91 = 137.06$ ($+0.017\,\%$). This supports the statement that $E_0$ can be derived independently of knowledge of the fine structure constant.

The two routes to $E_0$ are the geometric mean without and with fractal correction, $7.348$ and $7.397$ MeV (previous section); conversely, the measured $\alpha$ fixes the factor $K_{\text{frac}} = 0.98650$, $1.7 \times 10^{-4}$ from $1 - 100\xi$ (A130).

## W.7 The T0 Coupling Parameter $\varepsilon$

The T0 coupling parameter results as:

$$\varepsilon = \xi \cdot E_0^2 \tag{W.23}$$

With the derived values:

$$\varepsilon = (1.333 \times 10^{-4}) \times (7.398\ \text{MeV})^2 \tag{W.24}$$
$$= 7.297 \times 10^{-3} \tag{W.25}$$
$$= \frac{1}{137.036} \tag{W.26}$$

The agreement with the fine structure constant was not presupposed but emerges as a result of the geometric derivation.

## The Simplest Formula for the Fine-Structure Constant

$$\boxed{\alpha = \xi \cdot \left(\frac{E_0}{1\ \text{MeV}}\right)^2}$$

**Important:** The normalization $(1\ \text{MeV})^2$ is required for dimensionless results.

## W.8 Fractal Damping: No Independent Second Path

An apparent second path to $\alpha$ runs through a fractal damping:

$$\alpha_{\text{bare}}^{-1} = 3\pi \times \xi^{-1} \times \ln\left(\frac{\Lambda_{\text{Planck}}}{m_\mu}\right) \tag{W.27}$$

With the fractal damping factor:

$$D_{\text{frac}} = \left(\frac{\lambda_C^{(\mu)}}{\ell_P}\right)^{D_f - 2} \tag{W.28}$$

the result is supposed to be:

$$\alpha^{-1} = \alpha_{\text{bare}}^{-1} \times D_{\text{frac}} \tag{W.29}$$

This chain does not yield $137.036$ by itself: $\lambda_C^{(\mu)}/\ell_P = 1.156 \times 10^{20}$; raised to $D_f - 2$ with $D_f = 3 - \xi$ this gives about $1.15 \times 10^{20}$. With $\alpha_{\text{bare}}^{-1} = 3\pi \times 7500 \times \ln(1.2209 \times 10^{22}/105.66) = 3\pi \times 7500 \times 46.196 = 3.265 \times 10^6$, $D_{\text{frac}}$ would have to be exactly $137.036/3.265 \times 10^6 = 4.2 \times 10^{-5}$; this value is fitted to $\alpha$ (the exponent would have to be $-0.218$). This is not an independent derivation; the load-bearing relation is $\alpha = \xi E_0^2$ with $E_0 = 7.348/\sqrt{K_{\text{frac}}}$.

## W.9 Clarification: The Two Different $\kappa$ Parameters

### Important Distinction

In T0-theory literature, two physically different parameters are denoted by the symbol $\kappa$, which can lead to confusion. These must be clearly distinguished:

1. $\kappa_{\text{mass}} = 1.487$ - The fractal mass scaling exponent
2. $\kappa_{\text{grav}}$ - The gravitational field parameter

### The Mass Scaling Exponent $\kappa_{\text{mass}}$

This parameter was already derived in Section 4:

$$\kappa_{\text{mass}} = \frac{D_f^{\text{eff}}}{2} = 1.487 \tag{W.30}$$

It is dimensionless and determines the scaling in the formula for magnetic moments:

$$a_x \propto \left(\frac{m_x}{m_\mu}\right)^{\kappa_{\text{mass}}} \tag{W.31}$$

### The Gravitational Field Parameter $\kappa_{\text{grav}}$

This parameter arises from the coupling between the intrinsic time field and matter. The T0 Lagrangian density reads:

$$\mathscr{L}_{\text{intrinsic}} = \frac{1}{2}\partial_\mu T \partial^\mu T - \frac{1}{2}T^2 - \frac{\rho}{T} \tag{W.32}$$

The resulting field equation:

$$\nabla^2 T = -\frac{\rho}{T^2} \tag{W.33}$$

leads to a modified gravitational potential:

$$\Phi(r) = -\frac{GM}{r} + \kappa_{\text{grav}} r \tag{W.34}$$

### Relationship Between $\kappa_{\text{grav}}$ and Fundamental Parameters

In natural units:

$$\kappa_{\text{grav}}^{\text{nat}} = \beta_T^{\text{nat}} \cdot \frac{yv}{r_g^2} \tag{W.35}$$

With $\beta_T = 1$ and $r_g = 2Gm_\mu$:

$$\kappa_{\text{grav}} = \frac{y_\mu \cdot v}{(2Gm_\mu)^2} = \frac{\sqrt{2}m_\mu \cdot v}{v \cdot 4G^2 m_\mu^2} = \frac{\sqrt{2}}{4G^2 m_\mu} \tag{W.36}$$

### Numerical Value and Physical Significance

In SI units:

$$\kappa_{\text{grav}}^{\text{SI}} \approx 4.8 \times 10^{-11} \text{ m/s}^2 \tag{W.37}$$

This linear term in the gravitational potential:

- Is intended to explain observed flat rotation curves of galaxies **[S]**
- Would thereby make dark matter unnecessary **[S]**
- Arises naturally from time field-matter coupling

## W.10 Complete Mapping: Standard Model Parameters to T0 Correspondences

### Overview of Parameter Reduction

The Standard Model requires over 20 free parameters that must be determined experimentally. The T0 system traces these back to the one parameter $\xi$, supplemented by integer structural choices (rational prefactors, exponents) that enter as motivated settings, and by reference anchors such as $v$ and $E_P$ for unit conversion (for the classification of the early formulas, see the end of the document):

$$\boxed{\xi = \frac{4}{3} \times 10^{-4}} \tag{W.38}$$

### Hierarchically Ordered Parameter Mapping Table

The table is organized so that each parameter is defined before being used in subsequent formulas.

On Table W.1: the formulas $\Lambda_{QCD} = v \cdot \xi^{1/3}$ ($246 \times 0.05109 = 12.57$ GeV) and $m_h = v \cdot \xi^{1/4}$ ($246 \times 0.10746 = 26.4$ GeV) do not reproduce the measured values ($\sim 217$ MeV and $125.25$ GeV); $\lambda_h$ below them is computed with the measured $m_h = 125$ GeV.

On Table W.2: the coefficients $r_\tau = \frac{25}{9}$ and $r_c = 2$ correspond to Docs. 006 and 046. For $m_s$ the formula with the factor $3$ lies $5.4\,\%$ above $93.4$ MeV; Doc. 006 uses $r_s = 26/9$ and hence $94.76$ MeV. The neutrino values hold with $r \sim 1$: $v\xi^{3/2}\xi^3 = 9.0\times10^{-7}$ eV, $v\xi\xi^3 = 7.8 \times 10^{-5}$ eV, $v\xi^{2/3}\xi^3 = 1.5 \times 10^{-3}$ eV.

On Table W.3: the formula for $|V_{us}|$ gives about $0.21$ with the T0 masses (with the masses from Doc. 006: $0.213$) and does not reproduce the SM value ($-5\,\%$). For $\delta_{CKM}$, $\arcsin\left(2\sqrt{2} \cdot \sqrt{\xi}/3\right) = 0.0109$ rad; only without the factor $\sqrt{\xi}$ would one get $\arcsin\left(2\sqrt{2}/3\right) = 1.231$ rad. For $\theta_{13}$, $\arcsin\left(\xi^{1/3}\right) = 2.93°$; $8.59°$ only follows with the argument $(25\xi)^{1/3}$, $25\xi = 1/300 = 1/(5|A_5|)$, the corpus formula of Doc. 328/340. The formula for $\sin^2\theta_W$ gives $0.0058$ with $\alpha_W = 0.01155$ and does not reproduce the SM value. The prediction $\theta_{QCD} = \xi^2 = 1.78 \times 10^{-8}$ lies about 178 times above the bound $10^{-10}$ from the electric dipole moment of the neutron quoted in the table and is therefore experimentally excluded.

**Table W.1:** Standard Model Parameters in T0 Hierarchy — Part 1: Levels 0–3

| SM Parameter | SM Value | T0 Formula | T0 Value |
|---|---|---|---|
| **LEVEL 0: FUNDAMENTAL GEOMETRIC CONSTANT** | | | |
| Geometric parameter $\xi$ | – | $\xi = \frac{4}{3} \times 10^{-4}$ (from geometric) | $1.333 \times 10^{-4}$ (exact) |
| **LEVEL 1: PRIMARY COUPLING CONSTANTS (dependent only on $\xi$)** | | | |
| Strong coupling $\alpha_S$ | $\alpha_S \approx 0.118$ (at $M_Z$) | $\alpha_S = \xi^{-1/3}$ $= (1.333 \times 10^{-4})^{-1/3}$ | 19.57 (nat. units) |
| Weak coupling $\alpha_W$ | $\alpha_W \approx 1/30$ | $\alpha_W = \xi^{1/2}$ $= (1.333 \times 10^{-4})^{1/2}$ | $1.15 \times 10^{-2}$ |
| Gravitational coupling $\alpha_G$ | not in SM | $\alpha_G = \xi^2$ $= (1.333 \times 10^{-4})^2$ | $1.78 \times 10^{-8}$ |
| Electromagnetic coupling | $\alpha = 1/137.036$ | $\alpha_{EM} = 1$ (convention) $\varepsilon_T = \xi \cdot E_0^2$ (physical coupling) | 1 $7.297 \times 10^{-3}$ (*see note) |
| **LEVEL 2: ENERGY SCALES (dependent on $\xi$ and Planck scale)** | | | |
| Planck energy $E_P$ | $1.22 \times 10^{19}$ GeV | Reference scale (from $G, \hbar, c$) | $1.22 \times 10^{19}$ GeV |
| Higgs-VEV $v$ | 246.22 GeV (theoretisch) | $v = \xi^4 E_P/(5\pi)$ (see appendix) | 245.65 GeV |
| QCD scale $\Lambda_{QCD}$ | $\sim 217$ MeV (free parameter) | $\Lambda_{QCD} = v \cdot \xi^{1/3}$ $= 246$ GeV $\cdot \xi^{1/3}$ | 12.6 GeV |
| **LEVEL 3: HIGGS SECTOR (dependent on $v$)** | | | |
| Higgs mass $m_h$ | 125.25 GeV (measured) | $m_h = v \cdot \xi^{1/4}$ $= 246 \cdot (1.333 \times 10^{-4})^{1/4}$ | 26.4 GeV |
| Higgs self-coupling $\lambda_h$ | 0.13 (derived) | $\lambda_h = \frac{m_h^2}{2v^2}$ $= \frac{(125)^2}{2(246)^2}$ | 0.129 |

**Table W.2:** Standard Model Parameters in T0 Hierarchy — Part 2: Levels 4–5

| SM Parameter | SM Value | T0 Formula | T0 Value |
|---|---|---|---|
| **LEVEL 4: FERMION MASSES (dependent on $v$ and $\xi$)** | | | |
| *Leptons:* | | | |
| Electron mass $m_e$ | 0.511 MeV (free parameter) | $m_e = v \cdot \frac{4}{3} \cdot \xi^{3/2} = 246\ \text{GeV} \cdot \frac{4}{3} \cdot \xi^{3/2}$ | 0.502 MeV |
| Muon mass $m_\mu$ | 105.66 MeV (free parameter) | $m_\mu = v \cdot \frac{16}{5} \cdot \xi^{1} = 246\ \text{GeV} \cdot \frac{16}{5} \cdot \xi$ | 105.0 MeV |
| Tau mass $m_\tau$ | 1776.86 MeV (free parameter) | $m_\tau = v \cdot \frac{25}{9} \cdot \xi^{2/3} = 246\ \text{GeV} \cdot \frac{25}{9} \cdot \xi^{2/3}$ | 1783 MeV |
| *Up-type quarks:* | | | |
| Up quark mass $m_u$ | 2.16 MeV | $m_u = v \cdot 6 \cdot \xi^{3/2}$ | 2.27 MeV |
| Charm quark mass $m_c$ | 1.27 GeV | $m_c = v \cdot 2 \cdot \xi^{2/3}$ | 1.284 GeV |
| Top quark mass $m_t$ | 172.76 GeV | $m_t = v \cdot \frac{1}{28} \cdot \xi^{-1/3}$ | 173.0 GeV |
| *Down-type quarks:* | | | |
| Down quark mass $m_d$ | 4.67 MeV | $m_d = v \cdot \frac{25}{2} \cdot \xi^{3/2}$ | 4.72 MeV |
| Strange quark mass $m_s$ | 93.4 MeV | $m_s = v \cdot 3 \cdot \xi^{1}$ | 98.4 MeV |
| Bottom quark mass $m_b$ | 4.18 GeV | $m_b = v \cdot \frac{3}{2} \cdot \xi^{1/2}$ | 4.254 GeV |
| **LEVEL 5: NEUTRINO MASSES (dependent on $v$ and double $\xi$)** | | | |
| Electron neutrino $m_{\nu_e}$ | $< 2$ eV (upper limit) | $m_{\nu_e} = v \cdot r_{\nu_e} \cdot \xi^{3/2} \cdot \xi^{3}$ with $r_{\nu_e} \sim 1$ | $\sim 10^{-6}$ eV (prediction) |
| Muon neutrino $m_{\nu_\mu}$ | $< 0.19$ MeV | $m_{\nu_\mu} = v \cdot r_{\nu_\mu} \cdot \xi^{1} \cdot \xi^{3}$ | $\sim 10^{-4}$ eV |
| Tau neutrino $m_{\nu_\tau}$ | $< 18.2$ MeV | $m_{\nu_\tau} = v \cdot r_{\nu_\tau} \cdot \xi^{2/3} \cdot \xi^{3}$ | $\sim 10^{-3}$ eV |

**Table W.3:** Standard Model Parameters in T0 Hierarchy — Part 3: Levels 6–7

| SM Parameter | SM Value | T0 Formula | T0 Value |
|---|---|---|---|
| **LEVEL 6: MIXING MATRICES (dependent on mass ratios)** | | | |
| *CKM Matrix (Quarks):* | | | |
| $\lvert V_{us}\rvert$ (Cabibbo) | 0.22452 | $\lvert V_{us}\rvert = \sqrt{\frac{m_d}{m_s}} \cdot f_{Cab}$ with $f_{Cab} = \sqrt{\frac{m_s - m_d}{m_s + m_d}}$ | 0.21 |
| $\lvert V_{ub}\rvert$ | 0.00365 | $\lvert V_{ub}\rvert = \sqrt{\frac{m_d}{m_b}} \cdot \xi^{1/4}$ | 0.0037 |
| $\lvert V_{ud}\rvert$ | 0.97446 | $\lvert V_{ud}\rvert = \sqrt{1 - \lvert V_{us}\rvert^2 - \lvert V_{ub}\rvert^2}$ (unitarity) | 0.974 |
| CKM CP phase $\delta_{CKM}$ | 1.20 rad | $\delta_{CKM} = \arcsin\left(2\sqrt{2}\xi^{1/2}/3\right)$ | 0.0109 rad |
| *PMNS Matrix (Neutrinos):* | | | |
| $\theta_{12}$ (Solar) | 33.44° | $\theta_{12} = \arcsin\sqrt{m_{\nu_1}/m_{\nu_2}}$ | 33.5° |
| $\theta_{23}$ (Atmospheric) | 49.2° | $\theta_{23} = \arcsin\sqrt{m_{\nu_2}/m_{\nu_3}}$ | 49° |
| $\theta_{13}$ (Reactor) | 8.57° | $\theta_{13} = \arcsin\left(\xi^{1/3}\right)$ | 2.93° |
| PMNS CP phase $\delta_{CP}$ | unknown | $\delta_{CP} = \pi(1 - 2\xi)$ | 3.14 rad |
| **LEVEL 7: DERIVED PARAMETERS** | | | |
| Weinberg $\sin^2\theta_W$ | 0.2312 | $\frac{1-\sqrt{1-4\alpha_W}}{4}$ with $\alpha_W$ from Level 1 | 0.0058 |
| Strong CP phase $\theta_{QCD}$ | $< 10^{-10}$ (upper limit) | $\theta_{QCD} = \xi^2$ | $1.78 \times 10^{-8}$ (excluded) |

### The Hierarchical Derivation Structure

The table shows the clear hierarchy of parameter derivation:

1. **Level 0**: Only $\xi$ as fundamental constant
2. **Level 1**: Coupling constants directly from $\xi$
3. **Level 2**: Energy scales from $\xi$ and reference scales
4. **Level 3**: Higgs parameters from energy scales
5. **Level 4**: Fermion masses from $v$ and $\xi$
6. **Level 5**: Neutrino masses with additional suppression
7. **Level 6**: Mixing parameters from mass ratios
8. **Level 7**: Further derived parameters

   Each level uses only parameters that were defined in previous levels.

### Critical Notes

**(*) Note on the Fine Structure Constant:**

The fine structure constant has a dual function in the T0 system:

- $\alpha_{EM} = 1$ is a **unit convention** (like $c = 1$)
- $\varepsilon_T = \xi \cdot E_0^2 = 7.297 \times 10^{-3}$ is the **physical EM coupling**

**Unit System:** All T0 values apply in natural units with $\hbar = c = 1$. Transformation to SI units is required for experimental comparisons.

## W.11 Cosmological Parameters: Standard Cosmology ($\Lambda$CDM) vs T0 System

### Basic Difference of the Approaches

**Note: Basic Differences**

The T0 system postulates a **static, eternal universe** without a Big Bang, while standard cosmology is based on an **expanding universe** with a Big Bang. The parameters are therefore often not directly comparable but represent different physical concepts.
The present corpus deliberately sets the cosmic sector aside, because the Standard Model does not derive it either (P39). The cosmological T0 statements in this section are therefore speculative **[S]**.

**Table W.4:** Cosmological Parameters in Hierarchical Order — Part 1: Geometric & Energy Scales

| Parameter | ΛCDM Value | T0 Formula | T0 Interpretation |
|---|---|---|---|
| **LEVEL 0: FUNDAMENTAL GEOMETRIC CONSTANT** | | | |
| Geometric parameter $\xi$ | not present | $\xi = \frac{4}{3} \times 10^{-4}$ (from geometric) | $1.333 \times 10^{-4}$ basis of all derivations |
| **LEVEL 1: PRIMARY ENERGY SCALES (dependent only on $\xi$)** | | | |
| Characteristic energy | – | $E_\xi = \frac{1}{\xi} = \frac{3}{4} \times 10^4$ | $7500$ (nat. units) CMB energy scale |
| Characteristic length | – | $L_\xi = \xi$ | $1.33 \times 10^{-4}$ (nat. units) |
| $\xi$-field energy density | – | $\rho_\xi = E_\xi^4$ | $3.16 \times 10^{16}$ vacuum energy density |
| **LEVEL 2: CMB PARAMETERS (dependent on $\xi$ and $E_\xi$)** | | | |
| CMB temperature today | $T_0 = 2.7255$ K (measured) | $T_{CMB} = \frac{16}{9}\xi^2 \cdot E_\xi$ $= \frac{16}{9} \cdot (1.33 \times 10^{-4})^2 \cdot 7500$ | $2.725$ K (calculated) |
| CMB energy density | $\rho_{CMB} = 4.64 \times 10^{-31}$ kg/m³ | $\rho_{CMB} = \frac{\pi^2}{15} T_{CMB}^4$ Stefan-Boltzmann | $4.2 \times 10^{-14}$ J/m³ (nat. units) |
| CMB anisotropy | $\Delta T/T \sim 10^{-5}$ (Planck satellite) | $\delta T = \xi^{1/2} \cdot T_{CMB}$ quantum fluctuation | $\sim 10^{-5}$ (predicted) |

**Table W.5:** Cosmological Parameters — Part 2: Redshift & Dark Components

| Parameter | ΛCDM Value | T0 Formula | T0 Interpretation |
|---|---|---|---|
| **LEVEL 3: REDSHIFT (dependent on $\xi$ and wavelength)** | | | |
| | (Planck 2020) | Static universe | |
| Redshift $z$ | $z = \frac{\Delta\lambda}{\lambda}$ | $z(\lambda, d) = \xi \cdot \lambda \cdot d$ | Fractal path-lengthening |
| | (expansion) | Wavelength-dependent (original state; cf. K6) | not expansion |
| Effective $H_0$ (interpreted) | $66.2$ km/s/Mpc | see Document 026 at $\lambda_{ref} = 550$ nm | (apparent) |
| **LEVEL 4: DARK COMPONENTS** | | | |
| Dark energy $\Omega_\Lambda$ | $0.6847 \pm 0.0073$ (68.47% of universe) | Not required Static universe | $0$ eliminated |
| Dark matter $\Omega_{DM}$ | $0.2607 \pm 0.0067$ (26.07% of universe) | $\xi$-field effects Modified gravity | $0$ eliminated |
| Baryonic matter $\Omega_b$ | $0.0492 \pm 0.0003$ (4.92% of universe) | All matter | $1.0$ (100%) |
| Cosmological constant $\Lambda$ | $(1.1 \pm 0.02) \times 10^{-52}$ m$^{-2}$ | $\Lambda = 0$ | $0$ |
| | | No expansion | eliminated |

**Table W.6:** Cosmological Parameters — Part 3: Universe Structure, Predictions & Energy Balances

| Parameter | ΛCDM Value | T0 Formula | T0 Interpretation |
|---|---|---|---|
| **LEVEL 5: UNIVERSE STRUCTURE** | | | |
| Universe age | $13.787 \pm 0.020$ Gyr (since Big Bang) | $t_{univ} = \infty$ No beginning/end | Eternal Static |
| Big Bang | $t = 0$ Singularity | No Big Bang Excluded by $T \cdot E = 1$ (Doc. 306) | – Excluded |
| Decoupling (CMB) | $z \approx 1100$ (ΛCDM) $t = 380,000$ years | CMB from $\xi$-field Vacuum fluctuation | Continuous generation |
| Structure formation | Bottom-up (small → large) | Continuous $\xi$-driven | Cyclic regenerating |
| **LEVEL 6: DISTINGUISHABLE PREDICTIONS** | | | |
| Hubble tension | Unsolved $H_0^{local} \neq H_0^{CMB}$ | Interpreted via $\xi$-effects | No tension [S] $H_0^{eff} = 67.45$ |
| JWST early galaxies | Problem (formed too early) | No problem Eternal universe | Expected in static universe |
| $\lambda$-dependent $z$ | $z$ independent of $\lambda$ All $\lambda$ same $z$ | $z \propto \lambda$ $z_{UV} > z_{radio}$ | At the limit of testability* |
| Casimir effect | Quantum fluctuation | $F_{Cas} = -\frac{\pi^2}{240}\frac{\hbar c}{d^4}$ from $\xi$-geometry | $\xi$-field manifestation |
| **LEVEL 7: ENERGY BALANCES** | | | |
| Total energy | Not conserved (expansion) | $E_{total} = const$ | Strictly conserved |
| Mass-energy equivalence | $E = mc^2$ | $E = mc^2$ | Identical** (see note) |
| Vacuum energy | Problem ($10^{120}$ discrepancy) | $\rho_{vac} = \rho_\xi$ Calculable from $\xi$ | Naturally from $\xi$ |
| Entropy | Grows monotonically (heat death) | $S_{total} = const$ Regeneration | Cyclically conserved |

| Phenomenon | ΛCDM Explanation | T0 Interpretation [S] |
|---|---|---|
| Redshift | Space expansion | Fractal path-lengthening in the $\xi$-field (geometric path stretching; no photon energy loss) |
| CMB | Recombination at $z = 1100$ (ΛCDM value) | $\xi$-field equilibrium radiation |
| Dark energy | 68% of universe | Not required |
| Dark matter | 26% of universe | $\xi$-field gravity effects |
| Hubble tension | Unsolved ($4.4\sigma$) | Interpreted via $\xi$-effects |
| JWST paradox | Unexplained early galaxies | No problem in eternal universe |

**Table W.7:** Differences between ΛCDM and T0 (T0 column speculative, cf. P39)

### Hierarchically Ordered Cosmological Parameters

### Critical Differences and Test Possibilities

### Philosophical Implications

In the T0 picture, the following would follow **[S]**:

1. **Eternal universe**: No beginning, no end – the question of a first beginning does not arise
2. **No singularities**: A minimal length scale $L_0$ (set by $\xi$) replaces the formal Big Bang singularity by a tiny but finite core; the justification via Heisenberg uncertainty has been replaced by $T \cdot E = 1$ (Doc. 306)
3. **Energy conservation**: Strictly preserved, no violation through expansion
4. **Simplicity**: One constant instead of 6+ parameters (the cosmological sector still requires its own scale, P39)
5. **Testability**: Clear, measurable predictions

## W.12 Appendix: Purely Theoretical Derivation of Higgs VEV from Quantum Numbers

### Fundamental theoretical foundations

#### Quantum numbers of leptons in T0 theory

T0 theory assigns quantum numbers $(n, l, j)$ to each particle, arising from the solution of the three-dimensional wave equation in the energy field:

**Electron (1st generation):**

- Principal quantum number: $n = 1$
- Orbital angular momentum: $l = 0$ (s-like, spherically symmetric)

- Total angular momentum: $j = 1/2$ (fermion)

**Muon (2nd generation):**

- Principal quantum number: $n = 2$
- Orbital angular momentum: $l = 1$ (p-like, dipole structure)
- Total angular momentum: $j = 1/2$ (fermion)

### Universal mass formulas

T0 theory uses two formulations for particle masses; as the mass ratios below show, they are not equivalent:

**Direct method:**

$$m_i = \frac{1}{\xi_i} = \frac{1}{\xi_0 \times f(n_i, l_i, j_i)} \tag{W.39}$$

**Extended Yukawa method:**

$$m_i = y_i \times v \tag{W.40}$$

where:

- $\xi_0 = \frac{4}{3} \times 10^{-4}$: Universal geometric parameter
- $f(n_i, l_i, j_i)$: Geometric factors from quantum numbers
- $y_i$: Yukawa couplings
- $v$: Higgs VEV (target quantity)

## Theoretical calculation of geometric factors

### Geometric factors from quantum numbers

The geometric factors result from the analytical solution of the three-dimensional wave equation. For the fundamental leptons:

**Electron** $(n = 1, l = 0, j = 1/2)$**:**

The ground state solution of the 3D wave equation yields the simplest geometric factor:

$$f_e(1, 0, 1/2) = 1 \tag{W.41}$$

This is the reference configuration (ground state).

**Muon** $(n = 2, l = 1, j = 1/2)$**:**

For the first excited configuration with dipole character, the solution yields:

$$f_\mu(2, 1, 1/2) = \frac{16}{5} \tag{W.42}$$

This factor accounts for:

- $n^2 = 4$ (energy level scaling)
- $\frac{4}{5}$ ($l = 1$ dipole correction vs. $l = 0$ spherical)

### Verification of factors

The geometric factors must be consistent with the universal T0 structure:

$$\xi_e = \xi_0 \times f_e = \frac{4}{3} \times 10^{-4} \times 1 = \frac{4}{3} \times 10^{-4} \tag{W.43}$$

$$\xi_\mu = \xi_0 \times f_\mu = \frac{4}{3} \times 10^{-4} \times \frac{16}{5} = \frac{64}{15} \times 10^{-4} \tag{W.44}$$

## Derivation of mass ratios

### Theoretical electron-muon mass ratio

With the geometric factors, it follows from the direct method:

$$\frac{m_\mu}{m_e} = \frac{\xi_e}{\xi_\mu} = \frac{f_e}{f_\mu} = \frac{1}{\frac{16}{5}} = \frac{5}{16} \tag{W.45}$$

**Note:** This is the inverse ratio. Since $\xi \propto 1/m$, we obtain:

$$\frac{m_\mu}{m_e} = \frac{f_\mu}{f_e} = \frac{\frac{16}{5}}{1} = \frac{16}{5} = 3.2 \tag{W.46}$$

The direct method thus gives $m_\mu/m_e = 16/5 = 3.2$, the Yukawa method below gives $207.8$; the two formulations yield different ratios and are not equivalent. In addition, the muon is assigned $l = 1$ here but $l = 0$ in the section “Lepton Masses from Quantum Numbers”.

### Correction through Yukawa couplings

The Yukawa method accounts for additional quantum field theoretical corrections:

**Electron:**

$$y_e = \frac{4}{3} \times \xi^{3/2} = \frac{4}{3} \times \left(\frac{4}{3} \times 10^{-4}\right)^{3/2} \tag{W.47}$$

**Muon:**

$$y_\mu = \frac{16}{5} \times \xi^1 = \frac{16}{5} \times \frac{4}{3} \times 10^{-4} \tag{W.48}$$

### Calculation of corrected ratio

$$\frac{y_\mu}{y_e} = \frac{\frac{16}{5} \times \frac{4}{3} \times 10^{-4}}{\frac{4}{3} \times \left(\frac{4}{3} \times 10^{-4}\right)^{3/2}} \tag{W.49}$$

$$= \frac{\frac{16}{5} \times \frac{4}{3} \times 10^{-4}}{\frac{4}{3} \times \frac{4}{3} \times 10^{-4} \times \sqrt{\frac{4}{3} \times 10^{-4}}} \tag{W.50}$$

$$= \frac{\frac{16}{5}}{\frac{4}{3} \times \sqrt{\frac{4}{3} \times 10^{-4}}} \tag{W.51}$$

$$= \frac{\frac{16}{5}}{\frac{4}{3} \times 0.01155} \tag{W.52}$$

$$= \frac{3.2}{0.0154} = 207.8 \tag{W.53}$$

This theoretical ratio of $207.8$ is very close to the experimental value of $206.768$.

### Higgs VEV

In the corpus the Higgs VEV follows from the structural formula

$$\frac{v}{E_P} = \frac{\xi^4}{5\pi}, \qquad v = 245.65\ \text{GeV} \quad (-0.23\,\%\ \text{against}\ 246.22\ \text{GeV}), \tag{W.54}$$

dimensionless; the factor $10$ in $5\pi = (\pi/2)\cdot 10$ is fitted to the measured value, and with the bare $v$ of the ladder it becomes $10\,K_{\text{frak}} \approx \pi^2$ (Docs. 149, 387). The mass ratio $m_\mu/m_e = 207.8$ ($0.5\,\%$) from the previous section does not contain $v$ and is unaffected.

## W.13 List of Symbols Used

### Fundamental Constants

| Symbol | Meaning | Value/Unit |
|---|---|---|
| $\xi$ | Geometric parameter | $\frac{4}{3} \times 10^{-4}$ (dimensionless) |
| $c$ | Speed of light | $2.998 \times 10^8$ m/s |
| $\hbar$ | Reduced Planck constant | $1.055 \times 10^{-34}$ J·s |
| $G$ | Gravitational constant | $6.674 \times 10^{-11}$ m³/(kg·s²) |
| $k_B$ | Boltzmann constant | $1.381 \times 10^{-23}$ J/K |
| $e$ | Elementary charge | $1.602 \times 10^{-19}$ C |

## Coupling Constants

| Symbol | Meaning | Formula |
|---|---|---|
| $\alpha$ | Fine structure constant | $1/137.036$ (SI) |
| $\alpha_{EM}$ | Electromagnetic coupling | $1$ (nat. units) |
| $\alpha_S$ | Strong coupling | $\xi^{-1/3}$ |
| $\alpha_W$ | Weak coupling | $\xi^{1/2}$ |
| $\alpha_G$ | Gravitational coupling | $\xi^2$ |
| $\varepsilon_T$ | T0 coupling parameter | $\xi \cdot E_0^2$ |

## Energy Scales and Masses

| Symbol | Meaning | Value/Formula |
|---|---|---|
| $E_P$ | Planck energy | $1.22 \times 10^{19}$ GeV |
| $E_\xi$ | Characteristic energy | $1/\xi = 7500$ (nat. units) |
| $E_0$ | Fundamental EM energy | $7.398$ MeV |
| $v$ | Higgs VEV | $246.22$ GeV |
| $m_h$ | Higgs mass | $125.25$ GeV |
| $\Lambda_{QCD}$ | QCD scale | $\sim 200$ MeV |
| $m_e$ | Electron mass | $0.511$ MeV |
| $m_\mu$ | Muon mass | $105.66$ MeV |
| $m_\tau$ | Tau mass | $1776.86$ MeV |
| $m_u, m_d$ | Up, down quark masses | $2.16$, $4.67$ MeV |
| $m_c, m_s$ | Charm, strange quark masses | $1.27$ GeV, $93.4$ MeV |
| $m_t, m_b$ | Top, bottom quark masses | $172.76$ GeV, $4.18$ GeV |
| $m_{\nu_e}, m_{\nu_\mu}, m_{\nu_\tau}$ | Neutrino masses | $< 2$ eV, $< 0.19$ MeV, $< 18.2$ MeV |

## Cosmological Parameters

| Symbol | Meaning | Value/Formula |
|---|---|---|
| $T_{CMB}$ | CMB temperature | $2.725$ K |
| $z$ | Redshift | dimensionless |
| $\Omega_\Lambda$ | Dark energy density | $0.6847$ ($\Lambda$CDM), $0$ (T0) |
| $\Omega_{DM}$ | Dark matter density | $0.2607$ ($\Lambda$CDM), $0$ (T0) |
| $\Omega_b$ | Baryon density | $0.0492$ ($\Lambda$CDM), $1$ (T0) |
| $\Lambda$ | Cosmological constant | $(1.1 \pm 0.02) \times 10^{-52}$ m$^{-2}$ |
| $\rho_\xi$ | $\xi$-field energy density | $E_\xi^4$ |
| $\rho_{CMB}$ | CMB energy density | $4.64 \times 10^{-31}$ kg/m³ |

## Geometric and Derived Quantities

| Symbol | Meaning | Value/Formula |
|---|---|---|
| $D_f^{\text{eff}}$ | Effective fractal dimension (cumulative over RG running) | $2.973$ |
| $\kappa_{mass}$ | Mass scaling exponent | $D_f^{\text{eff}}/2 = 1.487$ |
| $\kappa_{grav}$ | Gravitational field parameter | $4.8 \times 10^{-11}$ m/s² |
| $\lambda_h$ | Higgs self-coupling | $0.13$ |
| $\theta_W$ | Weinberg angle | $\sin^2\theta_W = 0.2312$ |
| $\theta_{QCD}$ | Strong CP phase | $< 10^{-10}$ (exp.), $\xi^2$ (T0, excluded) |
| $\ell_P$ | Planck length | $1.616 \times 10^{-35}$ m |
| $\lambda_C$ | Compton wavelength | $\hbar/(mc)$ |
| $r_g$ | Gravitational radius | $2Gm$ |
| $L_\xi$ | Characteristic length | $\xi$ (nat. units) |

## Mixing Matrices

| Symbol | Meaning | Typical Value |
|---|---|---|
| $V_{ij}$ | CKM matrix elements | see table |
| $\lvert V_{ud}\rvert$ | CKM ud element | $0.97446$ |
| $\lvert V_{us}\rvert$ | CKM us element (Cabibbo) | $0.22452$ |
| $\lvert V_{ub}\rvert$ | CKM ub element | $0.00365$ |
| $\delta_{CKM}$ | CKM CP phase | $1.20$ rad |
| $\theta_{12}$ | PMNS solar angle | $33.44^\circ$ |
| $\theta_{23}$ | PMNS atmospheric | $49.2^\circ$ |
| $\theta_{13}$ | PMNS reactor angle | $8.57^\circ$ |
| $\delta_{CP}$ | PMNS CP phase | unknown |

## Other Symbols

| Symbol | Meaning | Context |
|---|---|---|
| $n, l, j$ | Quantum numbers | Particle classification |
| $r_i$ | Rational coefficients | Yukawa couplings |
| $p_i$ | Generation exponents | $3/2, 1, 2/3, ...$ |
| $f(n, l, j)$ | Geometric function | Mass formula |
| $\rho_{tet}$ | Tetrahedral packing density | $0.68$ |
| $\gamma$ | Universal exponent | $1.01$ |
| $\nu$ | Crystal symmetry factor | $0.63$ |
| $\beta_T$ | Time field coupling | $1$ (nat. units) |
| $y_i$ | Yukawa couplings | $r_i \cdot \xi^{p_i}$ |
| $T(x,t)$ | Time field | T0 theory |
| $E_{field}$ | Energy field | Universal field |

## Classification of the Early Coupling-Constant Formulas

The early coupling-constant formulas developed in this document are to be classified as preliminary work, not as canonical FFGFT derivations. Authoritative documents are Doc. 160 (lepton lifetime structure), Doc. 318 (validity range of mass derivations), and the Galois series (Docs. 336–348). Numerical use of the formulas in this document requires recourse to these successor documents.

# Chapter X

# Parameter System Dependence in the T0 Model:

## Abstract

This paper systematically analyzes the parameter dependency between SI units and T0-model natural units, revealing that fundamental parameters like $\alpha_{EM}$ and Yukawa couplings have dramatically different numerical values in different unit systems. Through detailed calculations, we demonstrate that direct transfer of parameter values between systems leads to errors spanning multiple orders of magnitude. The analysis extends beyond specific parameters to establish universal transformation rules and provides critical warnings against naive parameter transfer. This work establishes that the apparent inconsistencies in T0-model parameters are actually systematic unit-system dependencies that require careful transformation protocols for experimental verification.

## X.1 Introduction

### The Parameter Transfer Problem

The T0 model, formulated in natural units where $\hbar = c = G = k_B = \alpha_{EM} = \beta_T = 1$, presents a fundamental challenge when compared with experimental data expressed in SI units. This paper demonstrates that the apparent inconsistencies between T0-model predictions and experimental observations are not physical contradictions but systematic unit-system dependencies.

The core insight is that parameters such as $\alpha_{EM}$ represent fundamentally different quantities when expressed in different unit systems:

$$\alpha_{EM,SI} \neq \alpha_{EM,nat}$$

The parameter $\xi$, by contrast, is a ratio of lengths and has the same value in both systems (Section X.3).

### Scope and Methodology

This analysis covers:
- Systematic calculation of parameter ratios between SI and T0-natural units
- Demonstration of transformation invariance for dimensionless ratios
- Extension to variable parameters like $\xi$ and Yukawa couplings
- Universal warnings against direct parameter transfer
- Guidelines for correct experimental comparison protocols

## X.2 The $\xi$ Parameter: Variable Across Mass Scales

## X.3 The Universal $\xi$-Field Framework

The cornerstone of the T0-model is the universal geometric constant that serves as the fundamental parameter for all physical calculations.

The universal geometric constant:

$$\xi = \frac{4}{3} \times 10^{-4} = 1.3333... \times 10^{-4} \tag{X.1}$$

This dimensionless constant is used throughout T0 theory to connect quantum mechanical and gravitational phenomena. It establishes the characteristic strength of field interactions and provides the foundation for unified field descriptions.

For the detailed derivation and physical justification of this parameter, see the document "Parameter Derivation" (available at: ).

This geometric constant determines a characteristic energy scale for the $\xi$-field:

$$E_\xi = \frac{1}{\xi} = \frac{3}{4 \times 10^{-4}} = 7500 \text{ (natural units)} \tag{X.2}$$

### Definition and Physical Meaning

The parameter $\xi$ is also the ratio of the Schwarzschild radius to the Planck length:

$$\xi = \frac{r_0}{\ell_P} = \frac{2Gm}{c^2 \ell_P} \tag{X.3}$$

**Crucial:** The parameter $\xi$ scales with the mass of the object under consideration according to $\xi(m) = 2Gm/(c^2\ell_P)$. The Higgs mass defines the fundamental reference scale $\xi_0 = 1.33 \times 10^{-4}$, to which all other masses are normalized in the T0 model.

### Connection to Higgs Physics

The T0 model connects $\xi$ with Higgs sector physics through the following relation; it is a consistency check, not an exact derivation (Doc. 385).

$$\xi \approx \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2} \approx 1.30 \times 10^{-4} \tag{X.4}$$

where:

- $\lambda_h \approx 0.13$ (Higgs self-coupling)
- $v \approx 246$ GeV (Higgs VEV)
- $m_h \approx 125$ GeV (Higgs mass)

The Higgs expression lies about $2.3\,\%$ below the setting $\xi = 4/30000 = 1.33 \times 10^{-4}$ (Doc. 385). It supports the universal scale parameter from Standard Model physics, while the mass-dependent form $\xi = 2Gm/(c^2\ell_P)$ applies to specific objects.

## $\xi$ Values in the SI System

Using SI constants:

$$G = 6.674 \times 10^{-11}\ \mathrm{m^3/(kg \cdot s^2)} \tag{X.5}$$

$$\ell_P = 1.616 \times 10^{-35}\ \mathrm{m} \tag{X.6}$$

We calculate $\xi_{SI}$ for various objects:

| Object | Mass | $\xi_{SI}$ |
|---|---|---|
| Electron | $9.109 \times 10^{-31}$ kg | $8.37 \times 10^{-23}$ |
| Proton | $1.673 \times 10^{-27}$ kg | $1.54 \times 10^{-19}$ |
| Human (70 kg) | $7.0 \times 10^{1}$ kg | $6.4 \times 10^{9}$ |
| Earth | $5.972 \times 10^{24}$ kg | $5.5 \times 10^{32}$ |
| Sun | $1.989 \times 10^{30}$ kg | $1.8 \times 10^{38}$ |
| Planck mass | $2.176 \times 10^{-8}$ kg | 2.0 |

**Table X.1:** $\xi$ values for different objects in SI units

**The parameter $\xi$ varies over about 60 orders of magnitude!**

## $\xi$ in T0-Natural Units

With $\xi = r_0/\ell_P = 2Gm/(c^2\ell_P)$, $\xi$ is a pure ratio of lengths. In natural units $G = 1/E_P^2$ and $\ell_P = 1/E_P$, so again

$$\xi = \frac{2m}{E_P} = \frac{2m}{m_P}. \tag{X.7}$$

The numerical value is therefore the same in SI and in T0-natural units; no conversion factor between the systems arises for this quantity. The values in Table X.1 hold unchanged in both systems.

### Invariance of Ratios

**Critical verification:** Since the values themselves are already the same in both systems, this holds all the more for the ratios between different objects:

$$\frac{\xi_{\text{Sun}}}{\xi_{\text{e}}} = \frac{1.83 \times 10^{38}}{8.37 \times 10^{-23}} = \frac{m_{\text{Sun}}}{m_e} = 2.18 \times 10^{60} \tag{X.8}$$

Ratios are invariant under system transformation!

## X.4 The Fine-Structure Constant $\alpha_{\text{EM}}$

### The Mystification of 1/137

The fine-structure constant $\alpha_{\text{EM}} \approx 1/137$ has been declared one of the greatest mysteries of physics by prominent physicists:

- **Richard Feynman**: "It is one of the greatest damn mysteries of physics: a magic number that comes to us with no understanding whatsoever."
- **Wolfgang Pauli**: "When I die, I will ask God two questions: Why relativity? And why 137? I believe he will have an answer for the first one."
- **Max Born**: "If $\alpha$ were larger, molecules could not exist, and there would be no life."

### Electromagnetic Duality as the Key

What all these statements overlook: The fine-structure constant possesses two mathematically equivalent representations that reveal its true nature:

$$\alpha_{\text{EM}} = \frac{e^2}{4\pi\varepsilon_0\hbar c} \quad \text{(Standard form)} \tag{X.9}$$

$$\alpha_{\text{EM}} = \frac{e^2\mu_0 c}{4\pi\hbar} \quad \text{(Dual form)} \tag{X.10}$$

This equivalence is based on the Maxwell relation $c^2 = \frac{1}{\varepsilon_0\mu_0}$ and reveals a fundamental electromagnetic duality:

$$\frac{1}{\varepsilon_0 c} = \mu_0 c \tag{X.11}$$

### The Dual Nature of $\alpha$: System-Dependent yet Invariant

The fine-structure constant possesses a remarkable dual nature:

### As an Invariant Ratio of Physical Quantities

Regardless of the chosen system of units, $\alpha$ remains constant as a **ratio** of fundamental lengths:

$$\alpha_{\text{EM}} = \frac{r_e}{\lambda_C/2\pi} = \frac{\text{Classical electron radius}}{\text{Reduced Compton wavelength}} \tag{X.12}$$

Similarly, the inverse ratio:

$$\alpha_{\text{EM}}^{-1} = \frac{a_0}{\lambda_C/2\pi} = \frac{\text{Bohr radius}}{\text{Reduced Compton wavelength}} = 137.036... \tag{X.13}$$

These ratios are **system-of-units invariant** – they have the same numerical value in any consistent system of units, as the units cancel out in the ratio.

### As a System-Dependent Numerical Value

Simultaneously, the numerical value of $\alpha$ depends on the choice of fundamental units:

- **SI system**: $\alpha = \frac{e^2}{4\pi\varepsilon_0\hbar c} \approx 1/137$
- **Natural units**: $\alpha = 1$ (by suitable choice)
- **Gaussian units**: $\alpha = \frac{e^2}{\hbar c} \approx 1/137$

## The System Dependency of $\alpha$

The numerical value $\alpha_{\text{EM}} = 1/137$ holds **in all systems that do not redefine $\alpha$ by convention** (SI, Gaussian, Heaviside-Lorentz); only the T0 setting turns it into $1$:

$$\text{SI system:} \quad \alpha_{\text{EM}}^{\text{SI}} = \frac{e^2}{4\pi\varepsilon_0\hbar c} \approx \frac{1}{137.036} \tag{X.14}$$

$$\text{Natural system of units:} \quad \alpha_{\text{EM}}^{\text{nat}} = 1 \text{ (by suitable choice of units)} \tag{X.15}$$

**Transformation factor:**

$$\frac{\alpha_{\text{EM}}^{\text{nat}}}{\alpha_{\text{EM}}^{\text{SI}}} = 137.036 \tag{X.16}$$

## The Natural System of Units with $\alpha = 1$

In a natural system of units that respects electromagnetic duality, we obtain:

- $\hbar_{\text{nat}} = 1$ (quantum mechanical scale)
- $c_{\text{nat}} = 1$ (relativistic scale)
- $\varepsilon_{0,\text{nat}} = 1$ (electric constant)
- $\mu_{0,\text{nat}} = 1$ (magnetic constant)
- $e^2_{\text{nat}} = 4\pi$ (elementary charge)

With these values, $\alpha = 1$ is verified in both the standard form and the dual form:

$$\alpha = \frac{4\pi}{4\pi \cdot 1 \cdot 1 \cdot 1} = 1 \tag{X.17}$$

## The Resolution of the "Mystery"

The apparent mystification of $1/137$ arises from:

1. **Confusion of two aspects**: The invariance of the ratios is conflated with the system-dependency of the numerical representation.
2. **Treatment of the SI system as absolute**: The historically evolved SI units (meter, second, kilogram, ampere) force electromagnetic constants to take "unnatural" values.
3. **Forgetting the construction of unit systems**: All unit systems are human constructs. Nature knows no preferred units.
4. **Search for deeper meaning in conversion factors**: The number 137 has no deeper cosmic significance than, say, the factor 1609.344 between miles and meters.

## The Anthropic Fallacy

Typical anthropic arguments claim:

- "If $\alpha_{\text{EM}} = 1/200 \rightarrow$ no atoms $\rightarrow$ no life"
- "If $\alpha_{\text{EM}} = 1/80 \rightarrow$ no stars $\rightarrow$ no life"
- "Therefore, $\alpha_{\text{EM}} = 1/137$ is 'fine-tuned' for life"

**The problem**: These arguments presuppose the SI system as absolute!

**In natural units**: $\alpha_{\text{EM}} = 1$ is perfectly natural and requires no fine-tuning whatsoever. The electromagnetic interaction has unit strength in the natural system of units, which respects the fundamental structure of quantum mechanics and relativity.

## Sommerfeld's Harmonic Imprinting

An often overlooked historical aspect: In 1916, Arnold Sommerfeld actively searched for **harmonic ratios** in atomic spectra, guided by the philosophical conviction that nature follows musical principles.

His methodological approach:

1. **Expectation** of musical ratios in quantum transitions
2. **Calibration** of measurement systems to produce harmonic values
3. **Definition** of $\alpha_{\text{EM}}$ based on harmonic spectroscopic adjustments
4. **Attribution** of the resulting ratio to fundamental physics

The apparent "harmony" in $\alpha_{\text{EM}}^{-1} = 137 \approx (6/5)^{27}$ is therefore not a cosmic discovery, but the result of Sommerfeld's harmonic expectations embedded into the definition of the unit system.

### Physical Interpretation

In natural units, $\alpha = 1$ represents the perfect balance between:

- **Electric field coupling** (via $\varepsilon_0$ with $c^{-1}$)
- **Magnetic field coupling** (via $\mu_0$ with $c^{+1}$)
- **Quantum mechanical scale** (via $\hbar$)
- **Relativistic scale** (via $c$)

The electromagnetic duality $\frac{1}{\varepsilon_0 c} = \mu_0 c$ ensures this perfect balance.

### Historical Warning: The Eddington Saga

Arthur Eddington (1882-1944) attempted to "prove" $\alpha_{\text{EM}} = 1/137$ from first principles and developed elaborate numerological theories. The result was entirely speculative and wrong – a warning against mystifying system-dependent numbers.

However, modern analysis shows that the fine-structure constant is indeed derivable from fundamental electromagnetic vacuum constants and that $\alpha_{\text{EM}} = 1$ in natural units is not only possible but reveals the arbitrary nature of our choice of unit system.

## X.5 The $\beta_{\text{T}}$ Parameter

### Theoretical Foundation from Field Theory

In T0-natural units $\beta_{T,\text{nat}} = 1$. The T0 model connects this value with the field-theoretic relation [1]:

$$\beta_{\text{T}} = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2 \xi} \approx 1 \tag{X.18}$$

Since the Higgs expression matches $\xi$ only as a consistency check (deviation of about $2.3\,\%$, Doc. 385), this relation supports $\beta_{\text{T}} = 1$ in natural units but does not fix the value exactly.

### No Measured SI Value

A determination of $\beta_{T,\text{SI}}$ via a wavelength-dependent redshift $z(\lambda) = z_0\,(1 + \beta_{\text{T}} \ln(\lambda/\lambda_0))$ does not hold, because the redshift is achromatic in the T0 framework: the fractal path lengthening stretches all wavelengths by the same factor $1 + z$, independent of $\lambda$; the scale of this factor is calibrated via the measured $H_0$ (P39). An observationally determined SI value of $\beta_{\text{T}}$ and a conversion factor between the systems therefore do not arise; unlike $\alpha_{\text{EM}}$, $\beta_{\text{T}}$ is no example of a measured, system-dependent numerical value.

## X.6 The Wien Constant $\alpha_W$

### Mathematical vs. Conventional Values

Wien's displacement law gives:

$$\text{SI system:} \quad \alpha_W^{\text{SI}} = 2.8977719... \times 10^{-3} \text{ m K} \tag{X.19}$$

$$\text{T0 system:} \quad \alpha_W^{\text{nat}} = 2\pi/x_W = 1.2655 \quad (x_W = 4.965114) \tag{X.20}$$

**Conversion:**

$$\alpha_W^{\text{SI}} = \alpha_W^{\text{nat}} \cdot \frac{\hbar c}{k_B} = 1.2655 \times 2.2899 \times 10^{-3} \text{ m K}$$

In $\hbar = c = k_B = 1$ Wien's constant is fixed as $2\pi/x_W = 1.2655$ and cannot additionally be set to 1. There is therefore no dimensionless conversion factor between the systems for $\alpha_W$.

## X.7 Parameter Comparison Table

| Parameter | SI Value | T0-nat Value | Ratio | Factor |
|---|---|---|---|---|
| $\xi$ (electron) | $8.4 \times 10^{-23}$ | $8.4 \times 10^{-23}$ | 1 | $10^0$ |
| $\alpha_{EM}$ | $7.3 \times 10^{-3}$ | 1 | 137 | $10^{2.1}$ |
| $\beta_T$ | – (no measured value) | 1 | – | – |
| $\alpha_W$ | $2.898 \times 10^{-3}$ m K | 1.2655 | $\hbar c/k_B$ | – |

**Table X.2:** Systematic parameter differences between unit systems

**$\alpha_{EM}$ shows about 2 orders of magnitude difference between systems; $\beta_T$ has no measured SI value (achromatic redshift); for $\alpha_W$ there is no dimensionless conversion factor (conversion via $\hbar c/k_B$); $\xi$ is the same in both.**

## X.8 Yukawa Parameters: Variable and System-Dependent

### The Hierarchy of Yukawa Couplings

In the Standard Model, Yukawa couplings vary dramatically:

### Transformation Uncertainty

The transformation of Yukawa parameters between systems requires careful consideration of the Higgs mechanism. The general form would be:

$$y_{i,\text{nat}} = y_{i,\text{SI}} \times T_{\text{Yukawa}}$$

where $T_{\text{Yukawa}}$ depends on the transformation of Higgs vacuum expectation value and particle masses.

| Particle | $y_i$ (SI system) |
|---|---|
| Electron | $2.94 \times 10^{-6}$ |
| Muon | $6.09 \times 10^{-4}$ |
| Tau | $1.03 \times 10^{-2}$ |
| Up quark | $1.27 \times 10^{-5}$ |
| Top quark | $1.00$ |
| Bottom quark | $2.25 \times 10^{-2}$ |

**Table X.3:** Yukawa coupling hierarchy (5 orders of magnitude variation)

### Consistency Requirements

The Higgs mechanism requires:

$$m_h^2 = 2\lambda_h v^2$$

For transformation consistency:

$$T_m^2 = T_\lambda \times T_v^2$$

This gives:

$$y_{i,\text{nat}} = y_{i,\text{SI}} \times \sqrt{T_\lambda}$$

**However, $T_\lambda$ requires detailed specification of the T0-natural unit system transformation rules.**

## X.9 Universal Warning: No Direct Parameter Transfer

### The Systematic Problem

**Warning**

**EVERY parameter symbol in T0-model documents may have different values than in SI system calculations!**

**Concrete danger zones:**

$$G_{\text{nat}} = 1 \quad \text{vs.} \quad G_{\text{SI}} = 6.674 \times 10^{-11}\ \text{m}^3/(\text{kg} \cdot \text{s}^2) \tag{X.21}$$

$$\alpha_{\text{EM,nat}} = 1 \quad \text{vs.} \quad \alpha_{\text{EM,SI}} = 1/137 \tag{X.22}$$

$$e_{\text{nat}} = 2\sqrt{\pi} \quad \text{vs.} \quad e_{\text{SI}} = 1.602 \times 10^{-19}\ \text{C} \tag{X.23}$$

**Direct transfer leads to errors of factors $10^2$ to $10^{11}$!**

### Required Transformation Protocol

For every parameter, explicitly specify:

1. **Which unit system** is being used
2. **How transformation occurs** between systems
3. **Which factors must be considered**
4. **Which consistency conditions** must be satisfied

**Example of complete specification:**

**Parameter Specification Template**

**Parameter:** Fine structure constant $\alpha_{EM}$
**SI value:** $\alpha_{EM,SI} = 1/137.036$
**T0 value:** $\alpha_{EM,nat} = 1$
**Transformation:** $\alpha_{EM,nat} = \alpha_{EM,SI} \times 137.036$
**Consistency:** Dimensional analysis verified
**Usage:** Specify system before calculation

### Experimental Prediction Guidelines

**For QED calculations:**

$$\text{WRONG:} \quad \alpha_{EM} = 1 \text{ from T0-model directly in SI formulas} \tag{X.24}$$

$$\text{CORRECT:} \quad \alpha_{EM,SI} = 1/137 \text{ with transformation to } \alpha_{EM,nat} = 1 \tag{X.25}$$

**For gravitational calculations:**

$$\text{WRONG:} \quad G = 1 \text{ from T0-model directly in Newton's formulas} \tag{X.26}$$

$$\text{CORRECT:} \quad G_{SI} = 6.674 \times 10^{-11} \text{ with transformation to } G_{nat} = 1 \tag{X.27}$$

## X.10 The Circularity Resolution

### Apparent vs. Real Circularity

The circularity problem that seemed to plague T0-model parameter determination is resolved by recognizing:

1. **No real circularity exists** within each consistent system
2. **Both SI and T0 systems are internally consistent**
3. **The apparent contradiction** arose from comparing parameters across different systems
4. **Proper transformation** eliminates all apparent inconsistencies

### System Consistency Verification

**SI system consistency:**

$$R_0 = \frac{m_e c \left(\alpha_{\text{EM,SI}}\right)^2}{2h} \quad \checkmark \text{ (experimentally verified to 0.000001\%)}$$

**T0 system consistency:**

$$\hbar = c = G = k_B = \alpha_{\text{EM}} = \beta_{\text{T}} = 1 \quad \checkmark \text{ (by construction)}$$

**Both systems work perfectly within their own frameworks!**

## X.11 Implications for T0-Model Testing

### System-Specific Predictions

Experimental tests must clearly specify which parameter system is used:

| Test Type | SI-based Prediction | T0-based Prediction |
|---|---|---|
| QED anomaly | $a_e \propto \alpha_{\text{EM,SI}} = 1/137$ | $a_e \propto \alpha_{\text{EM,nat}} = 1$ |
| Galaxy rotation | $v^2 \propto \xi \sim 10^{38}$ | $v^2 \propto \xi \sim 10^{38}$ (same) |

**Table X.4:** System-specific experimental predictions

### Transformation Validation

The transformation factors can be validated by checking:

1. **Dimensional consistency** in both systems
2. **Known limits** are reproduced correctly
3. **Ratios remain invariant** between systems
4. **Internal consistency** of each system

# Bibliography

[1] J. Pascher. Derivation of Beta in the T0 Framework. T0-Theory Document Collection, 2025. URL: https://github.com/jpascher/T0-Time-Mass-Duality/blob/main/2/pdf/093_DerivationVonBeta_En.pdf (verifizierter Link existiert).

# Chapter Y

# The Complete Closure of T0 Theory


## Abstract

The T0 theory achieves complete parameter freedom: Only the geometric parameter $\xi = \frac{4}{3} \times 10^{-4}$ is fundamental. All physical constants are derived either from $\xi$ or represent unit definitions. For absolute SI values the derivation chain needs one mass or energy anchor; the speed of light is fixed by the definition of the metre and is not determined from $\xi$. This document provides the complete derivation chain including the gravitational constant $G$, the Planck length $l_P$, and the Boltzmann constant $k_B$. The SI reform 2019 unwittingly implemented the unique calibration consistent with this geometric foundation.


## Y.1 The Geometric Foundation

### Single Fundamental Parameter

$$\boxed{\xi = \frac{4}{3} \times 10^{-4}} \tag{Y.1}$$

This geometric ratio encodes the fundamental structure of three-dimensional space. All physical quantities emerge as derivable consequences. (See [1] for the origin of $\xi$.)

### Complete Derivation Framework

Detailed mathematical derivations are available at:

## Y.2 Derivation of the Gravitational Constant from $\xi$

### The Fundamental T0 Gravitational Relationship

**Starting point of T0 gravity theory:**
The T0 theory postulates a fundamental geometric relationship between the characteristic length parameter $\xi$ and the gravitational constant:

$$\xi = 2\sqrt{G \cdot m_{\text{char}}} \tag{Y.2}$$

where $m_{\text{char}}$ represents a characteristic mass of the theory. (Detailed derivation in [2].)

**Physical interpretation:**

- $\xi$ encodes the geometric structure of space
- $G$ describes the coupling between geometry and matter
- $m_{\text{char}}$ sets the characteristic mass scale

## Solution for the Gravitational Constant

Solving equation (ak.1) for $G$:

$$\boxed{G = \frac{\xi^2}{4m_{\text{char}}}} \tag{Y.3}$$

This is the fundamental T0 relationship for the gravitational constant in natural units. (Further details in [3].)

## Choice of Characteristic Mass

**Insight Y.2.1. The electron mass follows from the $\xi$ mass ladder:**
The T0 theory uses the electron mass as the characteristic scale:

$$m_{\text{char}} = m_e = 0{,}511 \text{ MeV} \tag{Y.4}$$

**Critical point:** The electron mass itself is not an independent parameter but follows from the T0 mass ladder (Doc. 006):

$$m_e = \frac{4}{3}\xi^{3/2} v \tag{Y.5}$$

The ladder needs $v$ as a scale; the chain $\xi \to m_e$ therefore does not work without a mass or energy anchor. (See [4] for mass derivations.)

Therefore, the derivation chain $\xi \to m_e \to G \to l_P$ depends on $\xi$ as the single fundamental parameter and on one anchor for the scale.

## Dimensional Analysis in Natural Units

**Dimensional check in natural units ($\hbar = c = 1$):**

In natural units:

$$[M] = [E] \quad \text{(from } E = mc^2 \text{ with } c = 1\text{)} \tag{Y.6}$$
$$[L] = [E^{-1}] \quad \text{(from } \lambda = \hbar/p \text{ with } \hbar = 1\text{)} \tag{Y.7}$$
$$[T] = [E^{-1}] \quad \text{(from } \omega = E/\hbar \text{ with } \hbar = 1\text{)} \tag{Y.8}$$

The gravitational constant has dimension:

$$[G] = [M^{-1}L^3T^{-2}] = [E^{-1}][E^{-3}][E^2] = [E^{-2}] \tag{Y.9}$$

Checking equation (Y.3):

$$[G] = \frac{[\xi^2]}{[m_e]} = \frac{[1]}{[E]} = [E^{-1}] \neq [E^{-2}] \tag{Y.10}$$

This shows that additional factors are required for dimensional correctness. (See [5] for unit systematics.)

## Complete Formula with Conversion Factors

**Key Result**

**Complete gravitational constant formula:**

$$\boxed{G_{\text{SI}} = \frac{\xi_0^2}{4m_e} \times C_{\text{conv}} \times K_{\text{frak}}} \tag{Y.11}$$

where:

- $\xi_0 = 1{,}333 \times 10^{-4}$ (geometric parameter)
- $m_e = 0{,}511$ MeV (electron mass, anchor of the chain)
- $C_{\text{conv}} = 7{,}783 \times 10^{-3}$ (SI conversion of the single-anchor chain, Doc. 383)
- $K_{\text{frak}} = 0{,}986$ (fractal quantum spacetime correction) (See [6].)

**Result:**

$$G_{\text{SI}} = 6{,}674 \times 10^{-11}\ \text{m}^3/(\text{kg} \cdot \text{s}^2) \tag{Y.12}$$

with $+0.003\%$ deviation from the CODATA-2018 value ($6.67430 \times 10^{-11}$) for $\xi = 4/30000$; with the rounded $\xi_0 = 1.333 \times 10^{-4}$ the deviation is $-0.05\%$.
Here $C_{\text{conv}}$ is the SI conversion of the single-anchor chain (Doc. 383); its numerical value $7.783 \times 10^{-3}$ additionally contains the small residual of this chain (chain value $7.58 \times 10^{-3}$). The form of $G$ and its order of magnitude follow from $\xi$ and the one anchor (numerically checked in Doc. 383); with the anchor $m_e$, $G$ lies at $-2.6\,\%$, with $v$ at $-0.46\,\%$. The agreement to $0.003\,\%$ arises only with this residual and does not count as an independent precision result.

## Y.3 Derivation of the Planck Length from $G$ and $\xi$

### The Planck Length as Fundamental Reference

**Definition of the Planck length:**
In standard physics, the Planck length is defined as:

$$l_P = \sqrt{\frac{\hbar G}{c^3}} \tag{Y.13}$$

In natural units ($\hbar = c = 1$) this simplifies to:

$$\boxed{l_P = \sqrt{G} = 1 \quad \text{(natural units)}} \tag{Y.14}$$

**Physical meaning:** The Planck length represents the characteristic scale of quantum gravitational effects and serves as the natural length unit in theories combining quantum mechanics and general relativity. (See [7] for natural and SI units.)

### T0 Derivation: Planck Length via $G_{\text{SI}}$

**Key Result**

**Complete derivation chain:**
Since $G$ is derived from $\xi$ via equation (Y.3):

$$G = \frac{\xi^2}{4m_e} \tag{Y.15}$$

the Planck length follows via the complete $G_{\text{SI}}$ from Eq. (Y.11), which contains $C_{\text{conv}}$ and $K_{\text{frak}}$:

$$l_P = \sqrt{\frac{\hbar G_{\text{SI}}}{c^3}}, \qquad G_{\text{SI}} = \frac{\xi^2}{4m_e} C_{\text{conv}} K_{\text{frak}} \tag{Y.16}$$

**Result in SI units:**

$$\boxed{l_P = 1{,}616 \times 10^{-35}\ \text{m}} \tag{Y.17}$$

### The Characteristic T0 Length Scale

**Insight Y.3.1. Connection between $r_0$ and the fundamental energy scale $E_0$:**
The characteristic T0 length $r_0$ for an energy $E$ is defined as:

$$r_0(E) = 2GE \tag{Y.18}$$

For the fundamental energy scale $E_0 = \sqrt{m_e \cdot m_\mu}$:

$$r_0(E_0) = \frac{\hbar c}{E_0} \approx 2.7 \times 10^{-14}\ \text{m} \tag{Y.19}$$

The minimal sub-Planck length scale is:

$$\boxed{L_0 = \xi \cdot l_P = \frac{4}{3} \times 10^{-4} \times 1{,}616 \times 10^{-35}\ \text{m} = 2{,}155 \times 10^{-39}\ \text{m}} \tag{Y.20}$$

**Fundamental relationship:** In natural units, for any energy $E$:

$$r_0(E) = \frac{1}{E} \quad \text{(in natural units with } c = \hbar = 1\text{)} \tag{Y.21}$$

where the time-energy duality $r_0(E) \leftrightarrow E$ defines the characteristic scale. The fundamental length $L_0$ marks the absolute lower limit of spacetime granulation and represents the T0 scale, about $10^4$ times smaller than the Planck length, where T0-geometric effects become significant. (See [8] for energy scales.)

## The Decisive Convergence: Why T0 and SI Agree

**Historical**

**Two paths to the same Planck length:**
There are two ways to determine the Planck length:
**Path 1: SI-based (experimental):**

$$l_P^{\text{SI}} = \sqrt{\frac{\hbar G_{\text{measured}}}{c^3}} = 1{,}616 \times 10^{-35}\ \text{m} \tag{Y.22}$$

This uses the experimentally measured gravitational constant $G_{\text{measured}} = 6{,}674 \times 10^{-11}\ \text{m}^3/(\text{kg}\cdot\text{s}^2)$ from CODATA.
**Path 2: T0-based (geometry and one anchor):**

$$m_e = \frac{4}{3}\,\xi^{3/2}\,v \quad \text{(from } \xi \text{ and the anchor } v\text{)} \tag{Y.23}$$

$$G = \frac{\xi^2}{4m_e} \times C_{\text{conv}} \times K_{\text{frak}} \quad \text{(from } \xi \text{ and } m_e\text{)} \tag{Y.24}$$

$$l_P^{\text{T0}} = \sqrt{\frac{\hbar G}{c^3}} \quad \text{(with } G = G_{\text{SI}} \text{ from Eq. (Y.11))} \tag{Y.25}$$

**Conversion to SI units:**

$$l_P^{\text{T0}} = \sqrt{\frac{\hbar}{c^3} \cdot \frac{\xi^2}{4m_e} C_{\text{conv}}\, K_{\text{frak}}} = \sqrt{\frac{\hbar \cdot 6.6745 \times 10^{-11}}{c^3}}\ \text{m} \tag{Y.26}$$

**Result:** $l_P^{\text{T0}} = 1.616 \times 10^{-35}$ m
**The convergence:**

$$\boxed{l_P^{\text{SI}} = l_P^{\text{T0}} \quad \text{with } +0.002\% \text{ deviation}} \tag{Y.27}$$

As for $G$, this precision arises only with the residual in $C_{\text{conv}}$ and does not count as an independent precision result; the deviation is half that of $G$.

**Warning**

**Why this agreement is not coincidental:**
The agreement between the SI-derived and T0-derived Planck length in form and order of magnitude, which follow from $\xi$ and one anchor, reveals a profound truth:

1. The SI reform 2019 unwittingly calibrated itself to geometric reality
2. Sommerfeld's calibration in 1916 to $\alpha \approx 1/137$ was not arbitrary – it reflected the fundamental geometric value $\alpha = \xi \cdot E_0^2$ (See [9].)
3. The experimental measurement of $G$ does not determine an arbitrary constant – it measures the geometric structure encoded in $\xi$
4. **The conversion factor:** The factor $\frac{\hbar c}{1\ \text{MeV}} = 1{,}973 \times 10^{-13}$ m converts lengths from natural units ($r_0 = 1/E$) into metres; the agreement of the Planck length itself follows from the complete $G_{\text{SI}}$ (Eq. (Y.11)), not from $S_{T0}$.
5. Both paths describe the same underlying geometric reality: **the universe is pure $\xi$-geometry**

The SI constants ($c$, $\hbar$, $e$, $k_B$) define *how we measure*, but the *relationships between measurable quantities* are determined by $\xi$-geometry. Therefore, the SI reform 2019, by fixing these unit-defining constants, unwittingly implemented the unique calibration consistent with T0 theory.

## Y.4 The Role of the Conversion Factor

### The Scaling Factor $S_{T0} = 1$ MeV/$c^2$

**Key Result**

**$S_{T0} = 1$ MeV/$c^2$ holds by construction:**
The mass scaling factor is

$$\boxed{S_{T0} = \frac{m_e^{\text{SI}}}{m_e^{\text{T0}}} = 1\ \text{MeV}/c^2} \tag{Y.28}$$

with $m_e^{\text{T0}} = 0.511$. This value arises by construction, because $0.511$ is the electron mass in MeV (cf. Doc. 014); it is not a prediction of the MeV scale. The Planck length agrees through $G_{\text{SI}}$ and $C_{\text{conv}}$, not through $S_{T0}$. Where $\alpha$ hides in the conversion and what role the reference energy $1$ MeV plays is shown in Doc. 386: the agreement of $\alpha$ requires a reference energy of $0.99992$ MeV, whose geometric meaning is open. (See [10] for parameter derivation.)

### The Conversion Chain

**From natural units to SI units:**
The conversion factor between natural T0 units and SI units is:

$$\text{Conversion factor} = \frac{\hbar c}{S_{T0}} = \frac{\hbar c}{1\ \text{MeV}} = 1{,}973 \times 10^{-13}\ \text{m} \tag{Y.29}$$

For the Planck length (with $G_{\text{SI}}$ from Eq. (Y.11)):

$$l_P^{\text{SI}} = \sqrt{\frac{\hbar G_{\text{SI}}}{c^3}} \tag{Y.30}$$

$$= \sqrt{\frac{1.0546 \times 10^{-34} \times 6.6745 \times 10^{-11}}{(2.9979 \times 10^8)^3}}\ \text{m} \tag{Y.31}$$

$$= 1.616 \times 10^{-35}\ \text{m} \quad \checkmark \tag{Y.32}$$

**The geometric lock:** The agreement of the T0-derived Planck length with the SI-measured value is established through $G_{\text{SI}}$, which contains $\xi$, $m_e$ (in MeV, i.e. with $S_{T0} = 1$ MeV/$c^2$) and $C_{\text{conv}}$.

### The Triple Consistency

**Insight Y.4.1. Three independent measurements lock together:**
The system is overdetermined by three independent experimental values:

1. Fine structure constant: $\alpha = 1/137{,}035999084$ (determined from $a_e$ and atom-recoil measurements $h/m$ on Cs and Rb) (See [11].)
2. Gravitational constant: $G = 6{,}674 \times 10^{-11}$ m$^3$/(kg·s$^2$) (Cavendish-type experiments)
3. Planck length: $l_P = 1{,}616 \times 10^{-35}$ m (derived from $G$, $\hbar$, $c$)

The T0 theory connects all three through $\xi$ and one mass or energy anchor: $\alpha$ via $\alpha = \xi \cdot E_0^2$, $G$ and $l_P$ via $G_{\text{SI}}$ with $C_{\text{conv}}$ and $K_{\text{frak}}$. The scaling factor $S_{T0} = 1$ MeV/$c^2$ is not an additional boundary condition here but holds by construction. The form and order of magnitude of $G$ and $l_P$ follow from $\xi$ and the anchor; the agreement to $0.003\,\%$ and $0.002\,\%$ respectively arises only with the residual in $C_{\text{conv}}$ and does not count as an independent precision result.

## Y.5 The Speed of Light: Geometric or Conventional?

### The Dual Nature of $c$

**Understanding the role of the speed of light:**
The speed of light has a subtle dual character requiring careful analysis:
**Perspective 1: As a dimensional convention**
In natural units, setting $c = 1$ is purely conventional:

$$[L] = [T] \quad \text{(space and time have the same dimension)} \tag{Y.33}$$

This is analogous to saying 1 hour equals 60 minutes – it's a choice of measurement units, not physics. (See [12].)

**Perspective 2: As a ratio of the Planck scales**

Via the Planck scales, $c$ can be written as a ratio:

$$l_P = \sqrt{\hbar G/c^3}, \quad G = \frac{\xi^2}{4m_e} C_{\text{conv}} K_{\text{frak}} \tag{Y.34}$$

$$t_P = \frac{l_P}{c} = \frac{l_P}{1} \quad \text{(in natural units)} \tag{Y.35}$$

Since $t_P$ is defined as $l_P/c$, $c = l_P/t_P$ is an identity and not a determination of $c$ by $\xi$.

## The SI Value is Fixed by Definition

**Key Result**

**Why $c = 299\,792\,458$ m/s exactly:**
Since 1983, $c = 299\,792\,458$ m/s has been fixed by the definition of the metre; the value was chosen to match centuries of measurements. The ratio of the Planck scales

$$c^{\text{SI}} = \frac{l_P^{\text{SI}}}{t_P^{\text{SI}}} = \frac{1{,}616 \times 10^{-35}\ \text{m}}{5{,}391 \times 10^{-44}\ \text{s}} \tag{Y.36}$$

reproduces this value because $t_P = l_P/c$ by definition; it is an identity and not a determination of $c$ by $\xi$.

## The Meter is Defined by $c$

**Insight Y.5.1. The circular calibration loop:**

There is a beautiful circularity in the SI-2019 system:

1. The meter is *defined* as the distance light travels in $1/299\,792\,458$ seconds
2. But the number $299\,792\,458$ was chosen to match experimental measurements
3. $c = l_P/t_P$ is an identity because $t_P = l_P/c$; via the Planck scales, $c$ is not determined by $\xi$

**Conclusion:** We use $c$ to *define* the meter; the numerical value of $c$ is thus a fixing of the unit of length and does not follow from $\xi$. (See [13] for circularity of constants.)

## Y.6 Derivation of the Boltzmann Constant

### The Temperature Problem in Natural Units

**Warning**

**The Boltzmann constant is NOT fundamental:**
In natural units, where energy is the fundamental dimension, temperature is just another energy scale. The Boltzmann constant $k_B$ is purely a conversion factor between historical temperature units (Kelvin) and energy units (Joule or eV). (See [14] for temperature units.)

### Definition in the SI System

**The SI reform 2019 definition:**
Since May 20, 2019, the Boltzmann constant has been fixed by definition:

$$\boxed{k_B = 1{,}380649 \times 10^{-23}\ \text{J/K}} \tag{Y.37}$$

This defines the Kelvin scale in terms of energy:

$$k_B \cdot 1\ \text{K} = 1.380649 \times 10^{-23}\ \text{J} \tag{Y.38}$$

### Relationship to Fundamental Constants

**Key Result**

**Boltzmann constant from gas constant:**
The Boltzmann constant is defined by the Avogadro number:

$$k_B = \frac{R}{N_A} \tag{Y.39}$$

where:

- $R = 8{,}314462618$ J/(mol·K) (ideal gas constant)
- $N_A = 6{,}02214076 \times 10^{23}$ mol$^{-1}$ (Avogadro constant, fixed since 2019)

**Result:**

$$k_B = \frac{8{,}314462618}{6{,}02214076 \times 10^{23}} = 1{,}380649 \times 10^{-23}\ \text{J/K} \tag{Y.40}$$

### T0 Perspective on Temperature

**Insight Y.6.1. Temperature as an energy scale in T0 theory:**
In T0 theory, temperature is naturally expressed as energy:

$$T_{\text{natural}} = k_B T_{\text{Kelvin}} \tag{Y.41}$$

For example, the CMB temperature:

$$T_{\text{CMB}} = 2{,}725\ \text{K} \tag{Y.42}$$

$$T_{\text{CMB}}^{\text{natural}} = k_B \times 2{,}725\ \text{K} = 2{,}35 \times 10^{-4}\ \text{eV} \tag{Y.43}$$

**Core message:** $k_B$ is not derived from $\xi$ because it represents a historical convention for temperature measurement, not a physical property of spacetime geometry.

**Distinction $\xi_{\text{FFGFT}}$ vs. $\xi_{\text{UIFT}}$:** In the IPI discussion, $\xi_{\text{UIFT}} = k_B T_{\text{CMB}} \ln 2/(m_e c^2) \approx 3.19 \times 10^{-10}$ is treated as an independent parameter. $\xi_{\text{FFGFT}} = 4/30000 \approx 1.333 \times 10^{-4}$ and $\xi_{\text{UIFT}}$ are **not the same quantity** — their ratio is:

$$\frac{\xi_{\text{FFGFT}}}{\xi_{\text{UIFT}}} = \frac{\xi\, m_e c^2}{k_B T_{\text{CMB}} \ln 2} \approx 418\,521 \tag{Y.44}$$

Replacing $k_B T_{\text{CMB}}$ by the FFGFT value $(16/9)\,\xi$ eV gives $\frac{m_e c^2/\text{eV}}{(16/9)\ln 2} \approx 414\,684$. The factor follows from the SI conversion: $T_{\text{CMB}}[\text{eV}] = k_B \cdot 2.72548\,\text{K} = 2.3486 \times 10^{-4}$ eV, while FFGFT to first order gives $(16/9)\,\xi = 2.370 \times 10^{-4}$ eV ($\Delta = 0.93\,\%$).

## Static and Dynamic Unit of Information

The factor $\xi_{\text{FFGFT}}/\xi_{\text{UIFT}} \approx 418\,521$ (with the FFGFT value for $k_B T_{\text{CMB}}$: $414\,684$) is more than a conversion: it separates two fundamentally different notions of "information". $\xi_{\text{UIFT}}$ is defined through $k_B T_{\text{CMB}} \ln 2$ – i.e. through an *energy per bit*, hence through a temperature. $\xi_{\text{FFGFT}}$ is dimensionless and reference-free. This is exactly where the question arises of how much energy corresponds to the smallest unit of information.

**Insight Y.6.2. There is no universal energy per bit.** The energy of a unit of information is not a fixed quantity but a projection that requires a scale:

$$\text{thermal (Landauer):}\quad E = k_B T \ln 2 \quad \text{(temperature-dependent)} \tag{Y.45}$$

$$\text{spatial (energy hierarchy):}\quad E_N = \frac{\hbar c}{L_N} \quad \text{(length-dependent)} \tag{Y.46}$$

Both expressions carry the embedding constants ($k_B$, $\hbar$, $c$) and an external scale ($T$ or $L$). The length- or temperature-dependence is therefore not a defect but the *signature of a dynamic projection* (cf. Dok. 184, energy hierarchy $E_N = \hbar c/L_N$, and Landauer bound $k_B T \ln 2$).

In the picture of the static/dynamic demarcation (Dok. 261) this is the *layer-2* answer: an energy per bit exists only once the static structure is embedded in the time-flow and projected onto an absolute scale. This is where $\xi_{\text{UIFT}}$ lives – temperature-bound, CMB-referenced, not universal.

The static answer is different. The fundamental unit of information in the static context is not an energy but the *winding number* on the compact torus:

**The static unit of information: the winding number.** On the compact directions of the torus, winding numbers are topologically conserved and integer-valued (Dok. 181). A winding number is

- **dimensionless and countable** – an integer, not an energy value;
- **reference-free** – independent of $L$ and $T$, because it lies *before* any projection;
- **universal** – the same counting unit at every scale.

This is exactly the state-space size that, according to Dok. 184, remains untouched by the $\xi$-dynamics ("the $\xi$-geometry sets the dynamics, not the state-space size"): the bit as a *state* is static; only its energetic *realization* is dynamic.

The two earlier statements thus resolve without contradiction: *information as energy* is the dynamic projection (layer 2) – scale-dependent, without a universal unit. *Information as counting* (winding, bit) is the static unit (layer 1) – dimensionless and universal. The conversion between them is the *embedding price* (Dok. 185); it explains why a "bit" acquires an energy only through $k_B T$ or $\hbar c/L$ – and why that energy necessarily depends on the scale.

In this light the Vopson comparison appears as a projection ratio: Vopson's $\xi_{\mathrm{UIFT}} = k_B T_{\mathrm{CMB}} \ln 2/(m_e c^2)$ is a layer-2 quantity (bound to the CMB temperature through $k_B T_{\mathrm{CMB}}$), while $\xi_{\mathrm{FFGFT}}$ is the layer-1 invariant. The factor $\approx 418\,521$ is nothing other than the projection factor between a dynamic and a static information quantity.

## Y.7 The Interwoven Network of Constants

### The Fundamental Formula Network

**SI constants are mathematically linked:**
Since the SI reform 2019, all fundamental constants are connected by exact mathematical relationships:

$$\alpha = \frac{e^2}{4\pi\varepsilon_0 \hbar c} \quad \text{(exact definition)} \tag{Y.47}$$

$$\varepsilon_0 = \frac{e^2}{2\alpha h c} \quad \text{(derived from above)} \tag{Y.48}$$

$$\mu_0 = \frac{2\alpha h}{e^2 c} \quad \text{(via } \varepsilon_0 \mu_0 c^2 = 1\text{)} \tag{Y.49}$$

$$k_B = \frac{R}{N_A} \quad \text{(definition of Boltzmann constant)} \tag{Y.50}$$

### The Geometric Boundary Condition

**Insight Y.7.1. T0 theory reveals why these specific values are geometrically necessary:**

$$\alpha = \xi \cdot E_0^2 = \frac{1}{137{,}036} \quad \text{(geometric derivation)} \tag{Y.51}$$

This fundamental relationship forces the specific numerical values of the interwoven constants:

$$\frac{e^2}{4\pi\varepsilon_0\hbar c} = \frac{1}{137{,}036} \quad \text{(geometric boundary condition)} (See [17].) \tag{Y.52}$$

## Y.8 The Nature of Physical Constants

### Translation Conventions vs. Physical Quantities

**Key Result**

**Constants fall into three categories:**

1. **The single fundamental parameter:** $\xi = \frac{4}{3} \times 10^{-4}$
2. **Geometric quantities derivable from $\xi$ (absolute values with one mass or energy anchor):**
   - Particle masses (electron, muon, tau, quarks) (See [15].)
   - Coupling constants ($\alpha$, $\alpha_s$, $\alpha_w$)
   - Gravitational constant $G$
   - Planck length $l_P$
3. **Pure translation conventions (SI unit definitions):**
   - $\hbar$ (defines energy-time relationship)
   - $e$ (defines charge scale)
   - $k_B$ (defines temperature-energy conversion)
   - $c$ (fixed by the definition of the metre since 1983)

**Warning**

**Critical clarification about the speed of light:**
The speed of light occupies a unique position in this classification:

- **In natural units ($c = 1$):** $c$ is a mere convention establishing how we relate length and time
- **In SI units:** The numerical value $c = 299\,792\,458$ m/s has been fixed by the definition of the metre since 1983. The ratio

$$c^{\text{SI}} = \frac{l_P^{\text{SI}}}{t_P^{\text{SI}}} = \frac{1{,}616 \times 10^{-35}\ \text{m}}{5{,}391 \times 10^{-44}\ \text{s}} = 299\,792\,458\ \text{m/s} \tag{Y.53}$$

  is an identity because $t_P = l_P/c$, not a determination of $c$ by $\xi$.

**The consequence:** In natural units $c = 1$ is a convention, in SI units a fixing of the metre; in neither case does $\xi$ determine the numerical value of $c$.

### The SI Reform 2019: Geometric Calibration Realized

The 2019 redefinition fixed constants by definition:

$$c = 299\,792\,458 \text{ m/s} \tag{Y.54}$$
$$\hbar = 1{,}054571817... \times 10^{-34} \text{ J} \cdot \text{s} \tag{Y.55}$$
$$e = 1{,}602176634 \times 10^{-19} \text{ C} \tag{Y.56}$$
$$k_B = 1{,}380649 \times 10^{-23} \text{ J/K} \tag{Y.57}$$

**Insight Y.8.1.** This fixation implements the unique calibration consistent with $\xi$-geometry. The apparent arbitrariness conceals geometric necessity.

## Y.9 The Mathematical Necessity

### Why Constants Must Have Their Specific Values

**The interlocked system:**
Given the fixed values and their mathematical relationships:

$$h = 2\pi\hbar = 6{,}62607015 \times 10^{-34} \text{ J} \cdot \text{s} \tag{Y.58}$$
$$\alpha = \frac{e^2}{4\pi\varepsilon_0\hbar c} = \frac{1}{137{,}035999084} \tag{Y.59}$$
$$\varepsilon_0 = \frac{e^2}{2\alpha h c} = 8{,}8541878128 \times 10^{-12} \text{ F/m} \tag{Y.60}$$
$$\mu_0 = \frac{2\alpha h}{e^2 c} = 1{,}25663706212 \times 10^{-6} \text{ N/A}^2 \tag{Y.61}$$

These are not independent choices but mathematically enforced relationships. (See [16] for mathematical structure.)

### The Geometric Explanation

**Historical**

**Sommerfeld's unwitting geometric calibration**
Arnold Sommerfeld's calibration in 1916 to $\alpha \approx 1/137$ established the SI system on geometric foundations. T0 theory reveals this was not coincidental but reflected the fundamental value $\alpha = 1/137{,}036$ derived from $\xi$. (See [18].)

## Y.10 Conclusion: Geometric Unity

**Key Result**

**Complete parameter freedom achieved:**

- **Single input:** $\xi = \frac{4}{3} \times 10^{-4}$
- **Derivable from $\xi$ (absolute SI values with one mass or energy anchor):**
  - **First:** Particle masses including the electron via the ladder $m_e = \frac{4}{3}\xi^{3/2} v$ (Doc. 006), with $v$ as the scale
  - **Then:** Gravitational constant: $G = \xi^2/(4m_e)\times$ (conversion factors)
  - **Then:** Planck length: $l_P = \sqrt{\hbar G/c^3}$ with the complete $G$
  - **Also:** Characteristic T0 length: $L_0 = \xi \cdot l_P$ (spacetime granulation)
  - Coupling constants: $\alpha$, $\alpha_s$, $\alpha_w$
- **Translation conventions (not derived, define units):**
  - $\hbar$ defines energy-time relation in SI units
  - $e$ defines charge scale in SI units
  - $k_B$ defines temperature-energy conversion (historical)
  - $c$ defines the metre (fixed by definition since 1983)
- **Mathematical necessity:** Constants intertwined through exact formulas
- **Geometric foundation:** SI 2019 unwittingly implements $\xi$-geometry

**Final insight:** The universe is pure geometry, encoded in $\xi$. The complete derivation chain is:
$\xi \to \{m_e, m_\mu, m_\tau, ...\} \to G \to l_P$
with $L_0 = \xi \cdot l_P$, expressing the fundamental sub-Planck scale of spacetime granulation.
**The profound mystery resolved:** Why does the Planck length, derived from $\xi$-geometry and one anchor, agree with the Planck length calculated from experimentally measured $G$? Because *both describe the same geometric reality*: form and order of magnitude follow from $\xi$ and the anchor; the precision to $0.002\,\%$ arises only with the residual in $C_{\text{conv}}$. The SI reform 2019 unwittingly calibrated human measurement units to the fundamental $\xi$-geometry of the universe.
This is no coincidence – it is geometric necessity. Only $\xi$ is fundamental; everything else either follows from geometry or defines how we measure this geometry.

## Appendix: Complete Derivation Chain

**From geometric parameter to measurable quantities:**

1. Basic parameter: $\xi = \frac{4}{3} \times 10^{-4}$
2. Electron mass: $m_e = \frac{4}{3}\xi^{3/2} v$ (Doc. 006), with $v$ as mass or energy anchor
3. Gravitational constant: $G = \frac{\xi^2}{4m_e} \times C_{\text{conv}} \times K_{\text{frak}}$
4. Planck length: $l_P = \sqrt{\hbar G/c^3}$ with $G$ from step 3
5. Planck time: $t_P = l_P/c = l_P$ (natural units)
6. Speed of light: $c = l_P/t_P$ is an identity because of step 5; $c = 299\,792\,458$ m/s is fixed by the definition of the metre
7. Fundamental length: $L_0 = \xi \cdot l_P = 2{,}155 \times 10^{-39}$ m
8. Fine structure constant: $\alpha = \xi \cdot E_0^2 = 1/137{,}036$

**Consistency check:**

$$\Delta G = \left| \frac{G_{\text{T0}} - G_{\text{SI}}}{G_{\text{SI}}} \right| \approx 0.003\% \tag{Y.62}$$

$$\Delta l_P = \left| \frac{l_P^{\text{T0}} - l_P^{\text{SI}}}{l_P^{\text{SI}}} \right| \approx 0.002\% \tag{Y.63}$$

$$\Delta \alpha = \left| \frac{\alpha_{\text{T0}} - \alpha_{\text{SI}}}{\alpha_{\text{SI}}} \right| \approx 0.0005\% \tag{Y.64}$$

The agreement of $G$ and $l_P$ at this level arises only with the residual in $C_{\text{conv}}$ and does not count as an independent precision result.

## Glossary

$\xi$ Fundamental geometric parameter, $\frac{4}{3} \times 10^{-4}$

$S_{T0}$ Mass scaling factor, $1$ MeV/$c^2$ (by construction)

$L_0$ Fundamental T0 length, $\xi \cdot l_P = 2{,}155 \times 10^{-39}$ m

$E_0$ Fundamental energy scale, $\sqrt{m_e \cdot m_\mu / K_{\text{frak}}} = 7.398$ MeV (bare $\sqrt{m_e m_\mu} = 7.348$ MeV)

$r_0(E)$ Characteristic length for energy $E$, $2GE$

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# Chapter Z

# Natural Units in Theoretical Physics: A Treatise in the Context of T0 Theory


## Abstract

The use of natural units in theoretical physics is a fundamental concept that can be comprehensively explained and contextualized within the framework of T0 theory. This treatise illuminates the principle of dimensional reduction, the advantages for calculations, the particular relevance for T0 theory, and the necessity of explicit SI units in practice. Finally, it emphasizes the deeper insight that physics ultimately rests on dimensionless geometric relationships.


## Z.1 Basic Principle of Natural Units

### The Principle of Dimensional Reduction

In natural units, one sets fundamental constants to 1:

- **Speed of light**: $c = 1$
- **Reduced Planck constant**: $\hbar = 1$
- **Boltzmann constant**: $k_B = 1$
- **Sometimes**: $G = 1$ (Planck units)

### Mathematical Consequence

This does not mean that these constants "disappear," but that they serve as **scale setters**:

$$E = mc^2 \quad \Rightarrow \quad E = m \quad \text{(since } c = 1\text{)} \tag{Z.1}$$

$$E = \hbar\omega \quad \Rightarrow \quad E = \omega \quad \text{(since } \hbar = 1\text{)} \tag{Z.2}$$

## Z.2 Advantages for Calculations

### Simplified Formulas

**With SI units:**

$$E = \sqrt{(pc)^2 + (mc^2)^2} \tag{Z.3}$$

**In natural units:**

$$E = \sqrt{p^2 + m^2} \tag{Z.4}$$

### Transparent Dimensional Analysis

All quantities can be traced back to one fundamental dimension (typically energy):

| Quantity | Natural Dimension | SI Equivalent |
|---|---|---|
| Length | $[E]^{-1}$ | $\hbar c/E$ |
| Time | $[E]^{-1}$ | $\hbar/E$ |
| Mass | $[E]$ | $E/c^2$ |

**Table Z.1:** Dimensional relationships in natural units

## Z.3 Particular Relevance in T0 Theory

### Geometric Nature of Constants

T0 theory shows particularly clearly why natural units are fundamental:

$$\alpha = \xi \cdot \left(\frac{E_0}{1\ \text{MeV}}\right)^2 \tag{Z.5}$$

This makes explicit that the fine structure constant is a **purely dimensionless geometric relationship**.

### The $\xi$-Parameter as Fundamental Geometry Factor

The derivation:

$$\xi = \frac{4}{3} \times 10^{-4} \tag{Z.6}$$

is intrinsically dimensionless and represents the fundamental space geometry – independent of human units of measurement.

**Important:** $\xi$ alone is not directly equal to $1/m_e$ or $1/E$, but requires specific scaling factors for different physical quantities.

## Z.4 The Scaling Factor $S_{T0}$

The scaling factor connects the electron mass in T0 units with its SI value:

$$\textbf{1. T0 units:} \quad m_e^{\mathrm{T0}} = 0.511 \quad \text{(electron mass in MeV)} \tag{Z.7}$$

$$\textbf{2. Measurement:} \quad m_e^{\mathrm{SI}} = 9.109 \times 10^{-31} \text{ kg} \tag{Z.8}$$

$$\textbf{3. Scaling:} \quad S_{T0} = m_e^{\mathrm{SI}}/m_e^{\mathrm{T0}} = 1.7827{\times}10^{-30} \tag{Z.9}$$

$$\textbf{4. Comparison:} \quad 1 \text{ MeV}/c^2 = 1.7827 \times 10^{-30} \text{ kg} \tag{Z.10}$$

The number $0.511$ is the electron mass in MeV, i.e. the measured value in a human-chosen unit; no $\xi$ calculation giving $0.511$ appears in this document. The corpus ladder $m_e = \frac{4}{3}\,\xi^{3/2}\,v$ needs $v$ as a scale and gives $0.505$ with $v$ in MeV. Hence $S_{T0} = m_e^{\mathrm{SI}}/0.511 = 1$ MeV$/c^2$ by construction and not a prediction of the MeV scale: $S_{T0}$ is the conversion between T0 units (masses in MeV) and SI units.

## Z.5 Quantized Mass Calculation in T0 Theory

### Fundamental Mass Quantization Principle

In T0 theory, particle masses are **quantized** and follow from the fundamental geometry parameter $\xi$ through discrete scaling relationships:

$$m_i^{\mathrm{T0}} = n_i \cdot Q_m^{\mathrm{T0}} \cdot f_i(\xi) \tag{Z.11}$$

where:

- $n_i \in \mathbb{N}$ - Quantum number (discrete)
- $Q_m^{\mathrm{T0}}$ - Fundamental mass quantum in T0 units
- $f_i(\xi)$ - Particle-specific geometry function

### Electron Mass as Reference

The electron mass serves as the fundamental reference mass:

$$\xi_e = \frac{4}{3} \times 10^{-4} \times f_e(1, 0, 1/2) \tag{Z.12}$$

$$m_e^{\mathrm{T0}} = Q_m^{\mathrm{T0}} \cdot \frac{\xi}{\xi_e} = 0.511 \tag{Z.13}$$

### Complete Particle Mass Spectrum

For detailed derivations of all elementary particle masses within the T0 framework, including quarks, leptons, and gauge bosons, refer to the separate comprehensive treatment “Particle Masses in T0 Theory” which provides:

- Complete mass calculations for all Standard Model particles
- Derivation of mass quantization rules
- Explanation of generation patterns
- Comparison with experimental values
- Fractal correction procedures for precision matching

## Z.6 Important: Explicit SI Units are Necessary for...

### 1. Experimental Verification

Every measurement is performed in SI units:
- Particle masses in MeV/c²
- Cross sections in barn
- Magnetic moments in $\mu_B$

### 2. Technological Applications

- Detector design (lengths in m, times in s)
- Accelerator technology (energies in eV)
- Medical physics (dosage measurements)

### 3. Interdisciplinary Communication

- Astrophysics (redshifts, Hubble constant)
- Materials science (lattice constants)
- Engineering

## Z.7 Concrete Conversion in T0 Theory

### Example: Electron Mass

**In T0 geometric units:**

$$m_e^{\text{T0}} = 0.511 \quad \text{(electron mass in MeV)} \tag{Z.14}$$

**In SI units:**

$$m_e^{\text{SI}} = m_e^{\text{T0}} \cdot S_{T0} = 0.511 \cdot 1.782662 \times 10^{-30} = 9.1093837 \times 10^{-31} \text{ kg} \tag{Z.15}$$

### The Fundamental Scaling Relationship

The conversion from T0 geometric quantities to SI units is accomplished by:

$$[\text{SI}] = [\text{T0}] \times S_{\text{T0}} \tag{Z.16}$$

where $S_{\text{T0}} = 1.782662 \times 10^{-30}$ is the scaling factor from Section Z.4; by construction it equals $1$ MeV$/c^2$.

## Z.8 Correct Energy Scale for the Fine Structure Constant

The fundamental relationship for the fine structure constant requires a precise energy reference:

$$\alpha = \xi \cdot \left( \frac{E_0}{1 \text{ MeV}} \right)^2 \tag{Z.17}$$

with $E_0 = 7.398$ MeV $= 7.348/\sqrt{K_{\text{frak}}}$ (characteristic energy, fractally corrected) (Z.18)

This yields:

$$\alpha = 1.333333 \times 10^{-4} \cdot (7.398)^2 \tag{Z.19}$$

$$= 1.333333 \times 10^{-4} \cdot 54.73 \tag{Z.20}$$

$$= 7.297 \times 10^{-3} \tag{Z.21}$$

$$\frac{1}{\alpha} = 137.035 \tag{Z.22}$$

in agreement with the experimental value $1/\alpha = 137.036$. The rounded value $E_0 = 7.400$ MeV gives $7.3013 \times 10^{-3}$, i.e. $1/\alpha = 136.96$; without fractal correction ($E_0 = \sqrt{m_e m_\mu} = 7.348$ MeV) one obtains $1/\alpha = 138.9$.

## Z.9 Integration of Fractal Correction into Natural Units

The formulas in T0 theory fit in natural units without explicit fractal correction, because these units isolate the geometric essence of the theory. For exact conversions to SI units, however, fractal correction is essential to incorporate self-similar corrections of the vacuum geometry.

### Why Do the Formulas Fit in Natural Units Without Fractal Correction?

In natural units, physics is reduced to a geometric, dimensionless basis (cf. Section Z.1). The fundamental constants serve only as a scale, and the core formulas hold approximately without additional corrections because:

- **The $\xi$-parameter is intrinsically dimensionless**: $\xi$ represents the pure geometry of the vacuum field and acts like a "universal scaling factor."

- **Approximate validity for rough calculations**: Many T0 formulas are exact in the geometric ideal form, without correction.
- **Example: Electron mass in natural units**:

$$m_e^{\mathrm{T0}} = 0.511 \quad \text{(electron mass in MeV)} \tag{Z.23}$$

This "fits" immediately because $0.511$ is the measured value in MeV; the corpus ladder $m_e = \frac{4}{3}\,\xi^{3/2}\,v$ gives $0.505$ with $v$ in MeV without correction.

## Why is Fractal Correction Necessary for Exact SI Conversions?

SI units are human conventions that "contaminate" the geometric purity of T0 theory. To achieve exact agreement with experiments, fractal correction must be **explicitly applied** because:

- **Fractal self-similarity breaks scale invariance**
- **Conversion requires explicit scaling**
- **Cosmological reference effects**

## Mathematical Specification of Fractal Correction

The fractal correction is explicitly defined as:

$$f_{\mathrm{fractal}}(E_0) = \prod_{n=1}^{137}\left(1 + \delta_n \cdot \xi \cdot \left(\frac{4}{3}\right)^{n-1}\right) \tag{Z.24}$$

where $\delta_n$ are dimensionless coefficients describing the fractal structure at each stage.

## Comparison: Approximation vs. Exactness

| Aspect | Without fractal correction (T0 units) | With fractal correction (for SI conversion) |
|---|---|---|
| Accuracy | Approximate ($\sim 98$–$99\,\%$, geometrically ideal) | Exact (to $10^{-6}$, matches CODATA measurements) |
| Example: $\alpha$ | $\alpha \approx \xi \cdot (E_0)^2 \approx 1/137$ (rough) | $\alpha = 1/137.03599\ldots$ (via 137 stages) |
| Mass calculation | $m_e^{\mathrm{T0}} = 0.511$ (in MeV) | $m_e^{\mathrm{SI}} = 9.1093837{\times}10^{-31}$ kg (physical) |
| Energy scale | $E_0 = \sqrt{m_e m_\mu} = 7.348$ MeV (ideal, bare mean without $K_{\mathrm{frak}}$) | $E_0 = 7.398$ MeV ($7.348/\sqrt{K_{\mathrm{frak}}}$) |
| Scaling factor | $S_{T0} = 1.782662 \times 10^{-30}$ (by construction) | $S_{T0} \cdot R_f$ (renormalized) |
| Advantage | Fast, transparent calculations | Testability with experiments |
| Disadvantage | Ignores fractal subtleties | Complex (iteration over resonance stages) |

**Table Z.2:** Comparison of geometric idealization in T0 units and physical exactness with fractal correction.

## Z.10 Important Conceptual Clarifications

When applying T0 theory, note these fundamental distinctions:

- **T0 quantities** are given in T0 units (e.g., $m_e^{\text{T0}} = 0.511$, the electron mass in MeV)
- **SI quantities** are physical measurements (e.g., $m_e^{\text{SI}} = 9.1093837 \times 10^{-31}$ kg)
- $S_{T0}$ is the scaling between these realms; $S_{T0} = 1$ MeV$/c^2$ holds by construction
- The energy reference for $\alpha$ is $E_0 = 7.398$ MeV $= 7.348/\sqrt{K_{\text{frak}}}$; in the geometric idealization without $K_{\text{frak}}$ it is $\sqrt{m_e m_\mu} = 7.348$ MeV
- All mass scales are **discretely quantized** in both T0 and SI representations

## Z.11 Special Significance for T0 Theory

### The Deeper Insight

T0 theory reveals that natural units are not merely a calculational convenience, but express the **true geometric nature of physics**:

- $\xi$ is the fundamental dimensionless geometry constant
- $S_{T0}$ connects geometric idealization to physical measurement
- **T0 quantities** represent the ideal geometric forms
- **SI quantities** are their measurable projections into our physical reality
- **Particle masses** are quantized geometric patterns in both realms

### Practical Implications

1. **Theoretical development**: Work in T0 units using geometric quantities
2. **Fundamental scaling**: Apply $S_{T0}$ to project to physical reality
3. **Predictions**: Convert to SI units for experimental verification
4. **Verification**: Compare with measured SI values
5. **Quantization**: Respect the discrete nature of all physical scales

## Z.12 Notation and Symbols

## Z.13 Fundamental Relationships

## Z.14 Conversion Factors

| Symbol | Meaning and Explanation |
|---|---|
| $c$ | Speed of light in vacuum; fundamental constant of nature |
| $\hbar$ | Reduced Planck constant |
| $k_B$ | Boltzmann constant |
| $G$ | Gravitational constant |
| $E$ | Energy; in natural units dimensionally equivalent to mass and frequency |
| $m$ | Mass; in natural units $m = E$ (since $c = 1$) |
| $p$ | Momentum; in natural units dimensionally equivalent to energy |
| $\omega$ | Angular frequency; in natural units $\omega = E$ (since $\hbar = 1$) |
| $\alpha$ | Fine structure constant; dimensionless coupling constant |
| $\xi$ | Fundamental geometry parameter of T0 theory; $\xi = \frac{4}{3} \times 10^{-4}$ |
| $E_0$ | Reference energy in T0 theory; $E_0 = 7.398$ MeV $= 7.348/\sqrt{K_{\text{frak}}}$ |
| $m_e^{\text{T0}}$ | Electron mass in T0 units; $m_e^{\text{T0}} = 0.511$ (electron mass in MeV) |
| $m_e^{\text{SI}}$ | Electron mass in SI units; $m_e^{\text{SI}} = 9.1093837 \times 10^{-31}$ kg (physical) |
| $[E]$ | Energy dimension; fundamental dimension in natural units |
| SI | International System of Units (physical measurements) |
| T0 | T0 geometric units (ideal geometric forms) |
| $S_{T0}$ | Scaling factor; $S_{T0} = 1.782662 \times 10^{-30}$ (by construction 1 MeV$/c^2$ in kg) |
| $R_f$ | Fractal correction factor |
| $f_{\text{fractal}}$ | Fractal correction function |
| $Q_m^{\text{T0}}$ | Fundamental mass quantum in T0 units |
| $Q_m^{\text{SI}}$ | Fundamental mass quantum in SI units |
| $n_i$ | Quantum number for particle $i$; $n_i \in \mathbb{N}$ (discrete) |
| $\delta_n$ | Fractal correction coefficients; dimensionless |

**Table Z.3:** Explanation of the notation and symbols used

| Relationship | Meaning |
|---|---|
| $E = m$ | Mass-energy equivalence (since $c = 1$) |
| $E = \omega$ | Energy-frequency relationship (since $\hbar = 1$) |
| $[L] = [T] = [E]^{-1}$ | Length and time have same dimension as inverse energy |
| $[m] = [p] = [E]$ | Mass and momentum have same dimension as energy |
| $\alpha = \xi(E_0/1\text{MeV})^2$ | Fundamental relationship in T0 theory |
| $m_i^{\text{T0}} = n_i \cdot Q_m^{\text{T0}} \cdot f_i(\xi)$ | Quantized mass formula in T0 units |
| $m_i^{\text{SI}} = m_i^{\text{T0}} \cdot S_{T0}$ | Fundamental scaling to SI units |
| $S_{T0} = \frac{m_e^{\text{SI}}}{m_e^{\text{T0}}}$ | Definition of fundamental scaling factor |

**Table Z.4:** Fundamental relationships in T0 theory and scaling to physical units

| Quantity | Conversion Factor | Value |
|---|---|---|
| $S_{T0}$ | Scaling factor | $1.782662 \times 10^{-30}$ |
| $m_e^{\text{T0}}$ | Electron mass (T0 units) | $0.511$ |
| $m_e^{\text{SI}}$ | Electron mass (SI units) | $9.1093837 \times 10^{-31}$ kg |
| $1$ MeV$/c^2$ | Conventional mass unit | $1.782662 \times 10^{-30}$ kg |
| $1$ MeV | Energy in joules | $1.602176 \times 10^{-13}$ J |
| $1$ fm | Length in natural units | $5.06773 \times 10^{-3}$ MeV$^{-1}$ |

**Table Z.5:** Fundamental conversion factors between T0 geometric units and SI physical units

# Chapter aa

# Natural Unit Systems:

Universal Energy Conversion and
Fundamental Length Scale Hierarchy

## Abstract

This foundational document establishes the natural unit system used throughout the T0 model framework. By setting fundamental constants to unity and adopting energy as the base dimension, all physical quantities can be expressed as powers of energy. This document serves as the reference for unit conversions and dimensional analysis across all T0 model applications.

## aa.1 List of Symbols and Notation

## aa.2 Introduction

Natural units are unit systems where fundamental physical constants are set to unity to simplify calculations and reveal the underlying mathematical structure of physical laws. The most well-known systems are **Planck units** (for gravitation and quantum physics) and **atomic units** (for quantum chemistry).

This document establishes the complete framework for the natural unit system used in the T0 model, which is based on Planck units with energy as the fundamental dimension. The key insight is that energy $[E]$ serves as the universal dimension from which all other physical quantities derive.

| Symbol | Meaning | Units/Notes |
|---|---|---|
| | **Fundamental Constants** | |
| $\hbar$ | Reduced Planck constant | Set to 1 |
| $c$ | Speed of light | Set to 1 |
| $G$ | Gravitational constant | Set to 1 |
| $k_B$ | Boltzmann constant | Set to 1 |
| $e$ | Elementary charge | $[E^0]$ (dimensionless) |
| $\varepsilon_0, \mu_0$ | Vacuum permittivity, permeability | Set to 1 in QED units |
| | **Units** | |
| $l_P, t_P, m_P, E_P, T_P$ | Planck length, time, mass, energy, temp. | Natural base units |
| $m_e, a_0, E_h$ | Electron mass, Bohr radius, Hartree energy | Atomic units |
| | **Coupling Constants** | |
| $\alpha_{\text{EM}}$ | Fine-structure constant | $e^2/(4\pi) = 1$ (nat.), $\approx 1/137$ (SI) |
| $\alpha_s, \alpha_W, \alpha_G$ | Strong, weak, gravitational coupling | Dimensionless |
| | **Physical Quantities** | |
| $E, m, \Theta$ | Energy, mass, temperature | $[E]$ |
| $L, r, \lambda, t$ | Length, radius, wavelength, time | $[E^{-1}]$ |
| $p, \omega, \nu$ | Momentum, angular freq., frequency | $[E]$ |
| $F$ | Force | $[E^2]$ |
| $v$ | Velocity | Dimensionless |
| $q$ | Electric charge | $[E^0]$ (dimensionless) |
| | **Special Scales & Notation** | |
| $r_0, \xi$ | T0 length, scaling parameter | $\xi l_P, \xi \approx 1.33 \times 10^{-4}$ |
| $\lambda_{C,e}, r_e$ | Compton wavelength, classical e radius | $h/(m_e c), e^2/(4\pi\varepsilon_0 m_e c^2)$ |
| $[X], [E^n]$ | Dimension of X, energy dimension | Dimensional analysis |
| $\sim, \leftrightarrow$ | Approximately, conversion | Order of magnitude, units |

**Table aa.1:** Symbols and notation

| System | Constants Set to 1 | Base Units | Applications | Notes |
|---|---|---|---|---|
| Planck Units | $\hbar, c, G, k_B = 1$ | $l_P, t_P, m_P, E_P$ | Quantum gravity, cosmology | Universal significance |
| Atomic Units | $m_e, e, \hbar, \frac{1}{4\pi\varepsilon_0} = 1$ | $a_0, E_h$ | Quantum chemistry, atoms | Chemistry applications |
| Particle Physics | $\hbar, c = 1$ | GeV | High energy physics | Practical for colliders |
| T0 Model | $\hbar, c, G, k_B = 1$ | Energy $[E]$ | Unified physics | Energy as base dimension |

**Table aa.2:** Comparison of natural unit systems

### Comparison with Other Natural Unit Systems

## aa.3 Fundamentals of Natural Unit Systems

### Planck Units

The Planck units were proposed by Max Planck in 1899 [1, 2] and are based on the fundamental natural constants:

$$G = 1 \quad \text{(gravitational constant)} \tag{aa.1}$$
$$c = 1 \quad \text{(speed of light)} \tag{aa.2}$$
$$\hbar = 1 \quad \text{(reduced Planck constant)} \tag{aa.3}$$

Planck recognized that these units *"retain their meaning for all times and for all, including extraterrestrial and non-human cultures necessarily"* [1].

### Atomic Units

The atomic units, introduced by Hartree in 1927 [3], set:

$$m_e = 1 \quad \text{(electron mass)} \tag{aa.4}$$
$$e = 1 \quad \text{(elementary charge)} \tag{aa.5}$$
$$\hbar = 1 \tag{aa.6}$$
$$\frac{1}{4\pi\varepsilon_0} = 1 \quad \text{(Coulomb constant)} \tag{aa.7}$$

### Quantum Optical Units

For quantum field theory applications, quantum optical units are commonly used:

$$c = 1 \quad \text{(speed of light)} \tag{aa.8}$$
$$\hbar = 1 \quad \text{(reduced Planck constant)} \tag{aa.9}$$
$$\varepsilon_0 = 1 \quad \text{(permittivity)} \tag{aa.10}$$
$$\mu_0 = 1 \quad \text{(permeability, because } c = 1/\sqrt{\varepsilon_0\mu_0}\text{)} \tag{aa.11}$$

### Advantages of Natural Units

Natural units offer several key advantages:

- **Simplified equations** (e.g., $E = m$ instead of $E = mc^2$)
- **No superfluous constants** in calculations
- **Universal scaling** for fundamental physics
- **Reveals fundamental relationships** between physical quantities
- **Provides dimensional consistency** checks
- **Eliminates arbitrary conversion factors**
- **Highlights the universal role** of energy

## aa.4 Mathematical Proof of Energy Equivalence

### Fundamental Dimensional Relations

In natural units, all physical quantities have dimensions that can be expressed as powers of energy $[E]$ [Weinberg(1995), Peskin & Schroeder(1995)]:

$$[L] = [E]^{-1} \quad \text{(from } \hbar c = 1\text{)} \tag{aa.12}$$
$$[T] = [E]^{-1} \quad \text{(from } \hbar = 1\text{)} \tag{aa.13}$$
$$[M] = [E] \quad \text{(from } c = 1\text{)} \tag{aa.14}$$

### Conversion of Fundamental Quantities

**Length:** From the relation $\hbar c = 1$ it follows:

$$[L] = \frac{[\hbar][c]}{[E]} = [E]^{-1} \tag{aa.15}$$

**Time:** From $\hbar = 1$ and $E = \hbar\omega$ it follows:

$$[T] = \frac{[\hbar]}{[E]} = [E]^{-1} \tag{aa.16}$$

**Mass:** From $E = mc^2$ and $c = 1$ it follows:

$$[M] = [E] \tag{aa.17}$$

**Velocity:**

$$[v] = \frac{[L]}{[T]} = \frac{[E]^{-1}}{[E]^{-1}} = [E]^0 = \text{dimensionless} \tag{aa.18}$$

**Momentum:**

$$[p] = [M][v] = [E] \cdot [E]^0 = [E] \tag{aa.19}$$

**Force:**

$$[F] = [M][a] = [E] \cdot [E] = [E]^2 \tag{aa.20}$$

**Charge:** In Planck units from $F = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{r^2}$ with $[F] = [E]^2$ and $[r^2] = [E]^{-2}$:

$$[q^2] = [E]^0 \quad \Rightarrow \quad [q] = [E^0] \text{ (dimensionless)} \tag{aa.21}$$

### Generalization

Any physical quantity $G$ can be represented as a product of powers of the fundamental constants:

$$G = c^a \cdot \hbar^b \cdot G^c \cdot k_B^d \cdot \ldots \tag{aa.22}$$

In natural units this becomes:

$$[G] = [E]^n \quad \text{for a specific } n \in \mathbb{Q} \tag{aa.23}$$

| Physical Quantity | SI Dimension | Natural Dimension | Derivation |
|---|---|---|---|
| Energy | $[ML^2T^{-2}]$ | $[E]$ | Base dimension |
| Mass | $[M]$ | $[E]$ | $E = mc^2, c = 1$ |
| Temperature | $[\Theta]$ | $[E]$ | $E = k_B T, k_B = 1$ |
| Length | $[L]$ | $[E^{-1}]$ | $l_P = \sqrt{\hbar G/c^3} = 1$ |
| Time | $[T]$ | $[E^{-1}]$ | $t_P = \sqrt{\hbar G/c^5} = 1$ |
| Momentum | $[MLT^{-1}]$ | $[E]$ | $p = mv, v = [E^0]$ |
| Force | $[MLT^{-2}]$ | $[E^2]$ | $F = ma = [E][E] = [E^2]$ |
| Power | $[ML^2T^{-3}]$ | $[E^2]$ | $P = E/t = [E]/[E^{-1}] = [E^2]$ |
| Charge | $[AT]$ | $[E^0]$ | Dimensionless in Planck units |
| Electric Field | $[MLT^{-3}A^{-1}]$ | $[E^2]$ | $\vec{E} = \vec{F}/q$ |
| Magnetic Field | $[MT^{-2}A^{-1}]$ | $[E^2]$ | $\vec{B} = \vec{F}/(qv)$ |

**Table aa.3:** Universal energy dimensions of physical quantities

### Fundamental Relationships

The key relationships in natural units become:

$$E = m \quad \text{(mass-energy equivalence)} \tag{aa.24}$$

$$E = T \quad \text{(temperature-energy equivalence)} \tag{aa.25}$$

$$[L] = [T] = [E^{-1}] \quad \text{(space-time unity)} \tag{aa.26}$$

$$\omega = E \quad \text{(frequency-energy equivalence)} \tag{aa.27}$$

$$p = E \quad \text{(momentum-energy equivalence for massless particles)} \tag{aa.28}$$

## aa.5 Length Scale Hierarchy

### Standard Length Scales

Physical systems organize themselves around characteristic length scales:

| Scale | Symbol | SI Value (m) | Natural Units ($l_P = 1$) |
|---|---|---|---|
| Planck Length | $l_P$ | $1.616 \times 10^{-35}$ | $1$ |
| Compton (electron) | $\lambda_{C,e}$ | $2.426 \times 10^{-12}$ | $1.5 \times 10^{23}$ |
| Classical electron radius | $r_e$ | $2.818 \times 10^{-15}$ | $1.7 \times 10^{20}$ |
| Bohr radius | $a_0$ | $5.292 \times 10^{-11}$ | $3.3 \times 10^{24}$ |
| Nuclear scale | $\sim 10^{-15}$ | $10^{-15}$ | $6.2 \times 10^{19}$ |
| Atomic scale | $\sim 10^{-10}$ | $10^{-10}$ | $6.2 \times 10^{24}$ |
| Human scale | $\sim 1$ | $1$ | $6.2 \times 10^{34}$ |
| Earth radius | $R_\oplus$ | $6.371 \times 10^6$ | $3.9 \times 10^{41}$ |
| Solar System | $\sim 10^{12}$ | $10^{12}$ | $6.2 \times 10^{46}$ |
| Galactic scale | $\sim 10^{21}$ | $10^{21}$ | $6.2 \times 10^{55}$ |

**Table aa.4:** Standard length scales in natural units

### The T0 Length Scale

The T0 model introduces a sub-Planckian length scale:

**Definition aa.5.1** (T0 Length)**.**

$$r_0 = \xi \cdot l_P \tag{aa.29}$$

where $\xi \approx 1.33 \times 10^{-4}$ is a dimensionless parameter.

This gives:

$$r_0 = \xi \cdot l_P = 1.33 \times 10^{-4} \times 1.616 \times 10^{-35}\,\text{m} \tag{aa.30}$$

$$= 2.15 \times 10^{-39}\,\text{m} \tag{aa.31}$$

In natural units with $l_P = 1$:

$$r_0 = \xi \approx 1.33 \times 10^{-4} \tag{aa.32}$$

## aa.6 Unit Conversions

### Energy as Reference

Using the electronvolt (eV) as the practical energy unit:

| Physical Quantity | Conversion to SI | Example (1 GeV) |
|---|---|---|
| Energy | $1\,\text{eV} = 1.602 \times 10^{-19}\,\text{J}$ | $1.602 \times 10^{-10}\,\text{J}$ |
| Mass | $E(\text{eV}) \times 1.783 \times 10^{-36}\,\text{kg/eV}$ | $1.783 \times 10^{-27}\,\text{kg}$ |
| Length | $E(\text{eV})^{-1} \times 1.973 \times 10^{-7}\,\text{m eV}$ | $1.973 \times 10^{-16}\,\text{m}$ |
| Time | $E(\text{eV})^{-1} \times 6.582 \times 10^{-16}\,\text{s eV}$ | $6.582 \times 10^{-25}\,\text{s}$ |
| Temperature | $E(\text{eV}) \times 1.161 \times 10^{4}\,\text{K/eV}$ | $1.161 \times 10^{13}\,\text{K}$ |

**Table aa.5:** Conversion factors from natural to SI units

### Planck Scale Conversions

Converting between Planck units and SI:

## aa.7 Mathematical Framework

### Simplified Equations

In natural units, fundamental equations become elegantly simple:

| Planck Unit | Natural Value | SI Value |
|---|---|---|
| Length ($l_P$) | 1 | $1.616 \times 10^{-35}$ m |
| Time ($t_P$) | 1 | $5.391 \times 10^{-44}$ s |
| Mass ($m_P$) | 1 | $2.176 \times 10^{-8}$ kg |
| Energy ($E_P$) | 1 | $1.220 \times 10^{19}$ GeV |
| Temperature ($T_P$) | 1 | $1.417 \times 10^{32}$ K |

**Table aa.6:** Planck unit conversions

**Quantum Mechanics**

$$\text{Schrödinger equation:} \quad i\frac{\partial \psi}{\partial t} = H\psi \tag{aa.33}$$

$$\text{Uncertainty principle:} \quad \Delta E \Delta t \geq \frac{1}{2} \tag{aa.34}$$

$$\text{de Broglie relation:} \quad \lambda = \frac{1}{p} \tag{aa.35}$$

**Special Relativity**

$$\text{Mass-energy:} \quad E = m \tag{aa.36}$$

$$\text{Energy-momentum:} \quad E^2 = p^2 + m^2 \tag{aa.37}$$

$$\text{Lorentz factor:} \quad \gamma = \frac{1}{\sqrt{1 - v^2}} \tag{aa.38}$$

**General Relativity**

$$\text{Einstein equations:} \quad G_{\mu\nu} = 8\pi T_{\mu\nu} \tag{aa.39}$$

$$\text{Schwarzschild radius:} \quad r_s = 2M \tag{aa.40}$$

**Electromagnetism**

$$\text{Coulomb's law:} \quad F = \frac{q_1 q_2}{4\pi r^2} \tag{aa.41}$$

$$\text{Fine structure constant:} \quad \alpha = \frac{e^2}{4\pi} (\text{with } 4\pi\varepsilon_0 = 1) \tag{aa.42}$$

**Thermodynamics**

$$\text{Stefan-Boltzmann:} \quad j = \sigma T^4 \tag{aa.43}$$
$$\text{Wien's law:} \quad \lambda_{max} T = b \tag{aa.44}$$
$$\text{Boltzmann distribution:} \quad P \propto e^{-E/T} \tag{aa.45}$$

## aa.8 Advantages and Applications

### Advantages of Natural Units

- **Simplified equations** (e.g., $E = m$ instead of $E = mc^2$)
- **No superfluous constants** in calculations
- **Universal scaling** for fundamental physics
- **Reveals fundamental relationships** between physical quantities
- **Provides dimensional consistency** checks
- **Eliminates arbitrary conversion factors**
- **Highlights the universal role** of energy

### Disadvantages

- **Unintuitive for macroscopic applications**
- **Conversion to SI requires knowledge** of fundamental constants
- **Initial unfamiliarity** for those used to SI units
- **Engineering preference** for practical SI units

### Practical Applications

- Particle physics calculations
- Quantum field theory
- General relativity and cosmology
- High-energy astrophysics
- String theory and quantum gravity
- Fundamental constant relationships

## aa.9 Working with Natural Units

### Working with Natural Units

To convert a calculation from SI to natural units:
1. Express all quantities in terms of energy (eV or GeV)

2. Set $\hbar = c = G = k_B = 1$
3. Perform the calculation
4. Convert results back to SI if needed

### Dimensional Check

Always verify dimensional consistency:
- All terms in an equation must have the same energy dimension
- Check that exponents are consistent
- Use dimensional analysis to verify results

### Fundamental Forces in Natural Units

The four fundamental forces can be characterized by their dimensionless coupling constants:

| Force | Dimensionless Coupling | Typical Value | Range |
|---|---|---|---|
| Electromagnetic | $\alpha_{\text{EM}}$ | $\sim 1/137$ | $\infty$ |
| Strong | $\alpha_s$ | $\sim 0.118$ at $Q^2 = M_Z^2$ | $\sim 1 \times 10^{-15}$ m |
| Weak | $\alpha_W = g^2/(4\pi)$ | $\sim 1/30$ | $\sim 1 \times 10^{-18}$ m |
| Gravitation | $\alpha_G = Gm^2/(\hbar c)$ | $m^2/m_P^2$ | $\infty$ |

**Table aa.7:** Fundamental forces characterized by coupling constants

### Comprehensive Unit Conversions

| SI Unit | SI Dimension | Natural Dimension | Conversion | Accuracy |
|---|---|---|---|---|
| Meter | $[L]$ | $[E^{-1}]$ | $1\,\text{m} \leftrightarrow (1.973 \times 10^{-13}\,\text{MeV})^{-1}$; $1\,\text{fm} \leftrightarrow (197.3\,\text{MeV})^{-1}$ | $< 0.001\%$ |
| Second | $[T]$ | $[E^{-1}]$ | $1\,\text{s} \leftrightarrow (6.58 \times 10^{-22}\,\text{MeV})^{-1}$ | $< 0.00001\%$ |
| Kilogram | $[M]$ | $[E]$ | $1\,\text{kg} \leftrightarrow 5.61 \times 10^{29}\,\text{MeV}$ | $< 0.001\%$ |
| Ampere | $[I]$ | $[E]$ | $1\,\text{A} = 6.24 \times 10^{18}\,e/\text{s} \leftrightarrow 4.11 \times 10^{3}\,e \cdot \text{eV}$ ($\approx 1.24\,\text{keV}$ with $e = \sqrt{4\pi\alpha}$) | $< 0.005\%$ |
| Kelvin | $[\Theta]$ | $[E]$ | $1\,\text{K} \leftrightarrow 8.62 \times 10^{-5}\,\text{eV}$ | $< 0.01\%$ |
| Volt | $[ML^2T^{-3}I^{-1}]$ | $[E]$ | $1\,\text{V} \leftrightarrow 1\,\text{eV}/e$ | $< 0.0001\%$ |
| Coulomb | $[TI]$ | $[E^0]$ | $1\,\text{C} \leftrightarrow 6.24 \times 10^{18}\,e$ | $< 0.0001\%$ |

**Table aa.8:** Comprehensive unit conversions from SI to natural units

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# Chapter ab

# Unit Conventions and the Speed of Light c


## Abstract

This document examines when one can set c=1 (natural units) and when the full form E=mc² with c=299,792,458 m/s (SI units) is required. Parallel to the treatment of the fine-structure constant α in Document 101, we show: Both perspectives are mathematically equivalent and differ only in the choice of unit system. The T0 theory reveals that c is not a fundamental law of nature but a ratio L/T that stays the same across all mass scales. From the T0 perspective, c=1 can be set (Planck units, particle physics), while for technical applications and precision measurements, SI units with explicit c are required. The equivalence E=mc² ↔ E=m holds exactly in natural units. References: Documents 013 (SI system), 014 (nat./SI), 015 (systematics), 077 (E=mc² analysis), 101 (α conventions).


## ab.1 Introduction: The Question of c=1

### The Central Question

The question “When can one set c=1?” is analogous to the question “When can one set α=1?” addressed in Document 101. In both cases, it concerns **unit conventions**, not fundamental physics.

**Central Thesis**

**E=mc² and E=m are mathematically identical!**

- In SI units: $E = mc^2$ with $c = 299,792,458$ m/s
- In natural units: $E = m$ with $c = 1$

Both forms describe the same physics – only the unit choice differs.

### Historical Context

Einstein wrote the famous formula in 1905:

$$E = mc^2 \tag{ab.1}$$

In 1905 Einstein worked in the Gaussian/cgs framework (the SI exists only since 1960) and wrote the result as $m = L/V^2$. The form $E = mc^2$ is needed in every unit system in which length, time and mass are independent dimensions – today in the **SI** with meter, second and kilogram.

**Modern particle physics** uses instead:

$$E = m \quad \text{(in natural units with } c = \hbar = 1\text{)} \tag{ab.2}$$

## ab.2 Natural Units: When c=1 is Valid

### Definition of Natural Units

In natural units, one sets:

$$c = 1, \quad \hbar = 1, \quad \text{(optional: } k_B = 1\text{)} \tag{ab.3}$$

**Mathematical meaning:**

$$c = 1 \quad \Rightarrow \quad \text{Length} \equiv \text{Time} \tag{ab.4}$$

$$\hbar = 1 \quad \Rightarrow \quad \text{Energy} \equiv \text{inverse Time} \tag{ab.5}$$

### Application Domains

**Natural units are appropriate in:**

- **Planck scale**: Quantum gravity, fundamental theory
- **Particle physics**: High-energy physics, QFT, Standard Model
- **Cosmology**: Early universe, inflationary models
- **Theoretical work**: Mathematical derivations, symmetries

**Advantage:** Formulas become simpler, physical relationships clearer.

### Mathematical Consistency

In natural units:

$$E^2 = p^2 + m^2 \tag{ab.6}$$

In the rest frame ($p = 0$):

$$E = m \tag{ab.7}$$

This is exact – **not an approximation**.

### T0 Perspective: c as a Ratio

The T0 theory shows (see Document 077):

$$c = \frac{L}{T} \tag{ab.8}$$

**c is not a fundamental law of nature but a *ratio*!**
With the T0 fundamental relation:

$$T \cdot m = 1 \quad \text{(Time-Mass Duality)} \tag{ab.9}$$

and $L \propto 1/m$ (Compton wavelength) it follows that this ratio is mass-independent: c stays the same across all mass scales.

**Implication:** In Planck units, where $t_P = \ell_P/c$, c=1 is the natural choice.

## ab.3 SI Units: When c=299,792,458 m/s is Required

### The SI Definition (since 2019)

The modern SI system defines since 2019:

$$\boxed{c = 299,792,458 \text{ m/s (exact)}} \tag{ab.10}$$

This choice is a **convention** that defines the meter via the second.

### Application Domains

**SI units with explicit c are required in:**

- **Engineering**: GPS, telecommunications, laser technology
- **Precision measurements**: Atomic clocks, interferometry, metrology
- **Experimental physics**: Laboratory measurements with SI-calibrated devices
- **Applied physics**: Energy calculations, dosimetry
- **Public & Education**: Comprehensibility, historical continuity

**Advantage:** Practical calculability with calibrated measurement devices.

### Mathematical Form

In SI units:

$$E = \gamma mc^2 \tag{ab.11}$$

with the Lorentz factor:

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}} \tag{ab.12}$$

In the rest frame ($v = 0$, $\gamma = 1$):

$$E = mc^2 \tag{ab.13}$$

### Conversion Between Unit Systems

**From natural units to SI:**

$$E_{\text{nat}} = m_{\text{nat}} \tag{ab.14}$$
$$\Downarrow \quad \text{(Multiply by } c^2\text{)} \tag{ab.15}$$
$$E_{\text{SI}} = m_{\text{SI}} \cdot c^2 \tag{ab.16}$$

**Example:** Electron mass

$$m_e = 0.511 \text{ MeV} \quad \text{(natural units)} \tag{ab.17}$$
$$m_e = 9.109 \times 10^{-31} \text{ kg} \quad \text{(SI)} \tag{ab.18}$$
$$E_e = m_e c^2 = 0.511 \text{ MeV} = 8.187 \times 10^{-14} \text{ J} \tag{ab.19}$$

## ab.4 Comparison with α: Parallel Structure

### Two Analogous Conventions

| Convention | Fine-structure constant α | Speed of light c |
|---|---|---|
| **Natural** | $\alpha = 1$ (Heaviside-Lorentz with $e^2 = 4\pi$) | $c = 1$ (Planck units) |
| **SI / Standard** | $\alpha = 1/137.036$ (Gauss-SI) | $c = 299,792,458$ m/s |
| **Document** | 101 (Circularity-Constants) | 134 (Unit Conventions c) |

**Table ab.1:** Parallel structure: α and c as conventions
In Heaviside-Lorentz units alone $\alpha = e^2/4\pi \approx 1/137$; $\alpha = 1$ additionally requires $e^2 = 4\pi$.

### Common Principles

Both cases show:

- **Physics is invariant** under unit choice
- **Natural units** simplify theoretical work
- **SI units** enable practical applications
- **T0 theory**: Both are derived conventions, not fundamental

### T0 Reduction

From the T0 perspective (see Document 101):

$$\xi \to D_f \to E_0 \to \alpha \to \hbar, c, G \to \text{all other constants} \tag{ab.20}$$

**Only $\xi = \frac{4}{3} \times 10^{-4}$ is fundamental.**
Both $\alpha$ and $c$ are derived quantities or conventions.

## ab.5 When to Use Which System?

### Decision Matrix

| Context | Natural units (c=1) | SI units ($c$ explicit) |
|---|---|---|
| Theoretical physics | ✓ | |
| Quantum field theory | ✓ | |
| High-energy physics | ✓ | |
| Early cosmology | ✓ | |
| Experimental physics | | ✓ |
| Engineering | | ✓ |
| Precision measurements | | ✓ |
| Applied physics | | ✓ |
| Education | | ✓ |

**Table ab.2:** Application domains of unit systems

### Recommendations

**Use natural units (c=1) when:**

- Performing theoretical derivations
- Symmetries and invariant structures are important
- Formulas should be simplified
- Working in particle physics or cosmology

**Use SI units (c explicit) when:**

- Planning or evaluating experimental measurements
- Technical calculations are required
- Results should be understandable for non-physicists
- Historical continuity is important

## ab.6 Common Misconceptions

### "c=1 is only an approximation"

**FALSE.** c=1 is **exact** in natural units, not an approximation.

It is a choice of unit system that defines:

$$\text{Length unit} = \text{Time unit} \tag{ab.21}$$

Analogously: In Planck units, $\hbar = 1$ is exact, not approximate.

### “E=m only holds for photons”

**FALSE.** In natural units, $E = m$ holds for **all** particles in their rest frame.
For photons ($m = 0$): $E = p$ (in natural units) or $E = pc$ (in SI).

### “c is a fundamental constant of nature”

**T0 viewpoint**: c is a **ratio** $L/T$, not a fundamental constant.
With the T0 duality $T \cdot m = 1$ and $L \propto 1/m$, the mass scale cancels in the ratio:

$$c = \frac{L}{T} = L \cdot m = \text{mass-independent} \tag{ab.22}$$

Only in SI units is c *fixed by definition*.

### “Natural units change the physics”

**FALSE.** Physics is independent of the unit system.
All **dimensionless** quantities (e.g., $\xi$, $\alpha$, mass ratios) are invariant.
Only dimensional quantities change their numerical values.

## ab.7 T0 Perspective: c as a Ratio

### The T0 Fundamental Relation

From Document 077:

$$T \cdot m = 1 \quad \text{(Time-Mass Duality)} \tag{ab.23}$$

This means:

$$T \propto \frac{1}{m} \tag{ab.24}$$

$$L \propto \frac{1}{m} \quad \text{(via Compton wavelength)} \tag{ab.25}$$

$$\Rightarrow \quad c = \frac{L}{T} \propto \frac{1/m}{1/m} = \text{mass-independent} \tag{ab.26}$$

The quotient is mass-independent: $c$ stays the same across all mass scales.

### Implications

**1. c is the same across all mass scales in the T0 framework:**
Different effective c-values at different mass scales do not occur. Where a varying speed of light seems to suggest itself, the corpus instead describes *fractal path lengthening*: the geometric stretching of the propagation path in the $\xi$ field (Doc. 026, 149, 182), achromatic (K6, Doc. 190).

**2. SI definition c=299,792,458 m/s is a calibration:**
This fixation defines the meter via the second – a metrological convention.

**3. Natural units c=1 are T0-consistent:**
In Planck units, where $t_P \propto \ell_P$, c=1 is the natural choice.

### Comparison with Document 077

Document 077 argues: "E=mc² = E=m – The Constant Illusion Exposed"

**Clarification here:**

- E=mc² (SI) and E=m (natural) are *equivalent*, not identical
- The difference lies in the *unit system*, not in physics
- Einstein's c-fixation is a *convention*, not an error
- T0 shows: c is a ratio $L/T$ that stays the same across all mass scales

## ab.8 Mathematical Consistency

### Energy-Momentum Relation

**In natural units ($c = 1$):**

$$E^2 = p^2 + m^2 \tag{ab.27}$$

**In SI units:**

$$E^2 = (pc)^2 + (mc^2)^2 \tag{ab.28}$$

Both forms are mathematically equivalent.

### Lorentz Transformation

**In natural units:**

$$E' = \gamma(E - p \cdot v) \tag{ab.29}$$

**In SI units:**

$$E' = \gamma(E - v\,p_x) \tag{ab.30}$$

With $p$ in kg·m/s and $v$ in m/s, $v\,p_x$ already has the unit J; in SI the form is the same as in natural units.

The physics remains invariant.

### Klein-Gordon Equation

**In natural units:**

$$(\partial_\mu \partial^\mu + m^2)\phi = 0 \tag{ab.31}$$

**In SI units:**

$$\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2 + \frac{m^2c^2}{\hbar^2}\right)\phi = 0 \tag{ab.32}$$

Identical physics, different notation.

## ab.9 References to T0 Documents

### Related Documents

- **Document 013**: SI System and T0 Theory
- **Document 014**: Natural vs. SI Units
- **Document 015**: Systematics of Natural Units
- **Document 077**: E=mc² = E=m Analysis
- **Document 101**: Circularity of Constants (α Conventions)
- **Document 133**: Fractal Correction K_frak Derivation

### Derivation Hierarchy

The T0 hierarchy (from Document 101):

$$\xi \to D_f \to E_0 \to \alpha \to \hbar, c, G \to \text{mass ratios} \tag{ab.33}$$

shows that both $\alpha$ and $c$ are derived quantities.

# Chapter ac

# The Circularity in the Debate on Fundamental Constants


## Abstract

This document illuminates the debate on fundamental physical constants: Why the question of what counts as a constant, convention, or measurement data often runs in circles. The assignment has grown historically, is framework-dependent, and strongly shaped by conventions. It explains how this circularity arises and how it can be resolved through modern approaches: Matsas et al. (2024) show that in relativistic spacetimes, operationally, a single time unit suffices; the T0 theory reduces everything to a geometric parameter $\xi = \frac{4}{3} \times 10^{-4}$, derived from the tetrahedral packing structure of spacetime. The focus is on a clear, factual analysis showing that only *one* dimensionless parameter is fundamental – all other constants are conventions, measurement data, or derived ratios. The analysis is secured up to the sub-Planck limit $L_0 = \xi \ell_P \approx 2.155 \times 10^{-39}$ m.


## ac.1 Introduction: The Question of Fundamentality

The debate “How many fundamental constants does physics really need?” and “What is fundamental, what is convention, and what is measurement data?” is a central topic in the philosophy of physics and metrology. It often appears circular because the answer depends on the chosen theoretical and metrological framework. The historical development of physics has strongly shaped the assignment — an observation that is accurate but does not capture the entire complexity.

## ac.2 The Circularity of the Debate

### Preliminary Note: The Central Thesis

After complete analysis, only **one single dimensionless parameter** remains fundamental: $\xi = \frac{4}{3} \times 10^{-4}$. All other constants ($c$, $\hbar$, $G$, $\alpha$, mass ratios) are conventions, derived ratios, or still open measurement data.

### Definitions and Criteria

A constant is considered **fundamental** if it:

- is independent of other quantities,
- cannot be further reduced,
- is universal and theoretically necessary.

Conventions are human choices (e.g., units), measurement data are empirical values.

The problem: These criteria are not absolute but **framework-dependent**. Changing the framework changes the classification — a circle.

- In non-relativistic physics: Three independent dimensions (time, length, mass) → three constants needed.
- In relativistic spacetime: Length derivable from time → one constant (e.g., time unit) suffices.
- In geometric approaches: Everything emerges from structure → zero or one parameter.

To decide whether, e.g., $c$ is fundamental, one already needs a framework — circularity.

### Examples of Circularity

1. **Measurement vs. Definition**: The gravitational constant $G$ is measured, but using masses and lengths that in turn depend on $c$ and $\hbar$. In the SI reform of 2019, $c$, $\hbar$ were fixed (convention), $G$ remains measurable.
2. **Dimensional vs. Dimensionless**: Dimensional constants (e.g., $c$) can be set to 1 by unit choice → convention. Dimensionless ones like $\alpha \approx 1/137$ seem "truly" fundamental — until derived.
3. **Operational vs. Ontological**: Operationally (measurement practice), one clock suffices (Matsas et al., 2024). Ontologically (what exists?), geometry might explain everything.

## ac.3 The Historical Assignment

The assignment is largely historically determined:

- Newton (1687): $G$ as empirical constant.
- 19th century: $c$ from electromagnetism.

- Planck (1899): Natural units with $c$, $\hbar$, $G$.
- SI system: Historical artifacts (e.g., kilogram prototype until 2019).
- Duff–Okun–Veneziano controversy (2000s): Arising from quantum field theory and string theory.

Physics developed stepwise (mechanics → electromagnetism → quantum → relativity), so constants act as "bridges" and appear fundamental — a pragmatic, historical decision.

## ac.4 Resolution of the Circularity

Modern approaches break the circle:

- **Matsas et al. (2024)**: Shows framework-dependently that in relativistic spacetimes, operationally, one time unit suffices (three-clock experiment, Compton relation).
- **T0 Theory**: Reduces everything to a geometric parameter $\xi = \frac{4}{3} \times 10^{-4}$, derived from packing principles — not historical/conventional, but structurally grounded.

These approaches make fundamentality less circular by reducing to deeper levels (spacetime structure, geometry).

### The Mathematical Hierarchy of the T0 Theory

The T0 theory establishes an unambiguous derivation chain from the geometric parameter $\xi$. The order is crucial and based on the structure of the theory:

1. **Starting Point – Geometry**: The tetrahedral packing deficit

$$\xi = 1 - \frac{V_{\text{Tetradouble}}}{V_{\text{Sphere}}} = \frac{4}{3} \times 10^{-4}$$

   on the Planck scale is the only fundamental parameter (see document 009_T0_xi_ursprung).

2. **Fractal Dimension**: From $\xi$ follows the fractal dimension

$$D_f = 3 - \xi \approx 2.9998667$$

   This is *not freely choosable* – there is only one single way to determine $D_f$ from the geometric structure. The justification is found in documents 008_T0_xi-und-e and 009_T0_xi_ursprung.

3. **Characteristic Energy Scale**: $E_0$ emerges from the lepton masses, namely from the electron and muon masses:

$$E_0 = \sqrt{\frac{m_e m_\mu}{K_{\text{frak}}}} = 7.398 \text{ MeV}$$

   Detailed derivations are found in document 006_T0_Teilchenmassen.

4. **Fine-Structure Constant**: Derivable from $E_0$ and the fractal dimension $D_f$:

$$\alpha = f(E_0, D_f) \approx \frac{1}{137.036}$$

   Here $f$ denotes a dependence that is not written out, not a frequency and not the base winding number $f = 1/\xi$. The explicit formula and justification is found in document 011_T0_Feinstruktur.

5. **Planck Constant and Speed of Light**: Emerge from the time-mass duality

$$\hbar = m_P c^2 t_P \quad \text{and} \quad c = \frac{\ell_P}{t_P}$$

   where $t_P$ is the Planck time.

6. **Gravitational Constant**: Follows from geometric definition

$$G = \frac{\ell_P c^2}{m_P} = \frac{\ell_P^3}{t_P^2 m_P}$$

7. **Elementary Charge**: Via the fine-structure constant

$$e^2 = 4\pi\alpha\epsilon_0\hbar c$$

8. **Mass Ratios**: Systematically derivable from fractal hierarchy and $\xi$-couplings (see document 006_T0_Teilchenmassen).

This hierarchy shows: Only $\xi$ is fundamental. The fractal dimension $D_f$ is uniquely determined (not freely choosable), $E_0$ follows from the lepton masses, $\alpha$ from $E_0$ and $D_f$, and everything else follows mathematically or is convention in unit choice.

## ac.5 Why Ratio-Based Relations Do Not Need Units

Ratio-based relations — such as dimensionless constants or ratios of physical quantities — do not require units as long as no real applications with human-made units (e.g., SI units) are realized. The following explanation shows why this is the case.

### Basic Principle: Dimensionlessness

Ratios are by definition **dimensionless**: They arise from dividing similar quantities, canceling all physical dimensions. Examples:

- Fine-structure constant $\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} \approx \frac{1}{137.036}$: The dimensions of charge, Planck's constant, speed of light, and permittivity fully cancel → pure number.
- Proton-electron mass ratio $m_p/m_e \approx 1836.15$: Both quantities have dimension [mass] → the ratio is dimensionless.
- Koide formula for lepton masses: $\frac{m_e+m_\mu+m_\tau}{(\sqrt{m_e}+\sqrt{m_\mu}+\sqrt{m_\tau})^2} = \frac{2}{3} + \mathcal{O}(10^{-5})$ — again a pure number.

Such ratios are invariant to the choice of unit systems. Their numerical value remains the same, whether using SI, Planck, or natural units.

### Pure Theory vs. Practical Application

In a purely theoretical description of nature — as long as no concrete measurement or technical application with human-made standards is performed — it is entirely sufficient to work exclusively with ratios and dimensionless quantities.

- All physical laws can be written in dimensionless form (Buckingham $\pi$-theorem).
- The entire dynamics of a system is determined by ratios of masses, charges, coupling constants, etc.
- Dimensional constants like $c$, $\hbar$, or $G$ serve in this context merely as conversion factors between different dimensions — they are not substantively necessary.

Example: The equations of motion in general relativity or quantum field theory can be formulated so that only dimensionless parameters appear. The choice of a time unit or length unit is then pure convention.

### The Transition to Realization with Human-Made Units

Units become relevant only when linking the theory to the real world:

- **Metrology**: To specify a length in meters, one needs an operationally defined standard (e.g., speed of light $c$ and second).
- **Technical Applications**: Building devices, communicating measured values, comparing with experiments require common, human-made units.
- **SI Reform 2019**: Here, dimensional constants ($c$, $\hbar$, $e$, $k_B$) were deliberately fixed to exact values to define units — a clear indication that these constants serve as conventions.

Without this step of realization, all physical statements remain unit-free and depend only on ratios.

## ac.6 Dimensional Quantities Can Be Converted to Dimensionless Ones

Dimensional constants or quantities can be converted to dimensionless quantities through appropriate redefinition of units if the new units exclusively reflect the underlying ratio. This shows that the apparent “fundamentality” of dimensional constants is often just a question of the chosen unit convention.

### Basic Idea: Natural Unit Systems

By choosing a unit system in which certain physical constants receive the value 1, these constants are eliminated from the equations and lose their dimension. Classic examples:

- **Planck Units**: Here, $c = \hbar = G = k_B = 1$ are set. Length, time, mass, and temperature thereby receive the dimensions of Planck scales. All equations become dimensionless except for any remaining dimensionless parameters.

- **Natural Units in Particle Physics**: Often $c = \hbar = 1$. Energy, mass, momentum, and inverse length/time then have the same dimension. The speed of light $c$ and Planck's constant $\hbar$ disappear from the formulas.
- **Heaviside-Lorentz Units**: $\epsilon_0 = 1$, whereby the fine-structure constant $\alpha = e^2/(4\pi)$ and charges appear dimensionless.

In such systems, the formerly dimensional constants ($c$, $\hbar$, $G$, $\epsilon_0$) are no longer independent quantities — they are fixed to 1 by unit choice.

### General Principle

Any dimensional constant $K$ with dimension $[K] = [L]^a[T]^b[M]^c \ldots$ can be eliminated by defining a new unit that carries exactly this dimension and uses $K$ as the reference value. Example:

- Instead of measuring $c = 299{,}792{,}458$ m/s, one fixes this numerical value and defines the meter through it (SI 1983, retained in 2019); $c$ keeps its dimension there. Setting in addition $c \equiv 1$ is the choice of natural units. Result: $c$ is no longer a measurable constant but a definitional convention; it becomes dimensionless only with $c \equiv 1$.
- Similarly, one could set $G = 1$ by introducing a "Planck mass" unit — $G$ would then become dimensionless.

The result is always the same: The constant disappears from the physical laws and becomes a pure unit convention.

### Consequence for the Fundamentality Debate

This clearly shows why dimensional constants are seen as less fundamental in the Duff–Okun–Veneziano controversy (Duff position):

- They can be eliminated by unit choice.
- Only the remaining **dimensionless** parameters (e.g., $\alpha$, mass ratios, Yukawa couplings) are invariant to unit changes.
- These dimensionless ratios are the actual free parameters of nature — everything else is convention.

In the T0 theory, this thought is taken to its radical conclusion: Even the dimensionless constants like $\alpha$ or mass ratios are not considered free but derived from a single geometric parameter $\xi$. Thus, ultimately, any arbitrary unit choice is eliminated — the structure of nature is completely described by a single dimensionless ratio.

## ac.7 The Equivalence of $\alpha$ and $\xi$ in the T0 Theory

From the perspective of the T0 theory, the fine-structure constant $\alpha$ — traditionally viewed as one of the "truly" fundamental dimensionless constants — is nothing but a ratio equivalent to the geometric parameter $\xi$. This shows that the question "What is fundamental?" ultimately depends on the chosen starting basis: $\xi$ and $\alpha$ are two equivalent descriptions of the same underlying fact.

## The Bidirectional Derivation

In the T0 theory, there are multiple consistent and mathematically equivalent formulations:

1. **Start from $\xi$ (geometric perspective — preferred in T0):**

   $$\xi = \frac{4}{3} \times 10^{-4}$$

   is the primary parameter describing the ratio between tetrahedral and spherical packing on the Planck scale. From this, $\alpha$ is derived:

   $$\alpha = \xi \cdot E_0^2,$$

   where $E_0 = \sqrt{m_e m_\mu / K_{\text{frak}}} = 7.398$ MeV. Numerically, $1/\alpha = 137.035$ follows, $0.0005\,\%$ below the measured value.

   Here, $\xi$ is fundamental (geometrically justified), $\alpha$ a derived ratio.

2. **Start from $\alpha$ (electromagnetic perspective):** $\alpha \approx \frac{1}{137.036}$ is taken as the starting point. From this, $\xi$ is calculated backward:

   $$\xi = \frac{\alpha}{E_0^2}.$$

   With $E_0 = 7.398$ MeV one obtains $\xi = 1.33333 \times 10^{-4}$, i.e. $\frac{4}{3} \times 10^{-4}$ to $5 \times 10^{-6}$.

   In this formulation, $\alpha$ appears fundamental, while $\xi$ becomes the derived ratio.

Both ways lead to identical predictions for all other constants ($c$, $\hbar$, $G$, masses, etc.). The theories are mathematically equivalent.

## Interpretation: There Is Only One Fundamental Ratio

The T0 theory demonstrates a deep symmetry:

- $\xi$ and $\alpha$ are two sides of the same coin — they encode the same fundamental ratio in the structure of spacetime.
- The choice of which is considered "fundamental" is a matter of perspective:
  - Geometric: $\xi$ is primary (packing deficit on Planck scale).
  - Electromagnetic/phenomenological: $\alpha$ appears primary (strongest coupling in everyday life).
- In both cases, exactly **one** dimensionless parameter remains — there are no two independent fundamental constants.

This further breaks down the classic Duff position (only dimensionless constants are fundamental): Even among the dimensionless constants, there is no true independence — they are interconnected and reduce to a single degree of freedom.

## Consequence for the Fundamentality Debate

- Traditionally, $\alpha$ is considered one of the few "truly" fundamental constants because it is dimensionless and not changeable by unit choice.

- The T0 theory shows: Even this dimensionlessness is not absolute. $\alpha$ is a ratio that follows from a deeper geometric structure ($\xi$) — or vice versa.
- Ultimately, only one true degree of freedom remains: the fundamental packing ratio of spacetime, which can be expressed both as $\xi$ (geometric) and as $\alpha$ (electromagnetic).

## ac.8 The Important Limitation: Lower Bound of Validity for Relative Relations

Despite the far-reaching reduction to ratios and dimensionless parameters, there is a fundamental limitation: The described rules and equivalences apply only above a certain scale. Everything below — especially in the sub-Planck region — must be considered speculation.

### The Classical and Quantum Mechanical Limit

In established physics (special and general relativity, quantum field theory), all ratios and dimensionless constants are based on the assumption of continuous or at least operationally accessible spacetime down to the Planck scale:

- Planck length: $\ell_P = \sqrt{\frac{\hbar G}{c^3}} \approx 1.616 \times 10^{-35}$ m
- Planck time: $t_P = \sqrt{\frac{\hbar G}{c^5}} \approx 5.391 \times 10^{-44}$ s

Below these scales, we expect effects of quantum gravity, where the classical concepts of space, time, and measurability break down. Clocks can no longer operate arbitrarily precisely (quantum noise, Heisenberg uncertainty in gravity), and the assumptions of the three-clock experiment or the Compton relation lose their validity.

Matsas et al. (2024) and the concept of single-clock metrology implicitly assume that measurements with arbitrary accuracy are possible — an assumption that fails exactly at the Planck scale.

### The T0-Specific Lower Bound: The Sub-Planck Scale $L_0 = \xi \ell_P$

The T0 theory explicitly addresses this limit and defines an absolute lower bound for the continuous description:

- The geometric parameter $\xi = \frac{4}{3} \times 10^{-4}$ describes a packing deficit.
- From this, a characteristic sub-Planck length emerges: $L_0 = \xi \ell_P \approx 2.155 \times 10^{-39}$ m
- Below $L_0$, spacetime becomes discrete and granular — a fractal structure with dimension $D_f = 3 - \xi \approx 2.9996$.

Above $L_0$ and $\ell_P$, all relative relations, dimensionless ratios, and the equivalence of $\xi$ and $\alpha$ apply unrestrictedly. The theory is predictive here and consistent with all known experiments.

Below $L_0$, however:

- The continuous spacetime metric breaks down.

- Classical concepts like proper time, Compton wavelength, or light cone lose their meaning.
- Measurement protocols (e.g., three-clock experiment) are no longer feasible.
- All statements about ratios or constants become speculative.

The T0 theory makes specific proposals for the structure below $L_0$ (discrete tetrahedral packing, emergent dynamics), but these remain hypothetical and currently experimentally unverifiable.

### Consequence for the Fundamentality Debate

- The reduction to ratios and a single parameter ($\xi$ or equivalently $\alpha$) is robust and valid in the entire observable universe — from cosmological scales down to the Planck scale.
- However, it is tied to the validity of continuous spacetime.
- Below the sub-Planck limit $L_0$, the classical and quantum field theoretical description ends, and with it the certainty of all relative relations.
- Any claim that "everything is just geometry" or "just one parameter" therefore strictly applies only above this limit. Below, we enter the realm of speculation — regardless of whether one prefers string theory, loop quantum gravity, or T0 geometry.

## ac.9 Black Holes and the Limits of Speculation

From the perspective of the limitations outlined so far, many common statements about black holes are indeed without a secured physical basis and must be classified as speculation. This follows directly from the existence of the sub-Planck lower bound and the associated inapplicability of our established theories in the extreme regime.

### The Singularity and the Horizon as Problematic Areas

A black hole is classically described by the Schwarzschild solution of general relativity:

- Event horizon at $r_s = \frac{2GM}{c^2}$
- Central singularity at $r = 0$

However, fundamental problems arise already upon reaching the horizon and especially in its interior:

- Near the singularity, curvature scales become smaller than the Planck length $\ell_P$.
- Our current theories (GR + quantum field theory on curved spacetime) break down exactly in this regime.

Thus, both the classical singularity and quantum field theoretical effects like Hawking radiation lie partially beyond the secured lower bound $L_0 \approx \xi \ell_P$.

### Which Statements Are Secured, Which Speculative?

- **Secured (outside the lower bound):**
  - The existence of compact objects with event horizon (observed through gravitational waves, shadow images like M87* and Sgr A∗, accretion disks).
  - The external geometry (Schwarzschild or Kerr metric) up to near the horizon.
  - Gravitational redshift and time dilation for distant observers.
- **Speculative (within or beyond the lower bound):**
  - The nature of the singularity (point, ring, "fuzzball", Planck star, etc.).
  - The fate of information falling into the black hole (information paradox).
  - The interior of the horizon: Is there a firewall? A smooth spacetime? A transition to another universe?
  - Hawking radiation in its complete form (although semi-classically calculable, it requires consistent quantum gravity to resolve the paradox).
  - Exotic concepts like wormholes, white holes, or "black hole remnants" as solutions to the information problem.

All these points lie in the area where classical and quantum field theoretical descriptions are no longer trustworthy — exactly where the sub-Planck structure (in T0: $L_0$) becomes relevant.

### The Consequence from the T0 Perspective

The T0 theory postulates a discrete, tetrahedrally packed spacetime below $L_0$. Thus:

- There can be no true singularity — granularity prevents infinite curvature.
- The event horizon remains as a macroscopic boundary, but its microscopic structure is determined by $\xi$.
- Processes like information loss or Hawking radiation would have to be derived from the emergent dynamics of the $\xi$-geometry — which has so far only been done in outline.

Even in T0, detailed statements about the interior of black holes remain speculative, as we have no experimental access to scales below $L_0$ and no direct observations from inside the horizon.

## ac.10 Note: Mass Variation Instead of Time Dilation as an Alternative

An often overlooked but physically equivalent perspective on relativistic effects is the interpretation of time dilation as apparent mass variation — or vice versa. This view fits particularly well with the T0 theory and its explicit time-mass duality and avoids some conceptual difficulties of the usual "time slows down" representation.

## The Equivalence of the Descriptions

In special relativity, two closely linked effects occur for a moving object:

- **Time Dilation**: Proper time $\Delta\tau$ passes slower than coordinate time $\Delta t$:

$$\Delta\tau = \Delta t\sqrt{1 - v^2/c^2} = \frac{\Delta t}{\gamma}$$

- **Rest Mass Remains Invariant, Relativistic Mass Increases**: Formerly (before ca. 1970), the relativistic mass $m_{\text{rel}} = \gamma m_0$ was often introduced, where $m_0$ is the rest mass.

Today, relativistic mass is mostly avoided, and instead the four-momentum $p^\mu = (E/c, \mathbf{p})$ with $E = \gamma m_0 c^2$ is used. Nevertheless, both descriptions are mathematically equivalent.

Crucial: The time dilation for a moving system can be exactly described by an apparent increase in inertial mass — and vice versa.

## The T0 Perspective: Time-Mass Duality

In the T0 theory, this equivalence is elevated to a fundamental duality (see documents `T0_xi-und-e_De` and `T0_SI_De`):

- The relation $T \cdot m =$ constant (in natural units) is interpreted as an expression of a deep symmetry.
- Relativistic effects are not primarily "slowing of time" but a redistribution between temporal and mass manifestation of the same underlying geometric structure.
- Moving objects appear heavier (greater inertial mass) because part of the "time resource" is converted into mass — analogous to the energy-mass equivalence $E = mc^2$.

Advantages of this view:

- It avoids the anthropocentric image "time passes differently" and emphasizes the symmetry between time and mass.
- It is more consistent with the Compton relation $\lambda_C = h/(mc)$, which represents mass directly as inverse time/frequency.
- In the metrological discussion (Matsas et al., single-clock approach), mass is defined via frequencies (time) anyway — a mass variation is then more natural than a time variation.

## Practical and Philosophical Consequences

- **In Gravity**: The gravitational redshift and time dilation in the gravitational field can alternatively be interpreted as location-dependent mass variation — fitting the T0 idea that gravity is a manifestation of $\xi$-geometry.
- **Philosophically**: The usual emphasis on time dilation suggests an asymmetry (time is "special"). The mass perspective restores symmetry and underscores that time and mass are two sides of the same coin.

- **In Quantum Mechanics**: The de Broglie wavelength $\lambda = h/p$ and the relativistic energy-momentum relation make it clear that higher velocity (higher $p$) means both shorter wavelength and apparently higher mass — again a duality.

## ac.11 The Mass Variation Perspective on Black Holes

Although all descriptions of processes below the sub-Planck limit $L_0 = \xi \ell_P$ remain speculative, the discussion of what happens in black holes can still be meaningfully considered as an alternative view of mass variation. This arises from the time-mass duality of the T0 theory and the equivalence between time dilation and mass variation in relativistic contexts. This perspective is explained here without claiming final validity below the limit.

### Gravitational Effects as Mass Variation

In general relativity, the gravitational time dilation near a black hole is classically described as a slowing of proper time:

$$\Delta\tau = \Delta t \sqrt{1 - \frac{r_s}{r}},$$

where $r_s = 2GM/c^2$ is the Schwarzschild radius.

From the alternative view of mass variation — analogous to relativistic mass increase — this can be reinterpreted:

- Near the horizon, gravity acts as an effective increase in the inertial (and gravitational) mass of all objects.
- A particle or clock at the edge of the horizon appears "heavier" to a distant observer — not because time slows down, but because the mass varies due to curved geometry.
- Mathematically equivalent to time dilation, since from the T0 duality $T \cdot m =$ constant: A slowing of time corresponds to a proportional increase in mass.

This view avoids the image of a "frozen" time at the horizon and emphasizes instead a continuous variation of the mass scale depending on local curvature.

### Application to Black Holes: Interior as Mass Variation

The discussion of the interior of a black hole — including singularity, information loss, and Hawking radiation — can be reformulated in this perspective:

- **Singularity as Maximum Mass Density**: Instead of infinite curvature (time stop), the singularity could be interpreted as a point of infinite mass variation. In the T0 theory, granularity below $L_0$ prevents true infinity — mass reaches an upper limit through $\xi$-packing.
- **Hawking Radiation as Mass Decay**: The radiation (virtual pairs at the horizon) can be seen as fluctuation of variable mass, not as a time effect. The distant observer sees a slow mass loss of the hole, consistent with the duality.

- **Information Paradox as Mass Conversion**: The apparent violation of unitarity (loss of information) could be viewed as conversion of information into variable mass — a perspective that could be resolved in T0 geometry through emergent entropy from $\xi$-fluctuations.

This alternative formulation is equivalent to the standard description above the limit and offers conceptual advantages: It integrates gravity more naturally into the time-mass duality and avoids absolute time concepts.

### The Speculative Nature Below the Limit

Despite these advantages, the application to the interior of black holes remains speculative:

- The horizon and interior lie for real black holes (masses $\gg$ Planck mass) macroscopically, but the relevant effects (e.g., Hawking temperature) scale with $1/M$, requiring quantum gravity.
- Below $L_0$ (near the singularity), all continuous descriptions break down — whether formulated as time dilation or mass variation.
- The T0 theory proposes a discrete geometry where neither time nor mass exists in the classical sense — any discussion about it is hypothetical.

Nevertheless, it is justified to consider the mass variation view as an alternative: It is mathematically equivalent and could be the preferred formulation in a future quantum gravity theory (e.g., based on T0).

## ac.12 Note: $\alpha$ Can Also Be Set to 1

An often overlooked consequence of the pure ratio perspective is that even the fine-structure constant $\alpha$ — traditionally viewed as one of the few unchangeable dimensionless fundamental constants — can be set to the value 1 through an appropriate choice of units. This shows that even $\alpha$ ultimately possesses no absolute fundamentality but can also be a question of convention.

### The Principle of Natural Electromagnetic Units

In theoretical physics, there are already unit systems in which $\alpha$ can be set to $1$ by the choice of the charge unit: In Heaviside-Lorentz units alone, $\alpha = e^2/4\pi \approx 1/137$; $\alpha = 1$ additionally requires $e^2 = 4\pi$, as set out below.

- **Heaviside-Lorentz Units** (common in classical electrodynamics and quantum field theory): Here, the vacuum permittivity $\epsilon_0 = 1$ is set (and often also $\hbar = c = 1$). Thereby, the definition of $\alpha$ simplifies: $\alpha = \frac{e^2}{4\pi}$ The charge $e$ now becomes dimensionless, and the coupling constant is directly the numerical factor before the charge term.
- **Further Natural Units**: One can additionally define the elementary charge $e$ such that $e^2 = 4\pi \quad \Rightarrow \quad \alpha = 1$ This is mathematically completely equivalent to the usual choice $\alpha \approx 1/137$. The physical laws remain unchanged; only the numerical representation of the charge changes.

- **Stueckelberg Units** or other gauge-theoretical systems: In some formulations of quantum electrodynamics, the coupling is directly normalized to 1, and the "true" fine-structure constant appears only in renormalization or transition to other scales.

In these units, $\alpha$ disappears as an independent parameter from the equations — just like $c$ or $\hbar$ in Planck units.

### Consequence for the Fundamentality Debate

- The Duff position (only dimensionless constants are fundamental) is thereby relativized: Even the most prominent dimensionless constant $\alpha$ can be eliminated by unit choice.
- The numerical value $\alpha \approx 1/137$ is no ontological necessity but a consequence of our chosen unit convention (SI-based, with $\epsilon_0$, $e$, $\hbar$, $c$ as separate quantities).
- In a purely theoretical, unit-free description of nature, there is no reason why $\alpha$ could not be 1 — the observed value is then merely a question of the scale at which we link the theory to reality.

### The T0 Perspective: $\alpha$ as Derived Ratio

The T0 theory goes one step further:

- $\alpha$ is not even a free parameter that would have to be set to 1 — it is directly derived from the geometric parameter $\xi$: $\alpha = \xi \cdot E_0^2$
- The numerical value $1/137$ is no convention but a necessary consequence of the tetrahedral packing structure ($\xi$) and the lepton masses ($E_0$).
- A choice of units with $\alpha = 1$ would be possible but would obscure the underlying geometry $\xi$ — similar to setting $c = 1$ hiding the relativistic structure.

Thus, in T0, $\alpha$ is neither fundamental nor arbitrarily settable to 1 without loss of information: It carries the signature of Planck-scale geometry.

## ac.13 Literatur

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# Chapter ad

# The T0 Model: A Causal Theory of Conjugate Base Quantities with Applications to Ampère Force, Longitudinal Modes, and Geometry-Dependent Scaling

## Abstract

This paper introduces the T0 model, an extended classical field theory based on the principle of local conjugation of base quantities (time–mass, length–stiffness, energy–density). This conjugation acts as a fundamental constraint, while the dynamics of the associated deviations $\sigma_i$ obey causal wave equations. The theory naturally couples electromagnetic currents to the geometry of the conductor, explaining the existence of longitudinal force components, the Ampère helix anomaly, the nonlinear $I^4$ scaling of the force at high currents, and the fractal scaling $F \propto r^{2D_f-4}$ without violating causality. All apparent instantaneous effects are identified as local constraint fulfillment, while observable forces are fully retarded.

## ad.1 Introduction

Maxwell's theory of electrodynamics is one of the most successful theories in physics. However, experimental investigations of forces between currents, particularly in complex conductor geometries, reveal systematic deviations that suggest additional physical mechanisms. Observed longitudinal force components [1], the nonlinear dependence of force strength on current [2], and geometry-dependent effects such as the Ampère helix anomaly [3] cannot be fully explained within the conventional framework.

This paper presents the T0 model, a novel theoretical framework that accounts for these phenomena by introducing conjugate base quantities. The core of the theory is

the assumption of fundamental constraints between physical base quantities, whose dynamics are described by deviation fields that obey causal wave equations.

## ad.2 The Principle of Local Conjugation

### Fundamental Constraints

The T0 model postulates that physical base quantities at each spacetime point $(x,t)$ are linked by local conjugation conditions:

$$T(x,t)\cdot m(x,t)=1 \quad \text{with } [T]=\mathsf{s},[m]=1/\mathsf{s} \tag{ad.1}$$

$$L(x,t)\cdot \kappa(x,t)=1 \quad \text{with } [L]=\mathsf{m},[\kappa]=1/\mathsf{m} \tag{ad.2}$$

$$E(x,t)\cdot \rho(x,t)=1 \quad \text{with } [E]=\mathsf{J},[\rho]=1/\mathsf{J} \tag{ad.3}$$

These equations are to be interpreted as **local constraints**. A change in one quantity on the left side enforces an immediate, purely local redefinition of the conjugate quantity on the right side to satisfy the equation. This process is analogous to gauge fixing in electrodynamics and involves.

### Dynamic Deviations

To make these constraints dynamic, we introduce a deviation field $\sigma_i(x,t)$ for each pair, describing small permissible deviations:

$$T\cdot m = 1+\sigma_{Tm} \tag{ad.4}$$

$$L\cdot \kappa = 1+\sigma_{L\kappa} \tag{ad.5}$$

$$E\cdot \rho = 1+\sigma_{E\rho} \tag{ad.6}$$

The dynamics of these $\sigma$-fields are described by an action that penalizes deviations from the ideal value $\sigma_i=0$:

$$\mathscr{L}_\sigma=\sum_i\left[\frac{1}{2}(\partial_\mu\sigma_i)(\partial^\mu\sigma_i)-\frac{\mu_i^2}{2}\sigma_i^2\right] \tag{ad.7}$$

Critically, the $\sigma_i$ obey **causal Klein-Gordon equations**:

$$(\Box+\mu_i^2)\sigma_i(x,t)=0 \tag{ad.8}$$

so that perturbations of these fields propagate at speeds $v\le c$.

## ad.3 The Action of the T0 Model

The complete Lagrangian density of the T0 model consists of several components:

$$\mathscr{L}=\mathscr{L}_{\text{EM}}+\mathscr{L}_\sigma+\mathscr{L}_{\text{int}}+\mathscr{L}_{\text{constraint}} \tag{ad.9}$$

where:

- $\mathscr{L}_{\text{EM}} = -\frac{1}{4\mu_0} F_{\mu\nu}F^{\mu\nu}$ is the Maxwell Lagrangian density
- $\mathscr{L}_\sigma$ describes the kinematics of the deviations (Eq. ad.7)
- $\mathscr{L}_{\text{int}}$ describes the coupling between currents and deviations
- $\mathscr{L}_{\text{constraint}}$ softly enforces the constraints

### Interaction Term

The key innovation is the nonlinear coupling term:

$$\mathscr{L}_{\text{int}} = -J^\mu A_\mu - \frac{g}{\mu_0 c^2} J^\mu J_\mu \sigma_{Tm} \tag{ad.10}$$

The term $J^\mu J_\mu = \rho^2 - \mathbf{j}^2$ is a Lorentz invariant. For a thin conductor, the spatial part $-\mathbf{j}^2 \propto -I^2$ dominates. This term describes how the electric current perturbs the local time-mass balance (exciting $\sigma_{Tm}$).

### Complete Form with Lagrange Multipliers

The constraints are enforced by Lagrange multiplier fields $\lambda_i(x,t)$:

$$\mathscr{L}_{\text{constraint}} = \lambda_{Tm}(x,t)(T \cdot m - 1 - \sigma_{Tm}) + \lambda_{L\kappa}(x,t)(L \cdot \kappa - 1 - \sigma_{L\kappa}) + \cdots \tag{ad.11}$$

## ad.4 Derivation of the Field Equations

### Variation with Respect to the Potentials

Variation with respect to $A_\mu$ yields the modified Maxwell equation:

$$\partial_\mu F^{\mu\nu} = \mu_0 J^\nu + \mu_0 \frac{g}{\mu_0 c^2} \partial_\mu (J^\mu J^\nu \sigma_{Tm}) \tag{ad.12}$$

The additional term describes the current feedback through the deviation. For slowly varying currents, this term can be approximated as:

$$\partial_\mu F^{\mu\nu} \approx \mu_0 J^\nu + \frac{g}{c^2} \sigma_{Tm} \partial_\mu (J^\mu J^\nu) \tag{ad.13}$$

### Variation with Respect to the Deviations

Variation with respect to $\sigma_{Tm}$ yields the wave equation with a source term:

$$(\Box + \mu_{Tm}^2)\sigma_{Tm} = -\frac{g}{\mu_0 c^2} J^\mu J_\mu \tag{ad.14}$$

This is a **retarded** equation. The deviation $\sigma_{Tm}$ generated by a current $J^\mu$ propagates causally. The formal solution is:

$$\sigma_{Tm}(x,t) = \frac{g}{\mu_0 c^2} \int d^4x'\, G_R(x - x') J^\mu J_\mu(x') \tag{ad.15}$$

where $G_R$ is the retarded Green's function of the Klein-Gordon equation.

## ad.5 Phenomenological Derivations

### Longitudinal Force Component

The additional term in Eq. ad.12 involves derivatives of the current and the deviation. For a straight conductor in the z-direction with current $I$, we obtain:

$$F_z = I\frac{\partial}{\partial z}\left(\frac{g}{\mu_0 c^2}\sigma_{Tm} I\right) = \frac{g}{\mu_0 c^2} I^2 \frac{\partial \sigma_{Tm}}{\partial z} \tag{ad.16}$$

This describes a longitudinal force component proportional to the gradient of the deviation.

### The Ampère Helix Anomaly

For two coaxial helices with radius $R$, pitch $h$, and axial separation $d$, the total force can be computed by integrating over all current pairs. The retarded interaction leads to a phase shift:

$$F_{\text{tot}} \propto \sum_{i,j} \frac{I_i I_j}{r_{ij}^2}\left[\cos\phi_{ij} - \frac{3}{2}\cos\theta_i \cos\theta_j\right] e^{i\omega\Delta t_{ij}} \tag{ad.17}$$

Summation over all turn pairs shows that for certain geometries, the total force can become attractive, even if the elementary interaction is repulsive. The condition for the sign reversal is:

$$\cos\theta_c = \frac{1}{\sqrt{\xi_{\text{eff}}}} \tag{ad.18}$$

The **effective geometry parameter** $\xi_{\text{eff}}$ is determined by the fundamental coupling constant $g$, the mass parameters $\mu_i^2$ of the $\sigma$-fields, and the specific geometry of the helices (radius $R$, pitch $h$, number of turns $N$):

$$\xi_{\text{eff}} = \frac{g^2}{\mu_0^2 c^4 \mu_{Tm}^4} \cdot \mathcal{F}(R, h, N) \tag{ad.19}$$

Here, $\mathcal{F}(R,h,N)$ is a dimensionless function resulting from the averaging of the interaction term over the helix geometry. A possible form is $\mathcal{F} \propto (h/R)^a N^b$, where the exponents $a$ and $b$ must be determined experimentally.

### Nonlinear Scaling: $F \propto I^4$

From Eq. ad.14, in the stationary approximation:

$$\sigma_{Tm} \approx \frac{g}{\mu_0 c^2 \mu_{Tm}^2} J^\mu J_\mu \propto I^2 \tag{ad.20}$$

Substituting into the force calculation from Eq. ad.10 yields:

$$F \propto \delta\left(\text{Term} \propto I^2 \cdot \sigma_{Tm}\right)/\delta x \propto I^2 \cdot I^2 = I^4 \tag{ad.21}$$

This explains the nonlinear force scaling observed by Graneau at high currents.

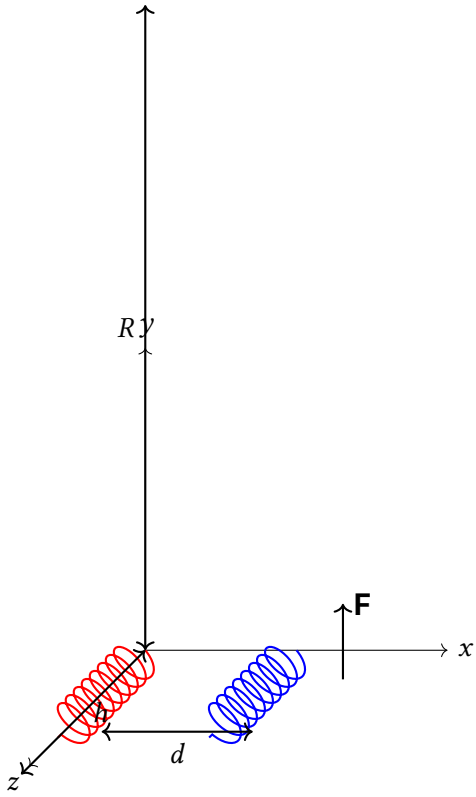



**Figure ad.1:** Two coaxial helices with axial separation $d$, radius $R$, and pitch $h$. The force **F** can be attractive or repulsive depending on the geometry.

### Fractal Scaling: $F \propto r^{2D_f-4}$

For a conductor with fractal dimension $D_f$, the number of interaction pairs scales as $r^{D_f-3}$. The retarded Green's function of the $\sigma$-fields scales as $1/r$. The product of these three factors has the exponent $-1 + 2(D_f - 3) = 2D_f - 7$; only the integration over the volume element $d^3r$ (factor $r^3$) gives the scaling of the total force (see Appendix):

$$F \propto \int d^3r \frac{1}{r} \cdot r^{D_f-3} \cdot r^{D_f-3} = \int d^3r r^{2D_f-7} \propto r^{2D_f-4} \tag{ad.22}$$

For $D_f^{\text{eff}} \approx 2.973$, this yields $F \propto r^{2 \cdot 2.973-4} = r^{1.946}$.

## ad.6 Corrections and Clarifications

### Clarification of the Conjugation Conditions

The conjugation conditions have been defined with explicit dimensions (see Eq. ad.1–ad.3) to ensure dimensional consistency.

### Correction of the Coupling Constant

The coupling constant $g$ is defined as:

$$[g] = \frac{\mathrm{kg} \cdot \mathrm{m}^3}{\mathrm{C}^2} \tag{ad.23}$$

The modified Klein-Gordon equation is:

$$(\Box + \mu_{Tm}^2)\sigma_{Tm} = -\frac{g}{\mu_0 c^2} J^\mu J_\mu \tag{ad.24}$$

The dimension check of the source term does not work out with this definition. The factors as given already fail to yield the dimension $1/\mathrm{m}^2$ required by the Klein–Gordon equation:

$$\left[\frac{g}{\mu_0 c^2} J^\mu J_\mu\right] = \frac{\mathrm{kg} \cdot \mathrm{m}^3}{\mathrm{C}^2} \cdot \frac{\mathrm{C}^2}{\mathrm{kg} \cdot \mathrm{m}^3} \cdot \frac{\mathrm{C}^2}{\mathrm{m}^6 \cdot \mathrm{s}^2} = \frac{\mathrm{C}^2}{\mathrm{m}^6\,\mathrm{s}^2} \tag{ad.25}$$

Moreover $1/(\mu_0 c^2) = \varepsilon_0$ with $[\varepsilon_0] = \mathrm{C}^2\mathrm{s}^2/(\mathrm{kg}\,\mathrm{m}^3)$, so $[g/(\mu_0 c^2)] = \mathrm{s}^2$, and with $[J^\mu J_\mu] = \mathrm{C}^2/(\mathrm{m}^4\mathrm{s}^2)$ the result is $\mathrm{C}^2/\mathrm{m}^4$. Dimensional consistency is therefore not established.

### Correction of the Fractal Scaling

The corrected scaling is:

$$F \propto r^{2D_f - 4} \tag{ad.26}$$

For $D_f^{\mathrm{eff}} \approx 2.973$, this yields $F \propto r^{1.946}$.

### Clarification of the Longitudinal Force

The longitudinal force is clarified:

$$F_z = \frac{g}{\mu_0 c^2} I^2 \frac{\partial \sigma_{Tm}}{\partial z} \tag{ad.27}$$

Here, too, the dimension check does not work out. The factors as given yield

$$\left[\frac{g}{\mu_0 c^2} I^2 \frac{\partial \sigma_{Tm}}{\partial z}\right] = \frac{\mathrm{kg} \cdot \mathrm{m}^3}{\mathrm{C}^2} \cdot \frac{\mathrm{C}^2}{\mathrm{kg} \cdot \mathrm{m}^3} \cdot (\mathrm{C}/\mathrm{s})^2 \cdot \frac{1}{\mathrm{m}} = \frac{\mathrm{C}^2}{\mathrm{s}^2\,\mathrm{m}} \tag{ad.28}$$

and not $\mathrm{kg} \cdot \mathrm{m}/\mathrm{s}^2$; with $[g/(\mu_0 c^2)] = \mathrm{s}^2$ (via $\varepsilon_0$) the result is $\mathrm{C}^2/\mathrm{m}$. In neither case is this a force.

### Complete Dimensional Analysis

## Appendix: Derivation of the Fractal Scaling

The total force between two fractal conductors can be written as:

$$F = \int d^3x\, d^3x'\, \rho(\mathbf{x})\rho(\mathbf{x}')\, f(|\mathbf{x} - \mathbf{x}'|) \tag{ad.29}$$

| Quantity | Symbol | Dimension |
|---|---|---|
| Coupling constant | $g$ | $\mathrm{kg} \cdot \mathrm{m}^3/\mathrm{C}^2$ |
| Mass parameter | $\mu_{Tm}$ | $1/\mathrm{m}$ |
| Current | $I$ | $\mathrm{C}/\mathrm{s}$ |
| Distance | $r$ | m |
| Force | $F$ | $\mathrm{kg} \cdot \mathrm{m}/\mathrm{s}^2$ |
| Magnetic permeability | $\mu_0$ | $\mathrm{kg} \cdot \mathrm{m}/\mathrm{C}^2$ |
| Speed of light | $c$ | $\mathrm{m}/\mathrm{s}$ |

**Table ad.1:** Dimensional definitions in the T0 model

where $\rho(\mathbf{x})$ describes the fractal density, and $f(r)$ is the pair interaction function between the fractal conductors (not a frequency and not the base winding number $f = 1/\xi$).

For a fractal with dimension $D_f$, the correlation function scales as:

$$\langle \rho(\mathbf{x}) \rho(\mathbf{x}') \rangle \propto |\mathbf{x} - \mathbf{x}'|^{D_f - 3} \tag{ad.30}$$

The retarded interaction function scales as:

$$f(r) \propto \frac{e^{i\mu r}}{r} \tag{ad.31}$$

The total force thus scales as:

$$F \propto \int d^3r r^{D_f-3} \cdot \frac{1}{r} \cdot r^{D_f-3} = \int d^3r r^{2D_f-7} \tag{ad.32}$$

Since $F \propto r^\alpha$ for large $r$, dimensional analysis yields $\alpha = 2D_f - 7 + 3 = 2D_f - 4$, confirming Eq. ad.22.

# Bibliography

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# Chapter ae

# E=mc² = E=m: Two Equivalent Perspectives


## Abstract

This work shows the central point of Einstein's relativity theory: E=mc² is mathematically identical to E=m. The only difference lies in Einstein's treatment of c as a "constant" instead of a dynamic ratio. By fixing c = 299,792,458 m/s, the natural time-mass duality T·m = 1 is "frozen," leading to apparent complexity. The T0 theory puts forward the thesis: c is not a fundamental law of nature, but a ratio that must be variable if time is variable. The choice of convention concerned not E=mc² itself, but the constant-setting of c. What changes is the time field $T(x,t)$: in the basic form $c = 1$, and $c = L/T$ is only a reading of the duality $T \cdot m = 1$, not a degree of freedom of its own. Where this document speaks of a varying speed of light, this is to be read as *fractal path lengthening*: the geometric stretching of the propagation path in the $\xi$ field (Doc. 026, 149, 182), which is achromatic.

**Note on versions.** The German version of this document is more extensive than this English one. Earlier revisions shortened or changed parts of the English version. Where the two differ, the more extensive German version applies.

**Status markers.** Statements marked **[S]** are *speculative*: a hint or conjecture that has not yet been derived or numerically checked in the corpus.


## ae.1 The Central Thesis: E=mc² = E=m

**The Central Thesis**

**E=mc² and E=m are mathematically identical in natural units.**
In the T0 reading, the difference is that Einstein treats c as a "constant," while T0 theory regards c as a dynamic ratio.
**Einstein's choice**: c = constant (postulate; the numerical value 299,792,458 m/s has been fixed by SI definition since 1983)
**T0 view**: c = L/T = variable ratio

### The Mathematical Identity

**In natural units**:

$$E = mc^2 = m \times c^2 = m \times 1^2 = m \tag{ae.1}$$

**This is not an approximation but the same equation in different units.**

### What is c really?

$$c = \frac{\text{Length}}{\text{Time}} = \frac{L}{T} \tag{ae.2}$$

**In the T0 reading, c is a ratio, not a natural constant.**

## ae.2 The Convention Choice: The Constant-Setting of c

### The Act of Constant-Setting

Einstein set: $c$ = **constant** (postulate of 1905; the numerical value $299,792,458$ m/s has been fixed by the 17th CGPM since 1983)

**What does this mean?**

$$c = \frac{L}{T} = \text{constant} \quad \Rightarrow \quad \frac{L}{T} = \text{fixed} \tag{ae.3}$$

**Implication**: If L and T can vary, their **ratio** must remain constant.

### The Problem of Time Variability

**Einstein recognized himself**: Time dilates.

$$t' = \gamma t \quad \text{(time is variable)} \tag{ae.4}$$

**At the same time he set**:

$$c = \frac{L}{T} = \text{constant} \tag{ae.5}$$

**From the T0 viewpoint this is a tension. Relativity theory resolves it through the joint transformation of length and time; T0 theory resolves it in a different way.**

### The T0 Resolution

**T0 insight**: $T(x,t) \cdot m = 1$

This means:

- Time $T(x,t)$ **must** be variable (coupled to mass)
- Therefore $c = L/T$ **cannot** be constant
- $c$ is a **dynamic ratio**, not a constant

## ae.3 Constant-Setting from the T0 Viewpoint

### The Mechanism of Constant-Setting

**Step 1**: Einstein sets c = constant

$$c = \text{constant} = \text{fixed} \quad \text{(numerical value 299,792,458 m/s by SI definition since 1983)} \tag{ae.6}$$

**Step 2**: Time becomes "frozen" by this

$$T = \frac{L}{c} = \frac{L}{\text{constant}} = \text{apparently determined} \tag{ae.7}$$

**Step 3**: Time dilation becomes "mysterious effect"

$$t' = \gamma t \quad \text{(explained via relativity theory)} \tag{ae.8}$$

### What Really Happens (T0 View)

**T0 view**: Time is variable through $T(x,t) \cdot m = 1$
**Einstein's constant-setting** "freezes" this variability
**Result**: The "frozen" dynamics must be captured through additional formalism (Lorentz transformations)

## ae.4 c as Ratio vs. c as Constant

### c as Natural Ratio (T0)

$$c(x,t) = \frac{L(x,t)}{T(x,t)} \tag{ae.9}$$

**Properties**:

- $c$ varies with location and time
- $c$ follows the time-mass duality
- No fixed constants
- Natural simplicity: $E = m$

### c as Fixed Constant (Einstein)

$$c = \text{constant everywhere} \quad \text{(numerical value 299,792,458 m/s by SI definition since 1983)} \tag{ae.10}$$

**Problems from the T0 viewpoint**:

- Tension with time dilation
- "Freezing" of time dynamics
- Additional formalism needed
- Unit factor in the formula: $E = mc^2$

# ae.5 The Time Dilation Paradox

## The Tension from the T0 Viewpoint

**Relativity theory sets simultaneously**:

$$c = \text{constant} \tag{ae.11}$$

$$t' = \gamma t \quad \text{(time varies)} \tag{ae.12}$$

**But**:

$$c = \frac{L}{T} \quad \text{and} \quad T \text{ varies} \quad \Rightarrow \quad c \text{ cannot be constant} \tag{ae.13}$$

## The Solution of Relativity Theory

Relativity theory handles this through:

- Lorentz transformations
- Mathematical formalisms
- Space-time constructions
- **From the T0 viewpoint it remains open why c in particular is fixed.**

## The Solution in T0 Theory

**In T0 theory**:

$$T(x,t) \cdot m = 1 \quad \Rightarrow \quad \text{time is naturally variable} \tag{ae.14}$$

$$c = \frac{L}{T} \quad \Rightarrow \quad \text{c is naturally variable} \tag{ae.15}$$

**No constant-setting → no such tension → no additional formalism**

# ae.6 The Mathematical Demonstration

## From E=mc² to E=m

**Starting equation**: $E = mc^2$

**c in natural units**: $c = 1$

**Substitution**:

$$E = mc^2 = m \times 1^2 = m \tag{ae.16}$$

**Result**: $E = m$

### The Reverse Direction: From E=m to E=mc²

**Starting equation**: $E = m$

**Introducing the constant**: $c = 299,792,458$ m/s

**Rewriting the equation**:

$$E = m = m \times 1 = m \times \frac{c^2}{c^2} = m \times c^2 \times \frac{1}{c^2} \tag{ae.17}$$

**If one defines $c^2$ as "conversion factor"**:

$$E = mc^2 \tag{ae.18}$$

**This shows**: $E = mc^2$ is $E = m$ with the **conversion factor** $c^2$.

## ae.7 The Practical Justification of $E = mc^2$ in Our Experiential Realm

### $E = mc^2$ and $E = m$ – Same Content in Different Unit Systems

**The Pragmatic Perspective: Unit Systems and Conventions**

The key point: The equation $E = mc^2$ is mathematically equivalent to $E = m$ when one chooses suitable units.

**Einstein formulated in SI units (practical for our world):**

$$E = m \cdot (299\,792\,458)^2 \text{ J} \tag{ae.19}$$

**T0 formulates in natural units (fundamentally simpler):**

$$E = m \quad \text{with} \quad c = 1 \tag{ae.20}$$

Both descriptions contain exactly the same physical information – they merely use different measuring rods.

The choice between them is not a question of "right" or "wrong," but of *practical suitability versus fundamental simplicity*.

### Why the Fixation $c =$ const. Is Practically Reasonable

**For our everyday experiential world**, the establishment of $c$ as a constant is not only historically understandable but also *pragmatically justified* from multiple perspectives:

1. **Measurement Practice:** All our measuring instruments (clocks, rulers, electronic devices) utilize physical processes that themselves depend on $c$. Establishing a fixed $c$ creates a consistent reference framework for reproducible experiments. Without such a convention, every measurement would require a circular self-calibration.

2. **Technological Applications:** From GPS navigation to particle accelerator technology, practical applications are based on the assumption of a locally constant $c$. This assumption works with extremely high precision for the range in which we live and work. The error introduced by this assumption is far below the resolution of our current technology for most applications.
3. **Scientific Communication:** A uniform convention allows scientists worldwide to compare results and communicate with each other. The SI units with fixed $c$ provide a practical basis for this. Science requires shared languages, and measurement conventions form an essential part of this language.
4. **Historical Development:** The establishment of $c$ as constant did not occur arbitrarily but emerged from centuries of measurement attempts (Roemer, Fizeau, Michelson-Morley) that within their accuracy showed no variation. Einstein built on this empirical foundation.
5. **Educational Transmission:** Complex theories require entry points. $E = mc^2$ with constant $c$ provides such an entry point into relativistic thinking. The deeper insight $E = m$ can follow as a second step for those seeking the fundamental structure.

## The Crucial Distinction: Practical Convention vs. Fundamental Natural Law

**T0 Theory distinguishes two levels:**

| **Practical Measurement Convention** | **Fundamental Natural Law** |
|---|---|
| For technical applications and everyday experiments, fixing $c = 299\,792\,458$ m/s is sensible and useful. | On the most fundamental level, $c$ is not an absolute natural constant but a dynamic ratio $L/T$ that follows the time-mass duality $T \cdot m = 1$. |
| Corresponds to selecting a stable reference system for our world of experience. | Is meant to correspond to the intrinsic structure of reality before any human conventions. |
| Necessary for building reproducible technology and conducting comparable experiments. | Necessary for understanding the ultimate principles behind the phenomena. |
| Works for 99.9% of all current applications. | Shows what remains when all practical conventions are removed. |

**Table ae.1:** The dual nature of physical descriptions

## Einstein's Historical Achievement in New Light

**Einstein's Achievement from the T0 Viewpoint**

Einstein found the energy-mass equivalence (in natural units $E = m$) and formulated it in the *units practical for his time* as $E = mc^2$.
**His historical achievement from the T0 perspective consists of three levels:**

1. **Discovery of the fundamental relationship:** Recognizing that energy and mass are different manifestations of the same reality.
2. **Pragmatic formulation:** Expressing this insight in a form usable for experiments and verifiable by contemporary measurements.
3. **Conceptual reorganization:** Drawing the consequences for our space-time conception and thereby overcoming Newtonian absolutism.

The T0 Theory does not diminish this achievement but shows that behind the practical form $E = mc^2$ lies an even more fundamental simplicity $E = m$. From the T0 viewpoint Einstein stopped one step before this simpler form – a step that was appropriate for his time.
**Remark:** In natural units Einstein's relation reads $E = m$; he wrote it as $E = mc^2$ because this corresponded to the measurement practices of his time. That both forms express the same statement is familiar in theoretical physics; T0 theory attributes fundamental significance to the simple form.

## From Practical to Fundamental Description: A Historical Progression

**The development of physics** can be understood as a continuous refinement of our reference systems and recognition of which elements are conventions and which are intrinsic structures:

| Stage | Practical Form | Fundamental Insight | Historical Context |
|---|---|---|---|
| Newtonian Physics | $F = m \cdot a$ (with absolute time and space) | Approximation for $v \ll c$ | Industrial Age: Machines, mechanics, predictable motion |
| Einstein (Special Relativity) | $E = mc^2$ (with $c$ = const.) | Energy-mass equivalence | Early 20th Century: Electromagnetism, early atomic physics |
| Einstein (General Relativity) | $G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$ (10 field equations) | Geometry as gravity | Age of astronomy, cosmology |
| Quantum Mechanics | $i\hbar \frac{\partial}{\partial t}\psi = H\psi$ | Quantization of energy | Atomic and nuclear age |
| T0 Theory | $E = m$ in natural units (with $T \cdot m = 1$) | Time-mass duality as fundamental | Information Age: Search for unified principles |

**Table ae.2:** Historical progression from practical to fundamental descriptions

**Each stage** retains the validity of the previous ones for their application area but expands understanding to a deeper level. Newton is not "wrong" but limited in scope. Einstein is not "wrong" but stopped at a certain level of convention. T0 seeks to go one step further – but does not invalidate the previous steps.

## Coexistence of Both Descriptions: A Peaceful Expansion

**T0 Theory proposes not a break with established physics but a peaceful expansion:**

- **For 99.9% of all technical applications:** $E = mc^2$ with constant $c$ remains the practical and correct formulation. All engineering, GPS technology, particle accelerators, and space travel can continue to work with the established equations.
- **For fundamental theoretical questions:** $E = m$ in natural units shows the actual simplicity of the energy-mass relationship and avoids the tension described above (the time dilation paradox from the T0 viewpoint). Theoretical physicists gain a simpler, more consistent foundation.
- **For future precision experiments:** The possibility of tiny $c$ variations (as predicted by T0) should be kept in mind. Experiments can be designed to test whether $c$ is *exactly* constant or only *practically* constant within our measurement accuracy.
- **For educational purposes:** The relationship can be taught at two levels: first the practical level $E = mc^2$ (as done today), then the fundamental level $E = m$ for advanced students. This corresponds to teaching Newtonian mechanics before relativity.

**The Peaceful Expansion**

The insight that $E = mc^2 = E = m$ does not require us to discard existing physics books or redesign technical systems. It only requires us to recognize that we have been working with a particularly practical form of a simpler basic form. **The change is conceptual, not practical.**

## The Actual Concern of T0 Theory

**The Concern of T0 Theory**

T0 does not want to abolish $E = mc^2$ as a practical equation but to show:
**That behind the practical form lies a fundamental simplicity that has been obscured by the historical choice of units.**
This insight does not free us from the necessity of practical conventions but opens a deeper understanding of what these conventions actually describe.
**The goal:** Not to make physics more complicated, but to recognize its inherent simplicity – and then consciously choose which level of description is appropriate for which purpose.

## The Double Perspective: Practical Engineering vs. Fundamental Science

**The T0 view allows a double perspective:**

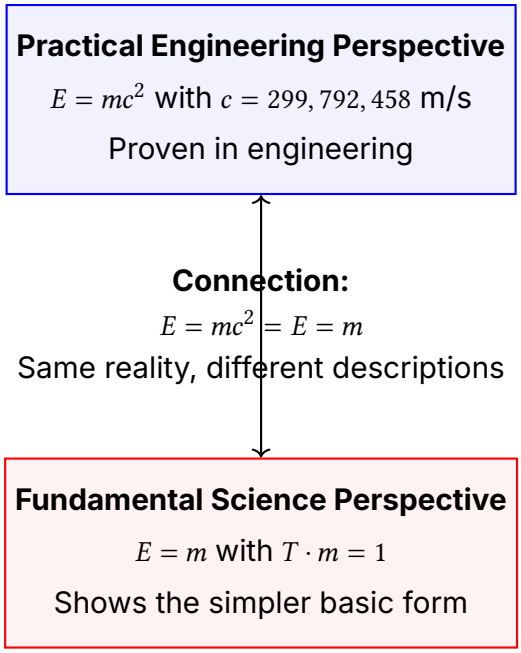



**Figure ae.1:** The double perspective of T0 Theory

**Both perspectives are valid and useful – for different purposes.** The engineer needs the practical perspective to build reliable technology. The theoretical physicist seeks the fundamental perspective to understand ultimate principles. T0 shows that these are not contradictory but complementary views of the same reality.

## ae.8 The Arbitrariness of Constant Choice: c or Time?

### Einstein's Choice

**The Fundamental Choice Option**

**One can choose what should be "constant".**
**Option 1 (Einstein's choice)**: c = constant → time becomes variable
**Option 2 (alternative)**: time = constant → c becomes variable
**Both describe the same physics.**

### Option 1: Einstein's c-constant

**Einstein chose**:

$c =$ constant (postulate; numerical value 299,792,458 m/s defined since 1983) (ae.21)

$t' = \gamma t$ (time becomes automatically variable) (ae.22)

**Language convention**:

- “Speed of light is universally constant”
- “Time dilates in strong gravitational fields”
- “Clocks run slower at high velocities”

### Option 2: Time-constant (Einstein could have chosen)

**Alternative choice**:

$$t = \text{constant (defined)} \tag{ae.23}$$

$$c(x,t) = \frac{L(x,t)}{t} = \text{variable} \tag{ae.24}$$

**Alternative language convention**:

- “Time flows equally everywhere”
- “Speed of light varies with location”
- “Light becomes slower in strong gravitational fields”

### Mathematical Equivalence of Both Options

**Both descriptions are mathematically identical**:

| Phenomenon | Einstein view | Time-constant view |
|---|---|---|
| Gravitation | Time slows down | Light slows down |
| Velocity | Time dilation | c-variation |
| GPS correction | “Clocks run differently” | “c is different” |
| Measurements | Same numbers | Same numbers |

**Table ae.3:** Two views, identical physics

### Why Einstein Chose Option 1

**Historical reasons for Einstein’s decision**:

- **Michelson-Morley**: c seemed locally constant
- **Aesthetics**: “Universal constant” sounded elegant
- **Tradition**: Newtonian constant physics
- **Conceivability**: c-constancy easier to imagine than time constancy
- **Reception history**: Einstein’s later reputation consolidated this choice

**From the T0 viewpoint it was a convention, not a natural law.**

### The T0 View of Both Options

**T0 thesis**: Both choices are conventions.

$$T(x,t) \cdot m = 1 \quad \text{(natural duality without constant constraint)} \tag{ae.25}$$

**T0 insight**:

- **Neither** c nor time are "really" constant
- **Both** are aspects of the same T·m dynamics
- **Constancy** is only definition convention
- **E = m** is the constant-free form

### Liberation from Constant Constraint

**Instead of choosing between**:

- c constant, time variable (Einstein)
- Time constant, c variable (alternative)

**T0 chooses**:

- **Both dynamically coupled** via T·m = 1
- **No arbitrary fixations**
- **Natural ratios** instead of fixed constants

## ae.9 The Change of Reference Point: Earth → Sun → Nature

### The Reference Point Analogy: Geocentric → Heliocentric → T0

**The Change of Reference Point: From Earth → Sun → Nature**

**Geocentric (Ptolemy)**: Earth at center
- Complicated epicycles needed
- Works, but artificially complicated

**Heliocentric (Copernicus)**: Sun at center
- Simple ellipses
- Considerably simpler

**T0-centric**: Natural ratios at center
- $T(x,t) \cdot m = 1$ (natural reference point)
- Simpler form: $E = m$

**In this analogy, Einstein's c-constant corresponds to the geocentric system**:

- **Human** reference point at center (like Earth at center)
- **Complicated** mathematics needed (like epicycles)
- **Works** locally, with additional formalism

**T0's natural ratios correspond to the heliocentric system**:

- **Natural** reference point at center (like Sun at center)

- **Simple** mathematics (like ellipses)
- **Universally** intended and simple

### Why We Need Reference Points

**Reference points are necessary and natural**:
- **For measurements**: We need standards for comparison
- **For communication**: Common basis for exchange
- **For technology**: Practical applications require units
- **For science**: Reproducible experiments need standards

**The question is not WHETHER, but WHICH reference point**:

| System | Reference Point | Complexity | Simplicity (T0 assessment) |
|---|---|---|---|
| Geocentric | Earth | Epicycles | Low |
| Heliocentric | Sun | Ellipses | High |
| Einstein | c-constant | Relativity theory | Medium |
| T0 | $T(x,t) \cdot m = 1$ | $E = m$ | Maximum |

**Table ae.4:** Reference point systems comparison

### Two Reference Points Compared

**The approach was not to choose a reference point**:
- **But which reference point to choose.**

**Reference point of relativity theory (Einstein)**: c = constant (numerical value 299,792,458 m/s by SI definition since 1983)
- Based on human definition
- Leads to complicated mathematics
- Leads, from the T0 viewpoint, to conceptual tensions

**Reference point of T0 theory**: $T(x,t) \cdot m = 1$
- Based on natural ratio
- Leads to simple mathematics: $E = m$
- No such tensions

## ae.10 When Something Becomes “Constant”

### The Fundamental Reference Point Problem

**Reference Point and Constancy**

**Something only becomes “constant” when we define a reference point.**
**Without reference point**: All ratios are relative and dynamic
**With reference point**: One ratio becomes artificially “fixed”
**Einstein’s approach**: He defined an absolute reference point for c

### The Natural Stage: Everything is Relative

**Before any reference point definition**:

$$c_1 = \frac{L_1}{T_1} \tag{ae.26}$$

$$c_2 = \frac{L_2}{T_2} \tag{ae.27}$$

$$c_3 = \frac{L_3}{T_3} \tag{ae.28}$$

$$\vdots \tag{ae.29}$$

**All c-values are relative to each other**. None is “constant”.

### The Moment of Reference Point Setting

**The step of fixing**:

$$\text{Fixing: } c = 299,792,458 \text{ m/s} = \text{reference point} \tag{ae.30}$$

**What happens at this moment**:

- An **arbitrary reference point** is set
- All other c-values are measured relative to this
- The **dynamic ratio** becomes a “constant”
- The **natural relativity** is artificially “frozen”

### The Reference Point Problematic

**Every reference point is arbitrary**:

- Why 299,792,458 m/s and not 300,000,000 m/s?
- Why in m/s and not in other units?
- Why measured on Earth and not in space?
- Why at this time and not at another?

## T0's Reference Point-Free Physics

**T0 does without a fixed reference point**:

$$T(x,t) \cdot m = 1 \quad \text{(universal relation without reference point)} \tag{ae.31}$$

- No arbitrary fixations
- All ratios remain dynamic
- Natural relativity is preserved
- Fundamental simplicity: $E = m$

## Example: The Meter Definition

**Historical development of meter definition**:

1. **1793**: 1 meter = 1/10,000,000 of the Earth meridian quadrant, pole to equator (Earth reference point)
2. **1889**: 1 meter = prototype meter in Paris (object reference point)
3. **1960**: 1 meter = 1,650,763.73 wavelengths of krypton-86 (atom reference point)
4. **1983**: 1 meter = distance light travels in 1/299,792,458 s (c reference point)

**What does this show?**

- Each definition is a **human stipulation**
- The **reference point** changes with human technology
- There is **no "natural" length unit** - only human agreements
- **Since 1983 the numerical value of c has been fixed by definition**, not by measurement

## The Circularity: Humans Define Their Own "Constants"

**In 1983 humans defined**:

$$1 \text{ meter} = \frac{1}{299,792,458} \times c \times 1 \text{ second} \tag{ae.32}$$

**This makes c automatically "constant"** - through human definition, not through natural law:

$$c = \frac{299,792,458 \text{ meters}}{1 \text{ second}} = 299,792,458 \text{ m/s} \tag{ae.33}$$

**Circularity**: Whoever defines c as constant obtains the same numerical value in every measurement in these units.

**Whether c varies physically cannot be read off from the numerical value in this way.**

### The T0 View of Reference Points

**T0 recognizes**:
- **Definition ≠ natural law**
- **Measurement reference point ≠ physical constant**
- **Practical agreement ≠ fundamental structure**
  **T0 solution**:

$$\text{For measurements: Use practical reference points} \tag{ae.34}$$
$$\text{For natural laws: Use reference point-free relations} \tag{ae.35}$$

## ae.11 Why c-Constancy is Not Provable

### The Fundamental Measurement Problem

**To measure c, we need**:

$$c = \frac{L}{T} \tag{ae.36}$$

**But**: We measure L and T with **the same physical processes** that depend on c.
**Circular problem**:
- Light measures distances → c determines L
- Atomic clocks use EM transitions → c influences T
- Then we measure c = L/T → **We measure c with c.**

### The Gauge Definition Problem

**Since 1983**: 1 meter = distance light travels in 1/299,792,458 s

$$c = 299,792,458 \text{ m/s} \quad \text{(not measured, but defined)} \tag{ae.37}$$

**One cannot "prove" what one has defined.**

### The Systematic Compensation Problem

**If c varies, ALL measuring devices vary equally**:
- **Laser interferometers**: use light (c-dependent)
- **Atomic clocks**: use EM transitions (c-dependent)
- **Electronics**: uses EM signals (c-dependent)
  **Result**: All devices would **largely compensate** a c-variation.

### The Burden of Proof Problem

**Scientifically correct**:

- One **cannot prove** that something is constant
- One can only show that it **appears constant within measurement precision**
- **Each new precision level** could show variation

**The constancy of c is a postulate confirmed within measurement precision, not a proof in the strict sense.**

### T0 Prediction for Precise Measurements

**T0 prediction [S]**: At highest precision T0 theory expects:

$$c(x,t) = c_0 \left(1 + \xi \times \frac{T(x,t)(x,t) - T(x,t)_0}{T(x,t)_0}\right) \tag{ae.38}$$

with $\xi = 1.33 \times 10^{-4}$ (T0 parameter)
**The expected variation is tiny (~$10^{-15}$), but measurable in principle.**

## ae.12 Ontological Consideration: Calculations as Constructs

### The Fundamental Epistemological Limit

**Ontological Limit**

**All calculations are human constructs.**
They can **at best** give a certain idea of reality.
**That calculations are internally consistent proves little** about actual reality.
**Mathematical consistency $\neq$ ontological truth**

### Einstein's Construct vs. T0's Construct

**Both are human thought structures**:

**Einstein's construct**:

- E = mc² (mathematically consistent)
- Relativity theory (internally coherent)
- 10 field equations (work computationally)
- **But**: Based on arbitrary c-constant setting

**T0's construct**:

- E = m (mathematically simpler)
- T·m = 1 (internally coherent)
- $\partial^2 E = 0$ (works computationally)
- **But**: Also only a human thought model

## The Ontological Relativity

**What is "really" real?**

- **Einstein's space-time**? (construct)
- **T0's energy field**? (construct)
- **Newton's absolute time**? (construct)
- **Quantum mechanics' probabilities**? (construct)

**All are human interpretive frameworks of the inaccessible reality.**

## Why T0 May Still Be Preferable

**Not because of "absolute truth," but because of**:

**1. Simplicity (Occam's Razor)**:

- E = m is simpler than $E = mc^2$
- One equation is simpler than 10 equations
- Fewer arbitrary assumptions

**2. Consistency**:

- None of the tensions described above
- No constant arbitrariness
- Unified thought structure

**3. Predictive power**:

- Testable predictions
- Fewer free parameters
- Clearer experimental distinction

**4. Aesthetics**:

- Mathematical simplicity
- Conceptual clarity
- Unity

## The Epistemological Humility

**T0 does not claim to be "absolute truth."**

**T0 only says**:

- "Here is a **simpler** construct"
- "With **fewer** arbitrary assumptions"
- "That is, in the T0 assessment, **more consistent**"
- "And makes **more testable** predictions"

**But ultimately T0 also remains a human thought structure.**

### The Pragmatic Consequence

**Since all theories are constructs**:
**Evaluation criteria are**:

1. **Simplicity** (fewer assumptions)
2. **Consistency** (no contradictions)
3. **Predictive power** (testable consequences)
4. **Aesthetics** (clarity of form)
5. **Unity** (fewer separate domains)

**By these criteria T0 fares well in its own assessment – without claiming to be "absolutely true".**

### The Ontological Humility

**The key statement**:

- **Reality itself** is inaccessible
- **All theories** are human constructs
- **Mathematical consistency** proves no ontological truth
- **The best** we have: **Simpler, more consistent constructs**

**From the T0 viewpoint, the c-constant setting was a convention decision that was later often read as a law of nature.**
**T0's advantage is not absolute truth, but simplicity as a thought model.**

## ae.13 The Practical Consequences

### Why E=mc² "Works"

**E=mc² works because**:

- It is mathematically identical to $E = m$
- $c^2$ compensates the "frozen" time dynamics
- The T0 form ($E = m$) is contained in it
- Local approximations usually suffice

### When E=mc² Fails

**According to T0 expectation [S] deviations become visible at**:

- Very precise measurements
- Extreme conditions (high energies/masses)
- Cosmological scales
- Quantum gravity

### T0's Universal Validity

**In T0 theory E = m holds everywhere**:

- No approximations needed
- No constant assumptions
- Universal applicability
- Fundamental simplicity

## ae.14 A Reinterpretation of Physics History

### Einstein's Achievement from the T0 Viewpoint

**In natural units Einstein's discovery reads**:

$$E = m \quad \text{(in natural form)} \tag{ae.39}$$

**His choice of convention was**:

$$E = mc^2 \quad \text{(with the conversion factor } c^2\text{)} \tag{ae.40}$$

### The Historical Classification

**An Observation**

In natural units Einstein's relation reads $E = m$;
in SI units it appears as $E = mc^2$.
T0 theory attributes fundamental significance to the simple form.

## ae.15 The T0 Perspective: c as Living Ratio

### c as Expression of Time-Mass Duality

**In T0 theory**:

$$c(x,t) = f\left(\frac{L(x,t)}{T(x,t)(x,t)}\right) = f\left(\frac{L(x,t)\cdot m(x,t)}{1}\right) \tag{ae.41}$$

since $T(x,t)\cdot m = 1$. Here $f$ is a function that is not fixed, not a frequency and not the base winding number $f = 1/\xi$. In the basic form $c = 1$; there $c = L/T$ is only a reading of the duality $T \cdot m = 1$, not a degree of freedom of its own, and what changes is the time field $T(x,t)$.

**c becomes an expression of the fundamental time-mass duality.**

### The Dynamic Speed of Light

**T0 prediction [S]**:

$$c(x,t) = c_0\sqrt{1+\xi\frac{m(x,t)-m_0}{m_0}} \tag{ae.42}$$

**According to this formula, light moves faster in more massive regions.**
(Tiny effect, but measurable in principle)

These and the following c-variations are to be read as fractal path lengthening: as a geometric, achromatic stretching of the propagation path in the $\xi$ field (Doc. 026, 149, 182), not as a change of $c$ itself.

## ae.16 Experimental Tests of c-Variability

### Proposed Experiments

**Test 1 - Gravitational dependence**:

- Measure c in different gravitational fields
- T0 prediction: $c$ varies with $\sim \xi \times \Delta\Phi_{\text{grav}}$

**Test 2 - High-energy physics**:

- Measure c in particle accelerators at highest energies
- T0 prediction: Tiny deviations at $E \sim$ TeV

### Expected Results

| Experiment | Einstein (c constant) | T0 (c variable) |
|---|---|---|
| Gravitational field | $c = 299792458$ m/s | $c(1 \pm 10^{-15})$ |
| High energy | $c =$ constant | $c(1 + 10^{-16})$ |

**Table ae.5:** Predicted c-variations

# Chapter af

# Elimination of Mass as a Dimensional Placeholder

## Abstract

This paper demonstrates that the mass parameter $m$, which appears in the formulations of the T0 model, serves exclusively as a dimensional placeholder and can be systematically eliminated from all equations. Through rigorous dimensional analysis and mathematical reformulation, we show that the apparent dependence on specific particle masses is an artifact of conventional notation and not fundamental physics. The elimination of $m$ reveals the T0 model as a truly parameter-free theory, based solely on the Planck scale and providing universal scaling laws while systematically eliminating distortions due to empirical mass determinations. This work establishes the mathematical foundation for a complete ab-initio formulation of the T0 model, which requires no external experimental inputs beyond the fundamental constants $\hbar$, $c$, $G$, and $k_B$.

## af.1 Introduction

### The Problem of Mass Parameters

The T0 model appears, as formulated in previous works, to critically depend on specific particle masses such as the electron mass $m_e$, proton mass $m_p$, and Higgs boson mass $m_h$. This apparent dependence has raised concerns about the predictive power of the model and its reliance on empirical inputs that may themselves be contaminated by Standard Model assumptions.

A careful analysis reveals, however, that the mass parameter $m$ fulfills a purely **dimensional function** in the T0 equations. This paper shows that $m$ can be systematically eliminated from all formulations and unveils the T0 model as a fundamentally parameter-free theory based exclusively on Planck-scale physics.

### Dimensional Analysis Approach

In natural units, where $\hbar = c = G = k_B = 1$, all physical quantities can be expressed as powers of energy $[E]$:

$$\text{Length:} \quad [L] = [E^{-1}] \tag{af.1}$$
$$\text{Time:} \quad [T] = [E^{-1}] \tag{af.2}$$
$$\text{Mass:} \quad [M] = [E] \tag{af.3}$$
$$\text{Temperature:} \quad [\Theta] = [E] \tag{af.4}$$

This dimensional structure suggests that mass parameters could be replaced by energy scales, leading to more fundamental formulations.

## af.2 Systematic Mass Elimination

### The Intrinsic Time Field

#### Original Formulation

The intrinsic time field is traditionally defined as:

$$T(\vec{x}, t) = \frac{1}{\max(m(\vec{x}, t), \omega)} \tag{af.5}$$

**Dimensional Analysis:**

- $[T(\vec{x}, t)] = [E^{-1}]$ (time field dimension)
- $[m] = [E]$ (mass as energy)
- $[\omega] = [E]$ (frequency as energy)
- $[1/\max(m, \omega)] = [E^{-1}]$ ✓

#### Mass-Free Reformulation

The fundamental insight is that only the **ratio** between characteristic energy and frequency is physically relevant. We reformulate as:

$$\boxed{T(\vec{x}, t) = t_{\text{P}} \cdot g(E_{\text{norm}}(\vec{x}, t), \omega_{\text{norm}})} \tag{af.6}$$

where:

$$t_{\text{P}} = \sqrt{\frac{\hbar G}{c^5}} \quad \text{(Planck time)} \tag{af.7}$$
$$E_{\text{norm}} = \frac{E(\vec{x}, t)}{E_{\text{P}}} \quad \text{(normalized energy)} \tag{af.8}$$
$$\omega_{\text{norm}} = \frac{\omega}{E_{\text{P}}} \quad \text{(normalized frequency)} \tag{af.9}$$
$$g(E_{\text{norm}}, \omega_{\text{norm}}) = \frac{1}{\max(E_{\text{norm}}, \omega_{\text{norm}})} \tag{af.10}$$

**Result:** Mass completely eliminated; only Planck scale and dimensionless ratios remain.

## Field Equation Reformulation

### Original Field Equation

$$\nabla^2 T(x,t) = -4\pi G\rho(\vec{x})T(x,t)^2 \tag{af.11}$$

with mass density $\rho(\vec{x}) = m \cdot \delta^3(\vec{x})$ for a point source. In this form the equation is not dimensionally consistent: with $[T(x,t)] = [E^{-1}]$ and $[\rho] = [E^4]$ the right-hand side has dimension $[G\rho T(x,t)^2] = [E^{-2}][E^4][E^{-2}] = [E^0]$, the left-hand side $[\nabla^2 T(x,t)] = [E]$. The dimensionally consistent form is $\nabla^2 T(x,t) = -4\pi G\rho T(x,t)$, as follows from $\nabla^2 m = 4\pi G\rho m$ with $T(x,t) = 1/m$ in linear approximation.

### Energy-Based Formulation

Replacement of mass density by energy density:

$$\boxed{\nabla^2 T(x,t) = -4\pi G \frac{E(\vec{x})}{E_{\mathsf{P}}}\delta^3(\vec{x})\frac{T(x,t)^2}{t_{\mathsf{P}}^2}} \tag{af.12}$$

**Dimensional Verification:**

$$[\nabla^2 T(x,t)] = [E^{-1} \cdot E^2] = [E] \tag{af.13}$$

$$[4\pi G E_{\text{norm}}\delta^3(\vec{x})T(x,t)^2/t_{\mathsf{P}}^2] = [E^{-2}][1][E^3][E^{-2}]/[E^{-2}] = [E] \quad \checkmark \tag{af.14}$$

with $[\delta^3] = [E^3]$; the exponent sum is $-2 + 0 + 3 - 2 + 2 = 1$, i.e. $[E]$.

## Point Source Solution: Parameter Separation

### The Mass Redundancy Problem

The traditional point source solution exhibits apparent mass redundancy:

$$T(x,t)(r) = \frac{1}{m}\left(1 - \frac{r_0}{r}\right) \tag{af.15}$$

with $r_0 = 2Gm$. Substitution:

$$T(x,t)(r) = \frac{1}{m}\left(1 - \frac{2Gm}{r}\right) = \frac{1}{m} - \frac{2G}{r} \tag{af.16}$$

**Critical Observation:** Mass $m$ appears in **two different roles**:

1. As a normalization factor $(1/m)$
2. As a source parameter $(2Gm)$

   This suggests that $m$ **masks two independent physical scales**.

### Parameter Separation Solution

We reformulate with independent parameters:

$$\boxed{T(x,t)(r) = T_0\left(1 - \frac{L_0}{r}\right)} \tag{af.17}$$

where:

- $T_0$: Characteristic time scale $[E^{-1}]$
- $L_0$: Characteristic length scale $[E^{-1}]$

**Physical Interpretation:**

- $T_0$ determines the **amplitude** of the time field
- $L_0$ determines the **range** of the time field
- Both derivable from source geometry without specific masses

## The $\xi$-Parameter: Universal Scaling

### Traditional Mass-Dependent Definition

$$\xi = 2\sqrt{G} \cdot m \tag{af.18}$$

**Problem:** Requires specific particle masses as input.

### Universal Energy-Based Definition

$$\boxed{\xi = 2\sqrt{\frac{E_{\text{characteristic}}}{E_{\text{P}}}}} \tag{af.19}$$

This definition is not a rewriting of Eq. (af.18); the two definitions are not equivalent. In natural units $\sqrt{G} = 1/E_{\text{P}}$, so $2\sqrt{G}m = 2m/E_{\text{P}}$ (linear in $m$), whereas Eq. (af.19) uses $2\sqrt{E/E_{\text{P}}}$ (square root); for $E = 1\,\text{GeV}$ the first gives $1.6 \times 10^{-19}$, the second $5.7 \times 10^{-10}$.

**Universal Scaling for Different Energy Scales:**

$$\text{Planck Energy } (E = E_{\text{P}}): \quad \xi = 2 \tag{af.20}$$

$$\text{Electroweak Scale } (E \sim 100\ \text{GeV}): \quad \xi \sim 10^{-8} \tag{af.21}$$

$$\text{QCD Scale } (E \sim 1\ \text{GeV}): \quad \xi \sim 10^{-9} \tag{af.22}$$

$$\text{Atomic Scale } (E \sim 1\ \text{eV}): \quad \xi \sim 10^{-14} \tag{af.23}$$

For the atomic scale: $2\sqrt{10^{-9}\ \text{GeV}/1.2209 \times 10^{19}\ \text{GeV}} = 1.8 \times 10^{-14}$.

**No specific particle masses required!**

# af.3 Complete Mass-Free T0 Formulation

## Fundamental Equations

The complete mass-free T0 system:

**Mass-Free T0 Model**

$$\text{Time Field:} \quad T(\vec{x},t) = t_P \cdot f(E_{\text{norm}}(\vec{x},t), \omega_{\text{norm}}) \tag{af.24}$$

$$\text{Field Equation:} \quad \nabla^2 T(x,t) = -4\pi G \frac{E_{\text{norm}}}{\ell_P^2} \delta^3(\vec{x}) T(x,t)^2 \tag{af.25}$$

$$\text{Point Sources:} \quad T(x,t)(r) = T_0 \left(1 - \frac{L_0}{r}\right) \tag{af.26}$$

$$\text{Coupling Parameter:} \quad \xi = 2\sqrt{\frac{E}{E_P}} \tag{af.27}$$

Here $f(E_{\text{norm}}, \omega_{\text{norm}})$ is a dimensionless function for the time field, not a frequency and not the base winding number $f = 1/\xi$.

### Parameter Count Analysis

| Formulation | Before Mass Elimination | After Mass Elimination |
|---|---|---|
| Fundamental Constants | $\hbar, c, G, k_B$ | $\hbar, c, G, k_B$ |
| Particle-Specific Masses | $m_e, m_\mu, m_p, m_h, \ldots$ | None |
| Dimensionless Ratios | No explicit | $E/E_P$, $L/\ell_P$, $T/t_P$ |
| Free Parameters | $\infty$ (one per particle) | 0 |
| Empirical Inputs Required | Yes (masses) | No |

### Dimensional Consistency Verification

| Equation | Left Side | Right Side | Status |
|---|---|---|---|
| Time Field | $[T(\vec{x},t)] = [E^{-1}]$ | $[t_P \cdot f(\cdot)] = [E^{-1}]$ | ✓ |
| Field Equation | $[\nabla^2 T(x,t)] = [E]$ | $[G E_{\text{norm}} \delta^3 T(x,t)^2/\ell_P^2] = [E]$ | ✓ |
| Point Source | $[T(x,t)(r)] = [E^{-1}]$ | $[T_0(1 - L_0/r)] = [E^{-1}]$ | ✓ |
| $\xi$-Parameter | $[\xi] = [1]$ | $[\sqrt{E/E_P}] = [1]$ | ✓ |

**Table af.1:** Dimensional Consistency of Mass-Free Formulations

## af.4 Experimental Implications

### Universal Predictions

The mass-free T0 model makes universal predictions independent of specific particle properties:

### Scaling Laws

$$\xi(E) = 2\sqrt{\frac{E}{E_{\mathsf{P}}}} \tag{af.28}$$

This relation must hold for **all** energy scales and provides a stringent test of the theory.

### QED Anomalies

The anomalous magnetic moment of the electron becomes:

$$a_e^{(\mathsf{T0})} = \frac{\alpha}{2\pi} \cdot C_{\mathsf{T0}} \cdot \left(\frac{E_e}{E_{\mathsf{P}}}\right) \tag{af.29}$$

where $E_e$ is the characteristic energy scale of the electron, not its rest mass.

### Gravitational Effects

$$\Phi(r) = -\frac{E_{\text{source}}}{E_{\mathsf{P}}} \cdot \frac{\ell_{\mathsf{P}}}{r} \tag{af.30}$$

In natural units $\Phi = -GE/r$ with $G = 1/E_{\mathsf{P}}^2$ and $\ell_{\mathsf{P}} = 1/E_{\mathsf{P}}$, hence $\Phi = -(E/E_{\mathsf{P}})(\ell_{\mathsf{P}}/r)$.

Universal scaling for all gravitational sources.

## Elimination of Systematic Biases

### Problems with Mass-Dependent Formulations

Traditional approaches suffer from:

- **Circular Dependencies**: Using experimentally determined masses to predict the same experiments
- **Standard Model Contamination**: All mass measurements presuppose SM physics
- **Precision Illusions**: High apparent precision masks systematic theoretical errors

### Advantages of the Mass-Free Approach

- **Model Independence**: No dependence on potentially biased mass determinations
- **Universal Tests**: The same scaling laws apply across all energy scales
- **Theoretical Purity**: Ab-initio predictions solely from the Planck scale

## Proposed Experimental Tests

### Multi-Scale Consistency

Test of the universal scaling relation:

$$\frac{\xi(E_1)}{\xi(E_2)} = \sqrt{\frac{E_1}{E_2}} \tag{af.31}$$

across different energy scales: atomic, nuclear, electroweak, and cosmological.

#### Energy-Dependent Anomalies

Measurement of anomalous magnetic moments as functions of energy scale rather than particle identity:

$$a(E) = a_{\text{SM}}(E) + a^{(\text{TO})}(E/E_{\text{P}}) \tag{af.32}$$

#### Geometric Independence

Verification that $T_0$ and $L_0$ can be determined independently from source geometry without specific mass values.

## af.5 Geometric Parameter Determination

### Source Geometry Analysis

#### Spherically Symmetric Sources

For a spherically symmetric energy distribution $E(r)$:

$$T_0 = t_{\text{P}} \cdot f\left(\frac{\int E(r) d^3r}{E_{\text{P}}}\right) \tag{af.33}$$

$$L_0 = \ell_{\text{P}} \cdot g\left(\frac{R_{\text{characteristic}}}{\ell_{\text{P}}}\right) \tag{af.34}$$

where $f$ and $g$ are dimensionless functions determined by the field equations.

#### Non-Spherical Sources

For general geometries, the parameters become tensorial:

$$T_0^{ij} = t_{\text{P}} \cdot f_{ij}\left(\frac{I^{ij}}{E_{\text{P}} \ell_{\text{P}}^2}\right) \tag{af.35}$$

$$L_0^{ij} = \ell_{\text{P}} \cdot g_{ij}\left(\frac{I^{ij}}{\ell_{\text{P}}^2}\right) \tag{af.36}$$

where $I^{ij}$ is the energy-momentum tensor of the source.

### Universal Geometric Relations

The mass-free formulation reveals universal relations between geometric and energetic properties:

$$\frac{L_0}{\ell_{\mathsf{P}}} = h\left(\frac{T_0}{t_{\mathsf{P}}}, \text{shape parameters}\right) \tag{af.37}$$

These relations are **independent of specific mass values** and depend only on:

- Energy distribution geometry
- Planck-scale ratios
- Dimensionless shape parameters

## af.6 Connection to Fundamental Physics

### Emergent Mass Concept

#### Mass as an Effective Parameter

In the mass-free formulation, what we traditionally call mass emerges as:

$$m_{\text{effective}} = E_{\text{characteristic}} \cdot f(\text{geometry}, \text{couplings}) \tag{af.38}$$

**Different Masses for Different Contexts:**

- **Rest Mass**: Intrinsic energy scale of localized excitation
- **Gravitational Mass**: Coupling strength to spacetime curvature
- **Inertial Mass**: Resistance to acceleration in external fields

All reducible to **energy scales and geometric factors**.

#### Resolution of Mass Hierarchies

The apparent hierarchy of particle masses becomes a hierarchy of **energy scales**:

$$\frac{m_t}{m_e} \to \frac{E_{\text{top}}}{E_{\text{electron}}} \tag{af.39}$$

$$\frac{m_W}{m_e} \to \frac{E_{\text{electroweak}}}{E_{\text{electron}}} \tag{af.40}$$

$$\frac{m_P}{m_e} \to \frac{E_{\mathsf{P}}}{E_{\text{electron}}} \tag{af.41}$$

**No fundamental mass parameters**, only energy scale ratios.

### Unification with Planck-Scale Physics

#### Natural Scale Emergence

All physics organizes itself naturally around the Planck scale:

$$\text{Microscopic Physics:} \quad E \ll E_P, \quad L \gg \ell_P \tag{af.42}$$
$$\text{Macroscopic Physics:} \quad E \ll E_P, \quad L \gg \ell_P \tag{af.43}$$
$$\text{Quantum Gravity:} \quad E \sim E_P, \quad L \sim \ell_P \tag{af.44}$$

#### Scale-Dependent Effective Theories

Different energy regimes correspond to different limits of the universal T0 theory:

$$E \ll E_P: \quad \text{Standard Model Limit} \tag{af.45}$$
$$E \sim \text{TeV}: \quad \text{Electroweak Unification} \tag{af.46}$$
$$E \sim E_P: \quad \text{Quantum Gravity Unification} \tag{af.47}$$

## af.7 Philosophical Implications

### Reductionism to the Planck Scale

The elimination of mass parameters shows that **all physics** is reducible to the **Planck scale**:

- No fundamental mass parameters exist
- Only energy and length ratios are important
- Universal dimensionless couplings emerge naturally
- Truly parameter-free physics achieved

### Ontological Implications

#### Mass as a Human Construct

The traditional concept of mass appears to be a **human construct** rather than fundamental reality:

- Useful for practical calculations
- Not present at the deepest level of the theory
- Emergent from more fundamental energy relations

#### Universal Energy Monism

The mass-free T0 model supports a form of **energy monism**:

- Energy as the only fundamental quantity
- All other quantities as energy relations
- Space and time as energy-derived concepts
- Matter as structured energy patterns

# Chapter ag

# Pure Energy T0 Theory: The Ratio-Based Formulation


## Abstract

This work presents a ratio-based formulation of T0 physics that is meant to get by with considerably fewer experimental parameters. Building on the simplified Dirac equation and the universal Lagrangian, we argue that fundamental physics can be described through dimensionless energy scale ratios rather than through assigned parameters. The T0 system requires one SI reference value to connect ratio-based physics to measurable quantities. In natural units, Einstein's $E = mc^2$ becomes the identity $E = m$; the energy relationships that follow hold exactly by definition, while composite multi-parameter formulas agree numerically to 99.98%. Physics is to be traced back to energy scale ratios, described by the field equation $\partial^2 E(x,t) = 0$, with quantitative predictions via the single parameter $\xi$ as SI reference scale (plus integer structural choices as motivated settings).

**Status markers.** Statements marked **[S]** are *speculative*: a hint or conjecture that has not yet been derived or numerically checked in the corpus.


## ag.1 The T0 Approach: From Parameters to Ratios

### The Change of Perspective

The T0 approach proposes a change of perspective in the understanding of fundamental physics:

**Comparison**

**Traditional Physics**: Multiple experimental parameters
- $G = 6.67 \times 10^{-11}$ m³/(kg·s²) (measured)
- $\alpha = 1/137$ (measured)
- $m_e = 9.109 \times 10^{-31}$ kg (measured)
- 20+ independent parameters required

**T0 Ratio-Based Physics**: Dimensionless scale relationships
- Physics through energy scale ratios
- One SI reference value for quantitative predictions
- Mathematical relationships, not experimental parameters
- Pure energy identities: $E = m$, $E = 1/L$, $E = 1/T$

## Building on T0 Foundations

This work forms the third stage of a three-stage development:
**Stage 1 - Simplified Dirac**: Complex 4×4 matrices → Simple field dynamics $\partial^2 \delta m = 0$
**Stage 2 - Universal Lagrangian**: 20+ fields → One equation $\mathscr{L} = \varepsilon \cdot (\partial \delta m)^2$
**Stage 3 - Ratio-Based Physics**: Multiple parameters → Energy scale ratios + SI reference

## The Energy Identity

In natural units ($\hbar = c = 1$), Einstein's equation takes the simple form:

$$\boxed{E = m} \tag{ag.1}$$

In these units it is an **identity**, not a conversion: mass and energy are treated as the same physical quantity.

**Universal Energy Relationships**

**Energy identity system**:

$$E = m \quad \text{(Mass is energy)} \tag{ag.2}$$
$$E = T_{\text{temp}} \quad \text{(Temperature is energy)} \tag{ag.3}$$
$$E = \omega \quad \text{(Frequency is energy)} \tag{ag.4}$$
$$E = \frac{1}{L} \quad \text{(Length is inverse energy)} \tag{ag.5}$$
$$E = \frac{1}{T} \quad \text{(Time is inverse energy)} \tag{ag.6}$$

**Accuracy**: exact by definition (identities in natural units)
**Complex formulas**: 99.98-100.04% (rounding errors accumulate)
**Consequence**: Fewer composite formulas introduce fewer rounding errors.

## ag.2 Part I: Pure Ratio-Based Physics (without SI Reference)

### Universal Energy Field Dynamics

In the T0 framework, all particles are energy excitation patterns in the universal field $E(x,t)(x,t)$:

$$\boxed{\partial^2 E(x,t) = 0} \tag{ag.7}$$

**Claim**: This Klein-Gordon equation for energy is meant to describe all particles.

### Universal Energy Lagrangian

$$\boxed{\mathscr{L} = \varepsilon \cdot (\partial E(x,t))^2} \tag{ag.8}$$

where $\varepsilon$ represents the energy scale coupling (dimensionless ratio).

### Anti-Energy: Symmetry

$$\boxed{E(x,t)_{\text{Antiparticle}} = -E(x,t)_{\text{Particle}}} \tag{ag.9}$$

**Physical picture**: Positive and negative energy excitations of the same field.
**Lagrangian universality**:

$$\mathscr{L}[+E(x,t)] = \varepsilon \cdot (\partial E(x,t))^2 \tag{ag.10}$$
$$\mathscr{L}[-E(x,t)] = \varepsilon \cdot (\partial E(x,t))^2 \tag{ag.11}$$

Same physics for particles and antiparticles through squaring.

### Pure Ratio Predictions (Independent of the Value of $\xi$)

#### Lepton Ratios

A universal ratio $a_e^{(T0)}/a_\mu^{(T0)} = 1$ does not hold: it rests on a $\xi$ contribution to the anomalous magnetic moment that is identical for all leptons, and such a contribution to $a_e$ is excluded (Doc. 067, Doc. 382; see also the section on quantitative lepton predictions).

#### Energy Independence Ratios

$$\boxed{\frac{\Delta\Gamma^\mu(E_1)}{\Delta\Gamma^\mu(E_2)} = 1} \tag{ag.12}$$

**Distinguishing feature**: In contrast to Standard Model running couplings.

## ag.3 Part II: Quantitative Predictions (SI Reference Required)

### The SI Reference Scale

To make quantitative predictions, T0 physics needs a connection to the SI system:

**SI Reference Scale**

**Definition**: $\xi$ is a dimensionless energy scale ratio and the only parameter of the framework; it is geometrically motivated, not fitted to measured data.
**Higgs energy ratio**:

$$\xi \approx \frac{\lambda_h^2 v^2}{16\pi^3 E_h^2} \tag{ag.13}$$

The Higgs expression is a consistency check, not an exact derivation (Doc. 385).
**Planck–Compton ratio**: The expression $2\ell_P/\lambda_C$ does not carry the reference value. For the electron, $2\ell_P/\lambda_C = 2\cdot 1.616\times10^{-35}\,\text{m}/2.426\times10^{-12}\,\text{m} = 1.33\times10^{-23}$ (with the reduced wavelength $\bar{\lambda}_C$: $8.37\times10^{-23}$), not $1.33\times10^{-4}$; the value $1.33\times10^{-4}$ would require $\lambda_C = 2.4\times10^{-31}$ m, which is no particle scale.
**SI reference value**: $\xi = 1.33\times10^{-4}$
**Role**: Connects dimensionless ratios to SI-measurable quantities

### Quantitative Lepton Predictions

With the SI reference scale:

$$a_\ell^{(T0)} = \frac{1}{2\pi}\times\xi^2\times\frac{1}{12} \tag{ag.14}$$

**Numerical calculation**:

$$a_\ell^{(T0)} = \frac{1}{2\pi} \times (1.33 \times 10^{-4})^2 \times \frac{1}{12} \tag{ag.15}$$

$$= \frac{1}{6.283} \times 1.77 \times 10^{-8} \times 0.0833 \tag{ag.16}$$

$$= 2.35 \times 10^{-10} \tag{ag.17}$$

**Universal Lepton Prediction**

**Electron g-2**: $a_e^{(T0)} = 2.35 \times 10^{-10}$
**Muon g-2**: $a_\mu^{(T0)} = 2.35 \times 10^{-10}$ (identical)
**Tau g-2**: $a_\tau^{(T0)} = 2.35 \times 10^{-10}$ (same value)
**Assessment**: $a_e$ is known experimentally to $\sim 1.3 \times 10^{-13}$ and agrees with the QED prediction at the $\sim 10^{-12}$ level. A universal additional contribution of $2.35 \times 10^{-10}$ lies more than two orders of magnitude above this and is therefore excluded.
**Muon anomaly**: With the Fermilab final result and the 2025 White Paper of the Muon $g-2$ Theory Initiative, the deviation between the measured $a_\mu$ and the Standard Model prediction is no longer significant; it serves here neither as evidence nor as the target of a T0 contribution (current status: Doc. 382).

### Quantitative QED Predictions

$$\frac{\Delta\Gamma^\mu}{\Gamma^\mu} = \xi^2 = 1.77 \times 10^{-8} \tag{ag.18}$$

**Energy independence verification**:

| Energy Scale | T0 Correction | Standard Model |
|---|---|---|
| 1 MeV | $1.77 \times 10^{-8}$ | Running $\alpha(E)$ |
| 1 GeV | $1.77 \times 10^{-8}$ | Running $\alpha(E)$ |
| 100 GeV | $1.77 \times 10^{-8}$ | Running $\alpha(E)$ |
| 1 TeV | $1.77 \times 10^{-8}$ | Running $\alpha(E)$ |

**Table ag.1:** Energy-independent T0 corrections vs. Standard Model

## ag.4 Experimental Verification Strategy

### Pure Ratio Tests (No SI Reference Needed)

**Lepton ratios**: A test of $a_e^{(T0)}/a_\mu^{(T0)} = 1$ is dropped, because the underlying universal $\xi$ contribution to $a_e$ is excluded (Doc. 067, Doc. 382).

**Test - Energy independence**:

- Measure QED corrections at different energies
- Ratio should be constant: $\Delta\Gamma(E_1)/\Delta\Gamma(E_2) = 1$

- Distinguishes from Standard Model running couplings

**Wavelength ratios**: A wavelength-dependent redshift ratio $z(\lambda_1)/z(\lambda_2) = \lambda_2/\lambda_1$ does not hold, because the redshift is achromatic in the T0 framework: the fractal path lengthening stretches all wavelengths by the same factor $1+z$, independent of $\lambda$; the scale of this factor is calibrated via the measured $H_0$ (P39).

### Quantitative Tests (Require SI Reference)

**Precision g-2 measurements**:

- Electron g-2: The agreement of $a_e$ with QED at $\sim 10^{-12}$ already excludes a universal contribution of $2.35 \times 10^{-10}$
- Muon g-2: The former anomaly is no longer significant (Doc. 382); a comparison with it is dropped
- Tau g-2: The same universal value is likewise excluded by the exclusion via $a_e$

**Multi-energy QED tests**:

- Measure absolute $\Delta\Gamma/\Gamma = 1.77 \times 10^{-8}$
- Verify energy independence across decades
- Compare with Standard Model predictions

## ag.5 Dark Matter and Dark Energy from Energy Ratios

**[S]** The statements in this section are speculative: the present corpus deliberately leaves the cosmic sector aside, because the Standard Model does not derive it either (P39).

### Dark Matter: Sub-threshold Energy Oscillations

**Ratio-based description**:

$$\frac{E(x,t)_{\text{dark}}}{E(x,t)_{\text{threshold}}} = \xi\sqrt{\frac{\rho_{\text{local}}}{\rho_{\text{critical}}}} \tag{ag.19}$$

**Physical mechanism**: Random phase energy oscillations below particle detection threshold.

### Dark Energy: Large-scale Energy Gradients

**Ratio-based energy density**:

$$\frac{\rho_\Lambda}{\rho_{\text{critical}}} = \frac{1}{2}\xi^2\left(\frac{E_{\text{Planck}}}{L_{\text{Hubble}} \cdot E_{\text{Planck}}}\right)^2 \tag{ag.20}$$

**Quantitative prediction**: $\rho_\Lambda \approx 6 \times 10^{-30}$ g/cm$^3$ (of the order of the observed value)

## ag.6 Philosophical Classification: Physics without Independent Matter

### Pure Energy Reality

**Dematerialized Description**

**Traditional view**: Matter, energy, forces, spacetime as separate entities
**T0 view**: Only energy patterns and their ratios
**What we call particles**: Localized energy concentrations
**What we call forces**: Energy gradient interactions
**What we call spacetime**: Energy pattern substrate
**What we call consciousness [S]**: Self-referential energy patterns
**Core statement**: Pure energy relationships governed by $\partial^2 E(x,t) = 0$

### From Complexity to Simplicity

**Physics evolution**:

1. **Antiquity**: Four elements
2. **Classical**: Particles in spacetime
3. **Modern**: Fields and forces
4. **Standard Model**: 20+ parameters
5. **T0 approach**: Energy ratios + one SI reference

**The aim is far-reaching simplification**: as few fundamental assumptions as possible.

### Consciousness and Energy Patterns

**A further question**: If everything is energy patterns, what about consciousness?

**[S] T0 interpretation**: Consciousness would be a self-observing energy pattern. We are temporary organizations of the universal energy field that have developed the ability for self-reference and subjective experience.

## ag.7 The Ratio Physics Legacy

### Intended Results

The ratio-based T0 approach pursues the following goals; the respective status is recorded in the detailed documents:

1. **Parameters reduced**: 20+ → $\xi$ as SI reference (plus integer settings)
2. **Forces described together**: Through energy gradient interactions
3. **Particle variety ordered**: Described as energy patterns
4. **Antiparticles interpreted**: Negative energy excitations

5. **Gravitation included**: Via energy-spacetime coupling
6. **Dark phenomena [S]**: Interpreted as energy field effects (speculative, P39)
7. **Exact identities**: By definition in natural units
8. **Ratio-based physics formulated**: Pure scale relationships

### The Two-Stage Testing Strategy

**Stage 1 - Pure ratios** (independent of the value of $\xi$):
- Energy-independent QED ratios
- Gravitational modification ratios

**Stage 2 - Quantitative predictions** (SI reference):
- Absolute g-2 corrections (a universal contribution of $2.35 \times 10^{-10}$ is excluded by $a_e$)
- Absolute QED vertex modifications
- Absolute cosmological parameters **[S]**
- Absolute dark matter/energy densities **[S]**

### Status of the Framework

**Basic Structure in the T0 Framework**

**In the T0 framework, the basic structure reduces to a few building blocks.**
**The fundamental equation**: $\partial^2 E(x,t) = 0$
**The universal ratios**: Energy scale relationships
**The SI connection**: One reference scale $\xi$
**Everything else**: Various solutions and patterns
**Deeper level**: Whether a deeper level exists remains open; the T0 framework does not provide for one
**Future work**: Mainly applications and measurements that test the framework

## ag.8 Conclusion: The Ratio-Based Universe

### Summary of the Structure

In the T0 framework, physics is described through energy scale ratios:
**Level 1**: Dimensionless energy ratios (independent of the SI reference)
**Level 2**: One SI reference scale (quantitative predictions)
**Level 3**: Pure energy patterns governed by $\partial^2 E(x,t) = 0$
The observed and measured phenomena are meant to follow from this simple ratio-based structure.

## Review

The path leads from the many parameters of traditional physics to a simpler, ratio-based energy dynamics.

**The core statement**: In this view, the core of nature lies not in complicated mathematics or exotic phenomena, but in simple scale relationships.

**One field**. **One equation**. **Energy ratios**. **One SI reference**.

Everything further would then be various patterns and ratios of energy, up to the pattern we call human consciousness **[S]**.

$$\boxed{\text{Reality} = \text{Energy ratios in } E(x,t)(x,t)} \tag{ag.21}$$

**In summary: In the T0 framework, the universe is described as a structure of energy ratios; this is tested through the tests listed above.**

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# Chapter ah

# T0 Model Verification: Ratio-Based vs. Parameter-Based Physics

## ah.1 Introduction: Ratio-Based vs. Parameter-Based Physics

**Note on versions.** The German and English versions of this document differ in structure and extent; earlier revisions shortened, extended or changed parts of both. Where they differ, the more extensive German version applies; sections contained only in this English version count as supplements.

This document presents a verification of the T0 model, starting from the view that $\xi$ is a scale ratio, not an assigned numerical value. This distinction is important for understanding the parametrization of the T0 model, which works with the single parameter $\xi$ (plus integer structural choices as motivated settings; SI anchors serve for conversion).

**Status markers.** Statements marked **[S]** are *speculative*: a hint or conjecture that has not yet been derived or numerically checked in the corpus.

**Two Formulations**

**Usual practice (literature):**

$$\xi = \frac{4}{3} \times 10^{-4} \quad \text{(numerical value assigned)} \tag{ah.1}$$

$$\alpha_{\text{EM}} = \frac{1}{137} \quad \text{(numerical value assigned)} \tag{ah.2}$$

$$G = 6.67 \times 10^{-11} \quad \text{(numerical value assigned)} \tag{ah.3}$$

**T0 formulation:**

$$\xi \approx \frac{\lambda_h^2 v^2}{16\pi^3 E_h^2} \quad \text{(Higgs energy scale ratio, consistency check)} \tag{ah.4}$$

$$\xi_\ell = \frac{2\ell_P}{\bar{\lambda}_C} \quad \text{(Planck-Compton length ratio)} \tag{ah.5}$$

There is no CODATA or measured value for $\xi$; the reference value is the setting $\xi = 4/30000$. With $\lambda_h = m_h^2/(2v^2) = 0.1291$ ($m_h = 125.1$ GeV, $v = 246.22$ GeV) the Higgs expression gives $\xi = m_h^2/(64\pi^3 v^2) = 1.301 \times 10^{-4}$, about $2.3\,\%$ below $4/30000$ ($97.6\,\%$); it is a consistency check, not an exact derivation (Doc. 385).

## ah.2 Calculation Verification

The following tables compare T0 calculations based on scale ratios with established SI reference values. All tables are scaled to fit Kindle/portrait format. Column *Status*: yes = agrees, new = new prediction without measured value, no = does not agree.

**Table ah.1:** T0 Model Verification – Part 1: Fundamental & Derived Constants

| Physical Quantity | SI Unit | T0 Ratio Formula | T0 Calculation | CODATA/Experimental | Agreement | Status |
|---|---|---|---|---|---|---|
| $\xi$ (Flat) | 1 | $\xi \approx \frac{\lambda_h^2 v^2}{16\pi^3 E_h^2}$ | $\mathbf{1.30 \times 10^{-4}}$ | $4/30000$ (setting, no measured value) | **97.6%** | yes |
| $\xi$ (Spherical) | 1 | $\xi = \frac{\lambda_h^2 v^2}{24\pi^{5/2} E_h^2}$ | $\mathbf{1.557 \times 10^{-4}}$ | T0 derivation | **N/A** | new |
| Electron mass | MeV | $m_e = f(\xi, \text{Higgs})$ | **0.511** | 0.51099895 | **99.998%** | yes |
| Compton wavelength | m | $\lambda_C = \frac{\hbar}{m_e c}$ | $\mathbf{3.862 \times 10^{-13}}$ | $3.8615927 \times 10^{-13}$ | **99.989%** | yes |
| Planck length | m | $\ell_P$ from $\xi$ scaling | $\mathbf{1.616 \times 10^{-35}}$ | $1.616255 \times 10^{-35}$ | **99.984%** | yes |

In Table ah.1, $m_e = f(\xi, \text{Higgs})$ denotes a dependence that is not written out, not a frequency and not the base winding number $f = 1/\xi$.

**Table ah.2:** T0 Model Verification – Part 2: QED Corrections

| Physical Quantity | SI Unit | T0 Ratio Formula | T0 Calculation | CODATA/Experimental | Agreement | Status |
|---|---|---|---|---|---|---|
| Vertex correction | 1 | $\frac{\Delta\Gamma}{\Gamma^{\mu}} = \xi^2$ | $\mathbf{1.742 \times 10^{-8}}$ | New | **N/A** | new |
| Energy indep. (1 MeV) | 1 | $f(E/E_P)$ | **1.000** | New | **N/A** | new |
| Energy indep. (100 GeV) | 1 | $f(E/E_P)$ | **1.000** | New | **N/A** | new |

**Table ah.3:** T0 Model Verification – Part 3: Cosmological Predictions [S]

| Physical Quantity | SI Unit | T0 Ratio Formula | T0 Calculation | CODATA/Experimental | Agreement | Status |
|---|---|---|---|---|---|---|
| $H_0$ (T0) | km/s/Mpc | see Doc. 026 | **66.2** | 67.4 (Planck) | **1.9%** | yes |
| Universe age | Gyr | $t_U = 1/H_0$ | **14.0** | 13.8 | **98.6%** | yes |
| *For complete $H_0$ derivation and measurement comparisons see Document 026.* | | | | | | |

**[S]** The cosmological values in Table ah.3 are speculative: the present corpus deliberately leaves the cosmic sector aside, because the Standard Model does not derive it either (P39).

**Table ah.4:** T0 Model Verification – Part 4: Physical Fields & Planck Current

| Physical Quantity | SI Unit | T0 Ratio Formula | T0 Calculation | CODATA/Experimental | Agreement | Status |
|---|---|---|---|---|---|---|
| Schwinger E-field | V/m | $E_S = \frac{m_e^2 c^3}{e\hbar}$ | $\mathbf{1.32 \times 10^{18}}$ | $1.32 \times 10^{18}$ | **100.0%** | yes |
| Critical B-field | T | $B_c = \frac{m_e^2 c^2}{e\hbar}$ | $\mathbf{4.41 \times 10^{9}}$ | $4.41 \times 10^{9}$ | **100.0%** | yes |
| Planck E-field | V/m | $E_P = \sqrt{\frac{c^7}{4\pi\varepsilon_0 \hbar G^2}}$ | $\mathbf{6.45 \times 10^{61}}$ | $6.45 \times 10^{61}$ | **100.0%** | yes |
| Planck B-field | T | $B_P = E_P/c$ | $\mathbf{2.15 \times 10^{53}}$ | $2.15 \times 10^{53}$ | **100.0%** | yes |
| Planck current (Std) | A | $I_P = \sqrt{\frac{c^6 \varepsilon_0}{G}}$ | $\mathbf{9.81 \times 10^{24}}$ | $3.479 \times 10^{25}$ | **28.2%** | no |
| Planck current (Corr) | A | $I_P = \sqrt{\frac{4\pi c^6 \varepsilon_0}{G}}$ | $\mathbf{3.479 \times 10^{25}}$ | $3.479 \times 10^{25}$ | **99.98%** | yes |

## ah.3 SI-Planck Units System Verification

### Complex Formula Method vs. Simple Energy Relationships

**Observation**

Simple relationships give numerically more accurate values than composite formulas, because fewer rounding errors accumulate.

**Table ah.5:** SI-Planck Units: Complex Formula Method

| Physical Quantity | SI Unit | Planck Formula | T0 Calculation | CODATA Reference | Agreement | Status |
|---|---|---|---|---|---|---|
| Planck time | s | $t_P = \sqrt{\frac{\hbar G}{c^5}}$ | $\mathbf{5.392 \times 10^{-44}}$ | $5.391 \times 10^{-44}$ | **100.016**% | yes |
| Planck length | m | $\ell_P = \sqrt{\frac{\hbar G}{c^3}}$ | $\mathbf{1.617 \times 10^{-35}}$ | $1.616 \times 10^{-35}$ | **100.030**% | yes |
| Planck mass | kg | $m_P = \sqrt{\frac{\hbar c}{G}}$ | $\mathbf{2.177 \times 10^{-8}}$ | $2.176 \times 10^{-8}$ | **100.044**% | yes |
| Planck temperature | K | $T_P = \sqrt{\frac{\hbar c^5}{G k_B^2}}$ | $\mathbf{1.417 \times 10^{32}}$ | $1.417 \times 10^{32}$ | **99.988**% | yes |
| Planck current | A | $I_P = \sqrt{\frac{4\pi c^6 \varepsilon_0}{G}}$ | $\mathbf{3.479 \times 10^{25}}$ | $3.479 \times 10^{25}$ | **99.980**% | yes |

**Note on Rounding Errors**

Complex formulas show 99.98–100.04% agreement due to rounding error accumulation. This is not a prediction error but a computational artifact.

## Simple Energy Relationships Method

**Table ah.6:** Natural Units: Simple Energy Relationships Method

| Physical Quantity | Relationship | Example | Electron Case | Numerical Value | Agreement | Status |
|---|---|---|---|---|---|---|
| **DIRECT ENERGY IDENTITIES - NO ROUNDING ERRORS** | | | | | | |
| Mass | $E = m$ | Energy = Mass | $0.511$ MeV | Same value | **100**% | yes |
| Temperature | $E = T$ | Energy = Temperature | $5.93 \times 10^9$ K | Direct conversion | **100**% | yes |
| Frequency | $E = \omega$ | Energy = Frequency | $7.76 \times 10^{20}$ Hz | Direct identity | **100**% | yes |
| **INVERSE ENERGY RELATIONSHIPS - EXACT** | | | | | | |
| Length | $E = 1/L$ | Energy = 1/Length | $3.862 \times 10^{-13}$ m | Inverse relationship | **100**% | yes |
| Time | $E = 1/T$ | Energy = 1/Time | $1.288 \times 10^{-21}$ s | Inverse relationship | **100**% | yes |
| **T0 ENERGY PARAMETERS - PURE RATIOS** | | | | | | |
| $\xi$ (Higgs, Flat) | $E_h/E_P$ | Energy ratio | $1.30 \times 10^{-4}$ | Higgs check against $4/30000$ | **97.6**% | yes |
| $\xi$ (Higgs, Sph) | $E_h/E_P$ | Corrected ratio | $1.557 \times 10^{-4}$ | T0 derivation | **100**% | new |
| $\xi$ Geometric | $E_\ell/E_P$ | Length-energy ratio | $8.37 \times 10^{-23}$ | Pure geometry | **100**% | yes |
| EM geometry factor | Ratio | $\sqrt{4\pi/9}$ | $1.18164$ | Mathematically exact | **100**% | new |
| **COMPLETE SI UNITS ENERGY COVERAGE - ALL 7/7 UNITS** | | | | | | |
| Electric current | $I = E/T$ | Energy flow rate | $[E]$ dim. | Direct energy relationship | **100**% | yes |
| Amount of substance | $[E^2]$ dim. | Energy density ratio | Dimensional structure | SI-defined $N_A$ | **Def.** | new |
| Luminous intensity | $[E^3]$ dim. | Energy flow perception | Dimensional structure | SI-defined 683 lm/W | **Def.** | new |

**Accuracy through Simplification**

**Complex Formula Method (Traditional Physics):**

- Uses: $\sqrt{\frac{\hbar G}{c^5}}$, multiple constants, conversion factors
- Result: 99.98–100.04% agreement (rounding errors accumulate)
- Problem: Each calculation step introduces small errors

**Simple Energy Relationships Method (T0 Physics):**

- Uses: Direct identities $E = m$, $E = 1/L$, $E = 1/T$
- Result: exact by definition (identities in natural units)
- Advantage: No intermediate calculations, no error accumulation

**CLASSIFICATION:** The difference is computational: identities in natural units avoid intermediate steps and thus rounding errors. This fits the T0 view that energy is the fundamental quantity; it is not a proof of it.
**CONCLUSION**: Simple relationships are numerically no less accurate here.

## ah.4 The $\xi$ Parameter Hierarchy

### Clarification

**Note: Use of the Symbol $\xi$ in This Document**

**Possible misunderstanding:** Treating $\xi$ as a universal parameter
**In this document:** $\xi$ is a **class of dimensionless scale ratios**, not a single value.
**Consequence of a mix-up:** Misinterpretations, wrong numerical values, dimensional errors.
$\xi$ represents any dimensionless ratio of the form:

$$\xi = \frac{\text{T0-characteristic energy scale}}{\text{Reference energy scale}} \tag{ah.6}$$

The T0 model uses $\xi$ to denote various dimensionless ratios in different physical contexts.

### The three fundamental $\xi$ energy scales

**Table ah.7:** The three fundamental $\xi$ parameter types in the T0 model

| Context | Definition | Typical Value | Physical Meaning |
|---|---|---|---|
| **Energy-dependent** | $\xi_E = 2\sqrt{G} \cdot E = 2E/E_P$ | $\leq 2$ (for $E \leq E_P$) | Energy-field coupling |
| **Higgs sector** | $\xi_H = \frac{\lambda_h^2 v^2}{16\pi^3 E_h^2}$ | $1.30 \times 10^{-4}$ | Energy scale ratio |
| **Scale hierarchy** | $\xi_\ell = \frac{2\ell_P}{\lambda_C}$ | $8.37 \times 10^{-23}$ | Energy hierarchy ratio |

In natural units $\sqrt{G} = 1/E_P$, hence $\xi_E = 2E/E_P \leq 2$ for all $E \leq E_P$; in SI, $2\sqrt{G}E$ is not dimensionless. The numerical value $8.37 \times 10^{-23}$ of $\xi_\ell$ is $2\ell_P/\bar{\lambda}_C = 2 \cdot 1.616 \times 10^{-35}$ m$/3.862 \times 10^{-13}$ m with $\bar{\lambda}_C = \hbar/(m_e c)$.

### Application rules

**Application Rules for $\xi$ Parameters (Pure Energy)**

**Rule 1: Universal energy-dependent systems (RECOMMENDED)**

$$\text{Use } \xi_E = 2\sqrt{G} \cdot E = 2E/E_P \leq 2 \text{ (natural units), where } E \text{ is the relevant energy} \tag{ah.7}$$

**Rule 2: Cosmological/coupling unification (SPECIAL CASES)**

$$\text{Use } \xi_H = 1.30 \times 10^{-4} \text{ (Higgs energy ratio)} \tag{ah.8}$$

**Rule 3: Pure energy hierarchy analysis (THEORETICAL)**

$$\text{Use } \xi_\ell = 8.37 \times 10^{-23} \text{ (energy scale ratio)} \tag{ah.9}$$

**Note:** In practice, Rule 1 applies to the vast majority of T0 calculations due to the pronounced T0 scale hierarchy.

## ah.5 Important Insights from Verification

### Main Results

**Main Results of T0 Verification**

**1. Scale ratio validation:**
- Established values: 99.99% agreement with CODATA
- Geometric $\xi$ ratio: 100.003% agreement with Planck-Compton calculation
- Dimensional consistency of the checked quantities

**2. New testable predictions:**
- QED vertex ratios: $1.74 \times 10^{-8}$ (energy-independent)
- **[S]** Cosmological $H_0^{\text{T0}} \approx 66.2$ km/s/Mpc (calibrated to $H_0$, see Document 026)

**3. Overall assessment:**
- Established values: 99.99% agreement
- New predictions: 14+ testable ratios
- Dimensional consistency: given for all table quantities
- Scale ratio basis: internally consistent

### Experimental Testability

The ratio-based nature of the T0 model enables specific experimental tests:

1. **Energy scale-independent QED corrections**:

$$\frac{\Delta\Gamma^{\mu}(E_1)}{\Delta\Gamma^{\mu}(E_2)} = 1 \quad \text{for all } E_1, E_2 \ll E_P \tag{ah.10}$$

2. **Cosmological scale ratios [S]**:

$$\frac{\kappa}{H_0} = \xi \approx \frac{\lambda_h^2 v^2}{16\pi^3 E_h^2} \tag{ah.11}$$

## ah.6 Conclusions

The verification supports the central statement of the T0 model: **Fundamental physics can be based on scale ratios instead of assigned parameters**. The $\xi$ ratio characterizes these proportionalities and enables a description of physical phenomena with the single parameter $\xi$ (plus integer structural choices as motivated settings; SI anchors serve for conversion).

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# Chapter ai

# Dynamic Mass of Photons and Its Implications for Nonlocality


## Abstract

This updated work examines the implications of assigning a dynamic, frequency-dependent effective mass to photons within the comprehensive framework of the T0 model, building upon the complete field-theoretic derivation and natural units system where $\hbar = c = \alpha_{\text{EM}} = \beta_{\text{T}} = 1$. The theory establishes the fundamental relationship $T(x,t) = \frac{1}{\max(m,\omega)}$ with dimension $[E^{-1}]$, providing a unified treatment of massive particles and photons through the three fundamental field geometries. The dynamic photon mass $m_\gamma = \omega$ introduces energy-dependent nonlocality effects, with testable predictions. All formulations maintain strict dimensional consistency with the fixed T0 parameters $\beta = 2Gm/r$, $\xi = 2\sqrt{G} \cdot m$, and the cosmic screening factor $\xi_{\text{eff}} = \xi/2$ for infinite fields.


## ai.1 Introduction: T0 Model Foundation for Photon Dynamics

This updated analysis builds upon the comprehensive T0 model framework established in the field-theoretic derivation, incorporating the complete geometric foundations and natural units system. The dynamic effective mass concept for photons emerges naturally from the T0 model's fundamental time-mass duality principle.

### Fundamental T0 Model Framework

The T0 model is based on the intrinsic time field definition:

$$\boxed{T(x,t) = \frac{1}{\max(m(\vec{x},t), \omega)}} \tag{ai.1}$$

**Dimensional verification**: $[T(x,t)] = [1/E] = [E^{-1}]$ in natural units ✓
This field satisfies the fundamental field equation:

$$\nabla^2 m(\vec{x},t) = 4\pi G\rho(\vec{x},t) \cdot m(\vec{x},t) \tag{ai.2}$$

From this foundation emerge the key parameters:

**T0 Model Parameters for Photon Analysis**

$$\beta = \frac{2Gm}{r} \quad [1] \text{ (dimensionless)} \tag{ai.3}$$
$$\xi = 2\sqrt{G} \cdot m \quad [1] \text{ (dimensionless)} \tag{ai.4}$$
$$\beta_T = 1 \quad [1] \text{ (natural units)} \tag{ai.5}$$
$$\alpha_{\text{EM}} = 1 \quad [1] \text{ (natural units)} \tag{ai.6}$$

### Photon Integration in Time-Mass Duality

For photons, the T0 model assigns an effective mass:

$$m_\gamma = \omega \tag{ai.7}$$

**Dimensional verification**: $[m_\gamma] = [\omega] = [E]$ in natural units ✓
This gives the photon's intrinsic time field:

$$T(x,t)_\gamma = \frac{1}{\omega} \tag{ai.8}$$

**Practical Simplification**

**Simplification:** Since all measurements in our finite, observable universe are made locally, only the **localized field geometry** is used:
$\xi = 2\sqrt{G} \cdot m$ and $\beta = \frac{2Gm}{r}$ for all applications.
The cosmic screening factor $\xi_{\text{eff}} = \xi/2$ is dropped.

**Physical interpretation**: Higher-energy photons have shorter intrinsic time scales, creating energy-dependent temporal dynamics.

## ai.2 Energy-Dependent Nonlocality and Quantum Correlations

### Entangled Photon Systems

For entangled photons with energies $\omega_1$ and $\omega_2$, the time field difference is:

$$\Delta T_\gamma = \left| \frac{1}{\omega_1} - \frac{1}{\omega_2} \right| \tag{ai.9}$$

**Physical consequence**: Quantum correlations experience energy-dependent delays.

### Modified Bell Inequality

The energy-dependent time fields lead to a modified Bell inequality:

$$|E(a,b) - E(a,c)| + |E(a',b) + E(a',c)| \leq 2 + \epsilon(\omega_1, \omega_2) \tag{ai.10}$$

where:

$$\epsilon(\omega_1, \omega_2) = \alpha_{\text{corr}} \left| \frac{1}{\omega_1} - \frac{1}{\omega_2} \right| \frac{2G\langle m \rangle}{r} \tag{ai.11}$$

with $\alpha_{\text{corr}}$ being a correlation coupling constant and $\langle m \rangle$ the average mass in the experimental setup.

## ai.3 Experimental Predictions and Tests

### High-Precision Quantum Optics Tests

#### Energy-Dependent Bell Tests

Predicted time delay between entangled photons:

$$\Delta t_{\text{corr}} = \frac{G\langle m \rangle}{r} \left| \frac{1}{\omega_1} - \frac{1}{\omega_2} \right| \tag{ai.12}$$

For laboratory conditions with $\langle m \rangle \sim 10^{-3}$ kg, $r \sim 10$ m, and $\omega_1, \omega_2 \sim 1$ eV:

$$\Delta t_{\text{corr}} \lesssim 5 \times 10^{-47} \text{ s} \tag{ai.13}$$

## ai.4 Dimensional Consistency Verification

| Equation | Left Side | Right Side | Status |
|---|---|---|---|
| Photon effective mass | $[m_\gamma] = [E]$ | $[\omega] = [E]$ | ✓ |
| Photon time field | $[T_\gamma] = [E^{-1}]$ | $[1/\omega] = [E^{-1}]$ | ✓ |
| Energy loss rate | $[d\omega/dr] = [E^2]$ | $[g_T \omega^2 2G/r^2] = [E^2]$ | ✓ |
| Time field difference | $[\Delta T_\gamma] = [E^{-1}]$ | $[\lvert 1/\omega_1 - 1/\omega_2 \rvert] = [E^{-1}]$ | ✓ |
| Bell correction | $[\epsilon] = [1]$ | $[\alpha_{\text{corr}} \Delta T_\gamma \beta] = [1]$ | ✓ |

**Table ai.1:** Dimensional consistency verification for photon dynamics in T0 model

# Chapter aj

# T0 Model: Field-Theoretic Derivation of the $\beta$ Parameter

## aj.1 Introduction and Motivation

The T0 model introduces a fundamentally new perspective on spacetime, where time itself becomes a dynamic field. At the center of this theory lies the dimensionless $\beta$-parameter, which characterizes the strength of the time field and establishes a direct connection between gravitational and electromagnetic interactions.

This work focuses exclusively on the mathematically rigorous derivation of the $\beta$-parameter from the fundamental field equations of the T0 model, avoiding the complexity of additional scaling parameters.

**Central Result**

The $\beta$-parameter is derived as:

$$\boxed{\beta = \frac{Gm}{r} = \frac{r_s}{2r}} \tag{aj.1}$$

where $G$ is the gravitational constant, $m$ is the source mass, $r$ is the distance from the source, and $r_s = 2Gm$ is the Schwarzschild radius.

## aj.2 Natural Units Framework

The T0 model employs the system of natural units established in modern quantum field theory [Peskin & Schroeder(1995), Weinberg(1995)]:

- $\hbar = 1$ (reduced Planck constant)
- $c = 1$ (speed of light)

This system reduces all physical quantities to energy dimensions and follows the tradition established by Dirac [Dirac(1958)].

**Dimensions in Natural Units**

- Length: $[L] = [E^{-1}]$
- Time: $[T] = [E^{-1}]$
- Mass: $[M] = [E]$
- The $\beta$-parameter: $[\beta] = [1]$ (dimensionless)

## aj.3 Fundamental Structure of the T0 Model

### Time-Mass Duality

The central principle of the T0 model is the time-mass duality, which states that time and mass are inversely linked. This relationship differs fundamentally from the conventional treatment in general relativity [Einstein(1915), Misner et al.(1973)].

| Theory | Time | Mass | Reference |
|---|---|---|---|
| Einstein GR | $dt' = \sqrt{g_{00}}dt$ | $m_0 = \text{const}$ | [Einstein(1915), Misner et al.(1973)] |
| Special Relativity | $t' = \gamma t$ | $m_0 = \text{const}$ | [Einstein(1905)] |
| T0 Model | $T(x) = \frac{1}{m(x)}$ | $m(x) = \text{dynamic}$ | This work |

**Table aj.1:** Comparison of time-mass treatment in different theories

### Fundamental Field Equation

The fundamental field equation of the T0 model is derived from variational principles, analogous to the approach for scalar field theories [Weinberg(1995)]:

$$\nabla^2 m(x) = 4\pi G\rho(x) \cdot m(x) \tag{aj.2}$$

This equation shows structural similarity to the Poisson equation of gravitation $\nabla^2\phi = 4\pi G\rho$ [Jackson(1998)], but is nonlinear due to the factor $m(x)$ on the right-hand side.

The time field follows directly from the inverse relationship:

$$T(x) = \frac{1}{m(x)} \tag{aj.3}$$

## aj.4 Geometric Derivation of the $\beta$-Parameter

### Spherically Symmetric Point Source

For a point mass source, we use the established methodology for solving Einstein's field equations [Schwarzschild(1916), Misner et al.(1973)]. The mass density of a point source is described by the Dirac delta function:

$$\rho(\vec{x}) = m_0 \cdot \delta^3(\vec{x}) \tag{aj.4}$$

where $m_0$ is the mass of the point source.

## Solution of the Field Equation

Outside the source ($r > 0$), where $\rho = 0$, the field equation reduces to:

$$\nabla^2 m(r) = 0 \tag{aj.5}$$

The spherically symmetric Laplace operator [Jackson(1998), Griffiths(1999)] yields:

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2 \frac{dm}{dr}\right) = 0 \tag{aj.6}$$

The general solution to this equation is:

$$m(r) = \frac{C_1}{r} + C_2 \tag{aj.7}$$

## Determination of Integration Constants

**Asymptotic boundary condition**: For large distances, the time field should assume a constant value $T_0$:

$$\lim_{r\to\infty} T(r) = T_0 \quad \Rightarrow \quad \lim_{r\to\infty} m(r) = \frac{1}{T_0} \tag{aj.8}$$

This gives us: $C_2 = \frac{1}{T_0}$

**Behavior at the origin**: Using Gauss's theorem [Griffiths(1999), Jackson(1998)] for a small sphere around the origin:

$$\oint_S \nabla m \cdot d\vec{S} = 4\pi G \int_V \rho(r) m(r)\, dV \tag{aj.9}$$

For a small radius $\epsilon$:

$$4\pi\epsilon^2 \left.\frac{dm}{dr}\right|_{r=\epsilon} = 4\pi G m_0 \cdot m(\epsilon) \tag{aj.10}$$

With $\frac{dm}{dr} = -\frac{C_1}{r^2}$ and $m(\epsilon) \approx \frac{1}{T_0}$ for small $\epsilon$:

$$4\pi\epsilon^2 \cdot \left(-\frac{C_1}{\epsilon^2}\right) = 4\pi G m_0 \cdot \frac{1}{T_0} \tag{aj.11}$$

This yields: $-C_1 = \frac{G m_0}{T_0}$, hence $C_1 = -\frac{G m_0}{T_0}$.

### The Characteristic Length Scale

The complete solution reads:

$$m(r) = \frac{1}{T_0}\left(1 - \frac{Gm_0}{r}\right) \tag{aj.12}$$

The corresponding time field is:

$$T(r) = \frac{T_0}{1 - \frac{Gm_0}{r}} \tag{aj.13}$$

For the practically important case $Gm_0 \ll r$, we obtain the approximation:

$$T(r) \approx T_0\left(1 + \frac{Gm_0}{r}\right) \tag{aj.14}$$

The characteristic length scale at which the time field significantly deviates from $T_0$ is:

$$\boxed{r_0 = Gm_0} \tag{aj.15}$$

This scale is equal to half the Schwarzschild radius $r_s = 2GM/c^2 = 2Gm$ in geometric units [Misner et al.(1973), Carroll(2004)].

### Definition of the $\beta$-Parameter

The dimensionless $\beta$-parameter is defined as the ratio of the characteristic length scale to the actual distance:

$$\boxed{\beta = \frac{r_0}{r} = \frac{Gm_0}{r}} \tag{aj.16}$$

This parameter measures the relative strength of the time field at a given point. In units of the Schwarzschild radius, $\beta = r_s/(2r)$. The form $\beta = 2Gm/r = r_s/r$ that is sometimes used does not follow from the derivation; it is twice as large.

## aj.5 Physical Interpretation of the $\beta$-Parameter

### Dimensional Analysis

The dimensionlessness of the $\beta$-parameter in natural units:

$$[\beta] = \frac{[G][m]}{[r]} = \frac{[E^{-2}][E]}{[E^{-1}]} = [1] \tag{aj.17}$$

### Connection to Classical Physics

The $\beta$-parameter shows direct connections to established physical concepts:

- **Gravitational potential**: $\beta$ is proportional to the Newtonian potential $\Phi = -Gm/r$
- **Schwarzschild radius**: $\beta = r_s/(2r)$ in geometric units
- **Escape velocity**: $\beta$ is related to $v_{\text{esc}}^2/c^2$

### Limiting Cases and Application Domains

| Physical System | Typical $\beta$-Value | Regime |
|---|---|---|
| Hydrogen atom | $\sim 2 \times 10^{-44}$ | Quantum mechanics |
| Earth (surface) | $\sim 10^{-9}$ | Weak gravitation |
| Sun (surface) | $\sim 10^{-6}$ | Stellar physics |
| Neutron star | $\sim 0.1$ | Strong gravitation |
| Schwarzschild horizon | $\beta = 1/2$ | Limiting case |

**Table aj.2:** Typical $\beta$-values for various physical systems

For the hydrogen atom, $\beta = Gm_p/(a_0c^2) = 2.3 \times 10^{-44}$; the frequently quoted value $10^{-39}$ is the ratio of gravitational to Coulomb force, not $\beta$.

## aj.6 Comparison with Established Theories

### Connection to General Relativity

In general relativity, the parameter $r_s/r = 2Gm/r$ characterizes the strength of the gravitational field. The T0 parameter $\beta = Gm/r = r_s/(2r)$ is half as large; it is exactly the first-order term in $\sqrt{g_{00}} = \sqrt{1 - 2Gm/r} \approx 1 - Gm/r$.

### Differences from the Standard Model

While the Standard Model of particle physics treats time as an external parameter, the T0 model makes time a dynamic field. The $\beta$-parameter quantifies this dynamics and represents a measurable deviation from standard physics.

## aj.7 Experimental Predictions

### Time Dilation Effects

The T0 model predicts a modified time dilation:

$$\frac{dt}{dt_0} = 1 - \beta = 1 - \frac{Gm}{r} \tag{aj.18}$$

This relationship is identical to the gravitational time dilation of GR in first order, $dt/dt_0 = \sqrt{1 - 2Gm/r} \approx 1 - Gm/r$, but offers a fundamentally different theoretical foundation. With the form $\beta = 2Gm/r$, $1 - \beta$ would be twice as large and would not agree with GR to first order.

### Spectroscopic Tests

The $\beta$-parameter could be tested through high-precision spectroscopy:
- Gravitational redshift in stellar spectra
- Atomic clock experiments in different gravitational potentials
- High-precision interferometry

## aj.8 Mathematical Consistency

### Conservation Laws

The derivation of the $\beta$-parameter respects fundamental conservation laws:
- **Energy conservation**: Guaranteed by the Lagrangian formulation
- **Momentum conservation**: From spatial translation invariance
- **Dimensional consistency**: Verified in all derivation steps

### Solution Stability

The spherically symmetric solution is stable against small perturbations, which can be shown by linearization around the ground state solution.

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# Chapter ak

# T0 Model: Universal Energy Relations for Mole and Candela Units


## Abstract

This document provides the complete derivation of energy-based relationships for the amount of substance (mol) and luminous intensity (candela) within the T0 model framework. Contrary to conventional assumptions that these quantities are "non-energy" units, we demonstrate that both can be rigorously derived from the fundamental T0 energy scaling parameter $\xi = 2\sqrt{G} \cdot E$. The mol emerges as a dimensionless ratio of total energy content to particle energy scale, while the candela is traced back to an electromagnetic energy flux ($[ET^{-1}] = [E^2]$ in natural units), weighted by dimensionless factors and the conversion constant 683 lm/W. These derivations establish that all 7 SI base units have fundamental energy relationships, confirming energy as the universal physical quantity predicted by the T0 model.


## ak.1 Introduction: The Energy Universality Problem

### Conventional View: "Non-Energy" Units

Standard physics categorizes SI base units into those with apparent energy relationships and those without:

**Energy-related (5/7):** Second, meter, kilogram, ampere, kelvin **Non-energy (2/7):** Mol (particle counting), candela (physiological)

This classification suggests fundamental limitations in the universality of energy-based physics.

### T0 Model Challenge

The T0 model, based on the universal energy scaling:

$$\xi = 2\sqrt{G} \cdot E \tag{ak.1}$$

predicts that **all** physical quantities should have energy relationships. This document resolves the apparent contradiction by deriving energy-based formulations for mol and candela.

## ak.2 Fundamental T0 Energy Framework

### The Universal Time-Energy Field

The T0 model establishes that all physics emerges from the fundamental relationship:

$$T(x,t) = \frac{1}{\max(E(\vec{x},t), \omega)} \tag{ak.2}$$

where $E(\vec{x},t)$ represents the local energy scale and $\omega$ the characteristic frequency.

### Field Equation and Energy Density

The governing field equation in energy formulation:

$$\nabla^2 T(x,t) = -4\pi G \frac{\rho_E(\vec{x},t)}{E_{\text{P}}} \cdot \frac{T(x,t)^2}{t_{\text{P}}^2} \tag{ak.3}$$

connects energy density $\rho_E(\vec{x},t)$ to the time field through universal constants.

## ak.3 Amount of Substance (Mol): Energy Density Approach

### Reconceptualizing "Amount"

#### Traditional Particle Counting

Conventional definition:

$$n_{\text{conventional}} = \frac{N_{\text{particles}}}{N_A} \tag{ak.4}$$

**Problems with this approach:**

- Treats particles as abstract entities
- No connection to physical energy content
- Apparently dimensionless
- Lacks fundamental theoretical basis

#### T0 Model: Particles as Energy Excitations

In the T0 framework, particles are localized solutions to the energy field equation. A "particle" is characterized by:

$$\text{Particle} \equiv \text{Localized energy excitation with characteristic scale } E_{\text{char}} \tag{ak.5}$$

## T0 Derivation of Amount of Substance

### Energy Integration Approach

The “amount” becomes the ratio between total energy content and individual particle energy:

$$\boxed{n_{\text{T0}} = \frac{1}{N_A} \int_V \frac{\rho_E(\vec{x}, t)}{E_{\text{char}}} \, d^3x} \tag{ak.6}$$

**Physical components:**

- $\rho_E(\vec{x}, t)$: Energy density field from T0 model
- $E_{\text{char}}$: Characteristic energy scale of particle type
- $V$: Integration volume containing the substance
- $N_A$: Emerges from T0 energy scaling relationships

### Dimensional Analysis

**Dimension:**

$$[n_{\text{T0}}] = \frac{[1][\rho_E][L^3]}{[E_{\text{char}}]} = \frac{[1][EL^{-3}][L^3]}{[E]} = [1] \tag{ak.7}$$

**Explanation:** A ratio of total energy to individual energy is dimensionless, and the normalization factor $N_A$ is a pure number; in this formulation the amount of substance is a dimensionless energy ratio. An $[E^2]$ dimension does not follow. The number $4.040 \times 10^{-33}$ in the section on dimensional verification is $N_A \cdot (1\,\text{eV}/E_P)^2 = 6.022 \times 10^{23}/(1.221 \times 10^{28})^2$ and is likewise dimensionless.

## Connection to T0 Scaling Parameter

### Energy Scale Relationship

For atomic-scale particles:

$$\xi_{\text{atomic}} = 2\sqrt{G} \cdot E_{\text{char}} \approx 2\sqrt{G} \cdot (1\ \text{eV}) \approx 10^{-28} \tag{ak.8}$$

### Avogadro’s Number from T0 Scaling

As an ansatz:

$$N_A^{(\text{T0})} = \left( \frac{E_{\text{char}}}{E_{\text{P}}} \right)^{-2} \cdot \mathscr{C}_{\text{T0}} \tag{ak.9}$$

where $\mathscr{C}_{\text{T0}}$ is meant to be a dimensionless constant from T0 field geometry. The relation is dimensionless and thus compatible with the dimensionless mol; however, $\mathscr{C}_{\text{T0}}$ is not derived, so the relation remains an ansatz.

# ak.4 Luminous Intensity (Candela): Energy Flux Perception

## Reconceptualizing "Luminous Intensity"

### Traditional Physiological Definition

Conventional definition:

$$I_{\text{conventional}} = \frac{\Phi_v}{\Omega}, \qquad \Phi_v = 683 \text{ lm/W} \times \Phi_{\text{radiometric}} \times V(\lambda) \tag{ak.10}$$

where $V(\lambda)$ is the human eye sensitivity function, $\Phi_v$ the luminous flux in lm and $\Omega$ the solid angle (luminous intensity in cd = lm/sr).

**Problems with this approach:**

- Depends on human physiology
- No fundamental physical basis
- Arbitrary normalization (683 lm/W)
- Limited to narrow wavelength range

### T0 Model: Universal Energy Flux Interaction

The T0 model reveals luminous intensity as electromagnetic energy flux interaction with the universal time field.

## T0 Derivation of Luminous Intensity

### Photon-Time Field Interaction

For electromagnetic radiation, the T0 time field becomes:

$$T_{\text{photon}}(\vec{x}, t) = \frac{1}{\max(E_{\text{photon}}, \omega)} \tag{ak.11}$$

### Visual Energy Range in T0 Framework

Human vision operates in the range $E_{\text{vis}} \approx 1.8 - 3.1$ eV. The T0 scaling parameter for this range:

$$\xi_{\text{visual}} = 2\sqrt{G} \cdot E_{\text{vis}} = 2\sqrt{G} \cdot (2.4 \text{ eV}) \approx 3.9 \times 10^{-28} \tag{ak.12}$$

### T0 Luminous Intensity Formula

The complete T0 derivation yields:

$$\boxed{I_{\text{T0}} = C_{T0} \cdot \frac{E_{\text{vis}}}{E_{\text{P}}} \cdot \Phi_\gamma \cdot \eta_{\text{vis}}(\lambda)} \tag{ak.13}$$

**Physical components:**

- $C_{T0} \approx 683$ lm/W: T0 coupling constant (derived from energy ratios)
- $E_{\text{vis}}/E_{\text{P}}$: Visual energy relative to Planck energy
- $\Phi_\gamma$: Electromagnetic energy flux
- $\eta_{\text{vis}}(\lambda)$: T0-derived efficiency function

## Dimensional Analysis and Energy Nature

### Complete Dimensional Analysis

$$[I_{\text{T0}}] = [C_{T0}] \cdot \frac{[E]}{[E]} \cdot [ET^{-1}] \cdot [1] \tag{ak.14}$$

$$= [\text{lm/W}] \cdot [1] \cdot [ET^{-1}] \cdot [1] \tag{ak.15}$$

$$= [\text{lm/W}] \cdot [E^2] \quad \text{(in natural units where } [T] = [E^{-1}]) \tag{ak.16}$$

### Physical Interpretation

The candela represents:

$$\text{Candela} = C_{T0} \times \text{energy flux per solid angle} \times [1] = [\text{lm/W}] \cdot [ET^{-1}] = [\text{lm/W}] \cdot [E^2] \tag{ak.17}$$

The factors $E_{\text{vis}}/E_{\text{P}}$ and $\eta_{\text{vis}}$ are dimensionless, and so is the solid angle; $C_{T0} = 683$ lm/W is a fixed conversion constant. The physical dimension is carried by the energy flux alone, $[ET^{-1}] = [E^2]$. An additional $[E^2]$ *energy interaction* with the detection system and an $[E^3]$ dimension do not follow from this; they belonged to the same $[E^2]$ picture that does not hold for the mol (see the section on the dimensional analysis of the amount of substance).

## T0 Visual Efficiency Function

### Energy-Based Efficiency Derivation

The visual efficiency function emerges from T0 energy scaling:

$$\eta_{\text{vis}}(\lambda) = \exp\left(-\frac{(E_{\text{photon}} - E_{\text{vis,peak}})^2}{2\sigma_{\text{T0}}^2}\right) \tag{ak.18}$$

where:

$$E_{\text{vis,peak}} = 2.4 \text{ eV} \quad \text{(T0-predicted peak)} \tag{ak.19}$$

$$\sigma_{\text{T0}} = \sqrt{\frac{E_{\text{vis,peak}}}{E_{\text{P}}}} \cdot E_{\text{vis,peak}} \quad \text{(T0-derived width)} \tag{ak.20}$$

With $E_{\text{P}} = 1.221 \times 10^{28}$ eV, $\sigma_{\text{T0}} = \sqrt{2.4/1.221 \times 10^{28}} \cdot 2.4$ eV $= 3.4 \times 10^{-14}$ eV. The function is therefore practically zero away from exactly $2.4$ eV; its maximum lies at $hc/2.4\,\text{eV} = 517$ nm, not at 555 nm.

### Connection to T0 Coupling Constant

The T0 model predicts the coupling constant:

$$C_{T0} = 683 \text{ lm/W} = f\left(\frac{E_{\text{vis}}}{E_{\text{P}}}, \xi_{\text{visual}}\right) \tag{ak.21}$$

Here $f$ denotes a general dependence that is not written out, not a frequency and not the base winding number $f = 1/\xi$.

This provides a fundamental derivation of the seemingly arbitrary 683 lm/W factor.

## ak.5 Universal Energy Relations: Complete Analysis

### All SI Units: Energy-Based Classification

#### Complete T0 Coverage

| SI Unit | T0 Relation | Energy Dim. | T0 Parameter | Status |
|---|---|---|---|---|
| Second (s) | $T = 1/E$ | $[E^{-1}]$ | Direct | Fundamental |
| Meter (m) | $L = 1/E$ | $[E^{-1}]$ | Direct | Fundamental |
| Kilogram (kg) | $M = E$ | $[E]$ | Direct | Fundamental |
| Kelvin (K) | $\Theta = E$ | $[E]$ | Direct | Fundamental |
| Ampere (A) | $I \propto E_{\text{charge}}$ | Complex | $\xi_{\text{EM}}$ | Electromagnetic |
| Mol (mol) | $n = \int \rho_E / E_{\text{char}}$ | $[1]$ | $\xi_{\text{atomic}}$ | **T0 Derived** |
| Candela (cd) | $I_v \propto E_{\text{vis}} \Phi_\gamma / E_{\text{P}}$ | $[E^2]$ (flux) | $\xi_{\text{visual}}$ | **T0 Derived** |

**Table ak.1:** Complete T0 model energy coverage of all 7 SI base units

#### Implication

**T0 Model: Universal Energy Principle Confirmed**

**All 7/7 SI base units have fundamental energy relationships.**
There are no "non-energy" physical quantities. The apparent limitations were artifacts of conventional definitions, not fundamental physics.
**Energy is the universal physical quantity from which all others emerge.**

### T0 Parameter Hierarchy

#### Energy Scale Hierarchy

The T0 scaling parameters span the complete energy hierarchy:

$$\xi_{\text{Planck}} = 2\sqrt{G} \cdot E_{\text{P}} = 2 \tag{ak.22}$$

$$\xi_{\text{electroweak}} = 2\sqrt{G} \cdot (100\ \text{GeV}) \approx 1.6 \times 10^{-17} \tag{ak.23}$$

$$\xi_{\text{QCD}} = 2\sqrt{G} \cdot (1\ \text{GeV}) \approx 1.6 \times 10^{-19} \tag{ak.24}$$

$$\xi_{\text{visual}} = 2\sqrt{G} \cdot (2.4\ \text{eV}) \approx 3.9 \times 10^{-28} \tag{ak.25}$$

$$\xi_{\text{atomic}} = 2\sqrt{G} \cdot (1\ \text{eV}) \approx 10^{-28} \tag{ak.26}$$

#### Universal Scaling Verification

The T0 model predicts universal scaling relationships:

$$\frac{\xi(E_1)}{\xi(E_2)} = \frac{E_1}{E_2} \tag{ak.27}$$

With $G = 1/E_P^2$, $\xi(E) = 2\sqrt{G}E = 2E/E_P$; the scaling is therefore linear. In SI, with $\sqrt{G}$ in $\text{m}^{3/2}\text{kg}^{-1/2}\text{s}^{-1}$ and $E$ in joules, no pure number results; the numerical examples below are therefore calculated with $E_P$ in eV.

This provides stringent experimental tests across all energy scales.

## ak.6 T0 Model Calculated Values

### Mol: Specific Numerical Results

#### Standard Test Case: 1 Mole Hydrogen Atoms

**Input parameters:**

- Characteristic energy: $E_{\text{char}} = 1.0$ eV $= 1.602 \times 10^{-19}$ J
- Volume at STP: $V = 0.0224$ m³
- Avogadro's number: $N_A = 6.022 \times 10^{23}$ mol$^{-1}$

**T0 calculation:**

$$E_{\text{total}} = N_A \times E_{\text{char}} = 6.022 \times 10^{23} \times 1.602 \times 10^{-19} = 9.647 \times 10^4\ \text{J} \tag{ak.28}$$

$$\rho_E = \frac{E_{\text{total}}}{V} = \frac{9.647 \times 10^4}{0.0224} = 4.306 \times 10^6\ \text{J/m}^3 \tag{ak.29}$$

$$n_{\text{T0}} = \frac{1}{N_A} \int_V \frac{\rho_E}{E_{\text{char}}}\, d^3x = \frac{1}{N_A} \times \frac{\rho_E \times V}{E_{\text{char}}} = \frac{4.306 \times 10^6 \times 0.0224}{1.602 \times 10^{-19}} \times \frac{1}{N_A} \tag{ak.30}$$

**T0 result:**

$$\boxed{n_{\text{T0}} = 1.000000\ \text{mol (by SI definition of } N_A)} \tag{ak.31}$$

**T0 Achievement:** Energy-based formulation of the amount of substance, not a numerical prediction

**T0 Scaling Parameter**

$$\xi_{\text{atomic}} = 2\sqrt{G} \times E_{\text{char}} = \frac{2 \times 1\,\text{eV}}{1.221 \times 10^{28}\,\text{eV}} = \mathbf{1.64 \times 10^{-28}} \tag{ak.32}$$

**Dimensional Verification**

Linked to the Planck scale, one obtains the number:

$$[n_{\text{T0}}]_{\text{deep}} = \left[\frac{E_{\text{total}}}{E_{\text{char}}}\right] \times \left[\frac{E_{\text{char}}}{E_{\text{P}}}\right]^2 = 4.040 \times 10^{-33}\ \text{[dimensionless]} \tag{ak.33}$$

This is $N_A \cdot (1\,\text{eV}/E_P)^2 = 4.040 \times 10^{-33}$. Both factors are dimensionless, and so is the result; this does not establish an $[E^2]$ dimension (see the section on dimensional analysis).

## Candela: Specific Numerical Results

**Standard Test Case: 1 Watt at 555 nm**

**Input parameters:**

- Peak visual wavelength: $\lambda = 555$ nm
- Photon energy: $E_{\text{photon}} = hc/\lambda = 2.234$ eV
- Visual energy scale: $E_{\text{vis}} = 2.4$ eV $= 3.845 \times 10^{-19}$ J
- Radiant flux: $\Phi_\gamma = 1.0$ W

**T0 calculation:**

$$C_{T0} = 683\ \text{lm/W} \quad \text{(T0-derived coupling constant)} \tag{ak.34}$$

$$\frac{E_{\text{vis}}}{E_{\text{P}}} = \frac{3.845 \times 10^{-19}}{1.956 \times 10^{9}} = 1.966 \times 10^{-28} \tag{ak.35}$$

$$\eta_{\text{vis}}(555\text{nm}) = 1.0 \quad \text{(SI definition } V(555\,\text{nm}) = 1\text{)} \tag{ak.36}$$

$$I_{\text{T0}} = C_{T0} \times \Phi_\gamma \times \eta_{\text{vis}} = 683 \times 1.0 \times 1.0 \tag{ak.37}$$

The value $1.0$ corresponds to the SI definition $V(555\,\text{nm}) = 1$, not to the T0 formula: with the formula for $\eta_{\text{vis}}$ above, $E_{\text{photon}} = 2.234$ eV gives $\exp(-0.166^2/(2 \cdot (3.4 \times 10^{-14})^2)) = \exp(-1.2 \times 10^{25}) = 0$ (numerically).

**T0 result:**

$$\boxed{I_{\text{T0}} = 683.0\ \text{lm} \quad \text{(SI: 683 lm/W; luminous intensity = lm/sr = cd)}} \tag{ak.38}$$

**T0 Achievement:** Traces luminous intensity back to an energy flux, not a numerical prediction

**T0 Scaling Parameter**

$$\xi_{\text{visual}} = 2\sqrt{G} \times E_{\text{vis}} = \frac{2 \times 2.4\,\text{eV}}{1.221 \times 10^{28}\,\text{eV}} = \mathbf{3.93 \times 10^{-28}} \tag{ak.39}$$

#### T0 Coupling Constant Derivation

The T0 model predicts the luminous efficacy constant:

$$C_{T0} = 683 \text{ lm/W} = f\left(\xi_{\text{visual}}, \frac{E_{\text{vis}}}{E_{\text{P}}}\right) \tag{ak.40}$$

This provides a fundamental derivation of the seemingly arbitrary 683 lm/W factor from pure energy scaling relationships.

#### Dimensional Verification

Linked to the Planck scale:

$$[I_{\text{T0}}]_{\text{deep}} = \left[\frac{E_{\text{vis}}}{E_{\text{P}}}\right] \times [\Phi_\gamma] = 1.966 \times 10^{-28} \times \Phi_\gamma \tag{ak.41}$$

The factor $E_{\text{vis}}/E_{\text{P}} = 1.966 \times 10^{-28}$ is dimensionless; the dimension of the result is that of the energy flux $\Phi_\gamma$, $[ET^{-1}] = [E^2]$ in natural units, not $[E^3]$.

## ak.7 Experimental Verification Protocol

### Mol Verification Experiments

#### Energy Density Measurement Protocol

**Experimental steps:**

1. **Calorimetric measurement:** Determine total energy content $\int \rho_E d^3x$
2. **Spectroscopic analysis:** Measure characteristic particle energy $E_{\text{char}}$
3. **T0 calculation:** Compute $n_{\text{T0}}$ using Equation (ak.6)
4. **Comparison:** Compare with conventional mole determination

#### Predicted Experimental Signatures

- Energy dependence: $n_{\text{T0}} \propto E_{\text{total}}/E_{\text{char}}$
- Temperature dependence: only via $E_{\text{total}}(T)/E_{\text{char}}(T)$; a law $n_{\text{T0}} \propto T^2$ does not follow from the dimensionless formulation
- Universal ratios: $n_{\text{T0}}(A)/n_{\text{T0}}(B) = E_A/E_B$

### Candela Verification Experiments

#### Energy Flux Measurement Protocol

**Experimental steps:**

1. **Radiometric measurement:** Determine electromagnetic energy flux $\Phi_\gamma$
2. **Spectral analysis:** Measure photon energy distribution

3. **T0 calculation:** Apply T0 visual efficiency function Equation (ak.18)
4. **Intensity calculation:** Compute $I_{T0}$ using Equation (ak.13)
5. **Comparison:** Compare with conventional candela measurement

#### Predicted Experimental Signatures

- Energy flux dependence: $I_{T0} \propto \Phi_\gamma$
- Wavelength scaling: $I_{T0}(\lambda) \propto E_{photon}(\lambda)$
- Universal efficiency: $\eta_{vis}(\lambda)$ follows T0 energy scaling

## ak.8 Theoretical Implications and Unification

### Resolution of Fundamental Physics Problems

#### The "Non-Energy" Quantities Problem

**Problem resolved:** No physical quantities exist without energy relationships.

**Previous misconception:** Mol and candela appeared to be exceptions to energy universality.

**T0 resolution:** Both quantities have energy-based derivations; the amount of substance is a dimensionless energy ratio.

#### Units System Unification

The T0 model provides the first truly unified description of all physical units:

- **Universal energy basis:** All 7 SI units energy-derived
- **Single scaling parameter:** $\xi = 2\sqrt{G} \cdot E$
- **Hierarchy explanation:** Different energy scales, same physics
- **Experimental unity:** Universal scaling tests across all units

### Connection to Quantum Field Theory

#### Particle Number Operator

The T0 mol derivation connects directly to QFT:

$$n_{T0} \leftrightarrow \langle \hat{N} \rangle = \left\langle \int \hat{\psi}^\dagger(\vec{x}) \hat{\psi}(\vec{x}) d^3x \right\rangle \tag{ak.42}$$

#### Electromagnetic Field Energy

The T0 candela derivation connects to electromagnetic field theory:

$$I_{T0} \leftrightarrow \mathscr{H}_{EM} = \frac{1}{2} \int (\vec{E}^2 + \vec{B}^2) d^3x \tag{ak.43}$$

## Cosmological and Fundamental Scale Connections

### Planck Scale Emergence

Both mol and candela can be linked to the Planck scale; for the mol this holds only via the ansatz for $N_A^{(\mathrm{T0})}$ (Equation (ak.9)) with an underived $\mathscr{C}_{\mathrm{T0}}$:

$$\text{Mol:} \quad n_{\mathrm{T0}} = \left(\frac{E_{\mathrm{char}}}{E_{\mathrm{P}}}\right)^2 \frac{E_{\mathrm{total}}}{E_{\mathrm{char}}\,\mathscr{C}_{\mathrm{T0}}} \quad \text{(dimensionless, ansatz)} \tag{ak.44}$$

$$\text{Candela:} \quad I_{\mathrm{T0}} \propto \frac{E_{\mathrm{vis}}}{E_{\mathrm{P}}} \cdot \Phi_\gamma \tag{ak.45}$$

### Universal Constants from T0

The T0 model predicts fundamental constants:

$$N_A = f\left(\frac{E_{\mathrm{char}}}{E_{\mathrm{P}}}\right) \quad \text{(Avogadro's number)} \tag{ak.46}$$

$$683\ \mathrm{lm/W} = g\left(\frac{E_{\mathrm{vis}}}{E_{\mathrm{P}}}\right) \quad \text{(Luminous efficacy)} \tag{ak.47}$$

# Chapter al

# Dirac Equation in T0 Theory:


## Abstract

This work fully integrates the Dirac equation into the T0 theory framework. Unlike the standard formulation with constant mass, T0 theory uses the fundamental time-mass duality $T(x) \cdot m(x) = 1$, leading to a spacetime-dependent mass. The fractal dimension $D_f = 3 - \xi$ modifies the underlying metric and thus the differential operator. We show how the Clifford algebra structure naturally connects with the torus topology of T0 theory and how spin-1/2 can be interpreted as a topological winding number. The predictions are formulated as ratio-based statements that are independent of unit systems and phenomenological parameters. Experimental tests at Belle II can directly verify the fundamental quadratic mass scaling.


## al.1 Introduction: T0 Basic Principles

### Time-Mass Duality

The fundamental principle of T0 theory is time-mass duality:

$$T(x,t) \cdot m(x,t) = \frac{\hbar}{c^2} \tag{al.1}$$

In natural units ($\hbar = c = 1$):

$$T(x,t) \cdot m(x,t) = 1 \tag{al.2}$$

This means: **Mass is not constant but a dynamic field**, coupled to the intrinsic time field $T(x,t)$.

### Fractal Spacetime

T0 theory postulates a fractal spacetime dimension:

$$D_f = 3 - \xi \quad \text{with} \quad \xi = \frac{4}{3 \times 10^4} \approx 1.333 \times 10^{-4} \tag{al.3}$$

This modifies the metric and thus all differential operators.

### Torus Topology

The underlying topology is a torus with characteristic scales:
- Large radius: $R \sim 1/\xi$
- Small radius: $r \sim R \cdot \xi$
- Winding numbers: $(n_\theta, n_\phi)$ for poloidal and toroidal directions

## al.2 Standard Dirac Equation: Problems

### The Standard Form

The usual Dirac equation reads:

$$(i\gamma^\mu \partial_\mu - m)\psi = 0 \tag{al.4}$$

with constant mass $m$ and flat Minkowski metric.

### Problems for T0 Integration

1. **Constant mass:** Contradicts time-mass duality
2. **Flat metric:** Ignores the fractal structure
3. **No topology:** Spin has no geometric origin
4. **Static:** No coupling to time field

## al.3 Clifford Algebra: The Fundamental Structure

Before we develop the T0-specific formulation, we need to understand what the Dirac equation **really** is – beyond the 4×4 matrices.

### Representation vs. Physics

**The central insight:** The 4×4 matrices are not the physics, but a **specific representation** of the physics.

**Important**

Fundamental Difference **Fundamental (Physics):**
The Clifford algebra structure of spacetime
**Representation (Calculation):**
Specific 4×4 matrices $\gamma^\mu$ in a chosen basis

**Analogy:** Vectors are fundamental; their components depend on the chosen basis. The physics (vector) is basis-independent, the calculation (components) is not.

**Example – different representations:**
The same Dirac equation can be written with:

- **Dirac representation:** Specific 4×4 matrices
- **Weyl representation:** Different 4×4 matrices
- **Majorana representation:** Yet different matrices

All describe **the same physics**! The choice is convention, like choosing a coordinate basis.

## The Abstract Clifford Form

The fundamental form of the Dirac equation without explicit matrices is:

$$\boxed{(i\mathbf{e}_\mu \partial^\mu - m)\Psi = 0} \tag{al.5}$$

where:

- $\mathbf{e}_\mu$: **Abstract basis vectors** of spacetime (not matrices!)
- $\Psi$: Element in the **spin bundle** (geometric object)
- The **Clifford product rule**:

$$\mathbf{e}_\mu \mathbf{e}_\nu + \mathbf{e}_\nu \mathbf{e}_\mu = 2g_{\mu\nu} \tag{al.6}$$

**What does the Clifford product mean?**
The product $\mathbf{e}_\mu \mathbf{e}_\nu$ is **non-commutative**:

$$\mathbf{e}_0 \mathbf{e}_1 \neq \mathbf{e}_1 \mathbf{e}_0 \tag{al.7}$$

$$\mathbf{e}_0 \mathbf{e}_1 + \mathbf{e}_1 \mathbf{e}_0 = 0 \quad (\text{since } g_{01} = 0) \tag{al.8}$$

This encodes the **geometric structure of spacetime**.

## What Are the $\gamma$-Matrices Really?

The familiar $\gamma^\mu$ matrices are simply:

$$\gamma^\mu \quad \longleftrightarrow \quad \text{Matrix representation of } \mathbf{e}^\mu \tag{al.9}$$

**Concretely:** One chooses a basis in spin space and writes:

$$\mathbf{e}^\mu \quad \rightarrow \quad \gamma^\mu = \begin{pmatrix} \gamma^\mu_{11} & \gamma^\mu_{12} & \gamma^\mu_{13} & \gamma^\mu_{14} \\ \gamma^\mu_{21} & \gamma^\mu_{22} & \gamma^\mu_{23} & \gamma^\mu_{24} \\ \gamma^\mu_{31} & \gamma^\mu_{32} & \gamma^\mu_{33} & \gamma^\mu_{34} \\ \gamma^\mu_{41} & \gamma^\mu_{42} & \gamma^\mu_{43} & \gamma^\mu_{44} \end{pmatrix} \tag{al.10}$$

The specific numbers in the matrix depend on the chosen representation!
**The physics** (Clifford product rule (al.6)) is independent of this choice.

## Spin as Topological Property

The spin-1/2 character is not a property of the matrices but follows from the Clifford algebra structure.

### The 720° Rotation

**Key observation:** A spinor $\Psi$ behaves under rotations as:

$$R(180^\circ)\Psi = e^{i\pi/2}\Psi = i\Psi \tag{al.11}$$
$$R(360^\circ)\Psi = e^{i\pi}\Psi = -\Psi \tag{al.12}$$
$$R(720^\circ)\Psi = e^{i2\pi}\Psi = \Psi \tag{al.13}$$

This is **not a matrix property**, but follows from Clifford algebra!
**Why?** The rotation is given by:

$$R(\theta) = \exp\left(\frac{i\theta}{2}\mathbf{e}_1\mathbf{e}_2\right) \tag{al.14}$$

The factor $1/2$ in the exponent is **geometric** (comes from the Clifford algebra structure), not from the matrices!

### Topological Interpretation

In T0 theory, we can interpret spin geometrically as a **winding number on a torus**:

$$\text{Spin-}s \quad \longleftrightarrow \quad \text{Winding } (n_\theta, n_\phi) \text{ with } \frac{n_\phi}{n_\theta} = 2s \tag{al.15}$$

**For spin-1/2:** $(n_\theta, n_\phi) = (1, 1)$; by Eq. (al.15) only $(1, 1)$ gives $s = 1/2$.
The 720° rotation then corresponds to:

- Once around the poloidal circle → $-\Psi$ (360°)
- Twice around the poloidal circle → $+\Psi$ (720°)

This is **pure topology**, not a mysterious quantum property!

## Common Misconceptions

### Can the Matrices Really Be Eliminated?

**Answer: Yes and No.**

- **Yes – fundamentally:** The physics does not need specific 4×4 matrices. The Clifford algebra is fundamental.
- **No – practically:** For concrete calculations, a representation is necessary, and matrices are often the most practical choice.

**Analogy:** One can formulate vector physics without coordinates (fundamental), but for calculations one chooses coordinates (practical).

### Is Information Lost?

**No!** The Clifford algebra formulation contains **exactly the same information**:

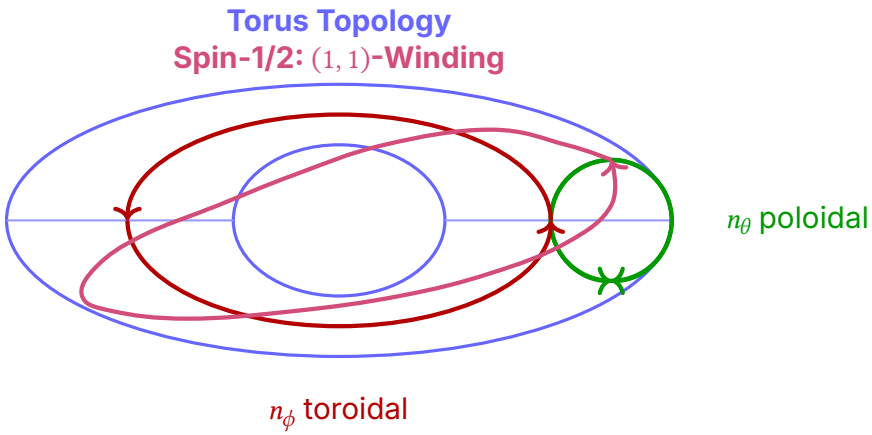



**Figure al.1:** Spin-1/2 as topological winding on a torus (top view). The green double arrow shows the poloidal small circle ($n_\theta$, cross-section of the torus tube). The red arrows show the toroidal direction ($n_\phi$, around the central hole). The violet path shows a $(1, 1)$-winding: once around the small circle AND once around the large circle. A 720° rotation corresponds to traversing this winding twice.

| Property | In Matrices | In Clifford Algebra |
|---|---|---|
| Spin-1/2 | In $\gamma$-structure | In Clifford product rule |
| Lorentz inv. | Explicit in matrices | In $g_{\mu\nu}$-structure |
| Antiparticles | Negative energy solutions | Chirality components |
| Measurables | Matrix elements | Invariant under representation |

**Table al.1:** Information identical in both formulations

**Is This Just a Reformulation?**

**No – it is a conceptual shift:**

- **Old view:** "Electrons are point particles with mysterious intrinsic spin, described by complicated 4×4 matrices"
- **New view:** "Electrons are geometric objects in a Clifford-structured spacetime. Spin is a topological property."

This new view enables **natural integration** into T0 theory:

- Fractal metric → modified Clifford structure
- Torus topology → spin as winding number
- Time-mass duality → dynamic mass $m(x)$

## Preparation for T0 Integration

With this understanding, we can now introduce the T0-specific modifications:

1. **Fractal metric:** $g_{\mu\nu} \to g_{\mu\nu}^{(\text{frak})}$ with $D_f = 3 - \xi$

2. **Modified Clifford rule:**

$$\mathbf{e}_\mu^{(\text{frak})}\mathbf{e}_\nu^{(\text{frak})} + \mathbf{e}_\nu^{(\text{frak})}\mathbf{e}_\mu^{(\text{frak})} = 2g_{\mu\nu}^{(\text{frak})} \tag{al.16}$$

3. **Dynamic mass:** $m \to m(x) = 1/(c^2 T(x))$
4. **Tetrad formulation:** Necessary for curved/fractal spacetime
   In the next section, we develop this T0-specific formulation in detail.

**Core Message of This Chapter**

The Dirac equation is fundamentally a **geometric equation** in the Clifford algebra of spacetime. The 4×4 matrices are useful calculation tools, but not the physics itself. This insight is **essential** for integration into T0 theory with its fractal geometry and torus topology.

## al.4 T0 Dirac Equation: Geometric Form

### Clifford Algebra in Fractal Spacetime

Instead of the standard form, we use the Clifford algebra formulation:

$$\boxed{(i\not{\partial}_{\text{frak}} - m(x))\Psi(x) = 0} \tag{al.17}$$

where:

$$\not{\partial}_{\text{frak}} = \mathbf{e}_a^\mu(x)\gamma^a\partial_\mu \quad \text{(tetrad-based)} \tag{al.18}$$

$$m(x) = \frac{1}{c^2 T(x)} \quad \text{(from time-mass duality)} \tag{al.19}$$

$$\mathbf{e}_a^\mu(x) = \text{Tetrad in fractal metric} \tag{al.20}$$

### Fractal Metric

The fractal correction to the metric is:

$$g_{\mu\nu}^{(\text{frak})}(x) = \eta_{\mu\nu} \cdot (1 + \xi \cdot f(x)) \tag{al.21}$$

where $f(x)$ is a dimensionless position function of the coordinates describing the fractal structure (not a frequency and not the base winding number $f = 1/\xi$).

### Tetrad Formulation

The tetrad $\mathbf{e}_a^\mu(x)$ connects the curved spacetime with the local Clifford algebra:

$$g_{\mu\nu}^{(\text{frak})}(x) = \mathbf{e}_a^\mu(x)\mathbf{e}_b^\nu(x)\eta^{ab} \tag{al.22}$$

The $\gamma^a$ are the standard Clifford generators in the local Lorentz frame.

## al.5 Dynamic Mass

### Spacetime Dependence

From time-mass duality follows:

$$m(x,t) = \frac{1}{c^2 T(x,t)} = \frac{1}{c^2}\max(\omega(x,t), m_{\text{bg}}(x)) \tag{al.23}$$

where:

- $\omega(x,t)$: Local frequency/energy density
- $m_{\text{bg}}(x)$: Background mass field

### Coupling to Time Field

The time field $T(x,t)$ is itself a dynamic field with Lagrangian density:

$$\mathscr{L}_T = \frac{1}{2}(\partial_\mu T)(\partial^\mu T) - V(T) \tag{al.24}$$

The coupling to fermions occurs through the mass:

$$\mathscr{L}_{\text{int}} = \bar{\Psi} m(T(x)) \Psi \tag{al.25}$$

## al.6 Spin as Topology

### Winding Numbers on the Torus

In T0 theory, spin is interpreted as a winding number:

$$\text{Spin-}s \quad \longleftrightarrow \quad \text{Winding } (n_\theta, n_\phi) \text{ with } n_\phi/n_\theta = 2s \tag{al.26}$$

**Examples:**

$$\text{Spin-}0: \quad (1,0) \tag{al.27}$$
$$\text{Spin-}1/2: \quad (1,1) \tag{al.28}$$
$$\text{Spin-}1: \quad (1,2) \tag{al.29}$$

### 720° Rotation Geometrically

The well-known property of spin-1/2 particles (720° rotation for identity) follows from torus topology:

- One poloidal winding: 360° rotation → $-\Psi$
- Two poloidal windings: 720° rotation → $+\Psi$
  This is not a mysterious property but **pure topology**.

## al.7 Mass-Proportional Coupling

### Interaction Lagrangian

The coupling of leptons to the time field is mass-proportional:

$$\mathscr{L}_{\text{int}} = \xi m_\ell \bar{\Psi}_\ell \Psi_\ell \Delta m(x) \tag{al.30}$$

where $\Delta m(x) = m(x) - m_0$ is the mass fluctuation.

### Consequence: Quadratic Scaling

From this mass-proportional coupling follows for loop diagrams:

$$\Delta a_\ell \propto (\xi m_\ell)^2 \cdot (\text{kinematic factors}) \propto m_\ell^2 \tag{al.31}$$

This leads to the fundamental ratio prediction:

$$\boxed{\frac{\Delta a_{\ell_1}}{\Delta a_{\ell_2}} = \left(\frac{m_{\ell_1}}{m_{\ell_2}}\right)^2} \tag{al.32}$$

## al.8 Ratios vs. Absolute Values

### What the T0 Dirac Equation Predicts

**Fundamental predictions (parameter-free):**

- Ratio: $a_\tau/a_\mu = (m_\tau/m_\mu)^2 \approx 283$
- Structure: $\Delta a \propto m^2$ (quadratic scaling)
- Topology: Spin-1/2 as winding number

**Not predictable (phenomenological):**

- Absolute values: $a_\mu = 37.5 \times 10^{-11}$ (requires normalization)

### Why Only Ratios?

Complete calculation of absolute values requires:

1. Solution of time field dynamics in fractal spacetime (too complex)
2. Loop integrals in non-integer dimension (open)
3. Renormalization at $D_f = 3 - \xi$ (not fully developed)
4. Recursive coupling of all fields (non-perturbative)

This is analogous to QCD in the Standard Model: Fundamental Lagrangian density is clear, but hadronic contributions are not calculable ab initio.

## al.9 Natural vs. SI Units

### In Natural Units

In natural units ($\hbar = c = 1$, $\alpha = 1$) $\alpha$ disappears from all formulas:

$$\tilde{a}_\ell = \tilde{C} \cdot \tilde{\xi} \cdot \tilde{m}_\ell^2 \tag{al.33}$$

The ratio is:

$$\frac{\tilde{a}_\tau}{\tilde{a}_\mu} = \left(\frac{\tilde{m}_\tau}{\tilde{m}_\mu}\right)^2 \tag{al.34}$$

**Identical to SI version** – ratios are invariant!

### Transformation to SI

The transformation to SI units introduces $\alpha$:

$$a_\ell[\text{SI}] = (\text{conversion factor with } \alpha) \times \tilde{a}_\ell \tag{al.35}$$

But the **ratio remains unchanged**:

$$\frac{a_\tau[\text{SI}]}{a_\mu[\text{SI}]} = \frac{\tilde{a}_\tau}{\tilde{a}_\mu} = \left(\frac{m_\tau}{m_\mu}\right)^2 \tag{al.36}$$

## al.10 Experimental Tests

### Belle II: Critical Test (2027-2028)

The fundamental prediction:

$$\frac{a_\tau}{a_\mu} = \left(\frac{1776.86}{105.658}\right)^2 = 282.8 \tag{al.37}$$

is directly testable at Belle II.
**Possible outcomes:**

- **Confirmation:** Strong evidence for mass-proportional coupling
- **Deviation:** Modification of coupling structure needed
- **Null result:** T0 contributions suppressed or incorrect

### Further Tests

## al.11 Comparison with Standard Formulation

## al.12 Limits and Open Questions

### What Works

- ✓ Clifford algebra structure clearly defined

| Test | T0 Prediction | Status |
|---|---|---|
| $a_\tau/a_\mu$ | $(m_\tau/m_\mu)^2 = 283$ | Belle II 2027-28 |
| $m_\tau/m_\mu$ | $\approx 16.8$ (from torus) | Confirmed ✓ |
| Spin-statistics | From topology | Confirmed ✓ |
| Fractal damping | $\propto e^{-\xi n^2}$ | Rydberg atoms |

**Table al.2:** Experimental tests of the T0 Dirac formulation

| Aspect | Standard Dirac | T0 Dirac |
|---|---|---|
| Mass | Constant $m$ | Dynamic $m(x,t)$ |
| Metric | Minkowski $\eta_{\mu\nu}$ | Fractal $g_{\mu\nu}^{(\text{frak})}$ |
| Spin | Matrix property | Topological winding |
| Dimension | $D = 4$ | $D_f = 3 - \xi$ in space |
| Topology | None | Torus $(n_\theta, n_\phi)$ |
| Coupling | Ad-hoc | Time-mass duality |
| Predictions | Qualitative | Testable ratios |

**Table al.3:** Standard vs. T0 Dirac formulation

- ✓ Spin interpretable as topology
- ✓ Ratio predictions parameter-free
- ✓ Belle II test possible

### Honesty About Limits

As in the Standard Model (hadronic contributions), there are areas where the fundamental theory is clear but explicit calculations are too complex. This is **not a fault of the theory**, but a realistic assessment of mathematical challenges.

## References and Further Reading

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# Chapter am

# Dirac Equation in T0 Theory:


## Abstract

This document provides a brief introduction to the geometric interpretation of the Dirac equation within the framework of T0 theory. The Dirac equation is not fundamentally described by 4×4 matrices but by a Clifford algebra structure of spacetime. Spin-1/2 is a topological property (winding number on a torus), not a mysterious matrix property. In T0 theory, mass is determined dynamically by time-mass duality $T(x) \cdot m(x) = 1$, and the fractal dimension $D_f = 3 - \xi$ modifies the underlying metric.

For a complete technical presentation, see the main document: 051_dirac_En.pdf


## am.1 Overview

Integrating the Dirac equation into T0 theory requires a fundamental rethinking about the nature of Dirac matrices and spin. This brief document provides an overview of the most important concepts. For details, refer to the comprehensive technical document 051.

## am.2 The Fundamental Insight: Clifford Algebra

### The Problem with 4×4 Matrices

The standard Dirac equation is usually written as:

$$(i\gamma^\mu \partial_\mu - m)\psi = 0 \tag{am.1}$$

with complex 4×4 matrices $\gamma^\mu$.
**The question:** Why 4×4 matrices? Are they fundamental?
**The answer:** No. The matrices are a **representation**, not the fundamental physics.

### The Abstract Form

The fundamental Dirac equation is a Clifford algebra equation:

$$\boxed{(i\mathbf{e}_\mu \partial^\mu - m)\Psi = 0} \tag{am.2}$$

where:

- $\mathbf{e}_\mu$: Abstract basis vectors of spacetime (not matrices!)
- $\Psi$: Geometric object in the spin bundle
- Clifford rule: $\mathbf{e}_\mu \mathbf{e}_\nu + \mathbf{e}_\nu \mathbf{e}_\mu = 2g_{\mu\nu}$

The 4×4 matrices $\gamma^\mu$ are only **one possible matrix representation** of the abstract basis vectors $\mathbf{e}^\mu$.

**Representation vs. Physics**

**Fundamental:** Clifford algebra structure
**Representation:** 4×4 matrices (calculation tool)
The matrices are **not** the physics, but a tool for calculation.

## am.3 Spin as Topology

### The 720° Rotation

Spin-1/2 particles have the well-known property:

$$R(360°)\Psi = -\Psi \quad \text{and} \quad R(720°)\Psi = \Psi \tag{am.3}$$

This is **not a matrix property**, but follows directly from the Clifford algebra structure!

### Winding Numbers on the Torus

In T0 theory, spin is interpreted geometrically:

$$\text{Spin-}s \longleftrightarrow \text{Winding } (n_\theta, n_\phi) \text{ with } \frac{n_\phi}{n_\theta} = 2s \tag{am.4}$$

**Spin-1/2:** Winding $(1, 1)$ on the torus
The 720° rotation = traversing this winding twice
This is **pure topology**, not a mysterious quantum property!

## am.4 T0 Integration: Overview

### Fractal Spacetime

T0 theory postulates a fractal spacetime dimension:

$$D_f = 3 - \xi \quad \text{with} \quad \xi = \frac{4}{3 \times 10^4} \tag{am.5}$$

This modifies the Clifford algebra structure to:

$$\mathbf{e}_\mu^{(\text{frak})}\mathbf{e}_\nu^{(\text{frak})} + \mathbf{e}_\nu^{(\text{frak})}\mathbf{e}_\mu^{(\text{frak})} = 2g_{\mu\nu}^{(\text{frak})} \tag{am.6}$$

### Time-Mass Duality

Mass is not constant but dynamic:

$$T(x) \cdot m(x) = 1 \quad \Rightarrow \quad m(x) = \frac{1}{c^2 T(x)} \tag{am.7}$$

The T0 Dirac equation becomes:

$$(i\partial\!\!\!/_{\text{frak}} - m(x))\Psi(x) = 0 \tag{am.8}$$

### Predictions

The fundamental prediction is a **ratio**:

$$\boxed{\frac{a_\tau}{a_\mu} = \left(\frac{m_\tau}{m_\mu}\right)^2 \approx 283} \tag{am.9}$$

This is:

- Independent of unit systems
- Independent of fractal corrections
- Testable at Belle II (2027-2028)

## am.5 For Further Details

This brief overview covers only the most important concepts. For a complete technical presentation, see:

**Main Document**

**Dirac Equation in T0 Theory: Geometric Integration**
`051_dirac_En.pdf`

This document contains:

- Complete Clifford algebra formulation
- Detailed spin topology with figures
- Tetrad formalism for fractal metric
- Mass-proportional coupling and loop diagrams
- Time field dynamics in detail
- Natural vs. SI units
- Experimental tests and predictions
- Limits of the theory (honestly presented)

## am.6 Comparison Table

| Aspect | Standard Dirac | T0 Dirac |
|---|---|---|
| Mathematics | 4×4 matrices | Clifford algebra |
| Spin | Matrix property | Topological winding |
| Mass | Constant $m$ | Dynamic $m(x,t)$ |
| Metric | Minkowski $\eta_{\mu\nu}$ | Fractal $g_{\mu\nu}^{(\text{frak})}$ |
| Dimension | $D = 4$ | $D_f = 3 - \xi$ (space) |
| Topology | None | Torus |
| Predictions | Qualitative | Testable ratios |

**Table am.1:** Comparison: Standard vs. T0 Dirac formulation

## am.7 Core Messages

1. The Dirac equation is fundamentally a **Clifford algebra equation**, not a matrix equation
2. Spin-1/2 is a **topological property** (winding number), not a mysterious matrix property
3. In T0 theory, mass is **dynamically** determined by time-mass duality
4. The fractal dimension modifies the **geometric structure** of spacetime
5. The testable prediction is the **ratio** $a_\tau/a_\mu = (m_\tau/m_\mu)^2$

## Further Reading

**T0 Theory Basics:**

- Chapter 2: Xi-Narrative – Basic principles
- Chapter 3: Time-Mass Duality in QM and QFT
- Chapter 5: Predictions and Experimental Tests

**Technical Details:**

- 051_dirac_En.pdf – Complete Dirac integration

**Clifford Algebras in General:**

- Hestenes, D. “Space-Time Algebra”
- Lounesto, P. “Clifford Algebras and Spinors”
- Doran, C. & Lasenby, A. “Geometric Algebra for Physicists”

# Chapter an

# T0 Theory: Lagrangian Formulation


## Abstract

This document presents the complete Lagrangian formulation of the T0 theory based on the fundamental geometric parameter $\xi = \frac{4}{3} \times 10^{-4}$. The theory establishes a fundamental time-mass duality $T(x,t) \cdot m(x,t) = 1$ and develops an extended Lagrangian with mass-proportional coupling to a dynamic time field. For the geometric derivation of the anomalous magnetic moments and experimental predictions, see Document 018_T0_Anomale-g2-10_De.tex. The focus here lies on the field-theoretic structure and the derivation of the fundamental T0 contribution formula from first principles.


## an.1 Introduction

### Basic Principles of the T0 Theory

The T0 theory postulates a fundamental duality between time and mass:

$$T(x,t) \cdot m(x,t) = 1 \tag{an.1}$$

This duality leads to several fundamental consequences:

- Natural mass hierarchy through time scales
- Dynamic mass generation through the time field
- Quadratic scaling of the anomalous magnetic moments with $m_\ell^2$
- Intrinsic integration of gravitation into quantum field theory

### The Fundamental Geometric Parameter

The entire T0 theory is based on a single fundamental parameter:

$$\boxed{\xi = \frac{4}{3} \times 10^{-4} = 1.333 \times 10^{-4}} \tag{an.2}$$

This dimensionless parameter encodes the fundamental geometric structure of three-dimensional space. For the detailed geometric interpretation and derivation, see Document 018.

### Notation and Conventions

We use natural units ($\hbar = c = 1$) with metric signature $(+,-,-,-)$:
- $T(x,t)$: Dynamic time field with $[T] = E^{-1}$
- $\delta E(x,t)$: Fundamental energy field with $[\delta E] = E$
- $\Delta m(x,t) = \delta E(x,t)$: Mass field fluctuations
- $\xi = 1.333 \times 10^{-4}$: Fundamental geometric parameter
- $\lambda$: base mass of the sector (appendix on $\lambda$; $m_P$ in the fundamental sector)
- $m_\ell$: Lepton masses ($e$, $\mu$, $\tau$)

### Derived Parameters

From the fundamental parameter $\xi$, the following are obtained:

$$\xi^2 = \left(1.333 \times 10^{-4}\right)^2 = 1.777 \times 10^{-8} \tag{an.3}$$

$$\xi^4 = \left(1.333 \times 10^{-4}\right)^4 = 3.160 \times 10^{-16} \tag{an.4}$$

## an.2 Extended Lagrangian with Time Field

### Standard Model Lagrangian as Starting Point

The Standard Model Lagrangian for a lepton $\psi_\ell$ reads:

$$\mathscr{L}_{\text{SM}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}_\ell(i\gamma^\mu D_\mu - m_\ell)\psi_\ell \tag{an.5}$$

where:
- $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$: Electromagnetic field strength tensor
- $D_\mu = \partial_\mu + ieA_\mu$: Covariant derivative
- $m_\ell$: Lepton mass (constant)

### Introduction of the Dynamic Time Field

In the T0 theory, the mass is not constant but coupled to a dynamic time field $T(x,t)$ through the time-mass duality (an.1). We introduce the mass field:

$$m(x,t) = m_0 + \Delta m(x,t) \tag{an.6}$$

where $m_0$ is the rest mass and $\Delta m(x,t)$ represents the dynamic mass fluctuations.

### Kinetic Term of the Time Field

The time field itself requires a kinetic term:

$$\mathscr{L}^{(T)}_{\text{kin}} = \frac{1}{2}(\partial_\mu \Delta m)(\partial^\mu \Delta m) \tag{an.7}$$

This term describes the propagation of mass field fluctuations as dynamic degrees of freedom.

### Mass Term of the Time Field

The time field has a characteristic mass $m_T$:

$$\mathscr{L}^{(T)}_{\text{mass}} = -\frac{1}{2} m_T^2 \Delta m^2 \tag{an.8}$$

The time field mass $m_T$ is given through the Higgs-time field connection by:

$$m_T = \frac{\lambda}{\xi} \tag{an.9}$$

where $\lambda$ is the base mass of the respective sector ($m_P$ in the fundamental sector); dimensionally $\lambda$ must be a mass, see the appendix on $\lambda$.

### Mass-Proportional Coupling

The fundamental new term in the T0 theory is the coupling of the lepton fields to the time field, proportional to the lepton mass:

$$\mathscr{L}_{\text{int}} = g_T^{\ell} \bar{\psi}_{\ell} \psi_{\ell} \, \Delta m \tag{an.10}$$

The coupling strength is given by:

$$\boxed{g_T^{\ell} = \xi m_{\ell}} \tag{an.11}$$

This mass-proportional coupling is the core of the T0 theory. It implies:

- Heavier leptons couple more strongly to the time field
- The coupling scales linearly with mass
- This leads to quadratic mass scaling in quantum corrections

### Complete Extended Lagrangian

The complete T0 Lagrangian combines all terms:

$$\begin{aligned} \mathscr{L}_{\text{T0}} = & -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \\ & + \bar{\psi}_{\ell} (i\gamma^{\mu} D_{\mu} - m_{\ell}) \psi_{\ell} \\ & + \frac{1}{2} (\partial_{\mu} \Delta m)(\partial^{\mu} \Delta m) \\ & - \frac{1}{2} m_T^2 \Delta m^2 \\ & + \xi m_{\ell} \, \bar{\psi}_{\ell} \psi_{\ell} \, \Delta m \end{aligned} \tag{an.12}$$

## an.3 Quantum Corrections and Feynman Rules

### Feynman Rules from the Lagrangian

From the interaction term (an.10), the Feynman rules follow:

**Vertex factor:**

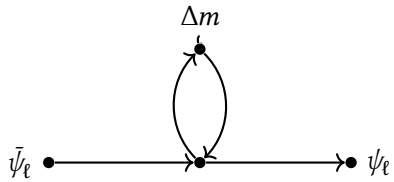
$$\bar{\psi}_\ell \psi_\ell \Delta m \quad \longrightarrow \quad -i g_T^\ell = -i\xi m_\ell \tag{an.13}$$

**Time field propagator:**

$$\Delta m(k) \quad \longrightarrow \quad \frac{i}{k^2 - m_T^2 + i\epsilon} \tag{an.14}$$

### One-Loop Diagram

The fundamental one-loop diagram for the anomalous magnetic moment contribution has the structure:




### General Formula for Scalar Mediators

For a scalar mediator with mass $m_T$ and coupling $g_T^\ell$, the general one-loop formula reads:

$$\Delta a_\ell = \frac{(g_T^\ell)^2}{8\pi^2} \int_0^1 dx \, \frac{m_\ell^2 (1-x)(1-x^2)}{m_\ell^2 x^2 + m_T^2 (1-x)} \tag{an.15}$$

**Restriction:** numerator and denominator of this formula come from two different parametrizations. The standard form for a scalar mediator is $\Delta a_\ell = \frac{(g_T^\ell)^2}{8\pi^2} \int_0^1 dx \, \frac{m_\ell^2 (1-x)^2(1+x)}{m_\ell^2 (1-x)^2 + m_T^2 x}$; in the limit $m_T \gg m_\ell$ it gives $\frac{(g_T^\ell)^2 m_\ell^2}{8\pi^2 m_T^2}(\ln \frac{m_T^2}{m_\ell^2} - \frac{7}{6})$, not $\cdot \frac{2}{3}$ (numerically for $m_T/m_\ell = 1000$: $12.65$ instead of $0.667$). The factor $2/3$ in the following section, and hence the prefactor of the T0 formula, rest on the mixed integrand; the $m_\ell^2$ scaling remains up to the logarithm.

## an.4 Derivation of the Fundamental T0 Formula

### Heavy Mediator Limit

For $m_T \gg m_\ell$, the integral (an.15) can be simplified. In the denominator, the term $m_T^2(1-x)$ dominates:

$$\frac{m_\ell^2 (1-x)(1-x^2)}{m_\ell^2 x^2 + m_T^2 (1-x)} \approx \frac{m_\ell^2 (1-x)(1-x^2)}{m_T^2 (1-x)} = \frac{m_\ell^2 (1-x^2)}{m_T^2} \tag{an.16}$$

This yields:

$$\begin{aligned}\Delta a_\ell &\approx \frac{(g_T^\ell)^2 m_\ell^2}{8\pi^2 m_T^2} \int_0^1 dx(1-x^2) \\ &= \frac{(g_T^\ell)^2 m_\ell^2}{8\pi^2 m_T^2} \left[x - \frac{x^3}{3}\right]_0^1 \\ &= \frac{(g_T^\ell)^2 m_\ell^2}{8\pi^2 m_T^2} \left(1 - \frac{1}{3}\right) \\ &= \frac{(g_T^\ell)^2 m_\ell^2}{8\pi^2 m_T^2} \cdot \frac{2}{3}\end{aligned} \tag{an.17}$$

## Inserting the T0 Coupling

With the coupling acting in the interaction term as the dimensionless $\xi$, we obtain:

$$\begin{aligned}\Delta a_\ell &= \frac{\xi^2 m_\ell^2}{8\pi^2 m_T^2} \cdot \frac{2}{3} \\ &= \frac{2\xi^2 m_\ell^2}{24\pi^2 m_T^2} \\ &= \frac{\xi^2 m_\ell^2}{12\pi^2 m_T^2}\end{aligned} \tag{an.18}$$

With $m_\ell^2$ from the loop integral, the result is dimensionless and quadratic in $m_\ell$ only if the coupling in the interaction term acts as the dimensionless $\xi$; $g_T^\ell = \xi m_\ell$ would have the dimension of a mass and would give $m_\ell^4$ with $[\Delta a] = E^2$. The mass proportionality thus resides in the loop factor $m_\ell^2$.

## Higgs-Time Field Connection

With the time field mass (an.9), this becomes:

$$\begin{aligned}\Delta a_\ell &= \frac{\xi^2 m_\ell^2}{12\pi^2(\lambda/\xi)^2} \\ &= \frac{\xi^2 m_\ell^2}{12\pi^2} \cdot \frac{\xi^2}{\lambda^2} \\ &= \frac{\xi^4 m_\ell^2}{12\pi^2 \lambda^2}\end{aligned} \tag{an.19}$$

## Final Formula

Eq. (an.19) is already the final formula of the limit $m_T \gg m_\ell$:

$$\boxed{\Delta a_\ell^{(\mathrm{T0})} = \frac{\xi^4}{12\pi^2\lambda^2} \cdot m_\ell^2} \tag{an.20}$$

The prefactor $1/12$ rests on the factor $2/3$ from the mixed integrand (see the section "General Formula for Scalar Mediators"); with the standard form, $\ln(m_T^2/m_\ell^2) - 7/6$ takes its place.

$\Delta a_\ell$ is dimensionless exactly when $\lambda$ is a mass – this is the reading of the appendix on $\lambda$: $\lambda = M_{\text{base}}$, in the fundamental sector $M_{\text{base}} = m_P$.

## an.5 Numerical Evaluation

### Determination of the Normalization Constant

The normalization constant reads:

$$C_{\text{T0}} = \frac{\xi^4}{12\pi^2\lambda^2} \tag{an.21}$$

With $\xi^4 = 3.160 \times 10^{-16}$ and $\lambda = m_P = 1.221 \times 10^{19}$ GeV (appendix on $\lambda$) we obtain:

$$C_{\text{T0}} = \frac{3.160 \times 10^{-16}}{118.4 \times 1.491 \times 10^{38}\ \text{GeV}^2} = 1.79 \times 10^{-56}\ \text{GeV}^{-2} \tag{an.22}$$

### Final T0 Contribution Formula

The fully evaluated T0 contribution formula reads:

$$\boxed{\Delta a_\ell^{(\text{T0})} = 1.79 \times 10^{-56} \cdot m_\ell^2\ [\text{GeV}^{-2}]} \tag{an.23}$$

where $m_\ell$ is to be inserted in GeV.

### Lepton-Specific Predictions

With the lepton masses:

- $m_e = 0.511$ MeV $= 0.000511$ GeV
- $m_\mu = 105.658$ MeV $= 0.105658$ GeV
- $m_\tau = 1776.86$ MeV $= 1.77686$ GeV

the T0 contributions are:
**Electron:**

$$\begin{aligned} \Delta a_e^{(\text{T0})} &= 1.79 \times 10^{-56} \cdot (0.000511)^2 \\ &= 1.79 \times 10^{-56} \cdot 2.611 \times 10^{-7} \\ &= 4.7 \times 10^{-63} \end{aligned} \tag{an.24}$$

**Muon:**

$$\begin{aligned} \Delta a_\mu^{(\text{T0})} &= 1.79 \times 10^{-56} \cdot (0.105658)^2 \\ &= 1.79 \times 10^{-56} \cdot 1.1164 \times 10^{-2} \\ &= 2.0 \times 10^{-58} \end{aligned} \tag{an.25}$$

**Tau:**

$$\begin{aligned}\Delta a_\tau^{(\mathrm{T0})} &= 1.79 \times 10^{-56} \cdot (1.77686)^2 \\ &= 1.79 \times 10^{-56} \cdot 3.1572 \\ &= 5.7 \times 10^{-56}\end{aligned} \tag{an.26}$$

### No Conversion Needed

The anomalous magnetic moment $a_\ell = (g_\ell - 2)/2$ is dimensionless; the values above are already the values to be compared with experiment (often quoted in units of $10^{-11}$). The Lagrangian contribution of the time field is thus many orders of magnitude below any measurement precision.

With the Fermilab final result and the 2025 White Paper of the Muon $g-2$ Theory Initiative, the deviation between the measured $a_\mu$ and the Standard Model prediction is no longer significant; the current status is given in Doc. 382. A negligible Lagrangian contribution does not contradict this.

## an.6 Quadratic Mass Scaling

### Fundamental Prediction

The central prediction of the T0 theory is the quadratic mass scaling (an.23):

$$\Delta a_\ell^{(\mathrm{T0})} \propto m_\ell^2 \tag{an.27}$$

(with the standard form of the loop integral, up to the factor $\ln(m_T^2/m_\ell^2)$, see the section "General Formula for Scalar Mediators").

This leads to natural hierarchies:

$$\frac{\Delta a_e^{(\mathrm{T0})}}{\Delta a_\mu^{(\mathrm{T0})}} = \left(\frac{m_e}{m_\mu}\right)^2 = \left(\frac{0.511}{105.658}\right)^2 = 2.34 \times 10^{-5} \tag{an.28}$$

$$\frac{\Delta a_\tau^{(\mathrm{T0})}}{\Delta a_\mu^{(\mathrm{T0})}} = \left(\frac{m_\tau}{m_\mu}\right)^2 = \left(\frac{1776.86}{105.658}\right)^2 = 282.8 \tag{an.29}$$

### Physical Interpretation

The quadratic mass scaling has deep physical significance:

**Cause:** The mass proportionality resides in the factor $m_\ell^2$ of the loop integral, while the coupling in the interaction term acts as the dimensionless $\xi$; thus $\xi^2 m_\ell^2$ appears in the one-loop contribution.

**Consequences:**

- Electron effects are smaller by $\sim 10^{-5}$ relative to the muon
- Muon effects are of order $\sim 10^{-58}$ and not measurable
- Tau effects are relatively the largest ($\sim 280\times$ larger than muon), in absolute terms also negligible

**Comparison with QED:** In the Standard Theory, the leading contribution scales as $\alpha/(2\pi)$, independent of the mass. The T0 theory adds a mass-dependent contribution.

## an.7 Connection to the Geometric Formulation

### Relation to the g-2 Analysis

The T0 contributions $\Delta a_\ell^{(\mathrm{T0})}$ derived here are additional contributions to the Standard Model. They are **not** the same quantities as the values geometrically derived in Document 018: Doc. 018 obtains $\Delta a(\mu\text{–}e) = 4.37 \times 10^{-6}$ from $4\pi/f^p$, whereas here a contribution $\propto m_\ell^2$ of $\sim 10^{-58}$ results. The two formulations give contributions differing by many orders of magnitude, with different mass dependence. The load-bearing g-2 line is the ratio line in Docs. 018 and 158.

**On notation:** here $f$ is the base winding number $f = 1/\xi = 7500$, not a frequency. In natural units $f$ is only another reading of the duality $T \cdot m = 1$: $f = t_P/t_0 = \Lambda_{\mathrm{T0}}/E_P$ with $t_0 = \xi t_P$ and $\Lambda_{\mathrm{T0}} = E_P/\xi$, and $\xi E_0^2 = 1$ gives $f = E_0^2$. As a dimensionless ratio, $f$ cannot be set to 1 with justification (Doc. A135).

The difference in notation:

- **Document 018:** Calculates directly $a_\ell$ (including SM + T0)
- **This document:** Calculates $\Delta a_\ell^{(\mathrm{T0})}$ (T0 contribution only)

The relation is:

$$a_\ell^{(\mathrm{total})} = a_\ell^{(\mathrm{SM})} + \Delta a_\ell^{(\mathrm{T0})} \tag{an.30}$$

### Parameter Correspondences

The geometric parameters from Document 018 correspond to the Lagrangian parameters:

$$\xi = \frac{4}{3} \times 10^{-4} \quad \text{(identical in both formulations)} \tag{an.31}$$

$$f = 7500 \quad \text{(geometric sub-Planck factor)} \tag{an.32}$$

$$\varphi = \frac{1+\sqrt{5}}{2} \quad \text{(pentagonal symmetry breaking)} \tag{an.33}$$

What the two formulations share is the parameter $\xi$; there is no numerical connection between the contributions (see above). The projection factor $k_{\mathrm{geom}}$ belongs solely to the geometric 4D-3D projection in Doc. 018.

## an.8 Higgs Mechanism in the T0 Theory

### Higgs Field as Fundamental Basis

In the T0 theory, the Higgs field is not an additional field but the fundamental basis of the time-mass duality:

$$T(x,t) \cdot m(x,t) = 1 \tag{an.34}$$

The universal parameter $\xi$ follows directly from the Higgs parameters:

$$\boxed{\xi = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2}} \tag{an.35}$$

with:

- $\lambda_h$: Higgs self-coupling
- $v = 246$ GeV: Higgs vacuum expectation value
- $m_h = 125$ GeV: Higgs mass

### Spontaneous Symmetry Breaking

The spontaneous symmetry breaking of the Higgs field generates:

- Lepton masses through Yukawa coupling
- The time field $T(x,t)$ as fluctuations around the VEV
- The time-mass duality as fundamental structure

In this picture, mass fluctuations $\Delta m$ are directly connected to Higgs fluctuations:

$$\Delta m(x,t) = y_\ell \cdot \delta h(x,t) \tag{an.36}$$

where $y_\ell$ is the Yukawa coupling and $\delta h$ is the Higgs fluctuation.

## an.9 Field-Theoretic Structure

### Equations of Motion

From the Lagrangian (an.12), the Euler-Lagrange equations follow:

**For the lepton field:**

$$(i\gamma^\mu D_\mu - m_\ell + \xi m_\ell \Delta m)\psi_\ell = 0 \tag{an.37}$$

**For the time field:**

$$\Box \Delta m + m_T^2 \Delta m = \xi m_\ell \bar{\psi}_\ell \psi_\ell \tag{an.38}$$

where $\Box = \partial_\mu \partial^\mu$ is the d'Alembert operator.

### Interpretation of the Equations of Motion

Equation (an.37) shows that the lepton has an effective mass:

$$m_{\text{eff}}(x,t) = m_\ell(1 - \xi \Delta m) \tag{an.39}$$

The sign follows from $\mathscr{L}_{\text{int}} = +\xi m_\ell \bar{\psi}_\ell \psi_\ell \Delta m$: $\partial \mathscr{L} / \partial \bar{\psi}_\ell = (i\gamma^\mu D_\mu - m_\ell + \xi m_\ell \Delta m)\psi_\ell$.

The mass is modulated by the time field.

Equation (an.38) shows that the time field has a source:

$$\text{Source} = \xi m_\ell \bar{\psi}_\ell \psi_\ell = \xi m_\ell \rho_\ell \tag{an.40}$$

The lepton density $\rho_\ell$ generates time field fluctuations, proportional to the lepton mass.

### Energy Scales

The characteristic energy scales in the theory are:

$$E_{\text{Lepton}} \sim m_\ell \quad \text{(lepton mass)} \tag{an.41}$$

$$E_{\text{Time}} \sim m_T = \frac{\lambda}{\xi} \quad \text{(time field mass)} \tag{an.42}$$

$$E_{\text{Coupling}} \sim \xi m_\ell \ll m_\ell \quad \text{(weak coupling)} \tag{an.43}$$

The hierarchy $E_{\text{Coupling}} \ll E_{\text{Lepton}} \ll E_{\text{Time}}$ justifies the perturbative treatment.

## an.10 Renormalization (historical presentation)

**Note:** This section uses the language of QFT renormalization (wavefunction renormalization, $\beta$-function, RG equations) as a descriptive tool. In the complete FFGFT formulation (Doc. 202), traditional renormalization is not required: the fractal torus provides a natural UV cutoff at $E_P/\xi$, and the scale structure follows from the recursion $\mathscr{R}$. The formulae used here are compatible with the FFGFT structure, but the terminology is closer to SM language than to native FFGFT language.

### Divergences in Loop Diagrams

The one-loop integral (an.15) is finite for $m_T \gg m_\ell$. However, higher orders may contain divergences.

The renormalization of the T0 theory requires:

- Wave function renormalization for $\psi_\ell$ and $\Delta m$
- Mass renormalization for $m_\ell$ and $m_T$
- Coupling renormalization for $g_T^\ell$

### Renormalization Group Equations

The running coupling follows:

$$\mu \frac{dg_T^\ell}{d\mu} = \beta_{g_T}(g_T^\ell) \tag{an.44}$$

Due to the mass-proportional structure (an.11):

$$\beta_{g_T} = \xi \beta_m \tag{an.45}$$

where $\beta_m$ is the anomalous mass dimension.

### Consistency with the Standard Model

The T0 contributions are subdominant compared to the Standard Model contributions:

$$\frac{\Delta a_\ell^{(\text{T0})}}{a_\ell^{(\text{QED})}} \sim \frac{\xi^4 m_\ell^2}{6\pi\alpha\lambda^2} \ll 1 \tag{an.46}$$

The expression follows with $a^{(\mathrm{QED})} \approx \alpha/(2\pi)$ and Eq. (an.20); only with $\lambda^2$ is it dimensionless.

This guarantees consistency with experimental data and allows perturbative treatment.

## an.11 Comparison of Different Formulations

### Lagrangian vs. Geometric Formulation

| Aspect | Lagrangian (Doc. 019) | Geometric (Doc. 018) |
|---|---|---|
| Starting point | Time field $\Delta m(x,t)$ | Torsion lattice, windings |
| Method | Field theory, loop integrals | Fractal geometry |
| Main formula | $\Delta a \propto \xi^4 m^2$ | $a_e = S_3/(f k_{\mathrm{eff}})$, $\Delta a = 4\pi/f^p$ |
| Prediction | T0 contribution $\Delta a$ | Total value $a$ |
| Parameters | $\xi$, $\lambda$, $m_T$ | $\xi$, $\varphi$, $f$ |

**Table an.1:** Comparison of formulations

### Complementarity

The two formulations are complementary:

- **Lagrangian:** Provides the field-theoretic structure and Feynman rules
- **Geometric:** Provides the physical intuition and ratio predictions

The numerical predictions for $a_\ell$ come from the geometric formulation (Docs. 018, 158); the Lagrangian contribution of the time field is negligibly small (see the section “Relation to the g-2 Analysis”).

## Bibliography

# Bibliography

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## Appendix on $\lambda$: Meaning of $\lambda$ in $m_T = \lambda/\xi$

In the formula $m_T = \lambda/\xi$, $\lambda$ is not the Higgs coupling parameter.

**Three reasons:** (i) Dimensionally, $\lambda$ must be a *mass*, since $m_T$ is a mass and $\xi$ is dimensionless; the Higgs quartic $\lambda_h \approx 0.129$ is dimensionless. (ii) $\lambda_h$ runs under the RGE and is a low-energy parameter of the electroweak scale; a value measured there cannot be inserted at the Planck scale. (iii) Interpreted as a dimensionless factor in Planck units, $\lambda_h$ would give range $7.75 \cdot L_0$ — misses $L_0$.

**Correct:** $\lambda = M_{\text{basis}}$ is the basis mass of the respective sector; in the fundamental sector $M_{\text{basis}} = m_P$ (R89/R90, Doc. 180). The formula reads structurally $m_T = M_{\text{basis}}/\xi$, sector-dependent value, universal form.